Subjunctive Conditionals and Time

A Defense of a Weak Classical Approach

Subjunctive Conditionals and Time

A Defense of a Weak Classical Approach

Johan Mårtensson

March 27, 2000

To Agneta,
Anna-Maria, Klara and Carl Johan.

Abstract: The subject matter of this thesis is the semantic and logical properties of subjunctive conditional sentences, or counterfactuals, as they are also called.

It is argued that current formal systems of conditional logic and semantics do not represent our intuitions about the logical properties of such sentences. In particular the principles

$$\text{(CSO)}\ ((A \mathrel{\Box\!\!\rightarrow} B) \wedge (B \mathrel{\Box\!\!\rightarrow} A)) \rightarrow ((A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C))$$

$$\text{(CV)}\ ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Diamond\!\!\rightarrow} B)) \rightarrow ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$$

are shown to be questionable from an intuitive point of view. The principle CV is contested, but CSO is a part of nearly every current axiomatization of conditional logic. It is argued that the inclusion of CSO stems from a certain conception of similarity or closeness between possible worlds inherent in those systems: Closeness between worlds is thought of as a relation fixed by the context of utterance. This assumption leads to the acceptance of the principle CSO. The main objective of this thesis is thus to show that this conception of closeness is unwarranted and that it leads to unintuitive results. Instead a certain *antecedent-relative* notion of closeness is proposed. This notion of closeness is defined in causal terms. The formal consequences of substituting this notion for the traditional non-relative notion of closeness are traced and the resulting formal semantics and its attendant axiomatic system are shown to possess desirable properties from an intuitive point of view.

©Johan Mårtensson, 1999

Distribution:
ACTA UNIVERSITATIS GOTHOBURGENSIS
Box 222
SE-405 30 GÖTEBORG
Sweden

ISBN 91-7346-365-5
ISSN 0283-2380

Printed in Sweden by
Kompendiet, Göteborg 1999

Preface

During the last three years I have devoted much time and energy to subjunctive conditionals, and I am afraid that, to a certain extent, I have required of some other people to do so too. Therefore I would like to thank, quite generally, all those at the Philosophy Department in Göteborg and elsewhere who have contributed—with their time, their intuitions, and their insight—to this work.

Some of them I would like to mention especially: First of all Per Lindström for being the chief representative at our Department of a certain philosophical approach, in which he excels, namely that of combining formal rigor with a sensitivity to profound philosophical issues. This thesis is inspired by this outlook, and I can only hope that it also exemplifies it in some measure, and that it may be instrumental in keeping it alive. Also, David Lewis who in replying to a question of mine, in effect put the question to me that I have tried to answer in this thesis, and whose work on subjunctive conditionals has been the obvious point of departure for my own thinking on this subject, and Staffan Carlshamre who supervised most of my early work and who was never slow in grasping what I groped for, and was thus invaluably stimulating. I would also like to thank Christian Bennet who took time to check and comment on most in the way of formulas that I produced, and Nils-Eric Sahlin who supported my project in the beginning. I would also like to mention Jonathan Bennett who bore with my questions and my comments on some of his views, and in replying to me and also in his writings provided some major philosophical inspiration for this work. I have also corresponded with Donald Nute whose work on the relations between different conditional logics has spared me much work, and with whom I discussed some thorny deductions, and Sven Ove Hansson who supplied valuable information on theories of belief revision.

Last, but of course not least—quite the contrary—I would like to thank my supervisor during the actual writing of the thesis, Dag Westerståhl, without whom this thesis would have remained in a rather disorganized state. He was always uncompromising in pointing out any muddle that I am sad to have brought before him, and

he always required that I clarified and worked out the formal consequences far beyond what I had initially intended, or thought myself capable of. In trying to meet the standards set by him, I feel I have actually grown beyond myself a bit.

Göteborg, September 1999 Johan Mårtensson

Contents

Introduction **1**

1 Semantics for Conditionals **7**
1.1 Conditionals of Various Kinds 10
1.2 Truth and Assertibility 12
1.3 Compositionality 15
1.4 Are Conditionals Truth-Functional? 20
1.4.1 Indicative Conditionals 20
1.4.2 Subjunctive Conditionals 22
1.5 Semantics 24
1.5.1 Extensional Semantics 27
1.5.2 Intensional Semantics 28
1.5.3 Hyperintensional Semantics 29
1.5.4 Class Selection Semantics 31
1.5.5 Sentence Set Revision 35
1.5.6 Relative Closeness Semantics 46

2 Closeness and Time **65**
2.1 Historical Conservativity 65
2.1.1 Modified Historical Conservativity 68
2.2 Historically Possible Facts and Events 71
2.3 Possible Worlds and Time 72
2.3.1 Genuine Historical Alternatives 78
2.3.2 Felonious Historical Alternatives 78
2.4 The Criterion of Late Departure 79
2.4.1 Simple Late Departure 79
2.4.2 S_L and Modified Historical Conservativity . . 82

2.4.3 S_L and Refined Historical Conservativity . . 84
2.4.4 Problems with Late Departure 85
2.4.5 Temporal Asymmetry 86
2.4.6 Lewis's Weights and Priorities 91
2.4.7 Lewis's Criterion and Late Departure 95
2.5 Future Similarity 99
2.5.1 Does Approximate Similarity Count? 99
2.5.2 The Causal Trail Criterion 102
2.5.3 Causal Trail and Late Departure. 104
2.5.4 Causal Trails and the Ramp 106
2.5.5 Criterion of Closeness for Non-Divisible Antecedents 108
2.5.6 Pure RCT-closeness. 111
2.6 Soundness of Axioms and Rules 111
2.6.1 The Principles RCEA, RT and CA with respect to LDCT- or RCT-orderings. 111
2.6.2 The Principles RCV and CV with respect to RCT-orderings. 118
2.7 Bennett's Causal Chain Criterion 119
2.8 Cotenability 124
2.8.1 Cotenability and Undercutting 125
2.8.2 Cotenability and the Despiteness Relation. . 128
2.8.3 Cotenability and the RCT-criterion 136
2.8.4 Undercutting and Antecedent Relativity . . . 137

3 Axiomatizations 143
3.1 Rules, Axioms and Logics 143
3.1.1 The Concept of Conditional Logic 143
3.1.2 Rules 144
3.1.3 Axiom schemata 146
3.1.4 Some Conditional Logics 147
3.2 The Canonical Semantics for Ψ 151
3.3 Class Selection Semantics 153
3.3.1 Soundness of Ck with respect to the Set of Class Selection Semantics 153
3.3.2 Completeness of Ck with respect to the Set of Class Selection Semantics 153

3.3.3 Half-Normal Logics and various Class Selection Semantics 155
3.4 Facts Revision Semantics 158
3.4.1 Soundness of PR with respect to the Set of Weak Generalized Facts Revision Semantics . 159
3.4.2 Completeness of PR with respect to the Set of Weak Generalized Facts Revision Semantics 160
3.4.3 Half Normal Extensions of PR and Weak Generalized Facts Revision Semantics 161
3.5 Antecedent Relative Closeness Semantics 164
3.5.1 Soundness of AR' with respect to the Set of ARRC Semantics 164
3.5.2 Completeness of AR' with respect to the Set of ARRC Semantics 164
3.5.3 Half-Normal Extensions of AR' and various ARRC-Semantics 172
3.6 Relative Closeness Semantics 177
3.6.1 Soundness of VW with respect to the Set of Relative Closeness Semantics 177
3.6.2 Completeness of VW with respect to the Set of Relative Closeness Semantics 177
3.6.3 Half-Normal Extensions of VW and various Relative Closeness Semantics 184

Summary **187**

Introduction

It can be said that formal semantic study of conditionals started about thirty years ago with [Stalnaker, 1968a].[1] Here for the first time an axiomatic system and a formal semantics was provided for conditional sentences. This paper inaugurated an extensive discussion going on in the seventies and eighties pertaining to a wide variety of issues concerning conditionals. A common presupposition in one important strand of this discussion was that the standard by which different conditional logics should be judged is the way we intuitively handle conditional sentences with respect to their truth value and logical relations to other sentences. Any proposed axiomatization and its attendant formal semantics was criticized by some theorist because it was felt to be vulnerable to some counterexample on intuitive grounds. In some cases this gave rise to new and modified axiomatizations and new formal semantics that were designed to steer free of some particular counterexample(s).

There is widespread agreement that the set of conditional sentences divides into at least two subsets of sentences with different logical and semantical properties. On the one hand *subjunctive conditionals*, and on the other hand *indicative conditionals*.

This thesis will be concerned exclusively with conditionals of the subjunctive variety, or "counterfactuals" as they are often called. I shall adopt the presupposition mentioned above that the criterion of success for any given axiomatic system is its agreement with our semantical intuitions. In this thesis I shall concentrate mainly on subjunctive conditionals that are used to express what had been the case given (what is taken to be) a certain historically possible

[1] See [Nute, 1984]

assumption.

A historical possibility is something that could have happened (but perhaps never did happen) at some point in the actual history. If determinism is true there are no non-actual historical possibilities in this sense. I don't know whether determinism is actually true, but our linguistic practices seem to evince that we take non-actual historical possibility seriously: We find it meaningful to speculate about things that we think could have happened in certain situations and of what would had happened as a result of that happening.[2] We express such thoughts by means of subjunctive conditionals like

> I went to work today but I could have decided to stay home. Had I done so, I wouldn't have been talking to you now.

One important, perhaps the most important, idea underlying and shaping most current formal semantics for subjunctive conditionals is that when we asses the truth value of a given conditional we should see whether the consequent clause of the conditional is true in a situation or world at which the antecedent clause of the conditional is true but which otherwise is as similar to the actual situation or world as possible. Hence the notion of relative similarity between possible worlds plays a central role in those systems. I shall adopt this approach myself, but I shall try to show that there is an assumption about similarity common to (most of) those systems which is unfounded, and may lead to unintuitive results, at least with respect to conditionals of the kind mentioned above. This assumption is that the same ordering of possible worlds can be used when evaluating different conditionals within the same linguistic context. Instead I shall try to argue that the relation of comparative similarity between worlds should not be held constant when

[2] There is a view found in Moore [1912] that the expression "could have", in typical contexts, is elliptical and that it needs to be supplemented with an if-clause such as in "could have if he had chosen", and that this expression should be interpreted as "should have if he had chosen". Hence it would seem that talk about what could have happened is after all perfectly compatible with determinism. For an insightful discussion and in my view convincing criticism of this idea see [Austin, 1956]

different conditionals are evaluated but should vary in a systematic way.

There have been theoretical endeavors pointing in this direction before. One of the most notable is John Pollock's similarity relation between worlds defined in terms of the notion of "undercutting". However, the full import of this seems to have gone largely unnoticed. Neither Pollock himself, nor his commentators, seem to have recognized that the similarity relation defined by him will often be different for conditionals with different antecedent, even within the same context, and thus this relation is *antecedent relative*.

I shall argue that something like the notion of undercutting is indispensable when defining relative similarity between possible worlds with the purpose of providing truth conditions for subjunctive conditionals, and therefore that the sought for relation of similarity may be different for different conditionals even within the same context. Now, certain axioms of conditional logic that are valid with respect to all semantics formulated in terms of similarity relations that do not vary in this way, are not necessarily valid with respect to semantics formulated in terms of antecedent relative orderings. I shall argue that some of those axioms should indeed be included in any intuitively correct conditional logic, but also that some of them, like the following

$$\text{(CSO)}\ ((A \mathrel{\Box\!\!\rightarrow} B) \wedge (B \mathrel{\Box\!\!\rightarrow} A)) \rightarrow ((A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C))$$

should not.

Also the following principle will be criticized from an intuitive point of view.

$$\text{(CV)}\ ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Diamond\!\!\rightarrow} B)) \rightarrow ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$$

This principle is valid with respect to all ordering semantics defined in terms of an ordering relation that relates any pair of possible worlds. CV is also disqualified by Pollock, and since his ordering relation is not connected in this sense it is not necessarily valid with respect to his semantics either. However, I shall argue that CSO is no more valid than CV from an intuitive point of view, and hence that it is not enough to assume that the ordering relation is not

connected; one has also to drop the assumption that it does not vary for different conditionals.

Instead of the notion of undercutting I shall use the (admittedly rather vague but) somewhat related notion of a *causal trail* when defining my species of antecedent relative ordering of possible worlds. I shall attempt to show that given certain, arguably rather natural, assumptions about this notion all the axioms that (I argue) are intuitively valid will also be valid with respect to semantics defined in terms of this ordering between worlds. At the same time I shall argue that principles like CSO and CV are not. I shall also suggest a way to define undercutting in terms of the notion of causal trail, and thus relate this notion not only to that of undercutting but also to that of *cotenability* introduced by Goodman [1946].

My case against the principles CSO and CV will rest heavily on certain examples that I hope will convince the reader. One of these examples (on p. 53 below) is such that I hope the reader will not only recognize that neither CSO nor CV is valid, but also appreciate why talk about causal trails of the antecedent may be relevant when closest antecedent-worlds are looked for with the purpose of evaluating subjunctive conditionals semantically.

The notions of a causal trail and of undercutting are causal notions, in the sense that they are both defined in terms of notions like cause, effect and causal law. If closeness between worlds must be understood to vary from one conditional to another in a way that involves causal notions then it seems that causality is conceptually prior to conditional truth. Certain theorists have thought that order to be reversed; They have assumed that causality can be defined in terms of conditional truth, which then in its turn would be definable in terms of a non-causal notion of similarity.[3] On the other hand, if I am right, causality is epistemically as well as conceptually prior to truth for conditionals, and hence there seems to be no hope for a non-circular analysis of causality in terms of conditional truth.

However, the question of the correct analysis of causality is not a concern of this thesis. I shall merely use this notion as a primitive with the objective of finding intuitively correct truth conditions for

[3]See for instance [Lewis, 1973a]

the kind of conditionals I have in mind, in order to be able to pass judgment on the intuitive suitability of different formal semantics for conditionals and the intuitive correctness of different axiomatic systems of conditional logic.

Much of Chapter 1 is concerned with expressing different formal semantical frameworks in similar terminology with the purpose of comparing them. This chapter starts with a short discussion on the methods used in this thesis, followed in Section 1.1, by a taxonomic discussion with the purpose of identifying the subject matter of the thesis. In Section 1.2 there is a short discussion of an alternative, epistemic approach to conditionals that will not be pursued further. After those preliminary considerations, in Section 1.3 follows a discussion of the notion of *compositionality*. Here a framework for classifying sentences with respect to this notion is suggested. This framework will receive a formal treatment in the first part of Section 1.5 (Subsections 1.5.1, 1.5.2 and 1.5.3). In Section 1.4 standard views on how conditionals should be classified with respect to this framework are presented, and discussed. This discussion is carried a bit further in Section 1.5.3.

Section 1.5 will be entirely devoted to a presentation and comparison of different formal semantics for conditionals found in the literature. These are the *class* (or world set) *selection semantics* (Subsection 1.5.4), semantics based on an operation of *sentence set revision* typical of cotenability theory (Subsection 1.5.5), the well known *relative closeness semantics* (Subsection 1.5.6), and a generalization of those, the *antecedent relative relative closeness semantics*. Those different kinds of formal semantics are positioned with respect to the framework of compositionality defined in Subsections 1.5.1 to 1.5.3, and there is also a discussion of the conditions under which different formal semantics are equivalent.

The whole of Chapter 3 is concerned with relating the different types of semantics defined in Section 1.5 to different axiomatizations of conditional logic. Proofs of soundness and completeness for axiomatic systems with respect to those different semantic types are presented. Those proofs will proceed by methods that are fairly standard within modal logic, hence no real novelty is claimed here. The purpose of this presentation is rather to provide a handy and

unified framework in which the special semantics presented in Chapter 2 can be related to its proper axiomatization, and in which this axiomatization can be related to other kinds of semantics.

All this is meant to be preambulatory with respect to the discussion on time and conditional truth conducted in Chapter 2, which I regard as the main contribution of this thesis. Chapter 2 starts with discussions of *historical conservativity*, and *historical possibility* in Sections 2.1 to 2.3 in which a formal framework for talking about possible worlds in their temporal dimension (temporal worlds) is presented. This framework is used in Section 2.4.1 to define a simple (preliminary) relation of closeness between such worlds (simple late departure), that is shown to define a relative closeness semantics. The rest of this section is devoted to a discussion of a theory that can be interpreted as a version of this kind of semantics, namely the semantics proposed in [Lewis, 1979].

In Section 2.5 it is argued that this relation of closeness fails on intuitive grounds and instead an alternative relation of closeness, defined in terms of causal notions, is proposed, and shown to define an antecedent relative relative closeness semantics of a special kind (causal trail semantics). In Section 2.6 this semantics is shown to possess desirable properties with respect to what axioms and rules of inference that are supported, and in Sections 2.7 and 2.8 it is compared to certain semantics (proposed by other authors) with which it shares important traits.

Chapter 1

Semantics for Conditionals

This thesis is about a certain class of conditional sentences, and, since sentences express thoughts, it is also indirectly about the kinds of hypothetical thinking expressed by those sentences. Below I shall attempt to define what kind of sentences I have in mind, but let me first say something about the objectives of this investigation and the methods I shall use.

In many cases we have rather strong intuitions about whether a certain conditional sentence is true or not. There are easily imagined situations in which we would all deem the following true:

(1) If he had let go of that stone it would have fallen to the ground.

This is also the case for instance with:

(2) If the tank is full you cannot pour more fuel in.

In other cases we are not so sure about the truth-value of a certain sentence even though we may think that it has a definite truth-value. Sometimes I don't know whether there is a coin in my pocket or not, so I don't know whether the sentence:

(3) If I had searched my pocket I would have found a coin.

is true or not. But this, of course, doesn't imply that I don't think that the sentence has a definite truth-value. In case there is no coin

in my pocket it is false; otherwise, it is true, and, since either is the case, the conditional is either true or false.

We also seem to have pretty strong intuitions about certain logical relations between conditional sentences among themselves and also with respect to sentences of other kinds. Here is a couple of examples that exemplify logical relations to sentences of other kinds. Since (1) is true we may conclude that either he hasn't let go of the stone or it has fallen, and, as we saw above, "There is a coin in my pocket" seems to imply (3).[1] And here is one exemplifying a logical relation between conditionals. If (3) is true and it is also true that had I searched your pocket I would have found a coin, then it seems certain that had I searched my pocket or your pocket I would have found a coin.

The aim of this thesis is to formulate intuitions such as these, with regard to a certain subclass of conditionals, in a systematic way. This will involve attempting to formulate truth-conditions in terms of a formal semantics, and testing these against such intuitions as we have. This has, of course, been done before, especially during the last 30 years, by many different investigators, like Lewis [1973b], Stalnaker [1968b], Nute [1980], and others. My objective is of course not to invent the wheel over again but rather to build upon this work, criticizing aspects of those theories that I think lead to counterintuitive consequences, and suggesting corrections that I think will yield a theory that better accords with our intuitions.

So my method will involve providing counterexamples, in the sense of sentences that intuitively have different truth-values from those accorded to them by a certain theory.[2] I shall of course help myself to counterexamples provided by others, but also present some of my own, and also in some instances interpret the counterexamples of others in a new way. The kind of formal semantics that I shall propose will, I hope, not only steer clear of those counterex-

[1] Strictly speaking we need also assume something like: "Had I searched my pocket I would have found any object in it at least as big as a coin"

[2] This is of course a simplification. A counterexample of this kind normally involves, not just one sentence, but a set of sentences that together describe a situation that appears to be possible, notwithstanding the circumstance that not all of those sentences may be true according to the semantics under investigation.

amples but also be motivated by considerations of how we in fact go about when evaluating conditional sentences, i.e., how we think when we think hypothetically. That is, I shall try to analyze what considerations enter our mind, and in what way, when we are deciding whether some conditional is true. This will be reflected in the semantic theory I shall propose.

I shall then try to identify a conditional logic for semantics of this kind, i.e., sets of axioms and rules of inference that are determined by the formal semantics in question, in the sense that all logical truths and no untruths may be deduced from the axioms by those rules. This logic is weaker than the well known logics VC of [Lewis, 1973b, Chap. 6] and C2 of [Stalnaker, 1968b, Sec. III] (see p. 148 below). The main thrust of my argument will be directed at the principles CSO and CV (p. 147 below) that are theorems of VC and C2.

The well known semantics that support those logics, like Stalnaker's world selection semantics [Stalnaker, 1968b, Sec. II] and Lewis's relative similarity (or system of spheres [Lewis, 1973b, Chap. 1]) semantics share a common presupposition, namely, that hypothetical situations or worlds may be ordered with respect to their relevance for evaluating conditionals regardless of what conditional is evaluated (at least within a given context). This presupposition forces the acceptance of CSO. If I am right in my criticism of CSO then the ordering of hypothetical situations must be different for different conditionals, and in that case the validity of CV is not guaranteed either. In fact, I think neither CSO nor CV should be valid in an intuitively correct semantics for subjunctive conditionals.

I shall investigate the result of substituting antecedent-relative orderings of worlds for the (absolute) orderings that are presupposed by the semantics I criticize. In support of a semantics of this kind I shall argue that we order hypothetical situations differently depending on what antecedent we are considering, in a systematic way, and that this accounts for the intuitive non-validity of CSO, and CV. I shall also identify the logics determined by semantics of this kind. In positioning these proposed logics within a framework of different conditional logics I build on the work of Nute and Chellas, among others. My main focus, however, is not on formal logical

work, but on philosophical arguments for and against systems that already exist and that have already been investigated by logicians.

1.1 Conditionals of Various Kinds

It is a common observation that there are fundamentally different kinds of conditionals. The following familiar pair[3] exemplifies this.

(4) If Oswald had not shot Kennedy then Kennedy would not have died in Dallas.

(5) If Oswald did not shoot Kennedy then Kennedy did not die in Dallas.

These two conditionals have different grammatical properties, and also different truth-values, it seems. Conditionals of these two grammatical types usually receive different treatment in the literature. Conditionals of the first kind are often called "counterfactuals" or "subjunctive conditionals", and those of the second kind "indicative conditionals".[4] Those of the first kind more often than not have false antecedent clauses. This is why they are called "counterfactuals". However, a false antecedent is by no means a necessary trait of such a "counterfactual". Most investigators in the field also consider "counterfactuals" with true antecedent and consequent clauses.[5] In other words, whether the antecedent is in accordance with fact or contrary to fact does not seem to matter when deciding whether a conditional is "counterfactual", but instead it seems that its grammatical mood is the distinguishing factor. Conditionals of the first kind are in the *subjunctive mood* whereas those of the second kind

[3]This is a slight modification of a well-known example due to Adams [1970].

[4]Cf. Lindström and Rabinowicz [1997, Section 2] who instead use the terms "ontic conditionals" for the first kind and "epistemic" or "doxastic conditionals" for the second. Lindström and Rabinowicz write that "[t]he grammatical difference between moods... seems... to be a symptom of a semantic distinction". According to them "[t]he epistemic conditionals have to do with hypothetical modifications of our *beliefs* about the world", and "the ontic conditionals represent the hypotheses concerning what would be the case if the *world itself* were different". For a similar view see [Gibbard, 1978, pp. 227–8].

[5]See for instance [Lewis, 1973b, Section 1.7]. Cf. footnote 36 on p. 41 below.

are in the *indicative mood*. I shall reserve the term "counterfactual" for those conditionals that actually have false antecedent clauses. In accordance with this convention there will be both counterfactual and factual (non-counterfactual) subjunctive, as well as indicative, conditionals.

There is a third distinction that does not necessarily coincide with the *grammatical* (subjunctive/indicative) one or the *semantic* (counterfactual/factual) one. This is between those conditionals that have antecedents contravening the beliefs of the speaker and those that do not. Assume there were a rule of language (one may doubt that there is) that in some sense would forbid subjunctive conditionals with true antecedents. Even conscientious application of such a rule would not necessarily bar conditionals with true antecedents but only such with antecedents not in accordance with the beliefs of the speaker. I shall call conditionals with antecedents that are contrary to the beliefs of the speaker *belief contravening* [Cf. Rescher, 1961], and the others *belief consistent*. This distinction I shall refer to as the *doxastic distinction*.

This leaves us with eight pigeonholes in which to put different conditionals. However, we shall only be concerned with subjunctive mood conditionals, which leaves us with four types that are different with respect to whether their antecedents accord with facts and/or the beliefs of the speaker, respectively. Neither the semantic nor the doxastic distinction seems to divide the class of conditionals into classes with similar semantic properties that differ from those of other classes. I mention those two distinctions merely because the name "counterfactuals", interpreted in one way, seems to suggest the former, and interpreted in another way, the latter distinction. On the other hand, I think the grammatical distinction does distinguish between essentially different types of conditionals. Going back to the conditionals (4) and (5) above, we notice that the only difference between them seems to be their grammatical mood.[6] Apart

[6]Bennett [1988], following Dudman [1981, 1983, 1984], argues that the distinction should not be drawn in this way, i.e., in terms of grammatical mood. Indicative mood conditionals concerning the future should be classified together with the subjunctive mood conditionals according toBennett and Dudman. Cf. however a revaluation of this in [Bennett, 1995]. Cf. also [Jackson, 1990].

from this, we would say that they have the same antecedent and consequent clauses. They seem to have different truth conditions and relations of entailment to other sentences, however. We would deem (5) false on account of the fact that Kennedy did in fact die in Dallas. On the other hand, to put this fact forward as a reason to reject (4) seems unreasonable. So truth conditions as well as logical relations to other sentences seem to be different for those two sentences.

Initially, I shall presuppose that all subjunctive conditionals have basically the same kind of truth conditions, and are susceptible to the same rules of inference. This is a natural and also common initial assumption. There might be reasons to revise this assumption, however. We may find good reasons to believe that a particular semantic theory gives the right answers with regard to a large class of subjunctive conditionals, even if there is a well defined subclass that does not "fit in". We could then argue that the conditionals in the subclass in question are essentially different and should be treated in a different way. Some investigators have argued in this way with respect to a subclass of subjunctive conditionals, as, for instance Nute [1980, Section 2.2] with respect to the class of subjunctive conditionals with disjunctive antecedents. It might turn out, eventually, that some such move is necessary,[7] but I think that initially, at least, we should try to resist the temptation to pick out some class of subjunctive conditionals for special treatment, and instead strive for a unified account of all subjunctive conditionals.

1.2 Truth and Assertibility

Truth is distinct from assertibility. That is, one must distinguish the question whether a sentence is warranted from the speaker's perspective, from the question whether it is true. It seems obvious that each of these conditions may obtain without the other.

Although the question of internal justification is very important, especially for epistemology, it will not be a part of the main focus of

[7] See, e.g., our distinction between divisible and non-divisible antecedents on p. 105 below.

this thesis. Instead the semantic question regarding the truth-value and the truth-conditions of subjunctive conditionals will be at the center of the investigation. I mention this distinction between semantic and epistemological questions because some theories of conditionals seem to be attempts to determine the criteria of rational assertibility rather than attempts to answer the semantic questions of truth and truth-conditions

Ramsey [1931],[8] for instance, suggested an approach to conditionals, that seems to be aimed at yielding assertibility conditions for conditionals:[9]

> If two people are arguing 'If p will q?' and are both in doubt as to p, they are adding p hypothetically to their stock of knowledge and arguing on that basis about q.

One simple—surely too simple—way to represent a person's belief state is by a set of sentences expressing the beliefs that constitute his belief state. Let's call this set K. We could then say that a certain conditional is assertible for a person, if the consequent C follows from K when A has been added to K. One notable difficulty[10] in implementing this proposal is the following: Any theory of conditionals needs to give an account of belief-contravening ones. Such conditionals have antecedents that are in a sense contradictory to the speaker's beliefs taken as a whole. If a conditional is belief-contravening in this way, its antecedent must in some sense contradict this set. One simple way to understand this is in terms of consistency; if the antecedent A is added to the belief set, the resulting set is inconsistent, i.e., $K \cup \{A\}$ is inconsistent. Of course, every sentence is deducible from an inconsistent set of sentences, so, in order to implement Ramsey's proposal on these lines, we must first modify the set K so as to obtain a set K' (a so called "contraction" of K with respect to A) which is such that $K' \cup \{A\}$ is not inconsistent.

[8] In [Ramsey, 1978, Section B].

[9] For discussions of this suggestion see [Chisholm, 1946, p. 297] and [Stalnaker, 1968a, Section II].

[10] One which Ramsey hedges against by assuming that his interlocutors "are in doubt as to p".

Similar considerations made Stalnaker [1968a, Section II] suggest the following revision of Ramsey's suggestion.

> First, add the antecedent (hypothetically) to your stock of beliefs; second, make whatever adjustments are required to maintain consistency (without modifying the hypothetical belief in the antecedent); finally, consider whether or not the consequent is then true.

So we should first find the contraction K' of K with respect to A and then see whether C is a logical consequence of $K' \cup \{A\}$. However, as is well known, there are non-equivalent ways of doing this and there doesn't seem to exist any straightforward way of choosing among these.[11] I shall not discuss this problem at this point since it is a general problem that in one form or another arises also for most semantic theories that will be discussed below.

Theories along these lines aiming to analyze conditionals in terms of Ramsey's suggestion[12] are often presented as theories of rational assertibility. However, there is an alternative approach that does not center on assertibility but rather on truth: Instead of thinking of the set K of sentences as a description of a belief state, we may think of it as the set of all true sentences.[13] If A is not true, $K \cup \{A\}$ is inconsistent. We then modify K in order to obtain some privileged $K' \subseteq K$ (the "contraction" of K with respect to A) which is compatible with A, and then check whether C is included in the set of all logical consequences of $K' \cup \{A\}$, which is called the "revision" of K with respect to A. We refer to this set by K_A. The "semantic" apparatus (in terms of "contractions", "revisions" and so forth) of these theories, is therefore susceptible to a semantic as well as the (intended) epistemological interpretation. When

[11] An early recognition of this problem is found in [Goodman, 1946, p. 14].

[12] E.g. theories like that in [Gärdenfors, 1988, Sec. 7] inspired by the AGM-theory of Alchourrón et al. [1985].

[13] For a discussion of an alternative interpretation on these lines see Hansson [1995, Sec. 3]. There is a problem with this simple account, however: We must choose whether we should think of the set K as already containing the true conditionals or not. If it does then we could simplify the truth conditions, if not, then perhaps we need to allow for iterated revision in the way that will be discussed briefly on p. 43 below.

given a suitable "semantic" interpretation one can show[14] that at least a subclass of these are reformulations of the more well known class selection semantics that will be discussed below. The essential formal difference between the "semantic" and the "epistemological" approach is that, whereas the set of all true sentences is a maximally consistent set which uniquely determines a possible world (the actual one), the set of sentences representing the belief-state of a person need not be, but is usually compatible with a number of possible worlds. This makes direct comparisons between theories of those different kinds somewhat problematic. But, as we shall later see, the same logics that are supported by the usual class selection and relative closeness semantics are also supported by belief revision "semantics", given that suitable conditions are put on the operation of revision.[15]

However, I see no reason why one should attempt to formulate assertibility conditions independently of truth conditions. A simple approach to assertibility that relates assertibility-conditions and truth-conditions is that a sentence is assertible for a person whenever he or she has sufficient reasons for thinking that it is true. If this is more or less correct then a theory of assertibility must depend upon some theory of truth, since we must, it seems, decide what it is for a sentence to be true before we can decide what is to be counted as reasons for thinking that it is true. Stalnaker [1968a, Section II] suggested something of this kind when he wrote:

> The problem is to make the transition from belief conditions to truth conditions; that is, to find a set of truth conditions for statements having conditional form which explains why we use the method we do use to evaluate them.

1.3 Compositionality

Sentences that have other sentences as parts I shall call *complex sentences*. With respect to any complex sentence one may ask whether

[14] See Theorem 1.14 on p. 41 below for proof.

[15] See footnote 38 on p. 45 and also Section 3.4.3.

there are any systematic relations between the semantic properties of the complex sentence in question, and of those sentences that are parts of it. The most obvious example of such relations are those between the truth values of complex sentences and those of their parts within propositional calculus, where the truth value of any complex sentence is functionally determined by the truth values of the sentences of which it is composed. Another way to put this is that substitution of a part doesn't affect the truth value of the whole as long as the replaced sentence has the same truth value as the one by which it is replaced. Complex sentences that have this property—like those composed by means of the truth functional connectives—I shall call *extensional complexes* or *extensional constructions*. And a semantic theory that distributes truth values in this way I shall call *extensional*.

Not all complex sentences are extensional in this way. Already Gottlob Frege [1891] noted that when sentences contain *indirect speech*, the truth value of the containing sentence does not depend functionally on the truth value of the sentence that is contained by it. For example assume that "Kopernicus thought that the planets move in circles" is a true sentence. Now "The planets move in circles" is false. If the sentence "Kopernicus thought that the planets move in circles" had been an extensional complex then also the sentence "Kopernicus thought that the earth is flat" would have been true, since, of course "The earth is flat" is false just as much as "The planets move in circles" is. Expressions like "Kopernicus thought that", "Napoleon doubted that" and so forth, are said to indicate an *intensional context*. Normally what follows such expressions are complete sentences as in the examples above. The sentences that follow such expressions are said to occur within an intensional context indicated by those expressions.

Frege stipulated that having a certain truth value should be understood as a special case of reference ("Bedeutung" in Frege's terminology) i.e., if sentences have a truth value at all, then they refer to their truth value.

Now, if a sentence or a word occurs within an intensional context, we have no guarantee that it may be replaced *salva veritate* by a sentences or word with the same normal reference. Frege dealt

with this by stipulating that when an expression stands within an intensional context it does not refer to the same thing as it does when not within such a context. Now it refers instead to what otherwise is its sense or meaning ("Sinn"). We need not determine at this point what Frege meant by "sense". It is sufficient to note that, according to Frege, different expressions that refer to the same thing, normally have different sense or meaning, and that if two expressions have the same meaning then they may be substituted *salva veritate* also within intensional contexts. As their normal meaning is to be regarded as their reference when they occur within intensional contexts, we can see how Frege thought that the principle of compositionality can be salvaged even with regard to this kind of construction. Even in this case the reference of the containing sentence, i.e., its truth value, is functionally determined by the reference of its parts, which—for those parts that stand within an intensional context—is what is normally their sense or meaning.

Sentences like those above that contain intensional contexts are obviously not extensional constructions in the sense defined above. There is, as we just noted, no functional relation between the truth value of the sentence that stands within the intensional context of the containing sentence and the truth value of the containing sentence. If Frege was right, however, there is such a relation between the sense of the sentence that stands within the intensional context of the containing sentence and the truth value of the containing sentence. In order to make this more precise we must of course decide what we mean by "sense" or "meaning" of a sentence. The terms "intension" and "propositional content" or simply "proposition" are often used when semantic properties of sentences, other than their truth value, are referred to. Often propositions are simply identified with sets of possible worlds. This must surely be a simplification if propositions are to be thought of as being what is said when sentences are uttered. The idea that the content of what is expressed by a sentence is captured by a set of possible worlds—namely the set of worlds at which the sentence in question is true—is attractive, though. However, we need not identify contents or propositions with sets of worlds to retain this idea: We may instead say that propositions are determined by, or correlated

to such sets. I shall understand this correlation to be a one-one correspondence between "senses" and some sets of worlds,[16] i.e. that for each "sense", "meaning" or "propositional content" of a sentence or each "proposition", there is one and only one set of possible worlds that determines the "sense", "meaning", "propositional content" or "proposition" in question, and that no set of worlds can determine two different "senses". Complex sentences in which sentences with the same propositional content, in this sense, may be substituted *salva veritate*, I shall call *intensional complexes* or *intensional constructions*. A semantic theory that distributes truth values in this way I shall call *intensional*.

Whether sentences containing indirect speech are intensional, in this sense, is an open question. One may for example doubt whether all people who believe that Sydney is the capital of Australia also believe that Sydney is the capital of Australia or that there is a largest prime, and vice versa. It may well be that the set-of-possible-worlds account of "sense" does not do justice to Frege's intuitions about sense, so that even though "Sydney is the capital of Australia" and "Sydney is the capital of Australia or there is a largest prime" are true at the same possible worlds they do not have the same sense or meaning. Following Stalnaker [see Nute, 1980, p. 30] I shall call complex sentences that are neither extensional nor intensional complexes *hyperintensional complexes*.

Subjunctive conditionals are regarded as complex sentences, in the sense defined above, by most investigators in the field. I shall do so as well. However, whether they are complex in this way is not an absolutely straightforward matter to decide. If we look at (4) above it seems that the best candidates for "sentencehood" among its parts are "Oswald had not shot Kennedy" and "Kennedy would not have died in Dallas", which do seem to be logically incomplete in some sense: They don't seem to be capable of having any truth value, as they stand. However, they are easily modified into sentences like (A) "Oswald did not shoot Kennedy" and (C) "Kennedy did not die in Dallas" that may be true or false. That this kind of modification is permissible is given further support if the sentence

[16] There need not be "senses" or propositions correlated to every conceivable set of worlds.

(6) If it had been the case that Oswald did not shoot Kennedy then it would have been the case that Kennedy did not die in Dallas.

is a proper translation of (4), and also by the view that the conditional (5) above is simply the indicative mood variant of the subjunctive mood conditional (4), and (5) contains (A) and (C) in a straightforward way. Whether this is a sensible way of extracting parts from subjunctive conditionals could certainly be contested; it is the conventional way, however, which means that it is normally presupposed when questions of the semantic properties of subjunctive conditionals arise. In accordance with this presupposition the sentence (A) is called the antecedent (clause) of the conditional (4) and (C) the consequent (clause) of the same conditional.

Notational Conventions There are different symbolic conventions for representing the connectives in conditional sentences: Stalnaker [1968b, p. 48] uses "$>$" as symbol for the connective in subjunctive conditionals and "$\supset$" in material conditionals. Those who use "$\supset$" for material conditionals also often use "$\rightarrow$" for indicative conditionals [see for instance Gibbard, 1978; Jackson, 1987], but some of those [like van Fraassen, 1978, p. 329] use "$\rightarrow$" as a generic symbol denoting the connective in any conditional, and others [like Pollock, 1976, p. 19] use it as a symbol for logical entailment. Lewis [1973b] also uses "$\supset$" for material conditionals, but unlike Stalnaker uses "$\Box\!\!\rightarrow$" in subjunctive conditionals of the kind we have spoken of, and "$\Diamond\!\!\rightarrow$" in what he calls "might-counterfactuals" [1973b, pp. 1–2], that is, conditionals of the kind:

(7) If Oswald had not shot Kennedy then Kennedy might not have died in Dallas.

which he formalizes: $A \Diamond\!\!\rightarrow C$. This counterfactual is usually regarded as the negation of

(8) If Oswald had not shot Kennedy then Kennedy would have died in Dallas.

(formalized $A \Box\!\!\rightarrow \neg C$). If this is generally correct we do not, strictly speaking, require two different symbols since, in that case "$\Diamond\!\!\rightarrow$"

is definable in terms of " $\Box\!\!\rightarrow$ " and "$\neg$". I shall for the most part assume that might-counterfactuals like (7) are negations of would-counterfactuals like (8), i.e. that $(A \Diamond\!\!\rightarrow C) =^{df} \neg(A \Box\!\!\rightarrow \neg C)$ [see Lewis, 1973b, p. 21]. However, I shall let notational simplicity give way to considerations of readability and still adoptLewis's dual symbolism. Like Nute [1984] (who uses "$\Rightarrow$" for indicative conditionals) I would like to reserve "$\rightarrow$" for the material implication so I shall instead use another symbol, " $i\!\!\rightarrow$ ', for the connective in indicative conditionals.

1.4 Are Conditionals Truth-Functional?

1.4.1 Indicative Conditionals

Are conditional sentences extensional complexes? Some investigators have answered in the affirmative, at least with regard to indicative conditionals; they have held that indicative conditionals have the truth conditions of the material conditional. Jackson in [1987, pp. 4–8] observes that $A\, i\!\!\rightarrow C$ seems to imply $A \rightarrow C$ since it is "a logically sufficient condition for the falsity of an indicative conditional... that its antecedent be true and its consequent be false" [see also Mackie, 1973, p. 74]. So $A\, i\!\!\rightarrow C$ seems to be at least as strong as $A \rightarrow C$. However, on the other hand, $A \rightarrow C$ doesn't seem to imply $A\, i\!\!\rightarrow C$ since otherwise e.g. $\neg A$ would have implied $A\, i\!\!\rightarrow C$ via $A \rightarrow C$, which doesn't seem to be the case.[17] If $\neg A$ is true then $A \rightarrow C$ as well as $A \rightarrow \neg C$ are true, but this doesn't seem to give us reason to believe that $A\, i\!\!\rightarrow C$ and $A\, i\!\!\rightarrow \neg C$ are true, so it seems that $A \rightarrow C$ doesn't imply $A\, i\!\!\rightarrow C$.

Jackson [1987, p. 5] argues otherwise, however. Consider the sentence "Either the butler or the footman did it". This seems to be an instance of normal truth functional (inclusive) disjunction, i.e., an instance of the formula $A \vee B$. Now if this is true then one "may infer that if the butler is innocent the footman is not", i.e. that $\neg A\, i\!\!\rightarrow B$ is true. This example seems to support the view that $\neg A \vee C$ logically implies $A\, i\!\!\rightarrow C$, but since $A \rightarrow C$ implies

[17] This is one of the so called paradoxes of material implication, another one being that C although it does imply $A \rightarrow C$ doesn't seem to imply $A\, i\!\!\rightarrow C$.

$\neg A \vee C$, $A \rightarrow C$ seems to imply $A \, i\!\!\rightarrow C$ after all. This leads [p. 70] to the conclusion that $A \, i\!\!\rightarrow C$ is true if and only if $A \rightarrow C$ is true. The strangeness of the inference "$\neg A$ therefore $A \, i\!\!\rightarrow C$" is explained (away) in terms of assertibility conditions.[18]

Also H. P. Grice [in Grice, 1967a, p. 58] held a similar view regarding the truth conditions of the indicative conditional. In [Grice, 1967b] his theory of conversational implicature is put forward: Interlocutors may be assumed to follow certain maxims of conversation following from the "Cooperative Principle" [p. 45]. Conversational implicature that q arises if a speaker "by (in, when) saying (or making as if to say) that p" blatantly fails to fulfill such a maxim, but is nevertheless "to be presumed to be observing [...] the cooperative principle" and "the supposition that he is aware that, or thinks that, q is required in order to make his saying or making as if to say p (or doing so in THOSE terms) consistent with this presumption" and also "the speaker thinks (and would expect the hearer to think that the speaker thinks) that it is within the competence of the hearer to work out, or grasp intuitively that the supposition that [the speaker thinks that p] is required." [pp. 49–50].

If somebody knows that $\neg A$ or that C and solely on those grounds utters $A \, i\!\!\rightarrow C$ he would be guilty of violating one of the maxims that, according to Grice, follow from the cooperative principle, namely, that one should make one's "contribution as informative as required (for the current purposes of the exchange)" [p. 45]. The hearer may assume that the speaker does not in fact violate those maxims, so therefore he may conclude that the speaker does not utter $A \, i\!\!\rightarrow C$ because he either thinks that $\neg A$ or that C, but on other grounds, presumably that he thinks that there would be some connection between the occurrence of A and C. Or as Mackie [1973, p. 76] puts it: "So if someone does say [$A \, i\!\!\rightarrow C$, (my notation throughout)], which on Grice's materialist theory is equivalent to [$A \rightarrow C$] and hence to [$\neg A \vee C$], it is a conversational implicature of his utterance that he does not know that [C], or that [$\neg A$] and hence that his assertion that [$\neg A \vee C$] must have non-truth-

[18] According to Jackson [1987, p. 70] the assertibility of $A \, i\!\!\rightarrow C$ equals the conditional probability $P(C/A)$ which is assumed to be high only when there is a connection between the occurrence of A and C.

functional grounds, that [A] must somehow be a good reason for [C]."

That indicative conditionals have the truth conditions of the material conditional may be disputed, however. In Jackson's butler-or-footman argument, above, it is presupposed that "Either the butler or the footman did it" is a truth functional disjunction. Jackson [1987, p. 7] reports that Mitchell [1962, p. 64] denied "the truth-functional treatment of 'not', 'or', and 'and' " in order to get around this argument.[19]

1.4.2 Subjunctive Conditionals

Be that as it may, there doesn't seem to be any reason for supposing that subjunctive conditionals have the truth conditions of the material conditional.[20] For instance, the "butler-or-footman" example seems not to be applicable in the subjunctive case: "Either the butler or the footman did it" doesn't seem to support the conclusion that if the butler had been innocent the footman would have done it, so $A \vee C$ seems to be compatible with $\neg(\neg A \mathrel{\Box\!\!\rightarrow} C)$.

There is also widespread agreement (see for instance Stalnaker [1968b, pp. 48–49], Lewis [1973b, pp. 31–35] and Nute [1980, p. 29]) that some rules of inference valid for the material conditional are invalid with regard to the subjunctive conditional. The rules most frequently mentioned in the literature are *transitivity*, *strengthening of antecedent* and *contraposition*.

Transitivity: Although the following is a valid rule of inference (for the material conditional):

$$\frac{A \rightarrow B, B \rightarrow C}{A \rightarrow C}$$

the following deduction seems to be invalid:

> If J. E. Hoover had been born a Russian, then he would have been a Communist.

[19] See also [Lewis, 1912]. For another example of criticism of this view see [Davis, 1979, p. 550–1].

[20] See Jackson [1987, p. 6] and Mackie [1973, p. 77].

If he had been a Communist, he would have been a traitor.

If he had been born a Russian, he would have been a traitor.

This is Stalnaker's classical (but perhaps somewhat dated) example [see Stalnaker, 1968b, p. 48] (in Lewis's [1973b, p. 33] version). So the following seems not to be a valid rule of inference:

$$\frac{A \mathrel{\Box\!\!\rightarrow} B, B \mathrel{\Box\!\!\rightarrow} C}{A \mathrel{\Box\!\!\rightarrow} C}$$

Strengthening of antecedent:[21] The following inference is, of course, valid for the material conditional:

$$\frac{A \rightarrow C}{(A \wedge B) \rightarrow C}$$

The next example [again from Stalnaker, 1968b, p. 48] indicates that it is not valid for subjunctive conditionals:

If this match were struck, it would light.

If this match had been soaked in water overnight *and* it were struck, it would light.

So

$$\frac{A \mathrel{\Box\!\!\rightarrow} C}{(A \wedge B) \mathrel{\Box\!\!\rightarrow} C}$$

seems not to be valid either. If $(A \wedge B) \mathrel{\Box\!\!\rightarrow} A$ is valid (as it should be),[22] transitivity implies strengthening of antecedent; equivalently: The invalidity of strengthening of antecedent implies the invalidity of transitivity also.

[21] Some call this "monotonicity". Nute [1980, p. 29] calls it "Weakening Antecedents".

[22] Arguably all instances of $A \mathrel{\Box\!\!\rightarrow} A$ are valid and $A \mathrel{\Box\!\!\rightarrow} B$ is implied by $A \mathrel{\Box\!\!\rightarrow} C$ whenever $C \rightarrow B$ is a logical truth. Those two assumptions imply that $(A \wedge B) \mathrel{\Box\!\!\rightarrow} A$ is valid.

Contraposition: Stalnaker's [1968b, p. 49] example is:

> If the U.S. halts the bombing, then North Vietnam will not agree to negotiate.
>
> ---
>
> If North Vietnam agrees to negotiate, then the U.S. will not have halted the bombing.

Stalnaker says with regard to the premise: "A person would believe that this statement is true if he thought that the North Vietnamese were determined to press for a complete withdrawal of U.S. troops. But he would surely deny the contrapositive." So also

$$\frac{A \mathrel{\Box\!\!\rightarrow} \neg C}{C \mathrel{\Box\!\!\rightarrow} \neg A}$$

seems to be an invalid inference, although the corresponding rule for the material conditional is valid. Since these three inference-patterns are valid for the material conditional in virtue of its truth conditions, subjunctive conditionals cannot have those truth conditions; indeed, the truth of $A \mathrel{\Box\!\!\rightarrow} C$ seems to be compatible with all distributions of truth values to A and C except the one in which A is true and C is false, which would constitute an obvious counterexample to the conditional in question.

We may therefore conclude that the truth of $A \rightarrow C$ is a necessary condition—but not sufficient—for the truth of $A \mathrel{\Box\!\!\rightarrow} C$, i.e., that $A \mathrel{\Box\!\!\rightarrow} C$ implies $A \rightarrow C$. Below (Section 3.1) when we discuss axiomatic theories of subjunctive logic we will encounter this requirement in the form of the following axiom (called MP):[23]

$$(A \mathrel{\Box\!\!\rightarrow} B) \rightarrow (A \rightarrow B)$$

1.5 Semantics

In order to state formally that subjunctive conditionals are not extensional complexes I shall use a semantic framework, by means of

[23] Names of axioms are due to Nute [1980, p. 128–129] and Chellas [1975].

which most of the proposed semantics found in the literature may be formulated, and compared with each other.

A *domain* is an ordered pair $\langle I, at\rangle$ such that I is a set and at is a function such that for each $w \in I$ and each atomic sentence A, $at(w, A) \in \{1, 0\}$. Instead of $at(w, A)$ we often write $at_w(A)$ or simply $w(A)$.[24] The elements of I we shall call *possible worlds*. For simplicity we shall refer to the domain $\langle I, at\rangle$ as I. Let L be the smallest set of sentences containing all atomic sentences[25] and closed under composition by the usual truth-functional connectives and the connective "$\Box\!\!\rightarrow$". We say that:

Definition 1.1 *S is a* semantics *for (a domain) I if it is a total function from L to $P(I)$ such that*

1. *If A is atomic then $S(A) = \{w \in I : w(A) = 1\}$.*
2. *$S(\neg A) = I - S(A)$.*
3. *$S(A \wedge B) = S(A) \cap S(B)$, and so on for the other connectives except "$\Box\!\!\rightarrow$".*

We call I the domain of S, and sometimes we shall write I_S. Let

$$w \models_S A$$

stand for $w \in S(A)$ and

$$\models_S A$$

stand for $S(A) = I$. If this is the case we say that A is *valid* with respect to S

We noted above that the propositions expressed by sentences are often identified with sets of possible worlds. Given this way of talking every semantics assigns a proposition to each sentence in L.

Notice that Definition 1.1 doesn't specify how $S(A \Box\!\!\rightarrow C)$ should be determined; indeed, the different semantics that will be discussed below differ exactly with respect to this. In some cases $S(A \Box\!\!\rightarrow C)$ will be defined by a clause of the following general form

[24] A domain is thus essentially a Kripke-style model.

[25] I assume that there are infinitely many atomic sentences.

$F(A, C, S(A), S(C))$

If F is a function from $L \times L \times P(I) \times P(I)$ to $P(I)$. we know that S exists and is unique, i.e., there is no semantics (for I) $S' \neq S$ such that also $S'(A \mathrel{\Box\!\!\rightarrow} C) = F(A, C, S'(A), S'(C))$. Most, but not all the semantics that will be discussed below are of this general type; some even fulfill the following stronger requirement:

Definition 1.2 *A semantics S for I is* compositional *if there is a function F from $P(I)^2$ to $P(I)$ such that $S(A \mathrel{\Box\!\!\rightarrow} B) = F(S(A), S(B))$.*

If this is the case we write S_F for S. A semantics is compositional if the proposition it assigns to a complex sentence is (functionally) determined by the propositions it assigns to the parts of this sentence. Functions from $P(I)^2$ to $P(I)$ we call *composition functions* or *c-functions*. Every binary truth function t (i.e. total function from $\{1, 0\}^2$ into $\{1, 0\}$), or *t-function* (for short), generates a c-function F_t in the following way:

For every $X, Y \subseteq I$; $F_t(X, Y) = \{w : (w \in X \cap Y \wedge t(1, 1) = 1) \vee (w \in X - Y \wedge t(1, 0) = 1) \vee (w \in Y - X \wedge t(0, 1) = 1) \vee (w \notin X \cup Y \wedge t(0, 0) = 1)\}$.

A c-function generated by a t-function in this way we call an *extensional function* or *e-function*. If F is an e-function we call S_F a *truth functional semantics* .

Consider, for example, the t-function t_{mat} with the value 1 for all arguments except $\langle 1, 0\rangle$. t_{mat} defines the e-function

$F_{t_{mat}}(S(A), S(B)) = \{w : w \in S(A) \cap S(B) \vee w \in S(B) - S(A) \vee w \notin S(A) \cup S(B)\} = I - (S(A) - S(B)) = S(A \rightarrow B)$.

This represents the case where subjunctive conditionals have the truth conditions of the corresponding material conditionals.

1.5.1 Extensional Semantics

We must establish that truth functional semantics thus defined are extensional in the sense described above. Let $Sub(A, B, C, D)$ iff D is the result of replacing one or more occurrences of A by B in C, where A, B, C and D are sentences [see Nute, 1980, p. 43].

We first give a formal definition of the notion of extensional complex: If S is a semantics for I

Definition 1.3 *C is an* extensional complex *under S iff for every $w \in I$, $w \models_S C \leftrightarrow D$ whenever $w \models_S A \leftrightarrow B$ and $Sub(A, B, C, D)$.*

That is, if some part A of C is replaced with a sentence B with the same truth value as A the resulting sentence D will have the same truth value as C. We also define the notion of extensional semantics:

Definition 1.4 *A semantics S for I is* extensional *iff for every $w \in I$, $w \models_S C \leftrightarrow D$ whenever $Sub(A, B, C, D)$ and $w \models_S A \leftrightarrow B$.*

So a semantics S is extensional if all $C \in L$ are extensional complexes under S. The following is easily shown by recursion on the complexity of formulas:

Proposition 1.1 *If S is a truth functional semantics then every $C \in L$ is an extensional complex under S.*

So the truth functional semantics defined by e-functions according to Def. 1.2 are extensional semantics according to Def. 1.4. The reverse is not true, however: If a semantics is extensional then for each world w there must be some unique truth table associated with the connective "$\Box\!\!\rightarrow$", and hence a t-function t_w, but we cannot conclude that $t_w = t_{w'}$ in every case where $w \neq w'$. For instance S might be such that $A \Box\!\!\rightarrow C$ has the truth conditions of $A \wedge C$ in all worlds w such that $w \models_S B$, and the truth conditions of $A \vee C$ in all other worlds of I.

So we cannot prove that each extensional semantics is truth functional in the sense defined above. However, there is a weaker

sense of truth functionality in which this may be proved: We say that t is an *rt-function* (for I) if t is a total function from $I \times \{1,0\} \times \{1,0\}$ into $\{1,0\}$. (Instead of $t(w,x,y)$ we write $t_w(x,y)$.) Given any domain I, each rt-function t for I defines a certain c-function F_t for I, which we shall call an *re-function* from $P(I)^2$ to $P(I)$, in the following way:

> For every $X, Y \subseteq I$; $F_t(X,Y) = \{w : (w \in X \cap Y \wedge t_w(1,1) = 1) \vee (w \in X - Y \wedge t_w(1,0) = 1) \vee (w \in Y - X \wedge t_w(0,1) = 1) \vee (w \notin X \cup Y \wedge t_w(0,0) = 1)\}$.

Obviously if t is an rt-function such that for all $x, y \in \{1,0\}$ and $w, w' \in I$ then $t_w(x,y) = t_{w'}(x,y)$ and t defines a certain re-function F_t, then F_t is not only an re-function but also an e-function defining a truth functional semantics. Generally the compositional semantics defined in terms of re-functions we shall call *world-relative truth functional (WRTF) semantics.* It is easily proved by recursion that

Proposition 1.2 *S is an extensional semantics iff S is a WRTF semantics.*

We noted above that there seems to be no reason to suppose that $F_{t_{mat}}$ or any other e-function defines a semantics that models our intuitions of the truth conditions of subjunctive conditionals, and since any re-function would have to assign a definite truth table to conditionals in the actual world there is no more reason to suppose that any such function would do this job better, i.e., there seems to be no WRTF semantics, and therefore no extensional semantics for a language that is intended to model the semantic properties of subjunctive conditionals.

1.5.2 Intensional Semantics

If subjunctive conditionals are not extensional complexes, they may still be intensional complexes, i.e., such that any subsentence may be replaced *salva veritate* by any sentence with the same propositional content as the subsentence in question:

Definition 1.5 *C is an* intensional complex *under S iff $\models_S C \leftrightarrow D$ whenever $\models_S A \leftrightarrow B$ and $Sub(A, B, C, D)$.*

That is, if some part A of C is replaced with a sentence B that is true at the same worlds as A then the resulting sentence D will be true at the same worlds as C. Obviously if every $C \in L$ is an intensional complex under S then S is an intensional semantics in the following sense:

Definition 1.6 *A semantics S is* intensional *iff $\models_S C \leftrightarrow D$ whenever $Sub(A, B, C, D)$ and $\models_S A \leftrightarrow B$.*

This means that given an intensional semantics parts of sentences may be replaced by logically equivalent sentences without changing the truth value of the containing sentence. Since logically equivalent sentences have the same truth value (at all worlds) *all extensional semantics are intensional* in this sense. We can also easily see that

Proposition 1.3 *A semantics is compositional iff it is intensional.*

A semantics is compositional, we recall, if the proposition it assigns to a complex sentence is (functionally) determined by the propositions it assigns to the parts of this sentence. Intensionality is a test of this: If parts of a given complex are replaced by other sentences expressing the same proposition then to say that the resulting complex will have the same truth value as the original complex at all worlds, is the same as to say that the proposition assigned to it is functionally determined by the propositions assigned to its parts. We shall consider different kinds of compositional semantics below, among which are Lewis's relative closeness semantics and what I shall call propositional class selection semantics, but let us first say some words about semantics that are not intensional.

1.5.3 Hyperintensional Semantics

Not all investigators have been satisfied with semantics for subjunctive conditionals that are intensional in the sense defined above.

One particular concern has been the fact that Strengthening of Antecedents is a valid rule of inference for all intensional semantics that validate the following schema:

$$\text{(SDA)}\ ((A \vee B) \mathrel{\Box\!\!\rightarrow} C) \rightarrow ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (B \mathrel{\Box\!\!\rightarrow} C)).$$

Proof. This is so because $S(A) = S(A \vee (A \wedge B))$ for all semantics S, so if S is intensional $S(A \mathrel{\Box\!\!\rightarrow} C) = S((A \vee (A \wedge B)) \mathrel{\Box\!\!\rightarrow} C)$. But if SDA is valid, $S((A \vee (A \wedge B)) \mathrel{\Box\!\!\rightarrow} C) \subseteq S((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$, so $S(A \mathrel{\Box\!\!\rightarrow} C) \subseteq S((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$, hence Strengthening of Antecedents is valid for S. $\dashv$

SDA is a very attractive principle, intuitively (see e.g. [Nute, 1980, Sect. 2.2] or [Ellis et al., 1977]), but at the same time Strengthening of Antecedents seems inadmissible;[26] indeed the inference rule:

$$\frac{A \mathrel{\Box\!\!\rightarrow} C}{(A \vee (A \wedge B)) \mathrel{\Box\!\!\rightarrow} C}$$

which is valid in all intensional semantics seems intuitively objectionable in its own right. Consider the following example:

> If he had jumped from that plane he would have killed himself.
>
> ---
>
> If he had jumped from that plane, or had jumped from that plane with a parachute he would have killed himself.

The classification of semantics into extensional, intensional, and hyperintensional, concerns the way in which each semantics is, or fails to be compositional. If we instead classify semantics with respect to the way in which truth conditions of conditional sentences

[26] Although SDA indeed seems attractive from an intuitive point of view, I shall nevertheless assume that it is not valid. I think that it may be the case that the grammatical connective 'or' as it occurs in antecedents of subjunctive conditionals does not typically express truth functional disjunction, but rather serves some other purpose (see [Fine, 1975], [Loewer, 1976] and [McKay and van Inwagen, 1977]); perhaps that of expressing the phenomenon of divisibility that will be discussed briefly on p. 105 below. The seeming validity of SDA may perhaps be explained in the way outlined in footnote 23 on p. 105 below. For a contrary view cf. e.g. [Ellis et al., 1977].

are established in each semantics respectively, we get a quite different classification that does not coincide with the first one. The methods of establishing or defining truth conditions for subjunctive conditionals that have received much attention are the following, that we shall discuss in turn: Class (or Set) Selection, Premise (or Facts) Revision, and finally Relative Closeness.

1.5.4 Class Selection Semantics

Some[27]—but not all—of the different semantics for subjunctive conditionals that we shall consider below will be formulated in terms of *class selection functions*. Given any set of possible worlds I:

Definition 1.7 *A* class selection function *for I is a total function from $L \times I$ to $P(I)$.*

Arguably, what we are doing when evaluating conditional sentences is that we imagine situations in which the antecedent of the conditional is true and then check whether the consequent is also true at those situations. This is what class selection functions are intended to model: If we wonder whether $A \mathrel{\Box\!\!\rightarrow} C$ is true at the world w we check whether C is true at all the worlds in $s(A, w)$ where s is some class selection function, i.e., we check whether $s(A, w) \subseteq S(C)$.

Definition 1.8 *If s is a class selection function for I and S is a semantics for I such that $S(A \mathrel{\Box\!\!\rightarrow} B) = \{w : s(A, w) \subseteq S(B)\}$ then S is a* class selection semantics *for I.*

The defining clause $\{w : s(A, w) \subseteq S(B)\}$ is of the form $F(A, S(B))$ and therefore if s is a class selection function then S exists and is unique, and hence for each class selection function s there exists exactly one semantics S defined in this way by means of s.

Not all class selection semantics are intensional, however. There may be a class selection semantics S defined by the class selection function s such that for some $A, B, C \in L$, $s(A, w) \subseteq S(C)$ and $s(B, w) \not\subseteq S(C)$ even in case $\models_S A \leftrightarrow B$. Then $w \in S(A \mathrel{\Box\!\!\rightarrow} C)$

[27] Cf., e.g. the world selection semantics of Stalnaker [1968b], that can easily be reformulated as a class selections semantics. See also [Nute, 1980, Ch. 3].

and $w \notin S(B \mathrel{\Box\!\!\rightarrow} C)$, and hence $S(A \mathrel{\Box\!\!\rightarrow} C) \neq S(B \mathrel{\Box\!\!\rightarrow} C)$, so $\not\models_S (A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C)$. But $Sub(A, B, A \mathrel{\Box\!\!\rightarrow} C, B \mathrel{\Box\!\!\rightarrow} C)$ so S is not an intensional semantics. However, all class selection semantics have a related (but weaker) property:

Definition 1.9 *A semantics S is* consequent intensional *iff* $\models_S (A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (A \mathrel{\Box\!\!\rightarrow} D)$ *whenever* $\models_S C \leftrightarrow D$.

Now it is easy to see that:

Proposition 1.4 *If S is a class selection semantics then S is consequent intensional.*

Proof. Assume that S is a class selection semantics defined by s, and that $\models_S C \leftrightarrow D$, so $S(C) = S(D)$. Then $s(A, w) \subseteq S(C)$ iff $s(A, w) \subseteq S(D)$, so $\models_S (A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (A \mathrel{\Box\!\!\rightarrow} D)$. $\dashv$

Indeed, there is an even stronger property shared by all class selection semantics, namely the so called Generalized Consequence Principle [cf. Pollock, 1976, p. 19–20]. If $\Gamma \subseteq L$ and S is a semantics for some set of possible worlds I. Let

$$S(\Gamma) = \{w \in I : \text{ If } B \in \Gamma \text{ then } w \in S(B)\}.$$

$S(\Gamma)$ is the set of worlds at which all the $B \in \Gamma$ are true (according to S). Also let

$$\Gamma \models_S B \text{ iff } S(\Gamma) \subseteq S(B).$$

Since $S(\emptyset) = I$,

$$\emptyset \models_S B \text{ iff } \models_S B.$$

Also, clearly

Proposition 1.5 $\Gamma \cup \{A\} \models_S B$ *iff* $\Gamma \models_S (A \rightarrow B)$.

This captures the usual model theoretical concept of logical consequence: B is a logical consequence of the set Γ iff B is true at all the worlds at which all the sentences in Γ are true. If S is a semantics for some set of possible worlds I we say that S satisfies GCP iff

Condition 1.1 (GCP) *For any $w \in I$, $A \in L$ and $\Gamma \subseteq L$; If for all $B \in \Gamma$, $w \models_S A \mathrel{\Box\!\!\rightarrow} B$ and $\Gamma \models_S C$ then $w \models_S A \mathrel{\Box\!\!\rightarrow} C$.*

Proposition 1.6 *If S is a class selection semantics then S satisfies GCP.*

Proof. Assume that S is a class selection semantics defined by s, $\Gamma \subseteq L$, and that $w \in I$ is such that for all $B \in \Gamma$, $w \models_S A \mathrel{\Box\!\!\rightarrow} B$, and $S(\Gamma) \subseteq S(C)$. Then $s(A, w) \subseteq S(\Gamma)$, since $s(A, w) \subseteq S(B)$ for all $B \in \Gamma$, so $s(A, w) \subseteq S(C)$, so $w \models_S A \mathrel{\Box\!\!\rightarrow} C$. ⊣

GCP implies consequent intensionality, but not the other way round. Lewis's relative closeness semantics—which we will discuss below—are intensional, and therefore consequent intensional, but some of them do not satisfy GCP, therefore the reverse of Prop. 1.4 is not true.

Now, under which conditions are class selection semantics intensional? We shall say that a class selection function s (defining the class selection semantics S) is *propositional* iff for every $w \in I$, $s(A, w) = s(B, w)$ whenever $S(A) = S(B)$. I.e., if s picks out the same set of worlds for A and B whenever S assigns the same proposition to A and B we say that s is propositional. A class selection semantics defined by a propositional class selection function I shall call a *propositional class selection semantics* .

Proposition 1.7 *All propositional class selection semantics are intensional semantics.*

Proof. Assume that S is a propositional class selection semantics defined in terms of the propositional class selection function s. We show that S is compositional and hence intensional:[28] Let

$$F(A, S(B)) = \{w : s(A, w) \subseteq S(B)\}$$

and let

[28] I am grateful to Dag Westerståhl for this elegant substitute for my original proof.

$$F'(G,H) = \begin{cases} F(A,S(B)) & \text{if } G = S(A) \text{ and } H = S(B), \\ & \text{for some } A, B \in L \\ \emptyset & \text{otherwise} \end{cases}$$

F' is well defined since $G = S(A) = S(A')$ and $H = S(B) = S(B')$ imply that $F(A, S(B)) = F(A', S(B'))$.

Now $S(A \mathrel{\Box\!\!\rightarrow} B) = F'(S(A), S(B))$ so S is compositional and hence intensional. $\dashv$

Are all intensional class selection semantics also propositional? Well, some subclasses are, at least: We say that a class selection function s is *finite* if for all $A \in L$ and w, $s(A, w)$ is finite, and that it is *selective* if

Definition 1.10 *for every $A \in L$ and $w, w', w'' \in I$ such that $w' \in s(A, w)$ and $w'' \notin s(A, w)$ there is $\phi \in L$ such that $w' \models_S \phi$ and $w'' \not\models_S \phi$.*

Proposition 1.8 *If S is defined by a finite and selective class selection function s that is not propositional then S is not an intensional semantics.*

Proof. Assume that S is defined by a finite and selective class selection function s that is not propositional. Then there are $A, B \in L$ and w such that $S(A) = S(B)$ and $s(A, w) \neq s(B, w)$. Assume that w' is such that $w' \in s(A, w)$ and $w' \notin s(B, w)$. Since $w' \notin s(B, w)$ and $s(B, w)$ is finite there is a finite set of formulas $F = \{A_1, A_2 \ldots, A_n\}$ such that if $w'' \in s(B, w)$ there is $A_k \in F$ such that $w'' \in S(A_k)$ and $w' \notin S(A_1 \vee A_2 \vee \ldots \vee A_n)$, since ($s$ being selective) each $w'' \in s(B, w)$ must differ from w' with regard to at least one sentence. Let $C = (A_1 \vee A_2 \vee \ldots \vee A_n)$. Then $w' \notin S(C)$, hence $s(A, w) \not\subseteq S(C)$, but since $s(B, w) \subseteq S(C)$, $w \models_S \neg(A \mathrel{\Box\!\!\rightarrow} C) \wedge (B \mathrel{\Box\!\!\rightarrow} C)$, so S is not intensional. $\dashv$

But this is not the only class of intensional class selection functions that must be propositional. We say that a class selection function f (defining the class selection semantics S_f for I) is *projective* if for each $A \in L$ and $w \in I$

there is $\Delta \subseteq L$ such that $f(A, w) = S_f(\Delta)$.

Proposition 1.9 *If f is projective and $f(A, w) \neq f(B, w')$ then there is $C \in L$ such that [$f(A, w) \subseteq S_f(C)$ and $f(B, w') \nsubseteq S_f(C)$] or [$f(A, w) \nsubseteq S_f(C)$ and $f(B, w') \subseteq S_f(C)$].*

Proof. Assume that $f(A, w) = S_f(\Delta)$, $f(B, w') = S_f(\Delta')$ and that $w'' \in f(A, w) - f(B, w')$. Then $w'' \not\models_{S_f} \Delta'$, and hence there is $C \in \Delta'$ such that $w'' \not\models_{S_f} C$, so $f(A, w) \nsubseteq S_f(C)$. But $f(B, w') \subseteq S_f(\Delta')$, so $f(B, w') \subseteq S_f(C)$. $\dashv$

Now it is easily seen that

Proposition 1.10 *If S_f is defined by a projective class selection function f that is not propositional then S_f is not an intensional semantics.*

Now we shall look at semantics formulated in a different way, an important subclass of which are equivalent to class selection semantics.

1.5.5 Sentence Set Revision

Revision of the Set of Facts

Chisholm in a seminal paper [1946] suggested a semantic theory of subjunctive conditionals of the following general form: The conditional $A \mathrel{\Box\!\!\rightarrow} C$ "is another way of saying: There is a statement p[29] such that p and $[A]$ entail $[C]$ and p is true" [p. 299]. Chisholm discusses at length the problem of identifying p. This discussion

[29] When speaking of "a statement p" Chisholm unnecessarily and, I think, unintentionally restricts himself to finite premise sets. In the following I shall speak a set P of true sentences, leaving open whether this set is finite or not.

is carried further in [Goodman, 1946]. An important constraint that Goodman recognizes is that from the set of true statements P "we must exclude statements logically incompatible with the antecedent" since otherwise P in conjunction with the antecedent would imply every consequent, and so every conditional with a false antecedent would be true.[30] Now we shall investigate some logical relationships between semantic theories of this form and class selection semantics.

If $\Gamma \subseteq L$, $A \in L$ let

$$\Gamma \models A \text{ iff } \Gamma \models_S A \text{ for every semantics } S$$

and let

$$Cn_S(\Gamma) = \{B \in L : \Gamma \models_S B\} \text{ and } Cn(\Gamma) = \{B \in L : \Gamma \models B\}.$$

For any semantics S for some set of possible worlds I and $w \in I$, let

$$Th_S(w) = \{B \in L : w \models_S B\},$$

and generally if $G \subseteq I$,

$$Th_S(G) = \{B \in L : \text{If } w \in G \text{ then } B \in Th_S(w)\}.$$

$Th_S(w)$ we call the *S-theory* of the world w, and it is the set of sentences true at the world w according to S, and if G is a set of worlds, $Th_S(G)$ is the set of sentences true at all the worlds in G according to S. Let

$$Th(w) = \{B \in L : \text{If } S \text{ is a semantics then } w \models_S B\}.$$

Obviously, for any S, $Th(w) \subseteq Th_S(w)$. Let

[30] Indeed, Goodman argues convincingly that we must make an even stronger demand: It is not enough that P is logically compatible with A in the sense that $P \cup \{A\}$ is consistent, all the statements in P must be *cotenable* with A. We shall return to the subject of cotenability in Section 2.8 below. The kind of semantics suggested by Goodman is covered by the present treatment since cotenability presupposes logical compatibility.

$$Th(G) = \{B \in L : \text{ If } w \in G \text{ then } B \in Th(w)\}.$$

It follows that $Th(G) \subseteq Th_S(G)$, for any $G \subseteq I$ and S. The following are true:

Proposition 1.11 **a** *For all* $\Gamma, \Delta \subseteq L$, *if* $\Gamma \subseteq \Delta$ *then for every* S, $S(\Delta) \subseteq S(\Gamma)$.

b *For all* $G \subseteq I$, $G \subseteq S(Th_S(G)) \subseteq S(Th(G))$.

c *For all* $\Gamma \subseteq L$, $Cn_S(\Gamma) = Th_S(S(\Gamma))$.

d *For all* $G \subseteq I$ *and* $\Gamma \subseteq L$, *if* $G \subseteq S(\Gamma)$ *then* $S(Th_S(G)) \subseteq S(\Gamma)$.

e *For all* $\Gamma, \Delta \subseteq L$, $S(\Gamma) \cup S(\Delta) \subseteq S(\Gamma \cap \Delta)$.[31]

f *For all* $\Gamma, \Delta \subseteq L$, $S(\Gamma) \cap S(\Delta) = S(\Gamma \cup \Delta)$.

We say that $\Gamma \subseteq L$ is *S-consistent* if $S(\Gamma) \neq \emptyset$, and that Γ is *S-compatible* with $A \in L$ if $\Gamma \cup \{A\}$ is S-consistent. Further Γ is *weakly S-compatible* with $A \in L$ if Γ is S-compatible with A or $\{A\}$ is not S-consistent. Let C_S be the set of S-consistent subsets of L.

Definition 1.11 *Given any semantics S for some set of possible worlds I a* premise contraction function *c for S is a total function from $L \times C_S$ to $P(L)$ such that for any $A \in L$ and $\Gamma \in C_S$, $c(A, \Gamma) \subseteq \Gamma$ and $c(A, \Gamma)$ is weakly S-compatible with A.*

There is at least one premise contraction function for each semantics since $\emptyset$ is weakly S-compatible with every $A \in L$. If c is a premise contraction function then for a given sentence A and a given world w, c provides a set of true sentences $c(A, Th_S(w))$ compatible with A that can be used for evaluating conditionals in the way suggested by Chisholm, viz. by checking whether $c(A, Th_S(w)) \cup \{A\}$ implies the consequent of the conditional. This idea is captured by the following definitions:

[31] The reverse of (e) is not true, however. Assume that $S(\{A \wedge B\}) \neq \emptyset$. $\{A \wedge B\} = \{\neg A, A \wedge B\} \cap \{\neg B, A \wedge B\}$, but then $S(\{\neg A, A \wedge B\}) = S(\{\neg B, A \wedge B\}) = \emptyset$.

Definition 1.12 *If c is a premise contraction function for a semantics S such that $S(A \Box\!\!\rightarrow B) = \{w : c(A, Th_S(w)) \cup \{A\} \models B\}$ then S is a* strong facts revision semantics *for I.*

Definition 1.13 *If c is a premise contraction function for a semantics S such that $S(A \Box\!\!\rightarrow B) = \{w : c(A, Th_S(w)) \cup \{A\} \models_S B\}$ then S is a* weak facts revision semantics *for I.*

These two definitions make use of different notions of logical consequence. There is on the one hand the weak requirement that according to the semantics in question the consequent is true at all worlds at which the premises are true. This weak requirement gives rise to a relatively strong notion of logical consequence. On the other hand there is the strong requirement that according to any semantics the consequent be true at all worlds at which the premises are true. This strong requirement, on the other hand, gives rise to the relatively weak standard notion of logical consequence of propositional calculus. The non-standard versions are stronger in the sense that they might validate arguments and sentences not validated by the weaker standard notion. I call the resulting semantics weak and strong respectively with reference to the relative strength of the requirements $c(A, Th_S(w)) \cup \{A\} \models_S B$ and $c(A, Th_S(w)) \cup \{A\} \models B$ occurring in the definitions of those semantics.

I shall not speculate what Chisholm and Goodman meant by locutions like "entail", and "imply". I simply note the possible ambiguity and offer those two definitions as different ways to understand what those investigators had in mind.

Notice that we have as yet no guarantee that there are facts revisions semantics (other than trivial ones). None of the defining clauses $\{w : c(A, Th_S(w)) \cup \{A\} \models B\}$ and $\{w : c(A, Th_S(w)) \cup \{A\} \models_S B\}$ is necessarily of the form $F(A, B, S(A), S(B))$ since $c(A, Th_S(w))$ cannot generally be assumed to be a function of $\langle A, B, S(A), S(B)\rangle$.[32] Hence we cannot prove that these semantics exist by establishing that they are inductively definable in the simple way we

[32] Of course c can be defined such as to insure this. Let for instance $c(A, \Gamma) = \emptyset$ for all A and Γ. This function defines a facts revision semantics such that for all $w \in I$ $w \models_S A \Box\!\!\rightarrow C$ iff $\models_S (A \rightarrow C)$.

have used up until now. However, Theorem 1.14 below establishes that some (but not necessarily all) premise contraction functions define strong facts revision semantics.

We shall show that weak facts revision semantics are (equivalent to) class selection semantics of a certain kind. In order to specify the kind in question we need the following definition: Let f be any class selection function for some set of possible worlds I, and S_f a class selection semantics defined in terms of f:

(FID) For all $A \in L$ and $w \in I$, $f(A, w) \subseteq S_f(A)$.

If f satisfies FID we shall (for simplicity) also say that S_f satisfies FID. Obviously S_f satisfies FID iff the following axiom (ID) is satisfied by S_f (See Prop. 3.24. For a list of axiom schemata see p. 146.)

$A \mathrel{\Box\!\!\rightarrow} A$

Consider also the condition

(FMP') For all $A \in L$ and $w \in I$, if $w \models_{S_f} A$ then $w \in f(A, w)$.

If S_f satisfies FMP' then (by Prop. 3.24 and S 3.25) for all $A, B \in L$, $\models_{S_f} (A \mathrel{\Box\!\!\rightarrow} B) \rightarrow (A \rightarrow B)$.[33] Since $A \mathrel{\Box\!\!\rightarrow} A$ and $(A \mathrel{\Box\!\!\rightarrow} B) \rightarrow (A \rightarrow B)$ obviously are desirable members of the class of logical truths, class selection semantics are usually made to satisfy FID and FMP'.

We say that a class selection function f for I is *uniform* if

for all $w, w' \in I$ and $A \in L$, $f(A, w) = f(A, w')$ whenever $Th_{S_f}(w) = Th_{S_f}(w')$.

Also if $f(A, w) \neq \emptyset$ whenever $\{A\}$ is S_f-consistent then we say that f is *nonempty* . We can now show that for any set of possible worlds I

[33]The reverse seems to require that we assume that f is finite and selective or projective in the sense defined on p. 34 above (For proof see Prop. 3.24 and Prop. 3.27 below).

Theorem 1.12 *If S is a weak facts revision semantics for I then S is a class selection semantics (for I) defined by a nonempty and uniform class selection function satisfying FID and FMP'.*[34]

Proof. Assume that S_c is a weak facts revision semantics for some set of possible worlds I defined by the premise contraction function c. Define

$$s(A, w) = S_c(c(A, Th_{S_c}(w)) \cup \{A\}).$$

s is obviously a class selection function for I. Let S_s be the class selection semantics defined by s. We show that $S_c = S_s$. Assuming that $S_c(A) = S_s(B)$ and $S_c(A) = S_s(B)$ it is enough to show that $S_c(A \mathrel{\Box\!\!\rightarrow} B) = S_s(A \mathrel{\Box\!\!\rightarrow} B)$, i.e., that

$$c(A, Th_{S_c}(w)) \cup \{A\} \models_{S_c} B \text{ iff } s(A, w) \subseteq S_s(B),$$

i.e., that $S_c(c(A, Th_{S_c}(w)) \cup \{A\}) \subseteq S_c(B)$ iff $s(A, w) \subseteq S_s(B)$. But this is obvious since we have assumed that $s(A, w) = S_c(c(A, Th_{S_c}(w)) \cup \{A\})$ and $S_c(B) = S_s(B)$.

s trivially satisfies FID since $s(A, w) = S_c(c(A, Th_{S_c}(w)) \cup \{A\}) \subseteq S_c(A) = S_s(A)$.

[34] [Hansson, 1995, pp. 22–23, footnote 13] offers what appears to be a non-inductive proof of a similar result. But although he relativizes truth as we have done, and talks of sentences being true according to (Definitions) (WS) and (D1) respectively, it seems he fails to make this distinction in the proof itself. (D1) is Hansson's version of our Def. 1.12 (or perhaps D. 1.13), and (WS) is essentially our Def. 1.7 and 1.8 (p. 31) plus the added requirements that the class selection function is nonempty and satisfies FID. If c is a premise contraction function and T the set of all truths, Hansson refers to $c(A, T)$ by T/A. In the proof he uses "$|W|$" as "the set of sentences that are true in [the world] W". However he does not tell us whether this should be understood as "true according to (D1)" or "true according to (WS)". If $|W|$ is the set of sentences that are true at W according to (WS) then it seems that the premise that there is no W such that $T/A \cup \{A, \neg C\} \subseteq |W|$ does not entitle him to infer that $T/A \cup \{A, \neg C\}$ is inconsistent in the sense that would be required for $A \mathrel{\Box\!\!\rightarrow} C$ to be true according to D1. If, on the other hand $|W|$ is the set of sentences that are true at W according to (D1), then "$[A \mathrel{\Box\!\!\rightarrow} C]$ is true according to (WS) iff $C \in |W|$ for all $W \in f(W_0, A)$", is no mere restatment of (WS) as Hansson seems to treat it. So it seems that this non-inductive proof does not go through. Our proof works, on the other hand, precisely because it is inductive and we use the 'weak' version of the facts revision truth condition for conditionals.

s is obviously uniform. Now we also show that s is nonempty: Assume that $\{A\}$ is S_s-consistent. Then $\{A\}$ is S_c-consistent, hence $c(A, Th_{S_c}(w)) \cup \{A\}$ is S_c-consistent, i.e., $S_c(c(A, Th_{S_c}(w)) \cup \{A\}) \neq \emptyset$, so $s(A, w) \neq \emptyset$.

Finally assume that $w \models_{S_s} A$. Then $w \models_{S_c} A$, so $c(A, Th_{S_c}(w)) \cup \{A\} \subseteq Th_{S_c}(w)$. Thus $w \in S_c(c(A, Th_{S_c}(w)) \cup \{A\}) = s(A, w)$, so s satisfies FMP'. ⊣

Every nonempty class selection function f (defining the semantics S_f) that satisfies FID also satisfies the following requirement:

> (FMOD) For all $A \in L$ and $w \in I$, if $f(\neg A, w) \subseteq S_f(A)$ then for all $B \in L$, $f(B, w) \subseteq S_f(A)$.[35]

i.e.,

Proposition 1.13 *If f is a nonempty class selection function satisfying FID then it also satisfies FMOD.*

Proof. Assume that $f(\neg A, w) \subseteq S_f(A)$. Then $f(\neg A, w) = \emptyset$ (by FID), so $S(\neg A) = \emptyset$ (by nonemptyness), so $S(A) = I$. Thus for any $B \in L$, $f(B, w) \subseteq S_f(A)$. ⊣

This means that any weak facts revision semantics may be reformulated as a class selection semantics that satisfies FID, FMP' and FMOD. Now on the other hand for any set of possible worlds I:

Theorem 1.14 *If S is a class selection semantics (for I) defined by a nonempty and uniform class selection function satisfying FID and FMP' then S is a strong facts revision semantics for I.*[36]

[35] It is easy to see that S_f satisfies FMOD iff $\models_{S_f} (\neg A \mathrel{\Box\!\!\rightarrow} A) \rightarrow (B \mathrel{\Box\!\!\rightarrow} A)$ (See Prop. 3.24).

[36] [Hansson, 1995, p. 23] provides a proof for a similar result. However he doesn't require that FMP' be satisfied by the class selection function. It appears that he instead assumes that the antecedent A is false at the actual world W_0. This assumption seems not to be accounted for. It is true that Hansson restricts

Proof. Assume that S_s is a class selection semantics for some set of possible worlds I defined by a nonempty and uniform class selection function s satisfying FID and FMP'. Define r thus:

$$r(A,\Gamma) = \begin{cases} Th_{S_s}(s(A,w)) \cap \Gamma & \text{if } \Gamma = Th_{S_s}(w), \text{ for some } w \in I, \\ \emptyset & \text{otherwise.} \end{cases}$$

Since s is uniform r is well defined.

Obviously $r(A,\Gamma) \subseteq \Gamma$ for each Γ. We must show that $r(A,\Gamma)$ is weakly S_s-compatible with A. This is trivial if $r(A,\Gamma) = \emptyset$, so assume that $\Gamma = Th_{S_s}(w)$ for some $w \in I$. Assume that $S_s(A) \neq \emptyset$. Then $s(A,w) \neq \emptyset$ since s is nonempty. If $w' \in s(A,w)$ then $w' \in S_s(A)$ by FID. Also if $B \in r(A, Th_{S_s}(w))$ then $B \in Th_{S_s}(s(A,w))$, so $w' \models_{S_s} B$. That is, $w' \in S_s(r(A, Th_{S_s}(w)))$. Thus, $w' \in S_s(A) \cap S_s(r(A, Th_{S_s}(w))) = S_s(r(A, Th_{S_s}(w)) \cup \{A\})$, by Prop. 1.11, and we have shown that r is a premise contraction function.

Let S_r be the (unique) semantics defined by the usual conditions plus $S_r(A \mathrel{\Box\!\!\rightarrow} B) = \{w : r(A, Th_{S_s}(w)) \cup \{A\} \models B\}$. We show that $S_r = S_s$; it then follows that S_s is a strong facts revision semantics (for I). We assume that $A, B \in L$ and show that $S_r(A \mathrel{\Box\!\!\rightarrow} B) = S_s(A \mathrel{\Box\!\!\rightarrow} B)$ i.e., that

$$r(A, Th_{S_s}(w)) \cup \{A\} \models B \text{ iff } s(A,w) \subseteq S_s(B).$$

Now $r(A, Th_{S_s}(w)) \cup \{A\} \models B$ iff $r(A, Th_{S_s}(w)) \models (A \to B)$ (by Prop. 1.5) iff $Th_{S_s}(s(A,w)) \cap Th_{S_s}(w) \models (A \to B)$.

Assume that $Th_{S_s}(s(A,w)) \cap Th_{S_s}(w) \models (A \to B)$. Then $Th_{S_s}(s(A,w)) \models (A \to B)$ so $S_s(Th_{S_s}(s(A,w))) \subseteq S_s(A \to B)$, and hence by Prop. 1.11, $s(A,w) \subseteq S_s(A \to B)$. Since $s(A,w) \subseteq S_s(A)$, $s(A,w) \subseteq S_s(B)$.

Now assume instead that $s(A,w) \subseteq S_s(B)$. Then $s(A,w) \subseteq S_s(A \to B)$. We must show that

$$Th_{S_s}(s(A,w)) \cap Th_{S_s}(w) \models (A \to B).$$

the subject matter to counterfactual conditionals [p. 14], and he has told me that by this he meant to exclude conditionals with true antecedents. I sympathize with this use of the term "counterfactual", and I am actually an accomplice in it (see p. 11 above), but since this is by no means the standard use of the term, an explanation would perhaps not have been entirely out of place.

If $w \models_{S_s} A$ then $Th_{S_s}(s(A,w)) \subseteq Th_{S_s}(w)$ (by FMP'), so $Th_{S_s}(s(A,w)) \cap Th_{S_s}(w) = Th_{S_s}(s(A,w))$, but since $s(A,w) \subseteq S_s(A \to B)$, $A \to B \in Th_{S_s}(s(A,w))$, so $Th_{S_s}(s(A,w)) \cap Th_{S_s}(w) \models (A \to B)$.

Assume instead that $w \not\models_{S_s} A$. Then $\neg A \in Th_{S_s}(w)$ so $(A \to B) \in Th_{S_s}(w)$, but also $(A \to B) \in Th_{S_s}(s(A,w))$ so $(A \to B) \in Th_{S_s}(s(A,w)) \cap Th_{S_s}(w)$. So in this case too $Th_{S_s}(s(A,w)) \cap Th_{S_s}(w) \models (A \to B)$. Thus $S_r = S_s$ and S_r is a strong facts revision semantics (for I). $\dashv$

From Theorem 1.12 and Theorem 1.14 it follows that

Proposition 1.15 *All weak facts revision semantics are strong facts revision semantics.*

Iterated Facts Revision

$Th_{S_s}(w)$ and also $r(A, Th_{S_s}(w)) = Th_{S_s}(s(A,w)) \cap Th_{S_s}(w)$ will contain sentences containing " $\Box\!\!\to$ ". One might be dissatisfied with this: It would surely be more satisfactory from an intuitive point of view, if we could instead start with the set of "normal", i.e., non conditional truths and then by means of revision define which conditionals are true and which not. One way would be to start with $Th(w)$ and define $r(A, Th(w)) = Th(s(A,w)) \cap Th(w)$ and then say that $S_r(A \Box\!\!\to B) = \{w : r(A, Th(w)) \cup \{A\} \models B\}$. But then $Th_{S_r}(w)$ will not contain all elements of $Th_{S_s}(w)$. Another way would be to allow for iteration: We would start with $Th(w) = Th_0(w)$ and generate a sequence $\{Th_0(w), Th_1(w), Th_2(w) \ldots\}$ such that for each n, $Th_{n+1}(w) = Cn(Th_n(w) \cup \Gamma_n)$ such that (roughly) Γ_n is the set of those conditionals that are supported by $Th_n(w)$ and the contraction function r, and the negations of those conditionals that are not thus supported. Finally we let $Th_{S_r}(w) = \cup\{Th_0(w), Th_1(w), Th_2(w) \ldots\}$. The semantics generated in this way we could call an *iterative facts revision semantics*, and it can be shown that each class selection semantics (satisfying the requirements above) is an iterative facts revision semantics.

Belief Revision

Above, in Definitions 1.12 and 1.13, we required that the arguments of the contraction function be sets comprising all truths at a certain world. Such sets are not only consistent but maximally consistent in the sense that if a sentence does not belong to the set in question then its negation belongs to it. For different reasons we might need an account that is more general in only requiring consistency of the arguments of the contraction function. One such reason is if we want to model what happens when a person having a set of beliefs Γ modifies this set in order to accommodate the antecedents of belief contravening conditionals. This is what is attempted by so called belief revision accounts of conditionals.[37]

It seems we cannot define a semantics for any arbitrary premise contraction function. At least, it seems we have no obvious way of relating worlds and premise sets that are not maximally consistent. One way one could try to do this is the following.

> If c is a premise contraction function for a semantics S such that $S(A \mathrel{\Box\!\!\rightarrow} B) = \{w : \text{if } S(c(A,\Gamma)) \cap S(A) \subseteq S(B) \text{ then } w \in S(\Gamma)\}$, then S is a belief revision semantics for I.

But this seems to be too strong: A conditional should be allowed to be true at a world without requiring that every set of premises that supports the conditional is true at this world. We could modify the definition thus:

> If c is a premise contraction function for a semantics S such that $S(A \mathrel{\Box\!\!\rightarrow} B) = \{w : \text{if } w \in S(\Gamma), \text{ then } S(c(A,\Gamma)) \cap S(A) \subseteq S(B)\}$, then S is a belief revision semantics for I.

Then we get another requirement that is also too strong, but in another way: It is obviously too much to require that every set of true sentences should support the conditional for it to be true.

[37] See discussion in Section 1.2 above and e.g. [Gärdenfors, 1988, Ch. 7].

Premise contraction functions are used in *belief revision* accounts of conditionals, like that of [Gärdenfors, 1988, Ch. 7]. The general strategy of such accounts is described in Section 1.2 above.

According to Gärdenfors [p. 84] " Lewis's systems [of spheres models] are not at all appropriate for modeling revision functions", presumably because they are "always centered around a *single* possible world". Instead systems like that of [Grove, 1986] are better suited. The essential difference between the models of Grove and Lewis is that in the former, and not the latter, a "system of spheres is centered on a belief set K or rather its representation $[K]$". The underlying problem, as I see it, is that in a truth-centered (i.e. semantic) account, making use of possible worlds, the notion of *the actual world* will play a prominent role, i.e., truth is always truth at a certain world. This is not the case in justification-centered or epistemological accounts, because a set of beliefs may be compatible with any world in a set of worlds determined by that set. So the notion of the actual world must give way for some notion of the set of "actual worlds", i.e. the set of those worlds where the set of actual beliefs are true. Grove's semantic system seems to be an adaptation of Lewis's system to this requirement.

I see no obvious way of directly comparing "semantics" like these with the ones we have discussed so far. We can however compare them in terms of which logics or axiomatizations that are determined by those different semantics respectively.[38]

[38]Depending on the formal requirements on the revision operation, different logics will be supported. This reflects the way in which different formal requirements on the class selection function yield different logics. For an example of this procedure see Section 3.4.3 below, for another see Gärdenfors [1978, Sec. 5] who by putting suitable—but arguably, from the point of view of belief revision, partially somewhat arbitrary—requirements on the revision operation, constructs a semantics that determines the logic VC, which is Lewis's official logic for subjunctive conditionals. In [1988, Sec. 7.2] he instead attempts (unsuccessfully, as far as I can see) to establish that VC follows from a set of minimal requirements that should be put on all belief revision operations. In [1978, p. 390] he uses the requirement (C7) that $K_A = K_B$ whenever $A \in K_B$ and $B \in K_A$, and proves that the schema (CSO) $((A \mathrel{\Box\!\!\rightarrow} B) \wedge (B \mathrel{\Box\!\!\rightarrow} A)) \rightarrow ((A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C))$ is valid iff the belief revision operation satisfies (C7). In [1988, p. 150] he thinks—referring back for proof to [1978]—that the requirement (K*6) that $K_A = K_B$ whenever $\vdash A \leftrightarrow B$ suffices for the validity of CSO. (K*6) is a simple requirement

Now we shall turn to a class of semantics defined in a quite different way. A subclass of these semantics will turn out to be (equivalent to) class selection semantics.

1.5.6 Relative Closeness Semantics

David Lewis doesn't use class selection functions for his well known semantics for subjunctive conditionals; indeed, he couldn't have, because—as we saw above—GCP, which implies the "limit assumption" [see Lewis, 1981, Sect. 6], is valid in all class selection semantics, and he questions the validity of this assumption.[39] Why does Lewis deny the validity of the limit assumption? We shall return to this question shortly.

Lewis formulates the limit assumption in terms of his relative closeness semantics, and we shall now turn to this kind of semantics which is, unlike class selection semantics, intensional without qualifications. We need some definitions:[40] If I is a set of possible worlds and for each $w \in I$ there is a set $R_w \subseteq I$ of worlds "accessible from w" then $\leq$ is an *rc-assignment* for I if $\leq$ is a total function from I to $P(I \times I)$ (instead of $\leq (w)$ we write $\leq_w$, $w' \leq_w w''$ instead of $\langle w', w'' \rangle \in \leq_w$, and $w' <_w w''$ instead of $w'' \not\leq_w w'$) such that for each $w \in I$ the following conditions obtain:

1. The relation $\leq_w$ is *transitive*; that is, whenever $w' \leq_w w''$ and $w'' \leq_w w'''$ then $w' \leq_w w'''$.

2. The relation $\leq_w$ is *strongly connected*; that is, for any worlds w' and w'' either $w' \leq_w w''$ or $w'' \leq_w w'$.

3. The world w is *self accessible*; that is, $w \in R_w$.

4. The world w is $\leq_w$*-minimal*; that is, for any world w', $w \leq_w w'$.[41]

of propositionality. CSO (in conjunction with $\vdash A \mathrel{\Box\!\!\rightarrow} A$) implies propositionality but not *vice versa*, as far as I know. To ensure the validity of CSO, which is included in Lewis's VC, the presumably stronger, but perhaps less obvious, requirement (C7) seems to be required.

[39] For a discussion of this assumption see Lewis [1973b, Section 1.4].

[40] I follow Lewis [1973b, Section 2.3] rather closely in the following definitions.

[41] Lewis [1973b, Section 2.3] requires strict minimality (i.e. that for any world

5. Inaccessible worlds are $\leq_w$-*maximal*; that is, if $w' \notin R_w$ then for any world w'', $w'' \leq_w w'$.

6. Accessible worlds are more similar to w than inaccessible worlds: If $w' \in R_w$ and $w'' \notin R_w$ then $w' <_w w''$.

Definition 1.14 *Given any rc-assignment (for I) $\leq$, $S_\leq$ is a* relative closeness semantics *for I if $S_\leq$ is a semantics such that $S_\leq(A \mathrel{\Box\!\!\rightarrow} B) = \{w : S_\leq(A) \cap R_w = \emptyset$ or there is $w' \in S_\leq(A)$ such that for all $w'' \leq_w w'$, $w'' \in S_\leq(A \rightarrow B)\}$.*

$S_\leq(A \rightarrow B) = I - (S_\leq(A) - S_\leq(B))$ so it easily seen that the defining clause is of the form $F(S_\leq(A), S_\leq(B))$. This means that $S_\leq$ is not only inductively definable but also compositional, so by Prop. 1.3,

Proposition 1.16 *If S is a relative closeness semantics then S is an intensional semantics.*

There is obviously some kind of *ceteris paribus* assumption at work when we evaluate subjunctive conditionals: When asking whether a conditional is true we should ask ourselves what would have been the case had the antecedent clause of the conditional obtained, everything else being as it actually is (i.e., *ceteris paribus*), as much as possible. One way to understand this assumption is in terms of *similarity* between possible worlds: We should check whether the consequent clause is true at those antecedent-worlds (i.e. worlds at which the antecedent clause is true) that are most similar to the actual world. Variants of this understanding lie behind most class selection semantics as well as Lewis's relative closeness (or relative similarity) semantics[see Lewis, 1981, Section 1].

$w' \neq w$, $w <_w w'$). The corresponding requirement when this semantics is put in terms of Spheres of Accessibility is called "Centering", by Lewis. This requirement guarantees the validity of the dubious principle (CS) $(A \wedge C) \rightarrow (A \mathrel{\Box\!\!\rightarrow} C)$. For criticism of this principle see [Bennett, 1974]. The present (weaker) requirement corresponds to what Lewis [1973b, p. 29] calls "Weak Centering" and only guarantees the validity of the plausible principle (MP) $(A \mathrel{\Box\!\!\rightarrow} C) \rightarrow (A \rightarrow C)$. I have made this replacement for reasons of generality (and plausibility). Cf. p. 184 below.

Class selection theorists assume that there are antecedent-worlds most similar (closest) to the actual world. Lewis questions this, while maintaining that we should check whether there is a set of most similar antecedent-worlds, in another sense, where the consequent is true, namely, whether there is a set of antecedent-worlds such that no antecedent-world where the consequent is false is as similar to the actual as any world in this set.

We said above that not all intensional semantics satisfy GCP. Now is the time to prove this:

Proposition 1.17 *There are relative closeness semantics that do not satisfy GCP.*

Proof. If A is an atomic member of L, let $\Delta_0 = \{B_0, B_1 \ldots\}$ be some non-repeating denumerable set of atomic members of L not containing A. Let $\Delta_1 = \Delta_0 - \{B_0\}$, and generally $\Delta_{n+1} = \Delta_n - \{B_n\}$. For each n define a function from L to $\{1, 0\}$ thus: $w_n(\phi) = 1$ iff $\phi = A$ or $\phi \in \Delta_n$. Let $W_\Delta = \{w_0, w_1, \ldots\}$. Let i be a function L to $\{1, 0\}$ such that $i(A) = 0$, and let $I = W_\Delta \cup \{i\}$. Obviously $w \notin I$ whenever $w(A) = 1$ and $w \notin W_\Delta$. Let for any $w \in I$, $R_w = I$. Define $\leq$ thus: if $w_n, w_m \in W_\Delta$ then $w_n \leq_i w_m$ iff $m \leq n$. Also for all $w \in W_\Delta$ let $i \leq_i w$ and $w \not\leq_i i$. Finally let for all $w \in W_\Delta$, $\leq_w = I \times I$.

$\leq$ is an rc-assignment such that $w_0 >_i w_1 >_i w_2 \ldots$, and for all $w \in W_\Delta$, $i <_i w$. Let $\Gamma = \{\neg B_0, \neg B_1 \ldots\}$. Then for each n, $i \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} \neg B_n$, since for all $w' \leq_i w_{n+1}$, $w' \in S_\leq(A \rightarrow \neg B_n)$. But $S_\leq(\Gamma) \subseteq S_\leq(\neg A)$ since if $w \notin S_\leq(\neg A)$ and $w \in I$ then $w = w_n$ for some n, but $w_n \notin S_\leq(\neg B_n)$, so $w_n \notin S_\leq(\Gamma)$. But at the same time $i \not\models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} \neg A$, so GCP is not satisfied by $S_\leq$. $\dashv$

An example of the same general kind as the counterexample to GCP provided by this proof is given by Lewis [1973b, p. 20] where he argues against what he calls the *limit assumption*, namely "the assumption that as we take [...] antecedent-worlds closer and closer to i [the actual world], we actually reach a limit: [...] the *closest* antecedent-world":

> [W]e have no right to assume that there always are [...] closest antecedent worlds. Suppose we entertain the counterfactual supposition that at this point [at this point in [Lewis, 1973b] there is actually a line just under one inch long] there appears a line more than one inch long. [...] There are worlds with a line $2''$ long; worlds presumably closer to ours with a line $1\frac{1}{2}''$ long; worlds presumably still closer to ours with a line $1\frac{1}{4}''$ long; worlds presumably still closer... But how long is the line in the *closest* worlds with a line more than one inch long? If it is $1+x''$ for any x however small, why are there not other still closer to ours in which it is $1+\frac{1}{2}x''$, a length still closer to its actual length? The shorter we make the line (above $1''$), the closer we come to the actual length; so the closer we come, presumably, to our actual world.

Define $\Delta_0 = \{B_0, B_1 \ldots\}$ thus: For each n let B_n be the sentence saying that the line is longer than $1\frac{1}{n+1}''$. Let A be the sentence saying that the line is longer than $1''$. Let $W_\Delta = \{w_0, w_1, \ldots\} \subseteq R_i$ be defined as in the proof above, so for each $w_n \in W_\Delta$ only A and $B_n, B_{n+1}, \ldots$ are true. Lewis's presumption is that these worlds are ordered as in the proof, namely: $w_0 >_i w_1 >_i w_2 \ldots$, and for all $w \in W_\Delta$, $i <_i w$. Hence for each w_n if $w' \leq_i w_n$ either $w' \not\models_{S_\leq} A$ or $w' \models_{S_\leq} \neg B_n$ so $w' \models_{S_\leq} (A \rightarrow \neg B_n)$, so for each n, $i \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} \neg B_n$. That is, at the actual world i, for any n, it is true that had the line been longer than $1''$ it would not have been longer than $1\frac{1}{n+1}''$.[42] But since there are $w \in R_i$ such that $w \models_{S_\leq} A$ (since $\{w_0, w_1, \ldots\} \subseteq R_i$), $i \not\models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} \neg A$, i.e.—as it should be—it is not true at the actual world that had the line been longer than $1''$ then it would not have been longer that $1''$, even though $\neg A$ is a logical consequence of $\Gamma = \{\neg B_1, \neg B_2 \ldots\}$,[43] so GCP is not satisfied.

[42] That Lewis's theory has untoward consequences of this kind was first brought to my attention by Per Lindström. See [Lindström, 1997].

[43] A logical consequence in the sense that given a semantics S for I such that there is no $w \in I$ such that $w(A) = 1$ and $w(B_n) = 0$ for all n then $S(\{\neg B_1, \neg B_2 \ldots\}) \subseteq S(\neg A)$.

However all relative closeness semantics have the following weaker property:

Condition 1.2 (CP) *For any* $w \in I$ *and* $B_1, B_2, \ldots, B_n, A, C \in L$*; If* $B_1 \wedge B_2 \wedge \cdots \wedge B_n \models_S C$ *and* $w \models_S A \mathrel{\Box\!\!\rightarrow} B_1 \wedge A \mathrel{\Box\!\!\rightarrow} B_2 \wedge \cdots \wedge A \mathrel{\Box\!\!\rightarrow} B_n$ *then* $w \models_S A \mathrel{\Box\!\!\rightarrow} C$.

This principle is a finite version of GCP. That it is satisfied by all relative closeness semantics will be proved below in connection with the discussion of ARRC semantics. All such semantics have this property and therefore the relative closeness semantics have this property, since all relative closeness semantics are ARRC semantics.

CP is implied by GCP since if $S((A_1 \wedge A_2 \wedge \cdots \wedge A_n) \rightarrow B) = I$ then $S(\{(A_1, A_2, \ldots, A_n)\}) \subseteq S(B)$. If we for any $w \in I$ assume that $w \models_S C \mathrel{\Box\!\!\rightarrow} A_1$, $w \models_S C \mathrel{\Box\!\!\rightarrow} A_2, \ldots$, $w \models_S C \mathrel{\Box\!\!\rightarrow} A_n$, then $w \models_S C \mathrel{\Box\!\!\rightarrow} B$ (by GCP) so $S(((C \mathrel{\Box\!\!\rightarrow} A_1) \wedge C \mathrel{\Box\!\!\rightarrow} A_2 \wedge \cdots \wedge C \mathrel{\Box\!\!\rightarrow} A_n) \rightarrow (C \mathrel{\Box\!\!\rightarrow} B)) = I$.

What is the relation between relative closeness semantics and class selection semantics? We need to define some further properties of class selection semantics before we can state this relationship. Let S_f be the class selection semantics defined in terms of the class selection function f.

> (FCSO) For all $A, B \in L$ and $w \in I$, if $f(A, w) \subseteq S_f(B)$ and $f(B, w) \subseteq S_f(A)$ then $f(A, w) = f(B, w)$.

It is easy to see that S_f satisfies FCSO iff for all $A, B, C \in L$, $\models_{S_f} ((A \mathrel{\Box\!\!\rightarrow} B) \wedge (B \mathrel{\Box\!\!\rightarrow} A)) \rightarrow ((A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C))$ (see Prop. 3.24). This axiom schema we call CSO.

> (FCV') For all $A, B \in L$ and $w \in I$ if $f(A, w) \not\subseteq S_f(\neg B)$ then $f(A \wedge B, w) \subseteq f(A, w)$.

Evidently if S_f satisfies FCV' then for all $A, B, C \in L$, $\models_{S_f} ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Diamond\!\!\rightarrow} B)) \rightarrow ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$ (see Prop. 3.24 and Prop. 3.25). This axiom schema we call CV.

If $S_{\leq}$ is the relative closeness semantics for I defined in terms of the relative closeness function $\leq$ then $S_{\leq}$ satisfies the *limit assumption* iff

Condition 1.3 *For any $i \in I$ and $A \in L$ such that $S_{\leq}(A) \neq \emptyset$ there is one $w \in S_{\leq}(A)$ such that for every $w' \in I$ if $w' <_i w$ then $w' \notin S_{\leq}(A)$.*

If a relative closeness semantics $S_{\leq}$ satisfies the limit assumption we can meaningfully refer to the set of A-worlds closest to a certain world i. Now

Theorem 1.18 *If S is a relative closeness semantics satisfying the limit assumption then S is a class selection semantics satisfying FID, FMOD, FCSO, FCV', FMP' and FCA'.*

This will be proved on page 59 below. Obviously, by Prop. 1.6,

Proposition 1.19 *If S is a relative closeness semantics satisfying the limit assumption then S satisfies GCP.*

So, if a relative closeness semantics satisfies the limit assumption we can always define a class selection function in terms of it and so get an equivalent class selection semantics. This class selection semantics will of course be intensional. The fact that it satisfies FCSO and FID suffices for this as we shall see below.

The requirements FID, FMOD, FCSO, FCV', FMP' and FCA', respectively guarantee that certain axioms are valid in class selection semantics satisfying those requirements.[44] However, we need not make a detour via class selection semantics in this way in order to state what axioms are valid in relative closeness semantics:

Theorem 1.20 *If S is a relative closeness semantics then S satisfies CP and all instances of the axiom schemata ID, MOD, MP, CSO, CV and CA are true in S.*

CA is the following axiom:

$$(((A \mathrel{\Box\!\!\rightarrow} B) \wedge (C \mathrel{\Box\!\!\rightarrow} B)) \rightarrow ((A \vee C) \mathrel{\Box\!\!\rightarrow} B)).$$

This also will be proved below in Section 1.5.6.

[44] See Prop. 3.24 and Prop. 3.25.

Relative Closeness Too Strong?

I shall now argue that there are reasons to suspect that not all of these axioms are compatible with our semantic intuitions regarding subjunctive conditionals. More specifically I shall argue that the axioms CSO and CV have such untoward consequences, from an intuitive point of view. At this point, however, I shall only level such criticism at the following consequence of CSO, which we call RCV:

$$((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Box\!\!\rightarrow} B)) \rightarrow ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C).$$

By using some notions and results that will be given in Chapter 3 we can see that

Proposition 1.21 *RCV is valid in all relative closeness semantics.*

Proof. By Prop. 3.11, RCV is a theorem of the logic VW (p. 148 below) which is sound with respect to all relative closeness semantics (See Section 3.6.1). ⊣

It is also valid in class selection semantics that fulfill the requirements corresponding to ID and CSO. Such semantics are also intensional:

Proposition 1.22 *If S_f is a class selection semantics satisfying FID and FCSO, then S_f is intensional and validates RCV.*

Proof. Assume that S_f is a class selection semantics satisfying FID and FCSO. By Prop. 3.11, RCV is a theorem of the logic $\langle\langle RCK; ID, CSO\rangle\rangle$ (See Section 3.1.1 below). By Prop. 3.18 and 3.24 this logic is sound with respect to S_f.

By Prop. 3.11 this logic is also normal. Hence by Prop. 3.4 and 3.6, S_f is intensional. ⊣

What is wrong with RCV (and consequently CSO) then? It has not been questioned before by anybody that I know of. However, a related principle:

(CV) $((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Diamond\!\!\rightarrow} B)) \rightarrow ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$

was criticized by Pollock [1976].[45] This principle is satisfied by all relative closeness semantics (as defined above),[46] and those class selection semantics defined by class selection functions f satisfying FCV'.

Above we showed that if f satisfies FID and FCSO then RCV is valid with respect to S_f. The following requirement on the class selection f implies the validity of RCV with respect to the semantics S_f defined in terms of f

(FRCV') For all $A, B \in L$ and $w \in I$, if $f(A, w) \subseteq S_f(B)$ then $f(A \wedge B, w) \subseteq f(A, w)$.

I shall criticize RCV and thereby also FRCV' on the ground that in cases where the occurrence of B is a causal consequence of the occurrence of A in every $w' \in f(A, w)$, FRCV' may result in unintuitive truth conditions. Let me take an example. Say that this is the case in the actual world i:

> There was a party yesterday, and neither Albert nor Betty went there. However, Betty sat at her window, feeling dejected and looking to see whether Albert would go, and had he gone she would also have, because of him, and this would have lifted her spirits considerably, so $i \models A \mathrel{\Box\!\!\rightarrow} B$ but also $i \models A \mathrel{\Box\!\!\rightarrow} C$, if C means that Betty was happy. However, Betty could have gone on her own accord also—although she didn't—and in that case she would have remained sad. Now what would have happened if both had gone? Can we be sure that Betty would have been happy, i.e. that $i \models (A \wedge B) \mathrel{\Box\!\!\rightarrow} C$?

[45] Pollock's argument is criticized by [Loewer, 1979]. For Pollock's answer to this criticism see [Pollock, 1981].

[46] Because according to the definition of rc-assignment all worlds are comparable (cf. the requirement that $\leq_w$ be strongly connected). If also partial orderings are admitted then RCV, but not CV is valid. For a definition of truth for relative closeness semantics for partially ordered domains of possible worlds see [Lewis, 1981, p. 230].

It seems to me that when asking what would have happened had both gone, we must consider not only the possibility that she goes because of Albert but also that she goes on her own accord (although he also happens to go, perhaps after she already left), and depending on which had been the case, she might have been happy, but she might also not have, i.e., it seems that $i \models (A \wedge B) \diamond\!\!\rightarrow C$ and $i \models (A \wedge B) \diamond\!\!\rightarrow \neg C$ so $i \not\models (A \wedge B) \Box\!\!\rightarrow C$. On the other hand, when we are asking what would have happened had Albert gone we need not consider the possibility that Betty goes on her own since she did in fact not go at all.[47] Although $f(A, i) \subseteq S(B)$, $f(A, i)$ does not contain any worlds where Betty goes on her own accord, but only such where she goes because of Albert (and is therefore happy) so $f(A, i) \cap S(B) \subseteq S(C)$. However, although also $f(A \wedge B, i) \subseteq S(B)$, $f(A \wedge B, i)$ should indeed contain worlds where she goes on her own (and is not happy), so $f(A \wedge B, i) \cap S(B) \subseteq S(C)$ should not be the case, and hence $f(A \wedge B, w) \not\subseteq f(A, w)$, i.e, FRCV' should not be satisfied.

An interesting question is whether all class selection semantics satisfying FRCV' satisfy FCSO. I don't know, but we can show that if a class selection semantics for subjunctive conditionals fulfills some very natural and uncontested requirements in addition to FRCV' then it satisfies FCSO. The requirements are the following:

(KI) For all $A, B \in L$, $f(A \wedge B, w) = f(B \wedge A, w)$.

This requirement implies the validity of $((A \wedge B) \Box\!\!\rightarrow C) \leftrightarrow ((B \wedge A) \Box\!\!\rightarrow C)$. I think no-one would want to allow class selection functions that do not have this property. The third property is the following:

(FRT') For all $A, B \in L$ and $w \in I$, if $f(A, w) \subseteq S_f(B)$ then $f(A, w) \subseteq f(A \wedge B, w)$.

This requirement is implied by FID and FCSO. It seems self evident in the sense that if all hypothetical situations we should be considering when we are wondering what would have happened had A been

[47] This is a consequence of the *ceteris paribus* presuppositions normally made when subjunctive conditionals are evaluated.

true are such that B is also true, then we should consider all these situations when we are wondering what would have happened had both A and B been true. FRT' implies the validity of the following schema (Prop. 3.24, and 3.25):

$$\text{(RT)}\ ((A \mathrel{\Box\!\!\rightarrow} B) \wedge ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)) \rightarrow (A \mathrel{\Box\!\!\rightarrow} C).$$

This is regarded as self evident by all investigators in the field,[48] so arguably these two properties are required of all satisfactory class selection semantics of subjunctive conditionals.

Now we show that

Proposition 1.23 *If f is a class selection function satisfying KI and FRT' in addition to FRCV', then f satisfies FCSO.*

Proof. Assume that f is a class selection function satisfying KI, FRT' and FRCV', that S_f is defined by f, that $f(A, w) \subseteq S_f(B)$, and $f(B, w) \subseteq S_f(A)$. Then by FRT' and FRCV' it follows that $f(A, w) = f(A \wedge B, w)$ and $f(B, w) = f(B \wedge A, w)$, and hence by KI, $f(A, w) = f(B, w)$. $\dashv$

It follows by Prop. 1.22 above that

Proposition 1.24 *If f is a class selection function satisfying KI, FID, FRT' and FRCV', then f is intensional.*

Above we described a situation that appeared to be such that intuitively $i \models A \mathrel{\Box\!\!\rightarrow} B$ and $i \models A \mathrel{\Box\!\!\rightarrow} C$ at the same time as $i \not\models (A \wedge B) \mathrel{\Box\!\!\rightarrow} C$. If this is the case then, as we have said, RCV is intuitively invalid. Of course, this situation is also such that $i \models \neg(A \mathrel{\Box\!\!\rightarrow} \neg B)$, i.e., $i \models (A \mathrel{\Diamond\!\!\rightarrow} B)$ so we can see that not only is RCV (and thereby CSO) falsified by this example, but also CV. We should therefore expect that an intuitively valid semantics for subjunctive conditionals should validate neither CSO nor CV.

[48] The cases where transitivity fails, i.e., when $w \models (A \mathrel{\Box\!\!\rightarrow} B) \wedge (B \mathrel{\Box\!\!\rightarrow} C)$ and $w \not\models A \mathrel{\Box\!\!\rightarrow} C$, are such that $f(A, w) \not\subseteq f(B, w)$ even though $f(A, w) \subseteq S_f(B)$. This is reasonable enough since the B-worlds most worthy of consideration may for instance be $(B \wedge \neg A)$-worlds. But it seems that among the $A \wedge B$-worlds most worthy of consideration must be the A-worlds most worthy of consideration, if they are all $A \wedge B$-worlds. Therefore $f(A, w) \subseteq f(A \wedge B, w)$ should be the case if $f(A, w) \subseteq S_f(B)$.

Antecedent Relativity

All relative closeness semantics are intensional and they satisfy ID and CSO and therefore also RCV.[49] Class selection semantics satisfy these requirements only if corresponding requirements are put upon the class selection function defining them. So it seems that if one wants to make room for SDA or/and doesn't accept RCV one has to leave the framework of relative closeness altogether, accept the limit assumption and opt for a class selection semantics defined by a non-propositional class selection function, or one that does not satisfy FRCV'. However, the notion that we should consider the closest or most similar worlds is not only intuitively appealing, but it also seems indispensable if we wish to have a method for selecting the class of worlds we should consider given a certain antecedent. Is there a way to retain this idea if one takes the criticism of RCV seriously? Well, there is: one can generalize the concept of relative closeness semantics by allowing that different closeness relations are used for the purpose of evaluating different conditionals. We shall look at one such generalization that has a certain intuitive appeal, and that we shall use later on. I shall call semantics of this kind *Antecedent Relative Relative Closeness Semantics* (or ARRC semantics for short).

If I is a set of possible worlds and for each $w \in I$ there is a set $R_w \subseteq I$ of worlds "accessible from w", then $\leq$ is an *arrc-assignment* for I if $\leq$ is a total function from $I \times L$ to $P(I \times I)$ (instead of $\leq (w, A)$ we write $\leq_w^A$, $w' \leq_w^A w''$ instead of $\langle w', w'' \rangle \in \leq_w^A$, and $w' <_w^A w''$ instead of $w'' \not\leq_w^A w'$) such that for each $A \in L$ and $w \in I$ the following conditions obtain:

1. The relation $\leq_w^A$ is *transitive*; that is, whenever $w' \leq_w^A w''$ and $w'' \leq_w^A w'''$ then $w' \leq_w^A w'''$.

2. The relation $\leq_w^A$ is *strongly connected*; that is, for any worlds w' and w'' either $w' \leq_w^A w''$ or $w'' \leq_w^A w'$.

3. The world w is *self accessible*; that is, $w \in R_w$.

[49] We note again that this is not only true of relative closeness semantics as defined above, i.e., with total orderings of the domain of worlds, but also of semantics with merely partial orderings of their domains.

4. The world w is $\leq_w^A$-*minimal*; that is, for any world w', $w \leq_w^A w'$.

5. Inaccessible worlds are $\leq_w^A$-*maximal*; that is, if $w' \notin R_w$ then for any world w'', $w'' \leq_w^A w'$.

6. Accessible worlds are more similar to w than inaccessible worlds: If $w' \in R_w$ and $w'' \notin R_w$ then $w' <_w^A w''$.

Definition 1.15 *Given any arrc-assignment (for I) $\leq$, $S_\leq$ is an* ARRC semantics *for I if $S_\leq$ is a semantics such that $S_\leq(A \Box\!\!\rightarrow B)$ $= \{w : S_\leq(A) \cap R_w = \emptyset$ or there is $w' \in S_\leq(A)$ such that for all $w'' \leq_w^A w'$, $w'' \in S_\leq(A \rightarrow B)\}$.*

It easily seen that the defining clause is of the form $F(A, S_\leq(A), S_\leq(B))$, so $S_\leq$ is inductively definable. Obviously if for all $w \in I$ and $A, B \in L$, $\leq_w^A = \leq_w^B$ then $\leq_w$ defines a relative closeness semantics. And if

$$\leq_w^A = \leq_w^B \text{ whenever } S_\leq(A) = S_\leq(B)$$

we say that $S_\leq$ is a *propositional* ARRC semantics. If $S_\leq$ is propositional the defining clause is of the form $F(S_\leq(A), S_\leq(B))$, so in this case $S_\leq$ is compositional. So by Prop. 1.3, we have

Proposition 1.25 *Every propositional ARRC semantics is intensional.*

All ARRC semantics are not intensional, however. We may illustrate this as follows: Let $S_\leq$ be an ARRC semantics for $I = \{i, w_1, w_2\}$ such that $i \models_{S_\leq} (\neg A \wedge \neg B \wedge \neg C)$, $w_1 \models_{S_\leq} (A \wedge B \wedge \neg C)$, $w_2 \models_{S_\leq} (A \wedge B \wedge C)$. Thus $S_\leq(A) = S_\leq(B) = \{w_1, w_2\}$. Let $i <_i^A w_1 <_i^A w_2$ and $i <_i^B w_2 <_i^B w_1$. Now if this is the case then $i \models_{S_\leq} A \Box\!\!\rightarrow \neg C$ and $i \models_{S_\leq} B \Box\!\!\rightarrow C$ so $S_\leq$ is not intensional.

All ARRC semantics (and so all relative closeness semantics) satisfy the principle CP (Condition 1.2 on page 50 above):

Proposition 1.26 *If S is an ARRC semantics then S satisfies CP.*

Proof. Assume that $(B_1 \wedge B_2 \wedge \cdots \wedge B_n) \models_{S_\leq} C$ and that $w \models_{S_\leq} (A \mathrel{\Box\!\!\rightarrow} B_1 \wedge A \mathrel{\Box\!\!\rightarrow} B_2 \wedge \cdots \wedge A \mathrel{\Box\!\!\rightarrow} B_n)$. Then if $S_\leq(A) \cap R_w = \emptyset$, trivially $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} C$, so we assume that $S_\leq(A) \cap R_w \neq \emptyset$. But then for each $1 \leq k \leq n$ there is $w_k \in S_\leq(A)$ such that for all $w'' \leq_w^A w_k$, $w'' \in S_\leq(A \rightarrow B_k)$. Let w_i be one of the $\leq_w^A$-closest to w of these worlds. If $w_j \leq_w^A w_i$ and $w_j \models_{S_\leq} A$ then $w_j \models_{S_\leq} (B_1 \wedge B_2 \wedge \cdots \wedge B_n)$, so $w_j \models_{S_\leq} C$, and so $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} C$. ⊣

ARRC semantics that satisfy an antecedent relativized version of the limit assumption are equivalent to a special class of class selection semantics: If $S_\leq$ is the ARRC semantics for I defined in terms of the arrc assignment $\leq$ then $S_\leq$ satisfies the *Antecedent Relative Limit (ARL) assumption* iff

Condition 1.4 *For any $i \in I$ and $A \in L$, if $S_\leq(A) \neq \emptyset$ then there is one $w \in S_\leq(A)$ such that for every $w' \in I$, if $w' <_i^A w$ then $w' \notin S_\leq(A)$.*

If $S_\leq$ is a relative closeness semantics the *limit assumption* is equivalent to the ARL assumption.

If a relative closeness semantics $S_\leq$ satisfies the ARL assumption we can meaningfully refer to the set of A-worlds A-closest to a certain world i. Thus we define

$$S_\leq(A)_i = \{w \in S_\leq(A) \cap R_i : \text{If } w' <_i^A w \text{ then } w' \notin S_\leq(A)\}.$$

By transitivity of $<_i^A$:

Lemma 1.27 *If $w \in S_\leq(A) - S_\leq(A)_i$ and $w' \in S_\leq(A)_i$ then $w' <_i^A w$.*

We show that

Lemma 1.28 *If $S_\leq$ satisfies the ARL assumption then $S_\leq(A)_w = \emptyset$ iff $S_\leq(A) \cap R_w = \emptyset$.*

Proof. From right to left is trivial. Assume that $S_{\leq}$ satisfies the ARL assumption and $S_{\leq}(A)_w = \emptyset$. If $S_{\leq}(A) = \emptyset$ then trivially $S_{\leq}(A) \cap R_w = \emptyset$, so we assume that $S_{\leq}(A) \neq \emptyset$. Then by the ARL assumption there is one $w' \in S_{\leq}(A)$ such that for every $w'' \in I$, if $w'' <_w^A w'$ then $w'' \notin S_{\leq}(A)$. If $S_{\leq}(A) \cap R_w \neq \emptyset$, then w' must be an element of $S_{\leq}(A) \cap R_w$ (since accessible worlds are more similar than inaccessible ones) and hence $S_{\leq}(A)_w \neq \emptyset$ which contradicts the assumption. Thus $S_{\leq}(A) \cap R_w = \emptyset$ ⊣

Also for any $G \subseteq I_{S_{\leq}}$,

Lemma 1.29 *If $S_{\leq}$ satisfies the ARL assumption and $S_{\leq}(A) \cap R_i \neq \emptyset$ then there is $w \in S_{\leq}(A)_i$ such that for all $w' \leq_i^A w$, $w' \in G$ iff there is $w'' \in S_{\leq}(A)$ such that for all $w' \leq_i^A w''$, $w' \in G$.*

Proof. From left to right is trivial. Assume that $S_{\leq}$ satisfies the ARL assumption, $S_{\leq}(A) \cap R_i \neq \emptyset$ and there is $w'' \in S_{\leq}(A) - S_{\leq}(A)_i$ such that for all $w' \leq_i^A w''$, $w' \in G$. Then by Lemma 1.28 and 1.27 there is $w \in S_{\leq}(A)_i$ such that $w <_i^A w''$, hence since $\leq_i^A$ is transitive and connex, for all $w' \leq_i^A w$, $w' \in G$. ⊣

Lemma 1.30 *If $S_{\leq}$ satisfies the ARL assumption, $\leq_w^A = \leq_w^B$, $S_{\leq}(B)_w \subseteq S_{\leq}(A)$ and $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ then $S_{\leq}(A)_w = S_{\leq}(B)_w$*

Proof. Assume that $\leq_w^A = \leq_w^B$, $S_{\leq}(B)_w \subseteq S_{\leq}(A)$ and $S_{\leq}(A)_w \subseteq S_{\leq}(B)$, and for contradiction that there is $w' \in S_{\leq}(A)_w - S_{\leq}(B)_w$. Then $w' \in S_{\leq}(B) - S_{\leq}(B)_w$. If $w'' \in S_{\leq}(B)_w$ then $w'' \in S_{\leq}(A)$, but also by Lemma 1.27, $w'' <_w^B w'$ and hence $w'' <_w^A w'$, so $w' \notin S_{\leq}(A)_w$ which is a contradiction. Hence $S_{\leq}(B)_w = \emptyset$ which again is a contradiction, so $S_{\leq}(A)_w = S_{\leq}(B)_w$. ⊣

Now we can prove the following result.

Theorem 1.31 *If S is an ARRC semantics satisfying the ARL assumption then S is a class selection semantics satisfying FID, FMOD and FMP'. If S is a relative closeness semantics S also satisfies FCSO, FCV' and FCA'.*

Proof. Assume that S is an ARRC semantics for I satisfying the ARL assumption. For $w \in I$ and $A \in L$ define

$$f(A, w) = S_{\leq}(A)_w.$$

f is a class selection function. Let S_f be the class selection semantics defined in terms of f. We show that $S_{\leq} = S_f$. Assuming that $S_{\leq}(A) = S_f(A)$ and $S_{\leq}(B) = S_f(B)$ we show that $S_{\leq}(A \mathrel{\Box\!\!\rightarrow} B) = S_f(A \mathrel{\Box\!\!\rightarrow} B)$, i.e. that

$$f(A, w) \subseteq S_f(B) \text{ iff } S_{\leq}(A) \cap R_w = \emptyset \text{ or there is } w' \in S_{\leq}(A) \text{ such that for all } w'' \leq_w^A w',\ w'' \in S_{\leq}(A \rightarrow B).$$

Now $f(A, w) \subseteq S_f(B)$ iff $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ iff $S_{\leq}(A)_w = \emptyset$ or there is $w' \in S_{\leq}(A)_w$ and $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ iff $S_{\leq}(A)_w = \emptyset$ or there is $w' \in S_{\leq}(A)_w$ such that for all $w'' \leq_w^A w'$, either $w'' \notin S_{\leq}(A)$ or $w'' \in S_{\leq}(B)$ (by Lemma 1.27) iff $S_{\leq}(A)_w = \emptyset$ or there is $w' \in S_{\leq}(A)_w$ such that for all $w'' \leq_w^A w'$, $w'' \in S_{\leq}(A \rightarrow B)$, and so by Lemma 1.28 and Lemma 1.29 we have proved that $S_{\leq}(A \mathrel{\Box\!\!\rightarrow} B) = S_f(A \mathrel{\Box\!\!\rightarrow} B)$, and hence that $S_{\leq} = S_f$.

Since $S_{\leq}(A)_w \subseteq S_{\leq}(A)$, $f(A, w) \subseteq S_{\leq}(A)$ so f satisfies FID. Assume that $f(\neg A, w) \subseteq S_f(A)$, i.e., that $S_{\leq}(\neg A)_w \subseteq S_f(A)$. Then $S_{\leq}(\neg A)_w = \emptyset$, so $S_{\leq}(\neg A) \cap R_w = \emptyset$, hence $R_w \subseteq S_{\leq}(A) = S_f(A)$. But $S_{\leq}(B)_w \subseteq R_w$, and hence $f(B, w) \subseteq S_f(A)$ so f satisfies FMOD. Finally, assume that $w \models_{S_f} A$. Since there is no $w' \in I$ such that $w' <_w w$, trivially $w \in S_{\leq}(A)_w$, i.e., $w \in f(A, w)$, so f satisfies FMP'.

Now assume in addition that $S_{\leq}$ is a relative closeness semantics, i.e., that for all $w \in I$ and $A, B \in L$, $\leq_w^A = \leq_w^B$. Assume that $f(A, w) \subseteq S_f(B)$ and $f(B, w) \subseteq S_f(A)$, i.e., that $S_{\leq}(B)_w \subseteq S_{\leq}(A)$ and $S_{\leq}(A)_w \subseteq S_{\leq}(B)$. Then by Lemma 1.30, $S_{\leq}(A)_w = S_{\leq}(B)_w$, so FCSO is satisfied by f. Next, if $f(A, w) \cap S_f(B) \neq \emptyset$, i.e., $S_{\leq}(A)_w \cap S_{\leq}(B) \neq \emptyset$, then $S_{\leq}(A \wedge B)_w \subseteq S_{\leq}(A)_w$. For assume that $w' \in S_{\leq}(A \wedge B)_w$ and for contradiction that $w' \notin S_{\leq}(A)_w$. Take any $w'' \in S_{\leq}(A)_w \cap S_{\leq}(B)$. Then (by Lemma 1.27) $w'' <_w^A w'$ and so $w'' <_w^{A \wedge B} w'$ so $w'' \notin S_{\leq}(A \wedge B)$, which contradicts the assumptions. Hence $f(A \wedge B, w) \subseteq f(A, w)$ so FCV' is satisfied.

$S_{\leq}(A \vee B)_w \subseteq S_{\leq}(A \vee B) = S_{\leq}(A) \cup S_{\leq}(B)$. Assume that $w' \in S_{\leq}(A \vee B)_w$. Then $w' \in S_{\leq}(A)$ or $w' \in S_{\leq}(B)$. Assume the the former. Now if $w'' <_w^{A \vee B} w'$ then $w'' \notin S_{\leq}(A \vee B)$. Since $<^{A \vee B} = <^A$, if $w'' <_w^A w'$ then $w'' \notin S_{\leq}(A)$, hence $w' \in S_{\leq}(A)_w$. Similarly if $w' \in S_{\leq}(B)$. Hence $S_{\leq}(A \vee B)_w \subseteq S_{\leq}(A)_w \cup S_{\leq}(B)_w$ so FCA' is satisfied. $\dashv$

So, if an ARRC semantics satisfies the ARL assumption we can always define a class selection function in terms of it and so get an equivalent class selection semantics that satisfies FID, FMOD and FMP', (and even FCSO, FCV' and FCA' if we start with a relative closeness semantics).

By Prop. 1.6 and Theorem. 1.31 we get

Proposition 1.32 *If S is an ARRC semantics satisfying the ARL assumption then S satisfies GCP.*

The requirements FID, FMOD, FCSO, FCV' and FMP', respectively, guarantee that certain axioms are valid in class selection semantics satisfying those requirements. However, we need not make a detour via class selection semantics in this way in order to state what axioms are valid in ARRC semantics:

Theorem 1.33 *If S is an ARRC semantics then S satisfies CP and all instances of the axiom schemata ID, MOD', MOD and MP are valid in S. If S is a relative closeness semantics also CSO, CV and CA are valid in S.*

Proof. Assume that $S_{\leq}$ is an ARRC semantics for some I. By Prop. 1.26, S satisfies CP.

ID Since $S_{\leq}(A \to A) = I$, if for some $w \in I$, $S_{\leq}(A) \cap R_w \neq \emptyset$ then there is $w' \in S_{\leq}(A)$ such that for all $w'' \leq_w^A w'$, $w'' \in S_{\leq}(A \to A)$, so $w \models_{S_{\leq}} A \mathrel{\Box\!\!\to} A$, and hence $\models_{S_{\leq}} A \mathrel{\Box\!\!\to} A$.

MOD' Assume for some $w \in I$ that $w \models_{S_{\leq}} A \mathrel{\Box\!\!\to} \neg A$. Then $S_{\leq}(A) \cap R_w = \emptyset$. Assume that $w' \in S_{\leq}(B) \cap R_w$. Then if $w'' \leq_w^B w'$, $w'' \in R_w \subseteq S_{\leq}(\neg A)$ so $w'' \in S_{\leq}(B \to \neg A)$, and

hence $w \models_{S_\leq} B \mathrel{\Box\!\!\rightarrow} \neg A$. Thus $\models_{S_\leq} (A \mathrel{\Box\!\!\rightarrow} \neg A) \rightarrow (B \mathrel{\Box\!\!\rightarrow} \neg A)$.

MOD This is shown as for MOD'.

MP Assume for some $w \in I$ that $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} B$, and further that $w \models_{S_\leq} A$. Then $S_\leq(A) \cap R_w \neq \emptyset$, so there is $w' \in S_\leq(A)$ such that for all $w'' \leq_w w'$, $w'' \in S_\leq(A \rightarrow B)$. Since $w \leq_w w'$, $w \in S_\leq(A \rightarrow B)$, so $w \models_{S_\leq} B$, and hence $w \models_{S_\leq} A \rightarrow B$.

Assume now that $S_\leq$ is a relative closeness semantics, i.e., that for all $A, B \in L$, $\leq^A = \leq^B$. (We simply write $\leq$ for any of those relations.)

CSO Assume for some $w \in I$ that $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} B$ and $w \models_{S_\leq} B \mathrel{\Box\!\!\rightarrow} A$. If $S_\leq(A) \cap R_w = \emptyset$ and $S_\leq(B) \cap R_w = \emptyset$ then trivially for any $C \in L$, $w \models_{S_\leq} (A \mathrel{\Box\!\!\rightarrow} C) \wedge (B \mathrel{\Box\!\!\rightarrow} C)$, so $w \models_{S_\leq} (A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C)$. Assume instead that $S_\leq(A) \cap R_w \neq \emptyset$. Then, since $\leq$ is connected, it is easy to see that there is $w_0 \in S_\leq(A) \cap S_\leq(B) \cap R_w$ such that for all $w'' \leq_w w_0$, $w'' \in S_\leq(A \leftrightarrow B)$. If there is $w_1 \in S_\leq(A)$ such that for all $w'' \leq_w w_1$, $w'' \in S_\leq(A \rightarrow C)$ then there is $w_2 \in S_\leq(A)$ such that for all $w'' \leq_w w_2$, $w'' \in S_\leq(B \rightarrow C)$ (simply let w_2 be the closest of w_0 and w_1) and also vice versa, so $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} C$ iff $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} C$, and hence $\models_{S_\leq} CSO$.

CV Assume for some $w \in I$ that $w \not\models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} \neg B$. Then $S_\leq(A) \cap R_w \neq \emptyset$. If $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} C$ then there is w_0 such that if $w'' \leq_w w_0$ then $w'' \in S_\leq(A \rightarrow C)$. Since $w \not\models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} \neg B$ there is some $w_1 \leq_w w_0$ such that $w_1 \in S_\leq(A \wedge B)$. Now if $w'' \leq_w w_1$ then $w'' \in S_\leq(A \rightarrow C)$, and hence $w'' \in S_\leq((A \wedge B) \rightarrow C)$, so $w \models_{S_\leq} (A \wedge B) \mathrel{\Box\!\!\rightarrow} C$.

CA Assume for some $w \in I$ that $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} C$ and $w \models_{S_\leq} B \mathrel{\Box\!\!\rightarrow} C$.

If $S(A \vee B) \cap R_w = \emptyset$ then trivially $w \models_{S_\leq} (A \vee B) \mathrel{\Box\!\!\rightarrow} C$, so we assume that $S_\leq(A \vee B) \cap R_w \neq \emptyset$.

1. If $S_\leq(A) \cap R_w \neq \emptyset$ and $S(B) \cap R_w = \emptyset$ then there is $w_1 \in S_\leq(A) \cap R_w$ such that for all $w'' \leq_w w_1$, $w'' \in$

$S_{\leq}(A \to C) \cap S_{\leq}(\neg B)$, and hence $w'' \in S_{\leq}((A \vee B) \to C)$, so $w \models_{S_{\leq}} (A \vee B) \mathrel{\Box\!\!\to} C$.

2. If also $S(B) \cap R_w \neq \emptyset$ then there is $w_1 \in S_{\leq}(A) \cap R_w$ such that for all $w'' \leq_w w_1$, $w'' \in S_{\leq}(A \to C)$, and there is $w_2 \in S_{\leq}(B) \cap R_w$ such that for all $w'' \leq_w w_2$, $w'' \in S_{\leq}(B \to C)$. Let w_3 be the closest to w of w_1 and w_2. $w_3 \in S_{\leq}(A \vee B)$ and if $w'' \leq_w w_3$ then $w'' \in S_{\leq}(A \to C)$ and $w'' \in S_{\leq}(B \to C)$ and hence $w'' \in S_{\leq}((A \vee B) \to C)$, so $w \models_{S_{\leq}} (A \vee B) \mathrel{\Box\!\!\to} C$.

$\dashv$

Class selection is often intuitively understood in terms of similarity:[50] The class selection function should pick out those worlds most similar to the actual one in which the antecedent is true. If we understand similarity in terms of rc-assignments (i.e. the relation of similarity is not dependent on which subjunctive conditional is being evaluated) then CSO, RCV and CV are forced upon us. The example on p. 53 above gave us reason to question the validity of RCV (and also of CV). If we on the other hand understand the relation of similarity to (partially) depend on which hypothetical assumption is being made, i.e., we use an antecedent relative similarity relation between worlds, then neither RCV nor CV is necessarily valid.

Below we shall discuss the relationship between similarity and time. We shall view possible worlds as having a temporal dimension. There is a discussion of subjunctive conditionals and time in [Lewis, 1973b, Sec. 3.3]. [Bennett, 1974] further elaborates on this theme, suggesting theories of similarity that are antecedent relative in the sense that only similarities, and dissimilarities, up to [p. 398] (or alternatively only at [p. 400][51]) the point in time to which the antecedent pertains are to be reckoned with, when comparing worlds. [Thomason and Gupta, 1978] provides a formal framework for such semantics. They favor a theory of the first kind, i.e., one in

[50] As for instance by [Stalnaker, 1968b, p. 45].

[51] This latter theory is further elaborated and defended in [Bennett, 1984], and then retracted (in favor of one similar to the one of [Lewis, 1979]) in his forthcoming [Bennett, 1998].

which only similarities and dissimilarities up to the time at which the antecedent pertains should count when comparing worlds. They call this principle of comparison "Past Predominance".[52] I shall argue later that although this is on the right track, it cannot be the whole story: Some, but not all, later similarities and dissimilarities should also count. Anyway, the semantics ofBennett, and Thomason and Gupta, yield ARRC semantics, since whenever different antecedents pertain to different points in time they will determine different relations of similarity, since different sets of similarities and dissimilarities should be counted (and ignored) in each case.[53]

[52]For further examples of past predominance theory see [Davis, 1979] and[Gibbard, 1978, pp. 227–8] As we shall see in Section 2.4.6 below [Lewis, 1979] arrives at a criterion of similarity yielding very much the same result (by utilizing the notions of a small and a big miracle) without relativizing similarity to the time of the antecedent.

[53]Also in[Pollock, 1981, p. 250] an antecedent-relative closeness relation is used for evaluating conditionals. This closeness relation is defined in terms of the notion of "undercutting". I shall discuss this at length in Section 2.8 below.

Chapter 2

Closeness and Time

Our world has a temporal dimension, and so normally do the worlds we should consider when evaluating subjunctive conditionals. The closeness relation that relative closeness semantics make use of is usually understood in terms of similarity: The worlds we should consider when evaluating the conditional $A \mathrel{\Box\!\!\rightarrow} C$ are the A-worlds most similar to our world. There is of course a general problem of weighing similarities against one another, that we have mentioned before. Now we shall consider the special question whether there is any systematic relationship between the temporal relations between events in a certain world and the relative weight that should be accorded to them when comparing this world to other worlds.

2.1 Historical Conservativity

A relatively early discussion of this topic is found in [Downing, 1959], where the author argues that the following principle,

$$\text{(EXP)}\ (B \wedge ((B \wedge A) \mathrel{\Box\!\!\rightarrow} C)) \rightarrow (A \mathrel{\Box\!\!\rightarrow} C),$$

called *the principle of exportation*, is valid provided that "[B] refers only to times earlier to those referred to in [A], or to the earliest time referred to in [A]" [p. 135]. EXP is not generally valid, as can be seen in the following example.

> Betty went to a party yesterday, but would have left immediately, had Albert shown up. Now, Albert has a good time if and only if Betty is present, so had both Betty and Albert been at the party then Albert would have enjoyed himself, but as we said, had he appeared, Betty would have left, so in that case he would not have had a good time.

This example illustrates that even if B is actually the case, we cannot generally conclude that it would have been the case had A been the case. However, we need to assume this in order to infer $A \mathrel{\Box\!\!\rightarrow} C$ from $B \wedge ((B \wedge A) \mathrel{\Box\!\!\rightarrow} C)$. In effect, Downing argues that in case the time of reference for B precedes that of A we can safely assume this, and therefore EXP is valid in these cases. He uses the following example, that has become famous, in support of this (I reword it a bit for simplicity):

> Jim and Jack quarreled yesterday. Jack will not forgive easily so had Jim asked Jack for help today then Jack would not have helped him. On the other hand Jim is a proud fellow and would only have asked for help had they not quarreled earlier. Hence had Jim asked Jack for help today, then they would not have quarreled yesterday, and so Jack, being normally quite generous, would have helped him.

Downing accepts the first line of argument that relies on EXP, and rejects the second. Indeed, it seems Downing is quite right in rejecting the second line of argument from an intuitive point of view. To take some other examples:

> Mr. and Mrs. Smith have been happily married for some years. If Mrs. Smith had filed for a divorce right now, she would have expressed an inexplicable wish to end a happy marriage, not a quite natural one to end an unhappy one, even though she would only have filed for divorce had their marriage been unhappy.

and

> Mr. Miller has \$ 1.000 on his bank account, and this is all he got. Had he withdrawn \$ 1.000 and spent it on women and horse-race then he would have had nothing left, even though he would only have done this had he been in possession of at least \$ 10.000

To argue (with his relatively non-frivolous character in mind) that had he withdrawn \$ 1.000 and spent it, he would have had at least \$ 9.000 left indeed seems to be a mistake. In these three examples EXP applies since (according to Downing's account) "[B] refers only to times earlier to those referred to in [A], or to the earliest time referred to in [A]".

We can modify the first of the last two examples in a way that indicates that it is indeed the time-relation that decides whether EXP should apply or not:

> Mr. and Mrs. Smith have been happily married for some years. If Mrs. Smith had filed for a divorce six months ago, an unhappy marriage would soon have been ended, (since, of course, their marriage would have become unhappy as a result of that very act) and not a happy one, notwithstanding the fact that their marriage is in fact happy.

so EXP does not apply, *and indeed, in this case B refers to times after the time to which A pertains.*

Downing also offers a modified version of his example [p. 129–130] (my rewording):

> Jim and Jack will quarrel at mid-day, so if Jim would have asked Jack for help this morning he would not have received help this afternoon. On the other hand, had Jim asked for help this morning Jim would not have quarreled with Jack at mid-day so he would have received help from generous Jack.

Here the first line of argument, that relies on EXP, seems invalid as it stands; again, in this case, B refers to times later than A does.

If ID (p. 39) and CP (p. 50) are satisfied, EXP implies the following *principle of conservativity*:

(CONS) $B \rightarrow (A \mathrel{\Box\!\!\rightarrow} B)$.

Downing thought that EXP should apply if "[B] refers only to times earlier to those referred to in [A], or to the earliest time referred to in [A]". We can now see that at least the second part of this must be a mistake: If A is counterfactual and refers only to a moment in time then $\neg A$ is true and refers to the same moment, but then Downing's theory would yield the undesired result $A \mathrel{\Box\!\!\rightarrow} \neg A$, and hence all such counterfactual suppositions would be subjunctively impossible (see p. 158 below). If we skip the last part of the requirement and say that EXP should be valid as long as B refers only to times earlier than those referred to in A we get a principle that I shall call *historical conservativity*.

2.1.1 Modified Historical Conservativity

The examples above certainly seem to support this principle, but I think one can also easily construct examples that tend to speak against it:

> I am sitting in my house and the snow has been falling all morning. In fact there are no footprints in the snow around my house, but had my friend appeared at my front door right now then there would have been footprints in the snow.

One can easily imagine circumstances in which this line of reasoning seems perfectly all right. Nevertheless if Downing is right it cannot be, it seems:

> If my friend had shown up at my front door right now and the footprints had not been there a short moment ago they would not have been there now either.

This line also seems all right, but by historical conservativity it can bee seen to contradict the previous one. Since B (that there

were no footprints up until now) pertains to a time earlier than A (that my friend shows up at my front door) historical conservativity implies that $A \mathrel{\Box\!\!\rightarrow} B$ is true, i.e. that had my friend shown up right now there would not have been any footprints up until now, and so $A \mathrel{\Box\!\!\rightarrow} C$ (Had my friend shown up at my front door now then there would not have been any footprints now) is true.[1] Let's look at another perhaps more convincing example of Jonathan Bennett's.[2] Bennett invites us to consider the following subjunctive conditional: "If the German army had reached Moscow in August 1941, they would have taken it." (We shall refer to this conditional by 'August $\mathrel{\Box\!\!\rightarrow}$ Capture'), and offers the following background facts:

> On June 22, 1941, Hitler invaded the Soviet Union; on December 6 of that year his troops reached the suburbs of Moscow but got no further, beaten back by Soviet armies and the Russian winter.

Bennett goes on to describe what would have been the situation in August 1941, had the Germans reached Moscow then, according to historical conservativity.

> "Here [...] is the situation in August 1941 [had the Germans reached Moscow then]: The German troops reach Moscow, and otherwise things in Russia are pretty much as they [in fact] were [...]—the layout of the Moscow streets, the weather, the durable aspects of the German and Russian national characters, the small number of Soviet troops (most of them fresh and inexperienced) in the immediate vicinity of Moscow,... Stop! Something has gone wrong here! Most of the listed similarities are all right, but not that last one. [In fact] Moscow did indeed contain few defenders in August 1941, but we will not let that fact into our evaluation of [August $\mathrel{\Box\!\!\rightarrow}$ Capture]. Most of us won't, anyway

[1] $A \mathrel{\Box\!\!\rightarrow} B$ together with $(A \wedge B) \mathrel{\Box\!\!\rightarrow} C$ yields $A \mathrel{\Box\!\!\rightarrow} C$ (by RT, p. 55).

[2] In his forthcoming paper [1998]. There are further examples in the literature with a similar import like e.g. in [Bowie, 1979, p. 487–8], [Weiner, 1979, p. 504–5] and [Davis, 1980].

> [...] most people will assume that if Hitler's troops had assaulted Moscow in August 1941 there would have been more Soviet troops in the vicinity than there actually were, and most would be battle-weary. The Soviet armies did in fact spend that summer and autumn falling back before the advancing Germans. If the Germans had reached Moscow by August, the retreating Russians would have been crowded back into its vicinity also."

If these intuitions are reliable, historical conservativity cannot be entirely correct.[3] What is then the difference between the examples that speak in favor and those that speak against historical conservativity? In the particular examples that we have been considering it seems to be the intuitive difference between what is merely psychologically incomprehensible, and what is outright physically impossible: We take it to be psychologically incomprehensible, but not physically impossible, that Jim should ask Jack for help given that they quarreled yesterday, that Mrs. Smith should file for a divorce given the happy state of their marriage, or that Mr. Miller should spend his last $ 1.000 on women and horse-race, but physically impossible that a German advance on Moscow should have gone unnoticed, or that my friend should have reached my front door leaving no trace behind him.

Perhaps we should try and modify Downing's original criterion to take account of this intuitive difference in some way. There are problems here, of course: Under determinism every counterfactual antecedent represents a physical impossibility when combined with all that was true before the time to which this antecedent pertains. Still we have a way of intuitively distinguishing what is impossible (like unnoticed German advances on Moscow, and traceless approaches by friends) from what is not. It seems that intuitive differences such as this one are built into the methods we use when

[3] Since it is seems true that had the Germans reached Moscow some day in August 1941, and the day before the number of defenders had been what it actually was (viz. very small), then at the day of the arrival the defenders would have been very few also.

deciding what facts to be conservative about and what facts to vary when deciding which worlds to consider.

We are now in a position to formulate a tentative principle of *modified historical conservativity*: If the set of past facts are, in an intuitive sense, physically compatible with the antecedent then EXP should be valid with regard to each of these facts, i.e., each of those facts should be conserved in the worlds we should consider for the purpose of evaluating conditionals with the antecedent in question. If these intuitions of what is physically compatible with the antecedent are incorrect then we will consider words that are in fact impossible, from the point of view of the physical laws of our world.

On the other hand, if the set of past facts are intuitively incompatible with the antecedent we cannot simply say that EXP should apply only to those past facts that does not seem to be incompatible with the antecedent, because what is compatible depends upon what other facts we choose to conserve: We choose to conserve the fact that our friend is unable to glide around in mid-air, or the fact that snow retains footprints, and abandon the fact that there were no footprints in the snow, rather than the other way round. Hence there seems to be a problem in cases like this of choosing which past facts to conserve and which to abandon when choosing worlds to consider when evaluating conditionals. This is, of course a special case of a more general problem that we have touched upon before (p. 14 above).

2.2 Historically Possible Facts and Events

In order to approach a solution to this problem it seems to me to be fruitful to think in terms of an evolving situation, and historical possibility. I shall give a couple of examples of historical possibilities in this sense: I sit in my home writing about historical possibility right now. However, I could have decided early this morning to go to my office at the University instead (at least I like to think I could), so I could have been there instead of here right now. Hitler could have decided to take on the Russians first instead of the Yugoslavians, and in that case perhaps the Germans could have reached Moscow

earlier than they in fact did.

The common denominator among these examples is the conception of the world being exactly as it really was up to a certain point in time. At that point something is imagined to have happened that (so we think) could have happened but never did: I decide to go to the University, which I did not do, Hitler decides to take on the Russians first, which he in fact did not do. These alternative events are what is historically possible in a narrow sense. Had the historically possible but unrealized event been realized then subsequent to that event the world would have been different from the way it actually is, at least in some respects. These differences represent unrealized facts[4] and events that are historically possible in a broader sense.

If determinism is true there are no historically possible facts or events in this sense. However, I think it is evident that our semantical intuitions are not on the side of determinism. For all we know, determinism may of course be true anyway. In that case, when we speak about unrealized events we speak about situations that involve violations of the laws that obtain at our world. Such violations we shall, following [Lewis, 1973b, pp. 75–76] call *miracles* from the point of view of the laws of our world. In order to facilitate the discussion I shall introduce a simple set theoretical apparatus which can be used to describe worlds as temporally extended objects, and their relations to each other.

2.3 Possible Worlds and Time

It seems natural to represent possible worlds in their temporal dimension (*temporal worlds*) by means of lines, each consisting of a continuum of points, each of which represents the world in question at a certain moment in time. We call those points *world-points*, or simply wp's. Each such line w is defined by an ordered pair

[4]To talk about *unrealized facts* may seem a contradiction in terms. I shall however, in order to avoid cumbersome expressions like "particular state of affairs" and the like, talk loosely about unrealized facts, merely possible facts, and facts of a (merely) possible world. If something is the case at a certain possible world w then I shall say that that is a fact of (or at) that world. When I speak of facts *simpliciter* I shall mean facts of the actual world.

$\langle D_w, <_w \rangle$ such that D_w is the set of wp's of the line in question and $<_w$ is the binary relation among the elements of D_w of being "to the left of" on the line in question. Let I_T be the set of temporal worlds. If $p_1 <_w p_2$ we say that p_1 is earlier than p_2 (and that p_2 is later than p_1) in w. For $p_1, p_2 \in D_w$ we write $p_1 \leq_w p_2$ instead of $p_2 \not<_w p_1$. If $p \in D_w$ it will sometimes be convenient to talk of the part of w that ends in p. We will refer to this initial segment of the temporal world w by w_p which we define thus: w_p is the ordered pair $\langle D_{w_p}, <_{w_p} \rangle$ such that $D_{w_p} = \{p' \in D_w : p' \leq_w p\}$ and $<_{w_p} = <_w \cap D^2_{w_p}$. Such initial segments of temporal worlds we shall call *subhistories* of that temporal world. For each temporal world w let $Sh(w)$ be the set of subhistories of w. Obviously

Proposition 2.1 *If $h, h' \in Sh(w)$ then $D_h \subseteq D_{h'}$ or $D_{h'} \subseteq D_h$, so $D_h \cap D_{h'} \neq \emptyset$.*

We assume that the same world point may be part of different temporal worlds, and say that two temporal worlds *overlap* if they share points in this way. If two temporal worlds overlap in such a way that they have a subhistory in common then we say that those temporal worlds are *co-commenced*. I.e., w and w' are co-commenced iff $Sh(w) \cap Sh(w') \neq \emptyset$.

It follows from the definition of subhistory that

Proposition 2.2 *If $h \in Sh(w) \cap Sh(w')$ and $p, p' \in D_h$, then $p \leq_w p'$ iff $p \leq_{w'} p'$.*

Now it is obvious that

Proposition 2.3

1. *If w and w' are temporal worlds, $D_{h'} \subseteq D_h$, and $h \in Sh(w) \cap Sh(w')$, then $h' \in Sh(w) \cap Sh(w')$.*
2. *If w and w' are temporal worlds, $p' \leq_w p$, and $w_p \in Sh(w) \cap Sh(w')$, then $w_{p'} \in Sh(w) \cap Sh(w')$.*

We show that

Proposition 2.4 *If w, w' and w'' are temporal worlds then either $Sh(w) \cap Sh(w'') \subseteq Sh(w) \cap Sh(w')$ or $Sh(w) \cap Sh(w') \subseteq Sh(w) \cap Sh(w'')$.*

Proof. Assume that $h \in Sh(w) \cap Sh(w'')$ and $h' \in Sh(w) \cap Sh(w')$. Then by Prop. 2.1, $D_h \subseteq D_{h'}$ or $D_{h'} \subseteq D_h$. If the former then, by Prop. 2.3, $h \in Sh(w) \cap Sh(w')$; if the latter, $h' \in Sh(w) \cap Sh(w'')$, This proves the proposition. ⊣

Proposition 2.5 *The relation of being co-commenced is reflexive, symmetrical and transitive.*

Proof.

1. This relation is obviously reflexive and symmetrical.

2. Assume that $h \in Sh(w) \cap Sh(w')$ and $h' \in Sh(w) \cap Sh(w'')$. By Prop. 2.1, $D_h \cap D_{h'} \neq \emptyset$. Let $d \in D_h \cap D_{h'}$. Then w'_d is a subhistory of both w and w'' so the relation is transitive.

⊣

Some temporal worlds will be such that they may be said to obey laws. In some cases two different temporal worlds may obey the same (set of) laws. We may say that a set of laws is *deterministic* iff every two temporal worlds that obey this set of laws are either identical or do not overlap (cf. the discussion in Sections 2.3.2 and 2.4.5).

We need a way of talking about time, different points of time in a world, time order and duration, in different worlds. There are lots of problems here: Talk about duration only seems to make sense with reference to a certain set of laws; so does it really make sense to compare periods of time between worlds that don't obey the same laws? Can we meaningfully refer to a point in time across worlds? Could we for instance have two temporal worlds w and w' otherwise identical except for the fact that everything in w' happens two seconds later than the corresponding event in w? It seems that reference to points in time across possible worlds presupposes some common point of reference at those different worlds and, of course, some common measure of duration. There is also the problem of defining contemporaneousness within a world if relativistic

laws prevail. We shall sidestep these problems, and associate with any temporal world w an embedding t_w from w into a fixed time-line, which we assume to be (isomorphic to) the line of real numbers. For any world point p of any temporal world w, t_w associates a certain time $t_w(p)$.

We have seen (Prop. 2.2) that if $h \in Sh(w) \cap Sh(w')$ then for all $d, d' \in D_h = D_{h'}$, $p \leq_w p'$ iff $p \leq_{w'} p'$. So we can safely put the following natural additional requirement on this function: If $h \in Sh(w) \cap Sh(w')$ then

$$\text{for all } d \in D_h,\ t_w(d) = t_{w'}(d).$$

I.e., if two worlds have a subhistory in common the wp's in this subhistory are accorded times in the same way by the time functions of both histories.

Also, if $P \subseteq D_w$ we let

$$t_w(P) = \{t_w(p) : p \in P\}.$$

So obviously,

$$\text{if } h \in Sh(w) \cap Sh(w'),\ t_w(D_h) = t_{w'}(D_h).$$

Hence if $h \in Sh(w)$ for some $w \in I_T$, we can define

$$t(h) = t_w(D_h).$$

If H is a set of subhistories we let

$$t(H) = \bigcup \{t(h) : h \in H\}.$$

Proposition 2.6 *If H, H' are sets of subhistories then $t(H) \subseteq t(H')$ whenever $H \subseteq H'$.*

Proof. Assume that $H \subseteq H'$ and that $t \in t(H)$. Then there is $h \in H$ such that $t \in t(h)$, but also $h \in H'$, so $t \in t(H')$. ⊣

Proposition 2.7 *If $\{H_i : i \in I\}$ is a set of sets of subhistories then*

$$t(\bigcup_{i \in I} H_i) = \bigcup_{i \in I} t(H_i).$$

Proof. $t \in t(\bigcup_{i\in I} H_i)$ iff $t \in \bigcup\{t(h) : h \in \bigcup_{i\in I} H_i\}$ iff there is $h \in \bigcup_{i\in I} H_i$ such that $t \in t(h)$ iff there is $i \in I$ and subhistory h such that $h \in H_i$ and $t \in t(h)$ iff there is $i \in I$ such that $t \in \bigcup\{t(h) : h \in H\}$ iff there is $i \in I$ such that $t \in t(H_i)$ iff $t \in \bigcup_{i\in I} t(H_i)$. ⊣

Proposition 2.8 *For any temporal world w, $t(Sh(w)) = t_w(D_w)$.*

Proof. $t \in t(Sh(w))$ iff $t \in \bigcup\{t(h) : h \in Sh(w)\}$ iff $t \in t(h)$ for some $h \in Sh(w)$ iff $t \in t_w(D_h)$ for some $h \in Sh(w)$ iff $t \in t_w(D_w)$. ⊣

We shall now relate this to the semantical framework that was introduced in Section 1.5 above. We want to be able to say not only that a sentence A is true at a certain world but also that a sentence pertains to a certain temporal part of that world. There are complexities and potential problems at this point: Some sentences seem to pertain only to one temporal point in a world whereas others pertain to several points. Some again seem to pertain to temporal stretches.[5] What about complex sentences built up from sentences that pertain to different points or stretches of a world? We shall simply say that there is a function T from L into the set of sets of times, indicating the time associated with each sentence of L.[6] It seems natural to require the following of this function:

[5] I shall leave open what times should be associated with logical or mathematical truths. One natural alternative is to associate the entire time-line, another is to associate the empty set with them.

[6] One consequence of this is that every sentence will be associated with the same time(s) at all worlds. This is of course not adequate as a final analysis. Many sentences are such that this requirement becomes impossible to implement. Take the sentence "Shortly after Hitler's death there was peace". At our world this sentence seems to be about some stretch of time in the second or third quarter of 1945, but what about a world at which Hitler died already in 1943? When fixing the time reference of such sentences the *de re*/*de dicto* distinction seems to come into play. If we interpret "Shortly after Hitler's death" *de re* then the time reference is not altered. However if it is interpreted *de dicto* then it might very well be altered. Nevertheless this idealization seems to be sufficient for the present discussion, and it certainly makes things a lot easier if we can simply talk of the time of A, without reference to a certain world.

1. $T(\neg A) = T(A)$
2. $T(A \wedge B) = T(A) \cup T(B)$
3. $T(A \mathrel{\Box\!\!\rightarrow} B) = T(A) \cup T(B)$.

I.e., a sentence pertains to the same temporal part of a world as its negation does, a conjunction pertains to those temporal parts of a world that its conjuncts do, and a conditional pertains to those temporal parts that its antecedent and consequent do.

We shall say that a semantics S for I_T is *temporally adequate* with respect to $A \in L$ if

> $w \models_S A$ iff $w' \models_S A$ whenever $T(A) \subseteq t(Sh(w) \cap Sh(w'))$.

That is, if two worlds are co-commenced and a certain sentence A only pertains to such temporal parts that they have in common then this sentence has the same truth value in each of those worlds.

We show that

Lemma 2.9 *If S is temporally adequate with respect to the atomic sentences then it is temporally adequate with respect to all sentences not containing $\Box\!\!\rightarrow$.*

Proof. We show this by recursion on the complexity of formulas. Assume that S is temporally adequate with respect to $A, B \in L$.

1. If $T(\neg A) \subseteq t(Sh(w) \cap Sh(w'))$, then $T(A) \subseteq t(Sh(w) \cap Sh(w'))$, and hence, since S is temporally adequate with respect to A, $w \models_S A$ iff $w' \models_S A$, so $w \models_S \neg A$ iff $w' \models_S \neg A$. Thus S is temporally adequate with respect to $\neg A$.

2. Assume that $T(A \wedge B) \subseteq t(Sh(w) \cap Sh(w'))$. Then $T(A) \subseteq t(Sh(w) \cap Sh(w'))$ and $T(B) \subseteq t(Sh(w) \cap Sh(w'))$, and hence, since S is temporally adequate with respect to A, B, $w \models_S A$ iff $w' \models_S A$ and $w \models_S B$ iff $w' \models_S B$, and so $w \models_S A \wedge B$ iff $w' \models_S A \wedge B$. So S is temporally adequate with respect to $A \wedge B$.

$\dashv$

2.3.1 Genuine Historical Alternatives

We say that two temporal worlds are *genuine historical alternatives* (to each other) if they obey the same set of laws and are co-commenced.[7]

The point of this definition is that it gives us a way of defining the concept of historical possibility. A *historical possibility* is simply some event or fact occurring in a temporal world that is a genuine historical alternative to this temporal world.

> A is *S-historically possible* at w if there is a genuine historical alternative w' to w such that $w' \models_S A$.

2.3.2 Felonious Historical Alternatives

As we observed above it may be the case that we take something to be a historical possibility even though it is not. In that case we are considering worlds that are perhaps co-commenced with the actual world but not obeying the same laws as it. Not every such world will do. If determinism is true, then, since I stayed home today, I could not—in a strict sense of "could"—have gone to the University this morning. In fact I stayed home, so the preceding facts together with the prevailing laws precluded anything other than that. If we are considering a world that is exactly as the actual one up to the point at which I decided to stay home, but at which I decided to go, then we have to allow for a miracle at that world, at that point in time, i.e., a violation, at *that world*, of the laws of *this world*. However, we do not wish that world to behave unlawfully after the point of (unlawful) departure from the actual world; rather we want it to obey the laws of the actual world from that point on. We should not be interested in worlds at which I, having decided to go to the University, make a psychokinetic trip there, possibly making it true that had I decided to go to the University I might had traveled there by telekinesis. We should only be interested in worlds at which, after the miracle, things behave according to the laws of this world. So what we are looking for is a temporal world that is co-commenced

[7] Of course if the laws are deterministic then the only genuine historical alternative to a world is the world itself.

with the actual world, at which there occurs some localized miracle (with respect to the laws of the actual world) at the point of departure from it, but which otherwise obeys the laws of the actual world.[8] I suspect there might be big trouble lurching behind seemingly innocuous phrases like "localized miracle" and "otherwise obeying laws", and it seems hard to provide formal definitions in this area. Anyway I shall call such worlds *felonious historical alternatives* to a certain world. Note that even though the relation of being a genuine historical alternative is symmetrical, w' may be a felonious historical alternative to w even though w is not a felonious historical alternative to w'. We don't even know whether the concept of natural law is meaningfully applicable to any description of what goes on at the world w'. It obeys the laws of w up to a certain point in time, it *almost* obeys these laws at that point and obeys them *entirely* thereafter.[9] Perhaps if one tried to formulate laws for w' one would have to make reference to the specific time at which the miracle occurs, and in that case one would, it seems, be justified in questioning whether such descriptions of what happens in w' qualify as laws.

In order not to get bogged down in these problems, we shall for a while pretend (or assume) that determinism is not true and that things could in a strict sense have turned out otherwise than they did, just as many of our linguistic practices seem to indicate.

2.4 The Criterion of Late Departure

2.4.1 Simple Late Departure

Let I be any subset of I_T. For each $w \in I$ let R_w be the subset of I of temporal worlds that are co-commenced with w.

We state the following simple criterion of closeness for elements of I: If $w, w', w'' \in I$, then $w' \leq_w w''$ iff

$$Sh(w) \cap Sh(w'') \subseteq Sh(w) \cap Sh(w').$$

[8] For thorough discussions of this topic see [Lewis, 1979], [Bennett, 1984] and [Bennett, 1998].

[9] As I indicated above, I'm not sure what sense, if any, the emphasized worlds have when qualifying obedience to laws.

This means that w' is as close to w as w'' is iff all the subhistories that w shares with w'' it also shares with w'. In this situation we will say that w' does not *fork off* earlier from w than w'' does. We shall call this criterion of closeness *Simple Late Departure* (*SLD-closeness*, for short), and we shall now show that it results in a certain relative closeness semantics that is temporally adequate with respect to all elements of L. First we show that

Proposition 2.10 *If $\leq$ is the relation of SLD-closeness then $\leq$ is an rc-assignment.*

Proof. Assume that $w \in I$.

1. Assume that $w', w'', w''' \in I$, $Sh(w) \cap Sh(w'') \subseteq Sh(w) \cap Sh(w')$ and $Sh(w) \cap Sh(w''') \subseteq Sh(w) \cap Sh(w'')$. Then $Sh(w) \cap Sh(w''') \subseteq Sh(w) \cap Sh(w')$, so $\leq_w$ is transitive.

2. By Prop. 2.4, $\leq_w$ is strongly connected.

3. w is co-commenced with itself, so $w \in R_w$. Hence w is self accessible.

4. Obviously, for every w', $Sh(w) \cap Sh(w') \subseteq Sh(w) \cap Sh(w)$, so $w \leq_w w'$; hence w is w-minimal.

5. Assume that $w' \notin R_w$. Then $Sh(w) \cap Sh(w') = \emptyset$ and hence for every $w'' \in I$, $Sh(w) \cap Sh(w') \subseteq Sh(w) \cap Sh(w'')$, i.e., $w'' \leq_w w'$.

6. Assume that $w' \in R_w$ and $w'' \notin R_w$. Then $Sh(w) \cap Sh(w') \not\subseteq Sh(w) \cap Sh(w'')$, i.e., $w'' \not\leq_w w'$, i.e., $w' <_w w''$.

$\dashv$

Lemma 2.11 *If S is a relative closeness semantics defined in terms of SLD-closeness and S is temporally adequate with respect to $A, B \in L$, then it is temporally adequate with respect to $A \Box\!\!\rightarrow B$.*

Proof. Assume that S is a relative closeness semantics defined in terms of SLD-closeness that is temporally adequate with respect to $A, B \in L$, and $T(A \mathrel{\Box\!\!\rightarrow} B) \subseteq t(Sh(w) \cap Sh(w'))$. Then $T(A) \subseteq t(Sh(w) \cap Sh(w'))$ and $T(B) \subseteq t(Sh(w) \cap Sh(w'))$. Now either $w \models_S A$ or $w \not\models_S A$.

1. Assume first that $w \models_S A$. Then $w' \models_S A$. Also, $w \models_S B$ iff $w' \models_S B$. Furthermore if $w'' \leq_w w$ then $T(B) \subseteq t(Sh(w) \cap Sh(w'')) = t(Sh(w))$, hence $w'' \models_S B$ iff $w \models_S B$, and likewise if $w'' \leq_{w'} w'$ then $T(B) \subseteq t(Sh(w') \cap Sh(w'')) = t(Sh(w'))$. Thus we see from Def. 1.14 and the assumptions of the case that $w \models_S A \mathrel{\Box\!\!\rightarrow} B$ iff $w' \models_S A \mathrel{\Box\!\!\rightarrow} B$.

2. Assume instead that $w \not\models_S A$. Then $w' \not\models_S A$. Assume also that $w \models_S A \mathrel{\Box\!\!\rightarrow} B$. Since w and w' are co-commenced, $R_w = R_{w'}$ (Prop. 2.5), so $S(A) \cap R_w = \emptyset$ iff $S(A) \cap R_{w'} = \emptyset$. We assume therefore that there is $w_0 \in S(A) \cap R_w$ such that if $w'' \leq_w w_0$ then $w'' \in S(A \rightarrow B)$. Also $w_0 \in S(A) \cap R_{w'}$. Assume that $w'' \leq_{w'} w_0$, i.e., that $Sh(w') \cap Sh(w_0) \subseteq Sh(w') \cap Sh(w'')$. Assume also that $w'' \in S(A)$ (otherwise trivially $w'' \in S(A \rightarrow B)$). Then $T(A) \not\subseteq t(Sh(w') \cap Sh(w''))$. Since $T(A) \subseteq t(Sh(w) \cap Sh(w'))$, $t(Sh(w) \cap Sh(w')) \not\subseteq t(Sh(w') \cap Sh(w''))$ and hence by Prop. 2.6, $Sh(w) \cap Sh(w') \not\subseteq Sh(w') \cap Sh(w'')$. Thus by Prop. 2.4, $Sh(w') \cap Sh(w'') \subseteq Sh(w) \cap Sh(w')$. Since also $T(A) \not\subseteq t(Sh(w) \cap Sh(w''))$, similarly $Sh(w) \cap Sh(w'') \subseteq Sh(w) \cap Sh(w')$. And since $w_0 \in S(A)$, similarly $Sh(w') \cap Sh(w_0) \subseteq Sh(w) \cap Sh(w')$ and $Sh(w) \cap Sh(w_0) \subseteq Sh(w) \cap Sh(w')$. Hence $Sh(w') \cap Sh(w'') = Sh(w) \cap Sh(w'')$ and $Sh(w') \cap Sh(w_0) = Sh(w) \cap Sh(w_0)$, from which it follows that $Sh(w) \cap Sh(w_0) \subseteq Sh(w) \cap Sh(w'')$, i.e., $w'' \leq_w w_0$, and hence by the assumption $w'' \in S(A \rightarrow B)$. Thus, $w' \models_S A \mathrel{\Box\!\!\rightarrow} B$. The other direction is similar.

$\dashv$

By Lemmas 2.9 and 2.11 we have by recursion on the complexity of formulas:

Theorem 2.12 *If S is a relative closeness semantics defined in terms of SLD-closeness then S is temporally adequate with respect all elements of L.*

Now, let I_L be the set of temporal worlds that obey the same laws as the actual world. If $w \in I_L$ we say that w is *lawful*. Let S_L be the relative closeness semantics for I_L defined by the relation of SLD-closeness.

It is reasonable to assume for $w, w' \in I_L$ that if $Sh(w) \subseteq Sh(w')$ then $w = w'$, i.e., that if w and w' obey the laws of the actual world and are different but co-commenced then they diverge at some point. Under this assumption, for every $w' \neq w$, $Sh(w) \cap Sh(w') \subset Sh(w) \cap Sh(w)$, so $w <_w w'$, hence w is *strictly* w-minimal. This means that (under this assumption) the relative closeness semantics defined in terms of this rc-assignment will not only support the logic VW but also the axiom CS, which means that it supports the logic VC which is Lewis's official logic.[10] This is the content of the next

Proposition 2.13 *If $\leq$ is an rc-assignment for some I such that for all distinct $w, w' \in I$, $w <_w w'$, then for all $A, B \in L$, $\models_{S_\leq} (A \wedge B) \rightarrow (A \mathrel{\Box\!\!\rightarrow} B)$. I.e., axiom CS is valid in $S_\leq$.*

Proof. Assume that $\leq$ is an rc-assignment for some I such that for all distinct $w, w' \in I$, $w <_w w'$, and that for some $w \in I$, $w \models_{S_\leq} A \wedge B$. Then if $w' \leq_w w$, $w' = w$, so $w' \models_{S_\leq} A \rightarrow B$, and hence $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} B$. $\dashv$

2.4.2 S_L and Modified Historical Conservativity

I will use S_L as a point of reference and departure in our discussions on subjunctive conditionals and time. First I shall investigate whether S_L is compatible with the principle of modified historical conservativity that was put forward tentatively on p. 71. This principle states, we recall, that if the set of facts that obtain earlier than the time to which the antecedent pertains are physically

[10] For definitions of axioms and conditional logics see Sections 3.1.3 and 3.1.4 below.

compatible with the truth of the antecedent, then EXP should be valid with regard to these facts. Let us make the assumption that intuitive physical compatibility reflects real physical compatibility in the sense that if an antecedent A is intuitively physically compatible with all facts obtaining at times earlier than the time of A, then there is a lawful temporal world, co-commenced with the actual world, which forks off from the actual world at the (beginning of) the time of A and at which A is true. Given this assumption modified historical conservativity seems to obtain according to S_L. Let us try to show this more formally.

Let T be a set of times and $Sh_T(w)$ be the set of subhistories of w that end before all the times of T. We define a relation $Comp$ between sentences, sets of times and lawful worlds such that if $A \in L$, T is a set of times, and $w \in I_L$ then $Comp(A, T, w)$ iff

> there is $w' \in I_L$ such that $w' \models_{S_L} A$ and $Sh_T(w) \subseteq Sh(w')$.

Given this definition, if w is the actual world, $Comp(A, T(A), w)$ says that the occurrence of A is physically compatible with all facts that obtain earlier than the time of A. It is now easy to see that modified historical conservativity obtains in these cases, but first we obtain a more general result.

If T and T' are sets of times we say that T is *earlier* than T' iff $t < t'$ whenever $t \in T$ and $t' \in T'$.

Proposition 2.14 *If T is a set of times, $Comp(A, T, w)$, $w \models_{S_L} B$, and $T(B) \subseteq t_w(D_w)$ is earlier than T, then $w \models_{S_L} A \mathrel{\Box\!\!\rightarrow} B$.*

Proof. Assume that $Comp(A, T, w)$, $w \models_{S_L} B$, and $T(B) \subseteq t_w(D_w)$ is earlier than T. We also assume that $R_w \cap S(A) \neq \emptyset$ (since otherwise trivially $w \models_{S_L} A \mathrel{\Box\!\!\rightarrow} B$).

Since $Comp(A, T, w)$ holds there is w_0 such that $w_0 \models_{S_L} A$ and $Sh_T(w) \subseteq Sh(w_0)$, and hence $Sh_T(w) \subseteq Sh(w_0) \cap Sh(w)$. Assume that $w' \leq_w w_0$, i.e, $Sh(w_0) \cap Sh(w) \subseteq Sh(w') \cap Sh(w)$. Hence $Sh_T(w) \subseteq Sh(w') \cap Sh(w)$. If $t \in T(B)$ then $t \in t_w(D_w)$, so there is $p \in D_w$ such that $t = t_w(p)$, but then $w_p \in Sh_T(w)$ and hence $t \in t(Sh_T(w))$. So $T(B) \subseteq t(Sh_T(w))$. Since $Sh_T(w) \subseteq$

$Sh(w_0) \cap Sh(w)$ we have $t(Sh_T(w)) \subseteq t(Sh(w_0) \cap Sh(w))$ (by Prop. 2.6), so $T(B) \subseteq t(Sh(w') \cap Sh(w))$. Since S_L is temporally adequate with respect to B, $w' \models_{S_L} B$, so $w' \models_{S_L} A \rightarrow B$, and hence $w \models_{S_L} A \mathrel{\Box\!\!\rightarrow} B$. ⊣

A corollary of this is the following

Proposition 2.15 *If $Comp(A, T, w)$ and $T(B) \subseteq t_w(D_w)$ is earlier than T, then $w \models_{S_L} (B \wedge ((B \wedge A) \mathrel{\Box\!\!\rightarrow} C)) \rightarrow (A \mathrel{\Box\!\!\rightarrow} C)$.*

Proof. Assume that $Comp(A, T, w)$ and $T(B)$ is earlier than T, and that $w \models_{S_L} B \wedge ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$. By Prop. 2.14, $w \models_{S_L} A \mathrel{\Box\!\!\rightarrow} B$, and by RT $w \models_{S_L} A \mathrel{\Box\!\!\rightarrow} C$. ⊣

A special case of Prop. 2.14 is of course

Proposition 2.16 *If $Comp(A, T(A), w)$, $w \models_{S_L} B$ and $T(B) \subseteq t_w(D_w)$ is earlier than $T(A)$, then $w \models_{S_L} A \mathrel{\Box\!\!\rightarrow} B$.*

This says that if the occurrence of A is physically compatible with all facts that precede the time of A then if B expresses one of those facts then $A \mathrel{\Box\!\!\rightarrow} B$ is true according to S_L, which is a counterpart of the principle of modified historical conservativity. However, Prop. 2.14 suggests a generalization of this principle, that we shall consider presently.

2.4.3 S_L and Refined Historical Conservativity

We call the following the principle of *refined historical conservativity*:

> If there is a time t such that the occurrence of A is physically compatible with all facts obtaining prior to t then if B expresses one of those facts, $A \mathrel{\Box\!\!\rightarrow} B$ is true.

Going back to our examples we can see that this seems to be a fairly reasonable suggestion. It was not physically possible for the Germans to reach Moscow unnoticed, or in no time at all, so

during some time before August 1941, not all facts are physically compatible with them reaching Moscow in August, but if there is a time, say at the end of March 1941, such that before that time, all facts are compatible with the counterfactual assumption we are considering, then indeed it seems reasonable to say with respect to those facts that they would have obtained even if the Germans had reached Moscow in August that year: If the Germans had reached Moscow in August 1941, Hitler would still have been elected Reichskanzler in 1933. He would still have invaded Poland, Austria, Bohemia. The Kristallnacht would still have occurred, etc.

2.4.4 Problems with Late Departure

The principle of refined historical conservativity, even if correct, is obviously of very limited applicability. Most conditionals do not have consequent clauses that refer to times at which all facts are compatible with the antecedent, so even if this principle reflects some of our intuitions we are a long way from a comprehensive theory of subjunctive conditionals. However there are problems with this principle and consequently with the criterion of simple late departure. In [Nute, 1980, p. 104] an example of John Pollock's bearing on this topic is discussed: "Suppose I left my coat in a classroom yesterday and was surprised to find it right where I left it today. Consider the antecedent, 'If my coat had been missing when I returned' ". We assume that during the time the coat was hanging in the classroom many people at various times had the opportunity and actually could have taken the coat. If we state this in terms of our SLD-closeness semantics it could be the case that we had several different temporal worlds departing from the actual one at different times within the interval during which the coat was hanging unattended in the classroom. At each of those worlds the coat is picked up by a different person a short while after its (the world's) departure from the actual world. Now, SLD-closeness will pick out the last such opportunity. If John was the last person who could have picked up the coat, being alone in the classroom only ten minutes before I found the coat, then a world in which John takes it may well be the SLD-closest world in which the coat is missing

when I return. SLD-closeness will therefore favor conditionals like, 'If my coat had been missing then John would have been the culprit (and not Lucy, Bob etc, who was in the classroom at different times before John)'. This of course is not a desirable outcome from an intuitive point of view. Instead in this case, it seems, we have to consider a number of different possible culprits, acting at different times, and not just the last one, and hence some worlds that depart from the actual one before the departure of the 'John takes it'-world should be regarded as equally close to the actual world as this one.

Several theorists have endorsed some version of closeness involving late departure[11] that is vulnerable to criticism of this kind. There are indeed considerations that seem to speak in favor of some version of late departure, so we shall return to this topic later, after we have discussed some issues pertaining to the question whether similarity after the time of departure should count or not.

2.4.5 Temporal Asymmetry

In his [1973b, pp. 72–77], Lewis touches on the subject of time when discussing how much weight should be given to similarities of particular fact between worlds as opposed to similarities with respect to what laws obtain. Lewis assumes for the sake of argument that the laws of this world are deterministic. He takes determinism to have the consequence that, if two worlds obey the same deterministic laws then they are either qualitatively identical throughout all of their history, or they are not qualitatively identical throughout any stretch of history. Now, what antecedent-worlds should be considered to be closest to the actual world? Are they worlds that obey the same deterministic laws as the actual world and consequently differ with respect to particular fact from the actual world throughout all of history? or perhaps worlds that allow for the truth of the antecedent and at the same time at least partially concurs with the actual world with respect to particular fact? Such worlds would have to contain "facts" that violate the laws of the actual world. Such violations Lewis calls *miracles*. He does not rule out the possibil-

[11] See for instance [Lewis, 1979], [Thomason and Gupta, 1978], and forthcoming [Bennett, 1998].

ity that we (should) regard antecedent-worlds with miracles in this sense as closer to the actual world than antecedent-worlds without miracles, as long as the miracles make room for widespread agreement with respect to particular fact. However, he does rule out the occurrence of miracles after the time of the antecedent, so the miracles that might be allowed must occur at the time of the antecedent, or before it. The reasons for this asymmetry should be fairly clear. Although Lewis doesn't appear to provide an argument for it, it seems his intuitions are on the side of some version of historical conservativity, and this, given determinism, requires miracles before or at the time of any counterfactual antecedent. On the other hand, the worlds we should consider will be different throughout all of time subsequent to the counterfactual antecedent, because the antecedent being true will have all sorts of consequences propagating throughout all of future time, and we don't want to annul those by allowing miracles after the time of the antecedent, since those counterfactual consequences are often the principal point of interest when making counterfactual assumptions.

This asymmetry may seem somewhat arbitrary from the point of view of similarity. Why should a world diverging form the actual world before the time of the antecedent be thought of as being more similar to the actual world than a world converging to the actual after the time of the antecedent, or for that matter, a world first diverging from the actual in order to allow for the antecedent and then converging to the actual after the antecedent? In other words, why should a divergence miracle[12] count less toward dissimilarity than non-identity with respect to particular fact throughout all of history, when a convergence miracle counts more than non-identity throughout all of the future? Lewis speculates [p. 76] that it might be "just [a] brute fact that we put more weight on earlier similarities of particular fact than on later ones", but he also tries to explain this asymmetry in terms of differences of comprehensiveness between divergence and convergence miracles respectively. Lewis's example is of the counterfactual assumption that the roulette wheel had stopped on red):

[12]The fairly self-explicatory terms *divergence miracle* and *convergence miracle* are used by Bennett in [1984, p. 62].

> But perhaps my standards are less discriminatory than they seem. For some reason—something to do with *de facto* or nomological asymmetries of time that prevail at i if i is a world something like ours—it seems to take less of a miracle to give us an antecedent-world exactly like i in the past than it does to give us one exactly like i in the future. For the first, all we need is one little miraculous shove, applied to the wheel at the right moment. For the second we need much more. All kinds of traces of the wheel's having stopped on red must be falsified. The rest position of the wheel; the distribution of light, heat and sound in the vicinity; the memories of the spectators—all must be changed to bring about a reconvergence of particular fact between the antecedent-world and i. One shove will not do; many of the laws of i must be violated in many ways at many places.

This way of dealing with the asymmetry in question is further elaborated in [Lewis, 1979]. In [Lewis, 1973b] it seems that Lewis wants to establish that worlds diverging from the actual world are in some intuitive sense more similar to the actual world than worlds that converge to the actual. This intuitive understanding of relative similarity is what explains our semantical judgments with regard to subjunctive conditionals, i.e., it seems he treats our intuitions of similarity (perhaps strengthened by argument) as primitive and our semantical judgments as derived from these. In his later work [1979] it instead seems that he regards our semantical judgments, or perhaps our intuitions of truth or falsity, as primitive and on the basis of these he attempts to determine how to weigh different respects of intuitive similarity against one another in order to arrive at a notion of comparative closeness that—given his relative closeness semantics—accords with those semantical intuitions.

It thus appears that the purpose of Lewis's project became somewhat redefined underway. It started as an attempt to *explain* our semantical intuitions in terms of our intuitions of similarity and a certain semantic theory formulated in terms of similarity—at least

others took that to be his intention[13]—and it ended up (much more defensively, it seems to me) in [Lewis, 1979] as an attempt to show that there is some relation of closeness that yields the intuitively right truth conditions by means of the theory. Here the requirement that closeness should be interpreted as intuitive similarity seems to have all but vanished. We shall return to this later discussion below, but it is important to see this redirection in order to understand why much early criticism of Lewis's theory was directed at the alleged unituitiveness of the similarity relations that must be assumed for Lewis's semantics to yield the intuitively correct answers, and how Lewis managed to sidestep this criticism by describing the purpose of his theory in a new way.

In [Bennett, 1974, Sect. 8], and [Fine, 1975, p. 452] one important form of such early criticism is put forward. This criticism relies on the fact that the relation of similarity must be asymmetric in the way discussed above, favoring divergence over convergence. Bennett first questions that our standards of similarity favor absence of big and widespread miracles over identity with regard to particular facts over long (perhaps infinite) stretches of time. Then he goes on to show that Lewis really needs to say that absence of even a tiny miracle is more worth than close similarity for all future time. Referring to Lewis's discussion in [Lewis, 1973b, pp. 71–72] Bennett considers the two conditionals:

(9) If Oswald had not killed Kennedy, someone else would have.

(10) If Oswald had not killed him, Kennedy would not have been killed.

Bennett assumes for the purpose of the discussion that the Warrenite contention that Oswald killed Kennedy without help and that there was no second killer waiting is correct. I shall quote from Bennett's discussion [p. 395] at some length because of its instructiveness:

> We Warrenites are inclined to deny [9] and to affirm [10], and so is Lewis. This commits him to something that is obviously false [...]. The truth of [10],

[13]See for instance [Bennett, 1974], and [Fine, 1975].

> by Lewis's theory, requires that some worlds in which no one kills Kennedy are more like the actual world than is *any* world in which Kennedy is killed by someone other than Oswald. [...T]his is incredible. Any possible world where Kennedy is not killed must differ enormously from the actual world, because the political and other consequences of that killing were so large and ramifying. On the other hand there is a possible world in which Kennedy is killed not by Oswald but by some other disaffected Dallas resident: that world differs from ours in certain events before the assassination, and in some of the legal and journalistic upshots; but those differences will be tiny when compared, by any reasonable standard, with the differences involved in Kennedy's not being killed at all. [...] This seems to me obvious, and yet Lewis denies it: "The worlds where Oswald did kill Kennedy, working alone, with no other killer waiting...are worlds to which worlds with no killing are closer than worlds with a different killer" [Lewis [1973b, p. 71]].

(We may imagine that the failure by Oswald to kill Kennedy as well as the killing of Kennedy by someone else are brought about by tiny miracles: a slip of Oswald's finger, a neuronal event in some Dallas policeman's brain causing him to fire at Kennedy.)

Lewis has already said that perfect match with respect to particular facts for infinite time (in one direction) is worth more than absence of a tiny miracle; now, this example of Bennett's seems to force Lewis to accept that close similarity with respect to particular facts for infinite time is not worth more than the absence of a tiny miracle. If somebody else had killed Kennedy then the future from that point in time might have been similar to the actual future, but not entirely identical. We shall see that in [1979] Lewis will stress this distinction and make it pivotal, but surely, in that case we have left intuitive similarity behind, as Lewis also seems to acknowledge.

2.4.6 Lewis's Weights and Priorities

Lewis's point of departure when discussing which weight should be given to different kinds of dissimilarity in [1979, pp. 43–48] is a similar example by Kit Fine [1975, p. 452]:

> The counterfactual "If Nixon had pressed the button there would have been a nuclear holocaust" is true or can be imagined to be so. Now suppose that there never will be a nuclear holocaust. Then that counterfactual is, on Lewis's analysis, very likely false. For given any world in which antecedent and consequent are both true it will be easy to imagine a closer world in which the antecedent is true and the consequent false. For we need only imagine a change that prevents the holocaust but that does not require such a great divergence from reality."

As in Bennett's example we have a subjunctive conditional that can be imagined to be true, with a consequent clause the truth of which would make a great difference to what is actually the case, and as in Bennett's example it seems that the antecedent-worlds that are most similar to the actual world, in an intuitive sense, falsify the conditional in question. In this case we need only imagine a world in which Nixon presses the button but in which the signal from the button dies out so there is no subsequent nuclear holocaust. The task Lewis sets himself is to define a relation of resultant similarity in terms of different respects of similarity weighed with respect to each other in a way that ensures that all such worlds will be weeded out from the ones most similar to the actual world. This will, according to Lewis still be "a relation of overall similarity—not because it is likely to guide our explicit judgments of similarity, but rather because it is a resultant, under some system of weights and priorities, of a multitude of relations of similarity in particular respects" [p. 43].

He assumes determinism and invites us to consider five different worlds w_0, w_1, w_2, w_3 and w_4. w_0 is a world that may or may not be the actual world at which Nixon does not press the button and at which there is no nuclear holocaust. w_0 is also "a world that

fits our worst fantasies of the button: there is such a button, it is connected to a fully automatic command and control system, the wired-in war plan consist of one big salvo, everything is in faultless working order, there is no way for anybody to stop the attack, and so on". Now, given that all this is the case at w_0, Lewis agrees that had Nixon pressed the button then there would have been a nuclear holocaust.

The worlds at which Nixon presses the button at t, that are worthy of consideration when evaluating the conditional in question fall in four different classes typified by the worlds w_1, w_2, w_3 and w_4 respectively. w_1 is like w_0 until "[s]hortly before t". It diverges from w_0 by means of a "tiny", "simple, localized" and "inconspicuous" divergence miracle (with respect to the deterministic laws obeyed by w_0). "Perhaps a few extra neurons fire in some corner of Nixon's brain", and as a result of this, Nixon presses the button. No more miracle occurs at w_1, so the nuclear holocaust takes place and as far as the surface of this earth is concerned w_0 and w_1 are very dissimilar from that time onwards.

w_2 is a miracle-free world obeying the laws of w_0, at which Nixon presses the button. Since the laws of w_0 are assumed to be deterministic, w_0 and w_2 do not coincide at all with respect to particular fact. Lewis believes that what I have called the principle of historical conservativity is correct: "[I]f $[T(C)]$ is earlier than $[T(A)]$, then such counterfactuals [i.e. $A \mathrel{\Box\!\!\rightarrow} C$] are true if and only if C is true" [p. 35].[14] Therefore, obviously, he would like his similarity relation to prefer w_1 to w_2. w_1 contains a tiny miracle (which should count toward dissimilarity) but matches w_0 perfectly with respect to particular fact for a very long stretch of time (which of course should count toward similarity), whereas w_2 is miracle-free (which should count toward similarity) but lacks perfect match with w_0 with respect to particular fact (which should count toward dissimilarity).

[14] Although he says this, which seems to indicate that he subscribes to an unqualified version of historical conservativity, he also at places speak of a "transition period" from the time of the miracle to the time of the antecedent during which conservativity is not valid, so it seems he actually subscribes to a version of historical conservativity similar to what I have called refined historical conservativity (p. 84).

In order for the similarity relation to prefer w_1 to w_2, perfect match for a considerable period of time should count more toward similarity than absence of miracles, in Lewis words: "a lot of perfect match is worth a little miracle".

w_3 starts out just like w_1, it—together with w_1—forks off from w_0 by a tiny divergence miracle, then just after Nixon has pressed the button another tiny miracle occurs that prevents Nixon's action from taking effect, the "fatal signal" gets derailed in some way so no rockets go off and there is no nuclear holocaust. We assume that w_3 and w_0 "remain very closely similar in matters of particular fact". However they are not identical after t. "Nixon's deed has left its mark on the world w_3".

> There are his fingerprints on the button. Nixon is still trembling, wondering what went wrong—or right. His gin bottle is depleted. The click of the button has been preserved on tape. Light waves that flew out the window, bearing the image of Nixon's's finger on the button, are still on their way into outer space. The wire is ever so slightly warmed where the signal current passed through it, And so on, and on, and on. The differences between w_3 and w_0 are many and varied, although none of them amounts to much.

The similarity relation we are seeking must prefer w_1 to w_3. w_1 has the advantage of containing only one tiny miracle whereas w_3 contains two. On the other hand w_3, but not w_1, is closely similar to w_0 for a considerable stretch of time after t. So absence of a miracle should count more toward similarity than a considerable period of close similarity with respect to particular fact: "close but approximate match of particular fact [...] is not worth even a little miracle".

Finally, the world w_4 begins like w_1 and w_3, but after Nixon has pressed the button at t there is a "widespread and complicated and diverse second miracle".

> It not only prevents the holocaust but also removes all traces of Nixon's button-pressing. [...T]he fatal signal

> vanishes, just as at w_3, but there is much more. The fingerprint vanishes, and the sweat returns to Nixon's fingertip. Nixon's nerves are soothed, his memories are falsified, and so he feels no need of the extra martini. The click on the tape is replaced by innocent noises. The receding light waves cease to bear their incriminating images. The wire cools down, and not by heating its surrounding in the ordinary way. And so on, and on, and on. Not only are there no traces that any human detective could read; in every detail of particular fact, however minute, it is just as if the button-pressing had never been.

In other worlds there is a convergence miracle at w_4 shortly after t, accomplishing reconvergence with w_0, so these worlds are "exactly alike again soon after t, and exactly alike forevermore." As we saw this miracle is "spread out and diverse".

> Because there are many different sorts of traces to be removed, and because the traces spread out rapidly, the cover-up job divides into very many parts. Each part requires a miracle at least on par with the small miracle required to prevent the holocaust, or the one required to get the button pressed in the first place. Different sorts of unlawful processes are needed to remove different sorts of traces[.]

w_1 should be counted as more similar than w_4 to w_0. An extra very long (infinite?) stretch of perfect match with w_0 speaks in w_4's favor, but absence of widespread and diverse miracles speaks in w_1's favor. Hence absence of widespread and diverse miracles should count more toward similarity than long (even infinite) stretches of perfect match, i.e., "prefect match of particular fact even through the entire future is not worth a big, widespread, diverse miracle."

Now Lewis finds himself in a position to formulate the following "system of weights and priorities" that should govern the similarity relation that together with his relative closeness semantics yield

correct truth conditions for subjunctive conditionals such as the one we have been considering [pp. 47–48]:

1. It is of the first importance to avoid big, widespread, diverse violations of law.
2. It is of the second importance to maximize the spatio-temporal region throughout which perfect match of particular fact prevails.
3. It is of the third importance to avoid even small, localized, simple violations of law.
4. It is of little or no importance to secure approximate similarity of particular fact, even in matters that concern us greatly.

1 has priority over 2 because w_4 should be counted as less similar than w_1 to w_0. 2 has priority over 3 because w_2 should be counted as less similar than w_1 to w_0, and 3 has priority over 4 because w_3 should be counted as less similar than w_1 to w_0.

2.4.7 Lewis's Criterion and Late Departure

Under determinism there are no genuine historical alternatives to the actual world. And of course w_1, which is the world favored by Lewis's system of weights and priorities, is instead a felonious historical alternative to w_0. The closeness relation defined by clauses 1–4 above does not define a version of SLD-closeness for several reasons. For instance, we cannot assume that (under the closeness relation defined by those clauses) there is some set I of temporal worlds such that for all $w, w', w'' \in I$, $w' \leq_w w''$ iff

$$Sh(w) \cap Sh(w'') \subseteq Sh(w) \cap Sh(w').$$

One problem is that we cannot assume that the relation of "being a felonious alternative to" is transitive. It may be the case that w' and w'' are felonious historical alternatives to some temporal world w and so should be included in R_w and therefore also in I, but, at the same time w'' is not a felonious historical alternative

to w'[15] and so should not be included in $R_{w'}$, and so R_w should not be identical to $R_{w'}$. But then R_w can't simply be the elements of I that are co-commenced with w. One might instead define R_w as the set of felonious historical alternatives to w. In this case we must also modify the definition of closeness in order to obtain an rc-assignment:

$w' \leq_w w''$ iff

$w'' \notin R_w$ or $[w' \in R_w$ and $Sh(w) \cap Sh(w'') \subseteq Sh(w) \cap Sh(w')]$.

It is easy to see that relative closeness defined in this way is indeed a species of rc-assignment. We call this *Felonious Late Departure* (*FLD-closeness*, for short). Let S_{FL} be the relative closeness semantics defined by this rc-assignment. One would like to be able to show that also this semantics is temporally adequate with respect to all elements of L. When proving that semantics defined in terms of SLD-closeness were temporally adequate (Lemma 2.11) we made critical use of the fact that under SLD-closeness $R_w = R_{w'}$ whenever $w' \in R_w$. This, as we have seen, we cannot assume with regard to FLD-closeness. I doubt that S_{FL} is temporally adequate but I have no proof of this conjecture.

One way to ensure temporal adequacy would be to assume indeterminism instead of determinism and only take genuine historical alternatives in consideration when evaluating conditionals. Lewis does this in his [Lewis, 1986b, Sect. D]. In this case we need not assume miracles in the worlds we are considering. Divergence is of course perfectly possible without miracles given indeterministic laws. Whether perfect convergence is possible under indeterminism seems to be an open question. It seems to depend on the particular nature or the indeterministic laws that obtain. It might be that we need to consider felonious alternatives even under indeterminism. Lawful divergence might be too sparse, for our semantical needs, under the indeterministic laws in question, but if we assume

[15] If w'' departs later from w than w' does, then it seems that w'' must contain two subsequent miracles with respect to the laws that obtain at w', the first at the point of departure from w' and the second at the point of departure from w. If this is the case w'' is not a felonious alternative to w'.

that we need only consider genuine historical alternatives when we evaluate conditionals then Lewis's system may be simplified thus:

1. It is of the first importance to maximize the spatio-temporal region throughout which perfect match of particular fact prevails.
2. It is of the second or no importance to secure approximate similarity of particular fact, even in matters that concern us greatly.

If miracle-free divergence, but not convergence, is possible (and sufficiently common), this system seems to yield a result similar to Lewis's original system under determinism. There is a miracle-free world w_1' diverging from the actual w_0' shortly before the time of the antecedent, at which Nixon presses the button and at which there is a subsequent nuclear holocaust. If the conditional in question is true at w_0 there should not be any miracle-free world w_3' also diverging from w_0' shortly before the time of the antecedent, at which Nixon presses the button and at which there is no subsequent nuclear holocaust, but even if there were worlds like w_3' more approximately similar to w_0' than worlds like w_1' are, we would not be forced to conclude that had Nixon pressed the button then there would not have been any nuclear holocaust, since we need not pay any attention to approximate similarity. Since we have assumed that miracle-free convergence is not possible there would not be any miracle free worlds like w_4' which diverges from w_0' at the same point as w_1' and w_3' but which reconverges (lawfully and perfectly) with w_0' after Nixon having pressed the button, so under this system worlds like w_1' seem to come out as the preferred ones.

Now, what if miracle-free reconvergence is possible? Lewis's attempts to deal with this possibility by means of the concept of a *quasi-miracle* [Lewis, 1986b, p. 60]. A quasi-miracle, although lawful, is a "remarkable coincidence". Lawful reconvergence would only be possible by means of such a quasi-miracle. Now quasi-miracles, although lawful, (might be seen to) detract from similarity as much as large miracles do because they "would be quite unlike the goings-on we take to be typical of our world". So even if there

are worlds like w'_4 with larger temporal regions of perfect match with w'_0 than w'_1 possesses, we may regard those as less similar to w'_0 than w'_1 is, since they contain quasi-miracles.

One problem, that Lewis mentions, with this suggestion is that conditionals like:

(11) If Nixon had pressed the button there would not have been any quasi-miracle.

seem to come out true, given this attempted solution to the problem. At the same time it seems that

(12) If Nixon had pressed the button there might have been a quasi-miracle.

is intuitively true. Under the normal "not-would-not" reading of might-counterfactuals, these both conditionals contradict each other and so cannot both be true. However, according to Lewis there is also a "would-be-possible" reading of might counterfactuals:

(13) If Nixon had pressed the button it would be that a quasi miracle is possible.

According to this reading of (12) it is compatible with (11) according to Lewis.

We shall not discuss this issue further. We turn instead to the question whether Lewis's system yields a version of FLD-closeness (or SLD-closeness, assuming indeterminism). The answer depends on what to make of clause 4 (2). If approximate similarity is of *no* importance then the closest A-worlds should simply be the alternatives (felonious or not, depending on the version) to the actual world at which A is true, that depart at the latest point from it, i.e. the FLD-closest (SLD-closest) A-worlds, so in this case Lewis's system seems to produce a species of FLD-closeness (SLD-closeness). If, however approximate similarity should count for something (i.e. it is of *little importance*) then we may have two alternatives w' and w'', to the actual world w, departing from it at the same point and thus equally close to it with respect to FLD-closeness (SLD-closeness). Yet w' might be more approximately similar (after the departure)

to w than w'' and thus, according to Lewis's system, closer to w than w'' is.

In either case Lewis's system seems to have the consequence that we should always only consider worlds that diverge (by a small miracle or lawfully) from the actual world at the latest possible point in time, and hence it will never be the case that we need to consider worlds that diverge at different points in time. This would seem to be at least problematic in the light of Pollock's example on p. 85 above.

2.5 Future Similarity

Barring large miracles or quasi-miracles, future similarity will always be approximate similarity at best, i.e., a world at which some counterfactual antecedent is true will not be perfectly similar to the actual one subsequent to the time of the antecedent if there is no widespread and diverse miracle, or a quasi-miracle, respectively.

2.5.1 Does Approximate Similarity Count?

As we have seen it is left open in clause 4 whether merely approximate similarity should count at all toward similarity. Lewis notes this and tells us that different cases come out differently and refers to [Tichý, 1976] and [Jackson, 1977] for examples that indicate that approximate similarity should not count at all, and to [Slote, 1978] for an example that goes the other way.[16] He says he would like to know why this is so. This is one version of what I regard as the central question of this thesis. We can modify Fine's example a little to get an example (similar to the one in [Tichý, 1976]) that seems to indicate that no weight should be given to approximate similarity. Imagine the following:

> At the actual world w_0 the infamous button was not in full working order at the time t we are considering, hence a certain firmness (in the act of pressing) would have been required to get it to work at all. Let w_p and

[16] This is the "coin tossing" example discussed on the next page.

> w_f be two worlds that diverge by tiny miracles from w_0 at the same time shortly before t—we imagine that different miraculous neuronal events at w_p and w_f respectively, give this outcome. At both worlds Nixon presses the button but at w_p he does this powerfully, the signal is off and there is a subsequent nuclear holocaust. At w_f however, Nixon only manages to apply a somewhat feeble push to the button and in this world there is no signal and no nuclear holocaust.

If approximate similarity is given any weight at all, then—with suitable additional assumptions like the one that Nixon did not press a second time at w_f—it seems that w_f is closer to w_0 than any world at which Nixon presses the button and there is a nuclear holocaust, which means that the conditional "If Nixon had pressed the button then there would not have been any nuclear holocaust" $N \mathrel{\Box\!\!\rightarrow} \neg H$ would come out true. The desired result, from an intuitive point of view, might very well instead be that there might have been and there might not have been a nuclear holocaust had Nixon pressed the button, i.e., that $N \mathrel{\Diamond\!\!\rightarrow} H$ and $N \mathrel{\Diamond\!\!\rightarrow} \neg H$ should come out true. However, if we don't give any weight to approximate similarity then we seem to get the right result, because then both w_p and w_f will be among the worlds most similar to w_0 at which Nixon presses the button at t. And since $w_p \models N \wedge H$ and $w_f \models N \wedge \neg H$ then $w_0 \not\models N \mathrel{\Box\!\!\rightarrow} \neg H$ and $w_0 \not\models N \mathrel{\Box\!\!\rightarrow} H$ and hence $w_0 \models (N \mathrel{\Diamond\!\!\rightarrow} H)$ and $w_0 \models (N \mathrel{\Diamond\!\!\rightarrow} \neg H)$.

On the other hand here is one example that seems to indicate that weight should after all be given to (some) approximate similarities [Bennett, 1984, p. 76].[17] This example seems to require that we assume indeterminism.

> At $[t_1]$ I bet that when the coin is tossed at $[t_2]$ it will come up heads; and in the upshot it does just that; but this is a purely chance event, with no causally sufficient prior conditions. Now consider the conditional "If I had bet on tails at $[t_1]$ I would have lost."

[17] A version of this important example is attributed to Sidney Morgenbesser in [Slote, 1978, footnote 33].

We symbolize this conditional with $T \mathrel{\Box\!\!\rightarrow} L$. Bennett reports that everyone he polled was "inclined to say that the conditional is true". We assume that there are lawful worlds that depart from w_0 (the actual world) shortly before t_1. Now, since at t_1 the outcome of the toss is not predetermined it seems that for every lawful $T \wedge L$-world w_1—at which I bet tails at t_1 and at t_2 the coin comes up heads and so I lose—there is another lawful world w_2 departing from w_0 at the same time as w_1 and at which I bet tails at t_1 and at t_2 the coin comes up tails, so I win, hence t_2 is a $T \wedge \neg L$-world. Now in order for $T \mathrel{\Box\!\!\rightarrow} L$ to be true at w_0 we must prefer worlds like w_1 to worlds like w_2 when comparing to w_0. w_1 and w_2 contain no miracles with respect to the laws of w_0 and they depart at the same time from w_0 and they will not coincide with w_0 after t so they have exactly the same amount of perfect match with w_0. Hence w_1 and w_2 will fare equally well with respect to clauses $1, 2$ and 3 above: Clauses 1 and 3 do not apply since no miracles occur at w_1 and w_2, and since both worlds match w_0 perfectly for the same stretch of time, they are also equal with respect to clause 2.

Why should we prefer worlds like w_1 to worlds like w_2?[18] Is it enough to say that approximate similarity should count in this case? It is true that w_1 is more similar to w_0 than w_2 is, in that at both w_0 and w_1, but not at w_2, the coin comes up heads. On the other hand at both w_0 and w_2, but not at w_1, I win. It thus seems

[18]It seems that Bennett [1984, p. 76] thinks he can get around this problem by the following modification: First divide the consequent, in this case "I would have lost" (L) into its constituent conjuncts, presumably "I would have bet on tails" (R) and "the coin would have come up heads" (S), then modify the criterion of truth in the following way: Replace "L is true at all the closest T-worlds", by "L is equivalent to the conjunction $R \wedge S$ such that R is true at all the closest worlds and S is true at some of those worlds and also at the actual world." (I have purged Bennett's account of some traits that seem to be inessential to the present discussion.) This suggestion appears to represent a major departure from the usual kinds of closeness criterion, and I must confess that I don't quite understand how to implement it. Unfortunately, Bennett doesn't discuss this suggestion further but says (in footnote 8) that this position is endorsed in his [1982, pp. 414–17]; however, I have not been able to find any endorsement, or argument for the theory in question on those pages. See however Section 2.7 below for a more recent, and in my view much more promising, attempt at a solution of this problem on the part of Bennett.

that we should count the first kind of agreement a similarity but not the second. Why is this so? Can we find any essential difference between the fact at w_1 that the coin comes up heads and the fact at w_2 that I win, which explains why agreement with respect to the first fact should count as a similarity but not the second.[19]

One obvious answer to this question is that agreement with respect to the final orientation of the coin should count as a similarity since this would not have been different had I betted differently, whereas the other agreement should be discounted since that would have been different. This is essentially a version of Goodman's cotenability criterion, which is plagued by circularity (see [Goodman, 1946, p. 16] and Section 2.8 below).

2.5.2 The Causal Trail Criterion

An alternative answer, that is not cast in terms of truth of subjunctive conditionals, is the following: That the coin comes up heads at w_1 should count towards similarity with w_0 because this is *not in the causal trail*[20] of me betting tails *at that world*. On the other hand that I win at w_2 should *not* count towards similarity with w_0

[19] Cf. the discussion in [Jackson, 1977, pp. 7,8]. Jackson asks a similar question at this point. He thinks his question can be answered by the criterion of closeness he presents [on p. 9]. According to this criterion the closest A-worlds are those that are identical to the actual world up to the time of the antecedent, as similar as possible to the actual at this point and lawful thereafter. For criticism of an aspect of this criterion see [Bennett, 1984, pp. 74–5]. Anyway, this criterion only seems plausible if determinism is assumed, which is not the case with respect to the present example. Incidentally, Jackson's criterion of closeness seems to be a species of antecedent relative relative closeness.

[20] The notion of *causal trail* will play an important role in what remains of this chapter; still, I shall not attempt to define it explicitly. Talk of a later event being in (or out of) the causal trail of some earlier event comes natural to me, and I hope (and trust) it will appear such to the reader too. I take it that we have an intuitive apprehension of such a relation between earlier and later events, and I shall maintain that we use distinctions between events defined in terms of this intuitive notion when evaluating subjunctive conditionals. The notion of causal trail should not be understood to require (or imply) determinism: My choosing a certain newspaper at the newsstand is clearly in the causal trail of me going there, which does not necessarily mean that my choosing this rather than another one was determined by some earlier events.

because this is *in the causal trail* of me betting tails *at that world.* We put this in negative terms thus: That the coin comes up tails at w_2 should count towards dissimilarity with w_0 because this is *not in the causal trail* of me betting tails *at that world.* On the other hand that I lose at w_1 should *not* count towards dissimilarity with w_0 because this is *in the causal trail* of me betting tails *at that world.*

It is possible to state this criterion in a more general way thus: If a conditional having A as antecedent is evaluated and A and C hold at w, then (intuitive) dissimilarity with respect to C should not count when assessing overall closeness iff

> the fact (at w) that C is in the causal trail of the fact (at w) that A.

I shall call this criterion the *causal trail criterion of similarity* (or simply the *CT-criterion*). We see immediately that this criterion is suitable for dealing with the previous example also: At w_p there is a nuclear holocaust so this world is very dissimilar in many respects to the actual world at which there was no such catastrophe, but all these dissimilarities are within the causal trail of Nixon pressing the button at that world, so those dissimilarities should not count towards dissimilarity, when evaluating the conditional in question, but of course dissimilarities not within the causal trail of Nixon's button-pressing should count: We should not consider worlds at which there are two moons orbiting earth, for example. Given this criterion it seems that w_p and w_f will be attributed the same degree of dissimilarity with respect to the actual world w_0, and both will be among the worlds closest to w_0, just as our semantical intuitions indicate.

On p. 53 above we presented a counterexample to the principle RCV. We shall now see what happens if we interpret this example in the light of the CT-criterion. We remember that there seemed to be two possibilities worthy of consideration, of both Albert and Betty going. One was that Betty went with Albert, the other was that Betty went on her own although Albert was also going. This means that under the closeness relation we are seeking there must be some world w_w where Betty goes with Albert and some world w_a where Betty goes alone among the $A \wedge B$-worlds closest to the actual when

evaluating the conditional $(A \wedge B) \mathrel{\Box\!\!\rightarrow} C$. On the other hand, w_w but not w_a should be among the closest A-worlds when $A \mathrel{\Box\!\!\rightarrow} B$ and $A \mathrel{\Box\!\!\rightarrow} C$ are evaluated. It is obvious that these requirements cannot be met by any single rc-assignment. When evaluating $A \mathrel{\Box\!\!\rightarrow} C$, w_w should be closer than w_a to w_0. When, on the other hand, evaluating $(A \wedge B) \mathrel{\Box\!\!\rightarrow} C$, w_w and w_a should both be on the same level of closeness with respect to w_0. This can now be seen to be quite natural given the CT-criterion (we assume that every fact at a world is in the causal trail of itself): When evaluating a conditional such as $(A \wedge B) \mathrel{\Box\!\!\rightarrow} C$, none of the facts that A or that B nor any of their consequences should count toward dissimilarity. On the other hand when evaluating e.g. $A \mathrel{\Box\!\!\rightarrow} C$ the fact that B (at a world) should indeed count toward dissimilarity if it is not in the causal trail of A. This is the case at w_a but not at w_w, so the fact that B at w_a but not at w_w should count toward dissimilarity and therefore w_w will come out closer than w_a to w_0 according to the CT-criterion.

2.5.3 Causal Trail and Late Departure.

One may combine the CT-criterion with other criteria of closeness like simple late departure, and hence use this criterion as an answer to the question left open in Lewis's fourth clause, i.e., as an answer to the question of when approximate similarity should count. This is also the way in which we have presented it. The CT-criterion would then be a way of choosing among the worlds that depart by small miracles or lawfully (I shall call such departures *smooth*) from the actual world at the latest possible point in time. As we have said, this would result in a version of late departure vulnerable to Pollock's counterexample on p. 85 above. E.g. if Lucy had the opportunity to take the coat, before John, we would not even consider the possibility that she took it, had it been missing, since there are worlds at which John takes the coat that depart smoothly from the actual later than the departure of any world at which Lucy takes it. This is, as we have said, a problem from an intuitive point of view. Therefore it would be desirable if the CT-criterion could be integrated into some other rule (than Lewis's) for deciding closeness. A rule that would respect our intuitions of historical conservativity,

but not carry the implication that we need only consider the latest departures.

In this case, it seems natural to say that we should not only consider one (or one type) of hypothetical situation, namely that of the coat missing, rather we should consider any of a number of different cases into which the hypothesis falls, so to speak, namely that of Lucy taking it at t_0, Bob taking it at t_1, and John taking it at t_2. I shall call such hypotheses *sub-cases*.[21] Only if the consequent is true at all the closest worlds representing those sub-cases then the counterfactual is true. We shall call conditionals with antecedents having subcases at a world w in this way, *divisible* at w, and those that don't, if any, *non-divisible* at w.[22]

If there are always ultimate sub-cases, then divisible conditionals seem to be analyzable in terms of non-divisible ones, the divisible ones being analyzed as conjuncts of non-divisible ones. If the antecedent "Someone took the coat" S is divisible into three different subcases "Lucy took the coat" L, "Bob took the coat" B and "John took the coat" J, it seems that for any C, $S \mathrel{\Box\!\!\rightarrow} C$ is true iff all of $L \mathrel{\Box\!\!\rightarrow} C$, $B \mathrel{\Box\!\!\rightarrow} C$ and $J \mathrel{\Box\!\!\rightarrow} C$ are true.[23]

We can put this more generally thus: An antecedent A is *divisible* at a world w if there are $A_1, \ldots A_n \in L$, such that $T(A_1), \ldots T(A_n) \subseteq T(A)$ and such that $A \mathrel{\Diamond\!\!\rightarrow} A_1, \ldots, A \mathrel{\Diamond\!\!\rightarrow} A_n$ are true at w and $A \mathrel{\Box\!\!\rightarrow} A_1, \ldots, A \mathrel{\Box\!\!\rightarrow} A_n$ are false at w. $A \wedge A_1, \ldots, A \wedge A_n$ are said to be *subcases* of A at w. If also $A \mathrel{\Box\!\!\rightarrow} A_1 \vee \ldots \vee A_n$ is true at w then $A \wedge A_1, \ldots, A \wedge A_n$ are all the subcases of A at w.

If $A \wedge A_1, \ldots, A \wedge A_n$ are all the subcases of A at w, then it seems natural to require the following:

[21] Cf. Jackson's discussion on *temporal precisification* in [1977, p. 10].

[22] Whether an antecedent has subcases, and which, if any, seems to be a function of the time of the antecedent which, if at all, is set for instance by the context of utterance, but also of what is the case at the world one is speaking about. E.g. if at world w there is only one possible time at which anybody could have taken the coat then at w the antecedent is not divisible.

[23] The reason why SDA seems to be valid may be that the grammatical connective 'or' in the antecedent of conditionals does not normally express the truth functional connective '$\vee$' but rather expresses a case of division. The "disjuncts" of the antecedent would thus represent the different subcases into which this antecedent would be divisible.

> For any $C \in L$, $A \mathrel{\Box\!\!\rightarrow} C$ is true at w iff all of $(A \wedge A_1) \mathrel{\Box\!\!\rightarrow} C$, ..., $(A \wedge A_n) \mathrel{\Box\!\!\rightarrow} C$ are true at w.

If a semantics conforms to this condition we say that it is *subcase-adequate*.

Presumably if A is a divisible antecedent at w there is always a (possibly infinite) list $B_1, B_2, \ldots$ of antecedents, non-divisible at w such that these are all the subcases of A at w.

2.5.4 Causal Trails and the Ramp

We have noted that in some cases the time of the divergence from the actual world cannot immediately precede the time of the antecedent, for the divergence to be smooth; there must be a transition period between the time of the divergence and the time of the antecedent. My friend is in his house and I in mine, but it could not have been the case that he was in his house up until a moment ago and at my door right now. Had he been at my door right now then he would perhaps have been on his way to me a moment ago, but certainly not at home. This means that a world diverging smoothly from the actual at which my friend is at my door right now cannot diverge from the actual just a moment ago but rather at a point in time somewhat farther off in the past—how far off depending on the location of my friends house, the available means of transportation, my friends intentions etc.

The stretch—of the antecedent-world we are considering—that starts with the point of departure from the actual world and ends with the time of the antecedent, is called the *ramp* in [Bennett, 1998]. Now, one important question is whether it is enough to disregard causal consequences of the antecedent as stated by the CT-criterion or whether consequents of some facts pertaining to the ramp should also be disregarded. Look at the following example:

> Say that at some places around my house there are (unknown to my friend) photo-cells connected to an alarm at the local police station notifying them about intruders. Now since the alarm has in fact not sounded it seems that worlds at which my friend is at my front door right

> now and at which the alarm has not gone off are intuitively more similar to the actual one than worlds at which my friend is at my front door and the alarm has sounded.

Had my friend shown up, might the alarm have gone off then, or not? I think our intuitions are on the side of an affirmative answer to this question. Now, consider the following worlds w_a and w_q. At both worlds my friend stands at my front door right now. At w_a my friend on his way to my front door happened to activate a photocell setting off the causal chain leading to the alarm sounding at the police station. At w_q my friend took a slightly different path to my front door thereby evading the photocell, so at w_q no alarm is sounding. In order for our semantical intuitions to be right here, there must be worlds like w_a among the closest to our world at which my friend shows up at my front door now. We imagine (not unrealistically, I think) that w_q is as similar to the actual world as w_a is with respect to circumstances other than whether the alarm sounds or not, so if any weight should be given to similarity with respect to the alarm, w_q seems to come out as more similar to the actual one than w_a would, which seems to yield counterintuitive truth conditions. The upshot of this is that dissimilarity with respect to whether the alarm goes off should be disregarded. The alarm sounding at w_a is not a causal consequence of the antecedent, namely that my friend stands at my door, so we cannot use the CT-criterion as it stands, rather it is a consequence of him *making his way* to my front door, at that world. It thus appears that we will have to amend the CT-criterion to be able to deal with this example. That the alarm sounds at w_a is a causal consequence, and so in the causal trail of my friend activating the photocell at that world. That fact, in its turn, was a part of the process of him ending up at my door. I shall call such "parts of the process" leading up to something, *preconditions* for that thing.

Everything in the causal trail of preconditions of the antecedent should not be treated in this way, however: That there is oxygen in the air at w_a before the time of the antecedent is a precondition for my friend showing up at my front door at that world, because

of course, he breathed and used oxygen on his way there. This and consequences of this should of course not be disregarded when comparisons to the actual world are made. There being oxygen in the air at that time was a precondition for the existence of aerobic life at that world after that time. Nevertheless we don't want to disregard every fact and event involving aerobic life subsequent to the time in question when comparisons are made to the actual world. Hence we need a criterion for selecting those preconditions of the antecedent occurring at the ramp, the consequences of which should be disregarded from those that should not be treated in this way. My suggestion is the following: We call a precondition of the antecedent *w-distinctive* if it is not a fact of the actual world w. Distinctive preconditions of the antecedent can only occur at the ramp, of course, since before the departure the antecedent world and the actual world coincide.

We are now in a position to state a causal-trail criterion that takes account of distinctive preconditions of the antecedent: If a conditional having A as antecedent is evaluated at the world w_0 and A and C hold at w, then (intuitive) dissimilarity with respect to C should not count when assessing overall closeness iff

> the fact (at w) that C is in the causal trail of A at w or of some w_0-distinctive precondition of the fact (at w) that A.

We call this criterion the refined causal trail criterion (or RCT-criterion for short).

2.5.5 Criterion of Closeness for Non-Divisible Antecedents

Since divisible conditionals are equivalent to conjunctions of non-divisible ones (at least this is what I have assumed) we need only a theory for the latter kind of subjunctive conditionals. I shall propose the following criterion of closeness: If we are evaluating a non-divisible subjunctive conditional with A as antecedent then $w' <_w w''$ iff

> w' and w'' genuine (or felonious) alternatives to w, and

1. w' diverges later from w than w'' does, or
2. w' and w'' diverge at the same point from w but w' is RCT-closer (with respect to A and w) to w than w'' is.

We call this the Late Departure Causal Trail ordering (or simply LDCT-ordering) of temporal worlds. Note that since the RCT-criterion is antecedent-relative, LDTC-ordering is a species of antecedent-relative ordering, in the sense that different antecedents might result in different orderings among worlds. It can also be seen to be a species of resultant similarity according to Lewis's four rules for weighing different respects of similarity on p. 95 above, and more precisely as an answer to the question implicit in rule 4, namely whether any weight should be given to merely approximate similarity.

We shall now see that given suitable extra assumptions there is an arrc-assignment definable in terms of the LDCT-ordering. The assumption is simply that even when certain facts of certain worlds are excluded from the comparison, the words may be ordered with respect to their resemblance to some world in a way that is reflexive, transitive and connected, and that each world is among the most similar to itself. I.e., we assume the following: If w, w', w'' and w''' are temporal worlds ($w' {}^{\mathrm{R}}\underline{\boxed{\mathrm{C}}}^{\mathrm{T}}{}^{A}_{w} w''$ stands for "w' is as RCT-close (with respect to A and w) to w as w'' is")

1. $w' {}^{\mathrm{R}}\underline{\boxed{\mathrm{C}}}^{\mathrm{T}}{}^{A}_{w} w'$
2. If $w' {}^{\mathrm{R}}\underline{\boxed{\mathrm{C}}}^{\mathrm{T}}{}^{A}_{w} w''$ and $w'' {}^{\mathrm{R}}\underline{\boxed{\mathrm{C}}}^{\mathrm{T}}{}^{A}_{w} w'''$ then $w' {}^{\mathrm{R}}\underline{\boxed{\mathrm{C}}}^{\mathrm{T}}{}^{A}_{w} w'''$
3. $w' {}^{\mathrm{R}}\underline{\boxed{\mathrm{C}}}^{\mathrm{T}}{}^{A}_{w} w''$ or $w'' {}^{\mathrm{R}}\underline{\boxed{\mathrm{C}}}^{\mathrm{T}}{}^{A}_{w} w'$
4. $w {}^{\mathrm{R}}\underline{\boxed{\mathrm{C}}}^{\mathrm{T}}{}^{A}_{w} w'$

Let $w'' {}^{\mathrm{R}}\boxed{\mathrm{C}}^{\mathrm{T}}{}^{A}_{w} w'$ stand for $w' {}^{\mathrm{R}}\not\boxed{\mathrm{C}}^{\mathrm{T}}{}^{A}_{w} w''$, $w' \underline{\boxed{\mathrm{LD}}}_{w} w''$ stand for $Sh(w) \cap Sh(w'') \subseteq Sh(w) \cap Sh(w')$, $w'' \boxed{\mathrm{LD}}_{w} w'$ stand for $w' \not\underline{\boxed{\mathrm{L}}}\mathrm{D}_{w} w''$, and $w' \overset{\mathrm{LD}}{=}_{w} w''$ for $Sh(w) \cap Sh(w') = Sh(w) \cap Sh(w'') \neq \emptyset$.

We also assume that the laws of the actual world are non-deterministic. In this way we can think of the relation of being an alternative as symmetric and transitive.

Let I_L be the set of lawful worlds obeying the non-deterministic laws of this world. The definition of the LDCT-ordering above now takes the following more precise form. For each $w \in I_L$ let R_w be the set of elements of I_L that are co-commenced with w. For each $w \in I_L$ and $A \in L$ we define $<^A_w$ thus: for $w', w'' \in I_L$, $w' <^A_w w''$ iff

1. $w' \sqsubseteq^{\text{LD}}_w w''$, or
2. $w' \overset{\text{LD}}{=}_w w''$ and $w' \,{}^{\text{R}}\sqsubseteq^{\text{CT}}{}^A_w w''$.

As usual we write $w'' \leq^A_w w'$ for $w' \not<^A_w w''$. We can now show that

Proposition 2.17 $\leq$ *defined in this way is an arrc-assignment for* I_L.

Proof.

1. Assume that $w'' \not<^A_w w'$ and $w''' \not<^A_w w''$. Then $w' \sqsubseteq^{\text{LD}}_w w''$ and $w'' \sqsubseteq^{\text{LD}}_w w'''$. Assume for contradiction that $w''' <^A_w w'$. Since $w' \sqsubseteq^{\text{LD}}_w w'''$, $w' \overset{\text{LD}}{=}_w w'''$ and $w''' \,{}^{\text{R}}\sqsubseteq^{\text{CT}}{}^A_w w'$, but then $w' \overset{\text{LD}}{=}_w w'' \overset{\text{LD}}{=}_w w'''$, so $w'' \,{}^{\text{R}}\not\sqsubseteq^{\text{CT}}{}^A_w w'$ and $w''' \,{}^{\text{R}}\not\sqsubseteq^{\text{CT}}{}^A_w w''$. Thus $w' \,{}^{\text{R}}\sqsubseteq^{\text{CT}}{}^A_w w'''$ (by transitivity of ${}^{\text{R}}\sqsubseteq^{\text{CT}}{}^A_w$), which is a contradiction. Hence $w''' \not<^A_w w'$, i.e., the relation $\leq^A_w$ is *transitive*.

2. Assume that $w' \not\leq^A_w w''$, i.e., that $w'' <^A_w w'$ and for contradiction that $w'' \not\leq^A_w w'$, i.e. $w' <^A_w w''$. Then by Prop. 2.4, $w' \overset{\text{LD}}{=}_w w''$, $w' \,{}^{\text{R}}\sqsubseteq^{\text{CT}}{}^A_w w''$ and $w'' \,{}^{\text{R}}\sqsubseteq^{\text{CT}}{}^A_w w'$, which is a contradiction (since ${}^{\text{R}}\sqsubseteq^{\text{CT}}$ is connected), hence for any worlds w' and w'' either $w' \leq^A_w w''$ or $w'' \leq^A_w w'$, so $\leq^A_w$ is *strongly connected*.

3. By definition of R_w, $w \in R_w$, i.e., the world w is *self accessible*.

4. For all $w' \in I_L$, $w' \not\sqsubseteq^{\text{LD}}_w w$ $(Sh(w) \cap Sh(w') \subseteq Sh(w) \cap Sh(w))$ and $w' \,{}^{\text{R}}\not\sqsubseteq^{\text{CT}}{}^A_w w$, hence $w' \not<^A_w w$, i.e., $w \leq^A_w w'$, so the world w is $\leq^A_w$*-minimal*.

5. Assume that $w' \notin R_w$ and that $w'' \in I_L$. Then $w' \not\sqsubseteq^{\text{LD}}_w w''$ and not $w' \overset{\text{LD}}{=}_w w''$ (since $\emptyset = Sh(w) \cap Sh(w') \subseteq Sh(w) \cap Sh(w'')$). Hence $w' \not<^A_w w''$, i.e., $w'' \leq^A_w w'$, so inaccessible worlds are $\leq^A_w$*-maximal*.

6. Assume that $w' \in R_w$ and $w'' \notin R_w$. Then $Sh(w) \cap Sh(w') \not\subseteq Sh(w) \cap Sh(w'')$ ($w' \boxed{\text{LD}}_w w''$), and so $w' <_w^A w''$, so accessible worlds are more similar to w than inaccessible worlds.

$\dashv$

Let S_{LC} be the semantics for I_L defined i terms of this arrc-assignment.

2.5.6 Pure RCT-closeness.

The reason LDCT-ordering cannot serve as a general criterion of closeness is that it incorporates the requirement that the closest A-worlds are among the A-worlds departing at the latest point in time from the actual world, thereby creating trouble with respect to Pollock's example (p. 85). Perhaps this requirement is unnecessary. After all, the main reason it was included was to ensure that the defined ordering would be of the type defined by Lewis's four rules on p. 95 above. What if the late departure requirement were simply skipped? If again for each $w \in I_L$ we let R_w be the set of elements of I_L co-commenced with w, it is easy to see that

Proposition 2.18 *$\boxed{\text{RCT}}$ is an arrc-assignment for I_L*

Let S_{RCT} be the ARRC semantics defined by $\boxed{\text{RCT}}$. It would be nice if S_{RCT} were subcase-adequate in the sense defined on p. 106 above, but I see no way of proving this.

2.6 Soundness of Axioms and Rules

2.6.1 The Principles RCEA, RT and CA with respect to LDCT- or RCT-orderings.

We shall now show that (given certain fairly natural assumptions about RCT-closeness) the principles RCEA, RT and CA[24] are valid

[24]See Sections 3.1.2 and 3.1.3 below

with respect to the ARRC semantics defined by these arrc-assignments, i.e., we shall show that they are valid with respect to S_{LC} and S_{RCT}.

If (given some semantics S) the fact (at w') that C is in the causal trail of A at w' or of some w-distinctive precondition of the fact (at w') that A, we shall say that C is *A-exempt* at w' with respect to w, in symbols:

$$C \in E_w^A(w').$$

If $w' \not\models A$ we let $E_w^A(w') = \emptyset$. Since $C \in E_w^A(w')$ only if C is a fact of w' it follows that $E_w^A(w') \subseteq Th_S(w')$.

If for some semantics S, $C \in E_w^A(w')$ or $C \in Th_S(w') \cap Th_S(w)$ we say that C is *A-non distinguished* at w' with respect to w, in symbols

$$C \in nD_w^A(w').$$

It follows that $nD_w^A(w') \subseteq Th_S(w')$ and that if $w' \not\models_S A$ then $nD_w^A(w') \subseteq Th_S(w)$. Since $nD_w^A(w') = E_w^A(w') \cup (Th_S(w') \cap Th_S(w))$, we have

Proposition 2.19 *If* $E_w^A(w') \subseteq E_w^B(w')$ *then* $nD_w^A(w') \subseteq nD_w^B(w')$.

Recall that (p. 58 above) for each $w \in I$ and $A \in L$,

$$S_{\leq}(A)_w = \{w' \in S_{\leq}(A) \cap R_w : \text{If } w'' <_w^A w' \text{ then } w'' \notin S_{\leq}(A)\}.$$

I.e., $S_{\leq}(A)_w$ is the set of accessible A-closest A-worlds to w.

We shall assume that S_{LC}, and S_{RCT} satisfy the ARL-assumption (the Antecedent Relative Limit assumption, page 58 above) and the following requirement: If S is S_{LC} or S_{RCT}:

Condition 2.1 *If* $w' \in S(A)_w$ *then* $Th_S(w') \subseteq nD_w^A(w')$.

This principle says that given any historically possible antecedent there are always historical alternatives to the actual world that only differ from it with respect to the antecedent in question, the process

leading to it, and facts within the causal trail of those. This principle could be taken as a consequence of the intuitive RCT criterion. However, it seems to have an existential import not warranted by the RCT-criterion: Why should any of the historical alternatives to the actual world w in which A is true be such that it only differs from w with respect to sentences that are elements of $E_w^A(w')$? In this respect, this principle is certainly questionable. Nevertheless, it is a handy way of translating rather vague talk about similarity under the causal trail criterion into a formal statement, the properties of which can be studied by formal methods. If the reader is still worried about this assumption it might perhaps instead be taken as a a contribution towards a definition of the imprecise and vague expressions "causal trail", "precondition of", etc.

We also assume the following trivial principle.

Condition 2.2 *If $Th_{S_{RCT}}(w') \subseteq nD_w^A(w')$ (with respect to S_{RCT}) and $w' \models_{S_{RCT}} A$ then $w' \in S_{RCT}(A)_w$.*

By Conditions 2.1 and 2.2, if S is S_{RCT} and $w' \models_S A$ then

Proposition 2.20 *$w' \in S(A)_w$ iff $Th_S(w') = nD_w^A(w')$.*

Hence if $Th_S(w') = nD_w^A(w')$ then w' is among the worlds RCT-closest to w with respect to A and w. Now by the definition of $nD_w^A(w')$ we see that for all $w \in I_L$ and $A \in L$, $Th_S(w) = nD_w^A(w)$ and hence w is minimal with respect to this ordering as we assumed above. Even if $w \models_S A$ there may be $w' \neq w$ such that $w' \models_S A$ and $Th_S(w') = nD_w^A(w')$ and hence $w \not<_w^A w'$ and so $w' \in S(A)_w$. If $w' \not\models_S B$ and $w \models_S B$, then $w \models_S A \wedge B$ but $w \not\models_S A \mathrel{\Box\!\!\rightarrow} B$. Therefore the axiom CS (p. 147 below) is not necessarily supported by S_{RCT}.

Condition 2.2 may fail to be satisfied by S_{LC}, however, since among the A-worlds satisfying this criterion there may be A-worlds departing earlier from the actual world than certain other A-worlds, so for S_{LC} we assume this modified but also trivial principle:

Condition 2.3 *If w' is among the A-worlds departing from w at the latest point and $Th_{S_{LC}}(w') \subseteq nD_w^A(w')$ (with respect to S_{LC}) then $w' \in S_{LC}(A)_w$.*

First we show that if S is S_{LC} or S_{RCT}, then for every $w \in I_L$ and $A, B, B' \in L$ we have

Proposition 2.21 *$S(A)_w \subseteq S(B)_w \cup S(B')_w$ iff $S(A)_w \subseteq S(B) \cup S(B')$ and for every $w' \in S(A)_w$, $nD^A_w(w') \subseteq nD^B_w(w')$ or $nD^A_w(w') \subseteq nD^{B'}_w(w')$.*

Proof. First assume that $S(A)_w \subseteq S(B)_w \cup S(B')_w$ and hence $S(A)_w \subseteq S(B) \cup S(B')$. Assume further that $w' \in S(A)_w$ so A is historically possible at w and $w' \in S(B)_w$ or $w' \in S(B')_w$. If $D \in nD^A_w(w')$ then $D \in Th_S(w')$ and since $w' \in S(B)_w$ or $w' \in S(B')_w$ (hence B or B' is historically possible at w), by Condition 2.1, $D \in nD^B_w(w')$ or $D \in nD^{B'}_w(w')$.

On the other hand assume that $S(A)_w \subseteq S(B) \cup S(B')$ and for every $w' \in S(A)_w$, $nD^A_w(w') \subseteq nD^B_w(w')$ or $nD^A_w(w') \subseteq nD^{B'}_w(w')$. Take $w'' \in S(A)_w$ (if $S = S_{LC}$ then w'' is among the latest departers); then $nD^A_w(w'') \subseteq nD^B_w(w'')$ or $nD^A_w(w'') \subseteq nD^B_w(w'')$. By Condition 2.1, $Th_S(w'') \subseteq nD^A_w(w'')$ and hence $Th_S(w'') \subseteq nD^B_w(w'')$ or $Th_S(w'') \subseteq nD^B_w(w'')$. Hence by Conditions 2.2 and 2.3, $w'' \in S(B)_w$ or $w'' \in S(B')_w$, so $S(A)_w \subseteq S(B)_w \cup S(B')_w$. $\dashv$

A corollary of Prop. 2.21 is the following

Proposition 2.22 *$S(A)_w \subseteq S(B)_w$ iff $S(A)_w \subseteq S(B)$ and for every $w' \in S(A)_w$, $nD^A_w(w') \subseteq nD^B_w(w')$.*

RCEA

For every $w, w' \in I_L$ and $A, B \in L$ we shall also assume the following principle:

Condition 2.4 *If S is S_{LC} or S_{RCT} and $\models_S A \leftrightarrow B$ then $E^A_w(w') = E^B_w(w')$.*

This doesn't seem too unreasonable given that the domain of S_{LC} and S_{RCT}, I_L, is the set of all lawful worlds: One could argue that if for all $w'' \in I_L$, $w'' \models_S A \leftrightarrow B$, then A and B must express the same fact, or facts that are nomically entailed by each other, in

which case, arguably, everything in the causal trail of the one would also be in the causal trail of the other.

By Prop. 2.19,

Proposition 2.23 *If S is S_{LC} or S_{RCT} and $\models_S A \leftrightarrow B$ then $nD^A_w(w') = nD^B_w(w')$.*

For $w \in I_L$ and $A, B \in L$, we also show the following:

Proposition 2.24 *If S is S_{LC} or S_{RCT} and $\models_S A \leftrightarrow B$ then $S(A)_w = S(B)_w$.*

Proof. Assume that S is S_{LC} or S_{RCT} and that $\models_S A \leftrightarrow B$. Then $S(A)_w \subseteq S(B)$. By Prop. 2.23, for all $w' \in I_L$, $nD^A_w(w') = nD^B_w(w')$, and hence by Prop. 2.22, $S(A)_w \subseteq S(B)_w$, and similarly in the other direction. ⊣

At p. 32 above we introduced the notion of consequent intensionality. Now we define the following related property.

Definition 2.1 *A semantics S is* antecedent intensional *iff $\models_S (C \mathrel{\Box\!\!\rightarrow} A) \leftrightarrow (D \mathrel{\Box\!\!\rightarrow} A)$ whenever $\models_S C \leftrightarrow D$.*

Proposition 2.25 *S_{LC} or S_{RCT} are antecedent intensional.*

Proof. If S is S_{LC} or S_{RCT} and $\models_S A \leftrightarrow B$ then, by Prop. 2.24, $S(A)_w = S(B)_w$, hence for any $C \in L$, $S(A)_w \subseteq S_{\leq}(C)$ iff $S(B)_w \subseteq S_{\leq}(C)$. It then follows by Prop. 3.73 below that $\models_S (A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C)$. ⊣

Also using Prop. 3.5 below, we get

Proposition 2.26 *RCEA is sound with respect to S_{LC} and S_{RCT}.*

RT

We assume the following principle

Condition 2.5 *If S is S_{LC} or S_{RCT} and $w' \models_S A \wedge B$ then $E_w^A(w') \subseteq E_w^{A \wedge B}(w')$.*

If the fact (at w') that C is in the causal trail of the fact (at w') that A then it must also be in the causal trail of the fact (at w') that $A \wedge B$, if B is also true at w'. On the other hand, for obvious reasons, there ought to be $A, B \in L$ and $w, w' \in I_L$ such that $E_w^{A \wedge B}(w') \not\subseteq E_w^A(w')$.

Hence, by Prop. 2.19 we get

Proposition 2.27 *If S is S_{LC} or S_{RCT} and $w' \models_S A \wedge B$ then $nD_w^A(w') \subseteq nD_w^{A \wedge B}(w')$.*

Also,

Proposition 2.28 *If S is S_{LC} or S_{RCT} then $S(A)_w \subseteq S(B)$ implies $S(A)_w \subseteq S(A \wedge B)_w$.*

Proof. Assume that $S(A)_w \subseteq S(B)$ (and hence $S(A)_w \subseteq S(A \wedge B)$) and $w' \in S(A)_w$. Then $w' \models_S A \wedge B$. By Prop. 2.27, $nD_w^A(w') \subseteq nD_w^{A \wedge B}(w')$. Hence by Prop. 2.22, $S(A)_w \subseteq S(A \wedge B)_w$.

$\dashv$

Hence S_{LC} and S_{RCT} satisfy the principle CRT' on p. 175 below, from which it follows by Prop. 3.74 and Prop. 3.75 below that

Proposition 2.29 *S_{LC} and S_{RCT} validate RT.*

CA

Conditions 2.4 and 2.5 imply the following principle.

Proposition 2.30 *If S is S_{LC} or S_{RCT} and $w' \models_S A$, then $E_w^{A \vee B}(w') \subseteq E_w^A(w')$.*

Proof. Assume that S is S_{LC} or S_{RCT} and $w' \models_S A$. Then $w' \models_S (A \vee B) \wedge A$ and hence by Condition 2.5, $E_w^{A \vee B}(w') \subseteq E_w^{(A \vee B) \wedge A}(w')$, and so by Condition 2.4, $E_w^{A \vee B}(w') \subseteq E_w^A(w')$. ⊣

Hence by Prop. 2.19,

Proposition 2.31 *If S is S_{LC} or S_{RCT} and $w' \models_S A$ then $nD_w^{A \vee B}(w') \subseteq nD_w^A(w')$.*

Furthermore,

Proposition 2.32 *If S is S_{LC} or S_{RCT} then $S(A \vee B)_w \subseteq S(A)_w \cup S(B)_w$.*

Proof. $S(A \vee B)_w \subseteq S(A) \cup S(B)$. Assume that $w' \in S(A \vee B)_w$. Then $w' \models_S A$ or $w' \models_S B$ and hence by Prop. 2.31, $nD_w^{A \vee B}(w') \subseteq nD_w^A(w')$ or $nD_w^{A \vee B}(w') \subseteq nD_w^B(w')$. Thus by Prop. 2.21, $S(A \vee B)_w \subseteq S(A)_w \cup S(B)_w$. ⊣

Hence S_{LC} and S_{RCT} satisfy the principle CCA' on p. 175 below. So again by Prop. 3.74 and Prop. 3.75 below,

Proposition 2.33 *S_{LC} and S_{RCT} validate CA.*

SDA

In this connection we must say something about SDA (p. 146) too. First we note that the reverse of Prop. 2.30 should not be true. We noted above that there should be $A, B \in L$ and $w, w' \in I_L$ such that $E_w^{A \wedge B}(w') \not\subseteq E_w^A(w')$. If we let $C = A \wedge B$ then by Condition 2.4, $E_w^C(w') \not\subseteq E_w^{C \vee A}(w')$. Let $D \in E_w^C(w') - E_w^{C \vee A}(w')$. If also $w \not\models_S D$ then $D \in nD_w^C(w') - nD_w^{C \vee A}(w')$. Hence $w \models_S C \diamond\!\!\rightarrow D$, but $w \not\models_S (C \vee A) \diamond\!\!\rightarrow D$, i.e., $w \models_S (C \vee A) \Box\!\!\rightarrow \neg D$. And in that case SDA is not satisfied.

2.6.2 The Principles RCV and CV with respect to RCT-orderings.

Right now I see no way of proving formally that causal trail semantics like S_{RCT} don't support the principles RCV and CV. Still, I think this to be the case and I shall try to work out a case for this in terms of the example on page 53 above: Say that $w \in I_L$ is the world at which Betty, in vain, sits at her window waiting for Albert. Let A express that Albert goes and B that Betty goes. We assume that (w and I_L are such that) for all $w' \in I_L$ such that $Th_S(w') \subseteq nD^A_w(w')$, $w' \models_S B$, i.e., that at all lawful worlds that differ from w only with respect to facts within the causal trail of A[25] (at those worlds), B is the case. Since $w \models_S \neg B$: if $Th_S(w') \subseteq nD^A_w(w')$ then $B \in E^A_w(w')$, i.e., B is within the causal trail of A at all those worlds that only differ from w with respect facts within the causal trail of A, that is, at all those worlds, Betty goes because Albert goes. Exactly the same goes for C, namely the sentence expressing that Betty is glad, since at those worlds Betty is glad ultimately because Albert goes. By Prop. 2.20, $S(A)_w \subseteq S(B)$ and $S(A)_w \subseteq S(C)$.

On the other hand we assume that there is a certain $A \wedge B$-world $w'' \notin S(C)$ that differs from w only with respect to facts within the causal trail of $A \wedge B$. This is a world at which Betty goes by herself and is not happy and at which Albert also goes. There are all sorts of $D \in L$ such that $w'' \models_S D$ and $w \not\models_S D$ and $D \notin E^A_w(w'')$, i.e., that are not within the causal trail of Albert's going at w'', such as e.g. the fact that Betty leaves before Albert walks past her window. Hence by Prop. 2.20, $w'' \notin S(A)_w$, but all these facts of w'' are within the causal trail of both going, i.e., $D \in E^{A \wedge B}_w(w'')$. By Condition 2.5 and Prop. 2.20, $w'' \in S(A \wedge B)_w$. Hence $S(A \wedge B)_w \not\subseteq S(C)$

Now we show that, given the assumptions regarding S_{RCT} above, we have

Proposition 2.34 *Neither CV nor RCV are validated by S_{RCT}.*

[25] This is of course just a convenient abbreviation for the more cumbersome: within the causal trail of A at w' or of some w-distinctive precondition of the fact (at w') that A.

Proof. Let S be S_{RCT}. By the assumptions above there are $A, B, C \in L$ and $w \in I_L$ such that $S(A)_w \subseteq S(C)$, $S(A)_w \subseteq S(B)$ (and $S(A)_w \not\subseteq S(\neg B)$) and $S(A \wedge B)_w \not\subseteq S(C)$. By Prop. 3.73 below it follows that $w \models_S A \mathrel{\Box\!\!\rightarrow} B$, $w \models_S A \mathrel{\Diamond\!\!\rightarrow} B$, and $w \models_S A \mathrel{\Box\!\!\rightarrow} C$, but $w \not\models_S (A \wedge B) \mathrel{\Box\!\!\rightarrow} C$, and hence $\not\models_S RCV$ and $\not\models_S CV$. ⊣

Also, given the assumptions above

Proposition 2.35 *CSO is not validated by S_{RCT}.*

Proof. Let S be S_{RCT}. ID and CP are valid with respect to S and hence if $w \models_S A \mathrel{\Box\!\!\rightarrow} B$ then $w \models_S A \mathrel{\Box\!\!\rightarrow} (A \wedge B)$. Since (also by ID and CP) $\models_S (A \wedge B) \mathrel{\Box\!\!\rightarrow} A$, by CSO, $w \models_S A \mathrel{\Box\!\!\rightarrow} C$ iff $w \models_S (A \wedge B) \mathrel{\Box\!\!\rightarrow} C$ whenever $w \models_S A \mathrel{\Box\!\!\rightarrow} B$, and hence $\models_S RCV$, so, by Prop. 2.34, $\not\models_S CSO$. ⊣

2.7 Bennett's Causal Chain Criterion

The causal trail criterion shares some important features with a criterion due to Jonathan Bennett.[26] The criterion in question is designed to take care of examples like the coin-tossing one on p. 100 above. Bennett observes (as we have also done) that the criterion we have called simple late departure (Sec. 2.4.1 above) is inadequate for dealing with this kind of example, at least if determinism is not assumed. We mentioned earlier that Lewis in [1979, p. 48] leaves the question open whether similarity after the time of departure should count at all towards overall similarity. This is because examples like the coin-tossing one seem to speak in favor of including some such future similarities, and examples like the doomsday-button one of p. 99 above speak in favor of excluding all such similarities from consideration.

[26] The criterion in question has not yet appeared in any publication of Bennett's. However, he has been kind enough to let me quote from some of his recent class notes on this subject.

As we have done, Bennett tries to define a criterion that gives the right answer with respect to both kinds of example. He doesn't present his criterion right away but starts by offering a provisional criterion of closeness that takes account of (too many or perhaps the wrong) future similarities. He calls this criterion PF (for *particular fact*). w_1 and w_2 are two temporal worlds that diverge at the same time TF from a world α. w_1 is closer to α than w_2 is if

1. w_1 is exactly like w_2 [and α] up to TF, and
2. both conform to the laws of α thereafter, and
3. they first become unlike in respect of one particular matter of post-TF fact which obtains at α and w_1 but not at w_2.

It seems that this criterion works as it should with respect to the coin-tossing example: At α I bet that the coin will come up heads and it does just that. At w_1 I bet that it will come up tails but it still comes up heads, and at w_2 I also bet that it will come up tails and it does so too. For the sake of the argument we assume that w_1, and w_2 are alike up to the time of tossing, at which w_2 departs from w_1 (and in a way also from α) in the coin coming up tails instead of heads. Hence w_1 should be regarded as closer than w_2 to α, which is the desired outcome.

On the other hand this criterion would e.g. favor worlds at which Nixon presses the button but no nuclear disaster ensues, so it seems that PF is inadequate with respect to examples of this type, but also with respect to the following example:

> We have a deeply and objectively random coin-tossing machine: pressing a button activates a device that fires a single photon towards two slits; its going through one slit causes the coin to be tossed so that it has to come down heads; similarly for the other slit and coming down tails. Joe activates this mechanism. He presses the button at TA and the coin comes down heads. Now consider the statement "If Susan had pressed the button at TA, the coin would have come down heads." Without PF added

> to it, Lewis's main theory makes that conditional false, because it selects as closest some A-worlds where the coin falls heads and some where it falls tails; Stalnaker's theory makes it indeterminate; neither makes it true, and I submit that that is as it should be. I count it against PF that it confers truth on that conditional.

In order to deal with this example Bennett offers the following modification of PF, that he calls MPF:

> If clauses (1)-(3) of PF obtain and (4) the fact mentioned in (3) obtains at w_1 through a causal chain the whole of which also occurs at α, then w_1 is closer to α than w_2 is.

Like PF this would handle the original coin tossing example, but also the example above:

> Of two worlds where Susan works the coin-tossing device, it comes down heads at w_1 and tails at w_2; it also falls heads at α. However, the causal chain leading to its coming down heads at w_1 includes a push by Susan on a button, while at α the button was pressed by Joe. So clause (4) is not satisfied by w_1 in this case, and the conditional is evaluated as false, which is the right answer.

This is as much as I know about Bennett's own use of the MPF criterion. Now, let's see how the doomsday-button example would be handled by MPF. Let w_1 be a world at which Nixon hits the button but at which, for some reason, there is no signal leaving the device, and therefore (like at α) no nuclear disaster; and w_2 one at which the signal is off, leading to such a disaster. w_1 and w_2 diverge from α immediately before the time Nixon hits the button in both those worlds and they part company right after that moment. We assume that the first respect in which they become different is whether there is any signal in a certain wire right after Nixon has pressed the ominous button. At w_1 like at α there is no such signal,

but at w_2 there is. The question is whether this similarity between w_1 and α should count towards similarity with α. According to the MPF criterion it boils down to the question whether this lack of signal (or current if you like) obtains at w_1 through a causal chain that also occurs at α or not. In order to get the right answer, this question must be answered in the negative. To get this result one might perhaps say that at w_1 the lack of signal is due to a dysfunction of the button whereas at α it didn't come about in that way, and hence at w_1 the absence of signal obtains through a causal chain not present at α. The question seems to be whether "lacks" and "absences" can be parts of causal chains. I am not sure what Bennett thinks on this subject and I don't really know what to say about it myself. I just wish to point out that if this is a problem it is not specific to Bennett's MPF criterion but it also affects the causal trail criterion:

> I bet on heads and since the coin ended up heads I won. Had I not betted at all then I would not have won. But presumably, the coin would still have ended up heads.

To account for this in terms of the causal trail criterion we must, it seems, talk about facts of a world that are in the causal trail of me not betting at all at this world.[27]

There are however other problems specific to the MPF criterion, I think. The first one is the problem of cross-world identity

[27] One way perhaps to avoid talking of causal trails of things not happening, would be to distinguish between *positive antecedents*, i.e., antecedents that state the occurrence of some event, and *negative* ones that state the absence of an event, like "If the lightning hadn't struck...", and apply the causal trail criterion as defined above only on conditionals with positive antecedents. With regard to conditionals with negative antecedents one could perhaps try something like the following: Instead of disregarding everything within the causal trail of the antecedent one should disregard everything within the causal trail (at the actual world) of the event at the actual world that the antecedent negates. So if the lightning struck at the actual world we should disregard everything within the causal trail of this when looking for most similar worlds at which it didn't strike. The causal trail criterion for negative antecedents would thus share some important traits with Pollock's undercutting criterion (Section 2.8.1 below). This should of course be regarded as a mere suggestion the consequences and ramifications of which have yet to be worked out.

for causal chains. What does it mean to say that two events in different possible worlds come about by the same causal chain? Does this imply that (numerically) the same causal chain occurs at both worlds? Or should one rather say that the same event is caused identically at two different possible worlds? Perhaps this problem is not insurmountable, but anyway we may note that it does not arise at all for the causal trail criterion, since causal trails at different worlds are not thus identified. My main reason, however, to prefer the causal trail criterion to some version of Bennett's MPF criterion is that it seems that this criterion might define a species of relative closeness between worlds that is connected and not antecedent relative, and hence the principles CV and CSO, that we have seen to be objectionable, seem to be supported.

The MPF criterion looks incomplete as it stands. One cannot help wondering what happens if w_1 and w_2 come out even with respect to their first difference but w_2 after this deviates from w_1 and α with respect to facts identically caused at both those worlds. One could answer this question in several different ways: One could disregard later differences and stick with the original MPF criterion; one could treat later differences on the same footing (whatever that would mean); one could give more weight to earlier ones than to later ones, or perhaps the opposite, etc.

The original MPF criterion seems to define a species of relative closeness that is not even transitive.[28] There are surely ways to remedy this but it looks like the natural outcome of such remedies are rc-assignments that define semantics that validate CSO and CV, and this, as I said, is my main reason for rejecting the MPF criterion.

[28]Say that w_1, w_2 and w_3 all fork off from α at the same point, w_1 and w_2 diverge from each other in a way that has w_2 closer than w_1 to α according to the MPF criterion, w_3 forks off from both w_1 and w_2 before they diverge from each other, but in a way that neither results in w_3 being closer to α than w_1 and w_2 nor in them being closer to α than w_3. Hence w_1 is as close as w_3 to α and w_3 is as close as w_2, but w_1 is not as close as w_2 to α, so this relation is not transitive.

2.8 Cotenability

In footnote 30 on page 36 above reference was made to the discussion on *cotenability* in [Goodman, 1946]. In this paper Goodman suggests a facts revision semantics for subjunctive conditionals. He observes that it is not enough to require that the contracted set S, of true sentences is logically compatible with the antecedent A. Goodman's famous example is as follows: We imagine a situation in which there is a certain dry, well made, match m surrounded by enough oxygen etc. The match is in fact not scratched so it does in fact not light. Goodman invites us to consider the following two subjunctive conditionals:

(14) If match m had been scratched, it would have lighted.

and

(15) If match m had been scratched, it would not have been dry.

It seems that the true sentence

(16) Match m did not light.

is perfectly compatible with the antecedent A: "m was scratched" and the further true sentences "m is well made", "Oxygen enough is present", etc. From this together with a suitable true law we can infer that

m was not dry,

and hence it seems that the conditional (15) is supported. In order to avoid this untoward result we must have some criterion by means of which we may exclude true sentences like (16) from the contracted set of true sentences S. Goodman suggests that sentences like (16) should be excluded because "they would not be true if A were true" [p. 14]. Goodman defines cotenability in those terms. A true "statement" should be excluded from S if it would not be true if A were, and hence it should be required that S be *cotenable* with A in the

sense that "it would not be the case that S would not be true if A were".[29]

If the requirement of cotenability is understood to present not only a necessary but also a sufficient condition on inclusion in the contracted set of true sentences S then it seems the resulting facts revision semantics will validate the principle CEM (see p. 147 below): If $A \mathrel{\Box\!\!\rightarrow} \neg C$ is not true and C is true then C is cotenable with A and so should be included in the contracted set S, but then obviously $S \cup \{A\} \models C$ and hence $A \mathrel{\Box\!\!\rightarrow} C$ is true. If on the other hand $\neg C$ is true then $\neg C$ is not cotenable with A since otherwise it would be included in S (and hence $A \mathrel{\Box\!\!\rightarrow} \neg C$ would be true after all) and hence $A \mathrel{\Box\!\!\rightarrow} \neg\neg C$ is true, so (by RCK) $A \mathrel{\Box\!\!\rightarrow} C$ is true in either case.

In footnote 9 on page 15 Goodman explicitly denies the validity of CEM, which means that he would be forced to bar some true sentences cotenable with the antecedent from entering the contracted set.

A perhaps more serious problem with Goodman's account (that he himself acknowledges on p. 16) is that the account involves us in "an infinite regressus or a circle; for cotenability is defined in terms of counterfactuals, yet the meaning of counterfactuals is defined in terms of cotenability", and hence "we can never explain a counterfactual except in terms of others, so that the problem of counterfactuals must remain unsolved".

2.8.1 Cotenability and Undercutting

There is therefore a need for a definition of cotenability that evades those difficulties. I shall suggest a way of defining cotenability in terms of the refined causal trail criterion. In doing this, I shall help myself to the notion of "undercutting" introduced by Pollock in [Pollock, 1976]. I quote from his discussion on Goodman's match-example on p. 77.

[29] It seems that Goodman presupposes that whenever a set of sentences S is not cotenable with A in this sense then there must be a member of S that would not be true if A were. In our discussion on GCP and the limit assumption on p. 48 above we saw that this is not necessarily the case.

> We are looking for a basis for choosing between two propositions Q and R when their conjunction is counter-implicated by P but neither proposition individually is counter-implicated by P. When do we preserve Q in preference to R? Sometimes the answer is, 'When part of the reason R is now true is that P is false'. To illustrate, consider the match case again. We preserve D ('The match is dry') in preference to $\neg L$ ('The match did not light'), because part of the reason the match did not light is that it was not struck, but this is not part of the reason it was dry. More precisely, there was a chain of antecedent circumstances implicating that the match was dry, and another chain implicating that that it did not light. These chains of circumstances represent the way in which it actually came about that the match was dry and did not light. But among the circumstances implicating the match's not lightning was the fact that it was not struck. If we remove that from the antecedent circumstances, the remaining circumstances are no longer sufficient to implicate that the match did not light.

Further, on p. 78.

> The above [...] example [...has] the characteristic that there are circumstances historically antecedent to R which implicate R and then others preceding those circumstances and implicating them, and so on. This chain of circumstances represents the way in which it actually came about that R was true. If we trace out this chain of historically antecedent circumstances, we find that they contain $\neg P$ so that by adopting the counterfactual hypothesis that P is true we 'undercut' the reason R is true. Then given a conflict between one proposition R which is undercut and another Q which is not, we preserve Q at the expense of R.

I quote this extensively because of the striking insights offered by this passage. Pollock continues

> In order for P to undercut R, it is not necessary for the historically antecedent circumstances implicating R to actually contain $\neg P$; more generally, P may counter-implicate some part of those circumstances.

In a later paper [Pollock, 1981, pp. 244–46], undercutting is defined on the same lines, but more explicitly, in terms of the notions of historic antecedent and nomic pyramid. Briefly, a *historic antecedent* of P (at the world α) is a set Γ of simple facts[30] (of α) preceding P such that Γ nomically implies P at α (i.e., $\Gamma \cup L_\alpha$ imply P, where L_α are the laws of α). A *nomic pyramid* of α is a set of simple facts (of α) that is closed under historical antecedents so that it contains the entire causal history of its members. Now, P *undercuts* Q at α iff Q is true at α and every nomic pyramid of α that contains Q nomically implies $\neg P$ at α.

Since Pollock is constructing a kind of relative closeness semantics, and not a facts revision semantics, he doesn't use undercutting to define a notion of cotenability. Undercutting is rather used by Pollock to identify those simple states of affairs we should be conservative about when looking for closest antecedent-worlds. Those states of affairs should obviously be the ones that are not undercut by the antecedent. If Pollock had chosen a facts revision framework instead he might have used undercutting to define cotenability instead: The facts cotenable with the antecedent would then have been precisely those facts not undercut by it. Above we mentioned two principal problems with Goodman's account of cotenability, namely that it appears to support CEM and that it is circular. The present version of cotenability doesn't seem to suffer from any of these drawbacks: It doesn't seem to entail the validity of CEM and most importantly it is not circular: Truth for subjunctive con-

[30] Pollock relies on a distinction between simple states of affairs, that are not logical compounds of other states of affairs, and such states of affairs that are. This is required for his definition of relative change (or "indexed difference"; See [p. 241]) to make sense.

ditionals is not defined in terms of subjunctive conditionals, on this version, but instead in terms of nomic relations.

However, one drawback of Pollock's definition of undercutting is that it appears to require that undercut states of affairs be causally determined by their antecedents, i.e., if a state of affairs is not causally determined it cannot be undercut by any antecedent. This presents a difficulty if we consider worlds that are not deterministic. In [Nute, 1981, p. 138] the following example to this effect is found.

> Suppose that protons sometimes emit photons when subjected to a strong magnetic field under certain circumstances. Let P be the state of affairs in which a certain proton is subjected to a strong magnetic field under the appropriate circumstances at time t and let Q be the state of affairs in which this same proton emits a photon shortly after t. Suppose that both P and Q obtain. Assuming that no other states of affairs nomologically relevant to Q obtain, we would intuitively say that $\neg P \mathrel{\Box\!\!\rightarrow} \neg Q$ is true, i.e., if the proton hadn't been subjected to the magnetic field at t, then it wouldn't have emitted a photon shortly after t.

Q has no historical antecedents (in Pollock's sense) since there is no set of simple facts Γ and of laws L such that $\Gamma \cup L$ implies Q, and hence Q cannot be undercut by $\neg P$. One should therefore be conservative with regard to Q when closest $\neg P$-worlds are identified, hence it seems $\neg P \mathrel{\Box\!\!\rightarrow} Q$ risks coming out true. It is evident, I think, that the notion of historical antecedent must be construed in a different way so that, in this case, P might be counted among the historical antecedents of Q.

2.8.2 Cotenability and the Despiteness Relation.

Michael Slote in [Slote, 1978] uses a somewhat different version of undercutting (that he calls "despiteness") in order to define his version of cotenability. Slote makes the interesting and I think important distinction between the *base-time* of a subjunctive conditional

and the *reference-time* of its antecedent clause. He explains the notion of base-time thus [p. 4].

> [W]hen we assert a tensed counterfactual, the conditions cotenable with the antecedent that are, along with the antecedent and laws, supposed to yield the consequent are implicitly thought of as existing all together at single time, rather than diffusely at different times. I shall call that time the *base-time* of the counterfactual[.]

The reference-time of the antecedent on the other hand is simply the time (or times) to which the antecedent in question pertains. On p. 76 above we defined a function T from L into the set of sets of times. The point of this definition was to model the relation between antecedent clauses and the times to which they pertain. If I understand Slote correctly this is roughly what he calls the reference-time of the antecedent.

Slote thinks that he needs this distinction in order to explain the contrast between the following subjunctive conditionals:

(17) If I were (ever) to jump out of this window, I would not hurt myself (since I would never do so without first putting out a safety net, and since I have every means and intention of putting out such a net by tomorrow).

and

(18) If I were now to jump out of this window, I would hurt myself (on the pavement below).

With regard to conditionals with present or future reference times, such as (17) and (18), the base-time is the time of utterance, i.e., the present. Slote explains the contrast between those two conditionals thus [p. 5]:

> For the moment, let us say that a tensed "would" counterfactual is true if and only if there are conditions existing at its base time (cotenable with its antecedent's

> being true with respect to its reference-time) which, together with the antecedent's being true with respect to its reference time and with certain laws, logically entail the consequent's being true with respect to *its* reference-time. [17] says, in effect, that there are present conditions of such a kind that, given laws and assuming that I shall jump at some time in the future, it follows that I shall not hurt myself immediately thereafter. And [17] can be true when the conditions mentioned between its parentheses obtain; for then the conditions obtaining at its base-time, the present, will include: my present witting reluctance to jump out of the window right away, my ability and intention to attach a net, and the absence of factors around with the potential to disturb my reluctance to jump out of the window without assuring my safety with a net.
>
> [18], on the other hand can be true in circumstances where [17] is, because there are conditions existing at its base-time, the present, and cotenable with its antecedent which, together with the truth of its antecedent with respect to its reference-time, the immediate future, and with certain laws, entail the truth of its consequent with respect to its reference time. Those conditions include the present absence of a net and the present hardness of the pavement below. But they do not include my present witting reluctance to jump immediately while being in circumstances where there is nothing with any tendency to destroy that reluctance. This lasts condition is not, presumably, cotenable with [18]'s antecedent; and this difference between [17] and [18] not only results from the difference in their reference-times, but is decisive for allowing them both to come out true despite their seeming incompatibility.

Now, why is it that my present witting reluctance to jump immediately while being in circumstances ... is cotenable with (17)'s antecedent but not with (18)'s antecedent? Slote presents a pro-

visional definition of cotenability of base-time conditions with antecedents at p. 14.

> [B]ase-time conditions b are cotenable with an antecedent just in case b's obtaining at the relevant base-time is nomically compatible with the antecedent.

This definition will later be amended by Slote in terms of the undercutting-like notion of despiteness. But we can already see how this provisional definition is supposed to do the required job: My witting reluctance to jump immediately is presumably nomically incompatible with my jumping right now and therefore not compatible with the antecedent of (18) but quite compatible with me jumping some (unspecified) time in the future, and hence cotenable with the antecedent of (18).

Slote also discusses Goodman's match-example, of which he offers two quite different accounts, only one of which utilizes this provisional notion of cotenability: Slote thinks that the base time of a conditional will not be later than the reference time of its antecedent. Presumably, it is the failure to light upon scratching, i.e., the state of not being light *after the time of scratching* that together with the law governing the behavior of matches and the conditions that it is scratched, that the match is well made and enough oxygen is present at the time of scratching, that yields as result that it is wet at the time of scratching. Since the base time is not later than the reference time of the antecedent, which is presumably the time of scratching, the state of not being light after the time of scratching cannot be included among the cotenable base time conditions, and hence the inference that purports to establish the illegitimate conditional is ruled out. In contrast, the condition that it is not wet is thought of as existing at the time of scratching, so this may be included among the cotenable base-time conditions and hence the inference supporting the legitimate conditional can be made.

Slote's other account of this utilizes of the notion of explanation. He argues on page 12 that

> [F]or appropriate true counterfactuals, the antecedent, cotenable conditions and laws must by themselves be

> capable of validly *explaining* the consequent[.][31]

This requirement is applied in the match-example thus:

> [...] since the match's being scratched and not lighting cannot be used to explain why the match is (was) not dry, our use of the concept of explanation enables us to account for the falsity of this counterfactual.

At this point (p. 15) Slote brings in his despiteness relation in order to deal with a further example[32] that he (correctly) thinks he cannot handle using the notions of explanation and cotenability defined in terms of base-time and logical compatibility. We are invited to

> [i]magine a branching electric circuit connecting a light to a power source with three independently controlled switches, k, l, and m, attached in such a way that the circuit is unbroken if and only if k is closed and at least one of l and m is also closed [This state is referred to by "k-and-l-or-m"]. [...] Suppose that k and l are closed and m is open, so that k-and-l-or-m obtains and the light is on. The counterfactual "if l were open, the light would still be on" is then, presumably, false. But both this counterfactual and its seemingly true *contrary* "if l would be open the light would be off", come out true on our previous account. For, on the one hand, k-and-l-or-m is an existent base-time condition nomically compatible with l's being open; and independently of whether l is open, k-and-l-or-m helps to explain *the the light's being on*. But, on the other hand, m's openness is another existent base-time condition nomically compatible with l's being open; and if we assume that l and m are both open it is easy to produce a valid explanation of *the light's being off*.

[31] Slote's notion of explanation is such that a person's height, the angle of the sun etc., can be said to explain the length of his shadow, but not the other way round.

[32] This example was provided by a reader for the *Philosophical Review*.

Slote observes that

> the relation between l's being open and m's being open is quite different from that between l's being open and k-and-l-or-m. Given the openness of switch m and other conditions (nomically compatible with l's being open) and laws that obtain in the situation described above, the further counterfactual assumption that l is open in no way compels us to say that m is open *despite* the fact that l is open. But given k-and-l-or-m and other appropriate conditions and laws, if we then counterfactually assume that l is open, we do have to say that k-and-l-or-m obtains—and the circuit is unbroken—*despite* the fact that l is open.

Slote thinks that "despiteness" is some kind of "causal-explanatory concept", and he offers the following "causal-explanatory gloss" of the concept:

> [A] base-time condition has a despiteness relation to the antecedent of a counterfactual if and only if there is a valid explanation (involving no extraneous elements) of the base-time condition in terms of the (internal) negation of the antecedent, other actual conditions nomically compatible with that negation, and laws.

I think that it is evident that there is a close affinity between this notion and undercutting as defined by Pollock. Pollock's requirement that *every* nomic pyramid that contains Q nomically implies $\neg P$ clearly serves the same purpose as Slote's requirement that the explanation not contain extraneous elements. A refinement in Pollock's account is that he doesn't require that the negation of the antecedent actually be a member of every nomic pyramid that contains Q, but merely that it be nomically implied by them. A more important difference is that Slote brings in the notion of explanation, that Pollock seems to think he can do without. Furthermore, it is not clear whether this requirement of explainability entails being

determined in the way entailed by Pollock's undercutting requirement. I shall not discuss the similarities and differences of those two notions further, but instead turn to the different roles they seem to play in their different theoretical settings respectively.

Undercutting plays a crucial and well integrated role in Pollock's theory, whereas Slote seems to throw in despiteness as an *ad hoc* solution to the k-and-l-or-m problem. One cannot but wonder if this notion couldn't be given a broader role in his theory. We have seen that Slote handles the match-example by means of the requirement that the consequent be explained by the antecedent and cotenable conditions, and the requirement that the cotenable conditions all exist at the time of scratching. In Pollock's case the explanatory weight with regard to this example is carried solely by the notion of undercutting. I believe, that the distinction between reference-time and base-time is important, but I think it is wrong to assume, as Slote does, that the base-time for past conditionals is the same as their reference time, i.e., the time of the antecedent. It would be better to treat future, present and past conditionals on the same footing and assume that the base time is always the present, i.e., the time of utterance. If this is assumed then the notion of base time cannot bear all the weight of solving the match-problem except when the reference time is the time of utterance. However, we have seen that the notion of cotenability defined in terms of undercutting or despiteness can do this job instead.

What are my reasons for thinking that the base-time should be the present also for past conditionals? The main reason is that I think it is required (for theories relying on the notion of base time) in order to deal with examples like the coin-tossing example above on p. 100. Slote is aware of a version of this example (see his footnote 33), but he admits being unable to handle it, and indeed he couldn't, given the requirement that the cotenable conditions all have to exist at the reference time, in this case the time of betting t_1. At t_1 there is no sufficient set of conditions to determine the outcome of the tossing at the later time t_2, so there is no basis for saying that the outcome of the tossing had been one rather than the other had I bet differently. Now, if we don't require that those conditions exist already at the reference time t_1 but instead that

they exist at the time of utterance we get a quite different outcome. I shall assume that all past and present occurrences can be said to exist at a given point in time, whereas at least some future ones may not (the past is closed but the future open). The outcome of the toss (heads) is then a condition existing at the time of utterance, as well as the fact that I won. In order for the conditional "If I had bet on tails at t_1 I would have lost" $T \mathrel{\Box\!\!\rightarrow} L$ to come out true, as it should, we must count the condition that the coin came up heads, but not the condition that I won as cotenable with the antecedent that I bet on tails. It is easy to see that this job is accomplished by the requirement that cotenable conditions not be undercut by the antecedent, or that they lack the despiteness relation to the antecedent, respectively. My winning, but not the coin coming up heads, is clearly undercut by, and is having a despiteness relation to my betting on tails, and hence should not be counted among the cotenable base time conditions.

Now, most of this could be handled by Pollock's theory alone without the notion of base-time. Pollock like most other theorists doesn't put any temporal requirements on cotenable conditions. However, I suspect that Slote's base-time requirement represents an important insight. This is shown by Slote's example sentences (17) and (18), that genuinely seem to require a base time requirement to be understood. There is also the following consideration: If we revert to the coin tossing example, I think there is a sense in which it could be said that at the time t_1 of betting the conditional $T \mathrel{\Box\!\!\rightarrow} L$ was not yet true, because at that time the coin tossing could still go either way. However, now (after t_2) $T \mathrel{\Box\!\!\rightarrow} L$ is true, since the coin actually came up heads. In the first case the base-time would be t_1 and hence the fact that the coin came up heads would neither be among the cotenable conditions, nor be derivable from them, and hence $T \mathrel{\Box\!\!\rightarrow} L$ would not be supported, whereas in the second case, since the base-time is in this case later than t_2, the fact that the coin came up heads would be included in the cotenable conditions, and hence $T \mathrel{\Box\!\!\rightarrow} L$ would be supported.

Although I think the notion of base-time would be valuable to pursue, I have not made room for it in my RCT-theory.

We shall now try to define undercutting and hence also cotenability in terms of the RCT-criterion.

2.8.3 Cotenability and the RCT-criterion

I shall suggest an alternative definition of undercutting in terms of the semantics S_{RCT} and our refined causal trail criterion. An advantage of this is that we get rid of the requirement of determinism inherent in Pollock's and perhaps also Slote's account.[33] On the other hand the RCT-criterion is defined in terms of the intuitive and admittedly somewhat hazy notion of a causal trail: If S is S_{RCT} and $w \models_S C$,

Definition 2.2 *C at w is* undercut *by A iff there is $w' \in S(A)_w$ such that $\neg C \in E^A_w(w')$.*

One may use this notion of undercutting as a criterion of cotenability in the following way: R is cotenable with P iff R is true and R is not undercut by P, and cotenability, we remember, should serve as a criterion for deciding which true sentences to retain in the contracted set of true sentences.

We shall now see that (a variant of) our refined causal trail semantics S_{RCT} is a strong facts revision semantics defined by a premise contraction function having this property.

We have assumed that S_{RCT} satisfies the ARL-assumption. By Theorem 1.31, S_{RCT} is a class selection semantics defined by the class selection function f such that for all $A \in L$ and $w \in I_L$, $f(A, w) = S_{RCT}(A)_w$. Since f satisfies FID and FMP, by Theorem 1.14 if f is nonempty S_{RCT} is a strong facts revision semantics S defined by the premise contraction function

$$c(A, Th_S(w)) = Th_S(w) \cap \bigcap_{w' \in S(A)_w} Th_S(w').$$

Now f may not be nonempty. It may be the case that $S_{RCT}(A) \neq \emptyset$ and for some $w \in I_L$, $S_{RCT}(A) \cap R_w = \emptyset$, and in that case

[33] I.e., the requirement that undercut facts be causally determined by earlier facts.

$S_{RCT}(A)_w = f(A, w) = \emptyset$. In order to get around this we define a restricted variant of S_{RCT}. It is easy to see (by the assumptions about $\text{R}\underline{\overline{\text{C}}}\text{T}$ on p. 109) that if $I \subseteq I_L$, $\text{R}\underline{\overline{\text{C}}}\text{T}$ is an arrc-assignment for I. For some $w \in I_L$ let I_w be the set of lawful worlds co-commenced with w and let for any $w' \in I_w$, $R_{w'} = I_w$. $\text{R}\underline{\overline{\text{C}}}\text{T}$ is an arrc-assignment for I_w. Let S^w_{RCT} be the ARRC semantics for I_w defined by $\text{R}\underline{\overline{\text{C}}}\text{T}$.

We show that whenever S is S^w_{RCT} (for some $w \in I_L$) and $w \models_S C$,

Proposition 2.36 *C at w is undercut by A iff $C \notin c(A, Th_S(w))$.*

Proof. Assume that C at w is undercut by A. Then by Def. 2.2 there is $w' \in S(A)_w$ such that $\neg C \in E^A_w(w')$, hence $\neg C \in Th_S(w')$. Thus $C \notin \bigcap_{w' \in S(A)_w} Th_S(w')$ and so $C \notin c(A, Th_S(w))$.

Assume on the other hand that $w \models_S C$ and that C at w is not undercut by A. Then by Def. 2.2, for all $w' \in S(A)_w$, $\neg C \notin E^A_w(w')$, and hence by Prop. 2.20 and the definition of $nD^A_w(w')$, $\neg C \notin Th_S(w')$, so $C \in Th_S(w')$ and thus $C \in c(A, Th_S(w))$. $\dashv$

We say that C is *cotenable* with A at w if $w \models_S C$ and C is not undercut by A at w. By Prop. 2.36, obviously whenever S is S_{RCT} and $w \models_S C$,

Proposition 2.37 *C is cotenable with A at w iff C at w is not undercut by A iff $C \in c(A, Th_S(w))$.*

2.8.4 Undercutting and Antecedent Relativity

Pollock [1981, p. 249] (and in his earlier [1976, pp. 42–3] identifies the logic

$$\langle\langle \text{RRCK, RCEA, RCE; CC, RCV, MP, CA, CS} \rangle\rangle^{34}$$

as the one supported by his semantics. He calls this logic SS and in [1976, p. 43] he conjectures that it is complete in the sense that it "contains as theorems all true principles regarding simple subjunctives that can be formulated in this language". On the other hand in [Nute, 1980, p. 130] SS is defined as

[34] For the meaning of this symbolism see Section 3.1 below.

$$\langle\langle \text{RCEC, RCK; ID, MOD, MP, CSO, CA, CS} \rangle\rangle.$$

In what follows I shall refer to the logic defined by Pollock as SS' and the one defined by Nute as SS. Now

Proposition 2.38 *SS is a normal, entailment preserving*[35] *extension of SS'*

Proof. Assume that $\vdash_{SS} A \leftrightarrow B$. Hence by ID and RCK, $\vdash_{SS} A \mathrel{\Box\!\!\rightarrow} B$ and $\vdash_{SS} B \mathrel{\Box\!\!\rightarrow} A$, and hence by CSO for any $C \in L$, $\vdash_{SS} (A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C)$, so SS is closed under RCEA, Since SS is also closed under RCK it is normal.

Since SS is normal and contains ID by Prop. 3.9 below, it is entailment preserving. By definition SS contains the axioms MP, CA, CS. By Prop. 3.10, SS contains CC, and by Prop. 3.11, it contains RCV. ⊣

By Prop. 3.10, SS' is normal, but whether it is also an extension of SS depends on whether MOD and CSO (since ID is obtained by RCE) are contained in SS'.

Pollock says [1976, p. 43] that Lewis's VC[36] is equivalent to SS'+CV. Since VC contains CSO, this is true only if SS'+CV also contains CSO. Since VC is SS+CV, I shall assume that this possibly stronger logic is what Pollock meant to define.

On the other hand, on Pollock's account every counterfactual antecedent undercuts its internal negation, hence even in cases where all the closest A-worlds are B-worlds there may be closest $A \wedge B$-worlds that are not among those. Hence CRCV' and CCV' (p. 175 below) may not be satisfied. If some of those closest $A \wedge B$-worlds that are not closest A-worlds fail to verify some $C \in L$ that is verified by all the closest A-worlds then CRCV and CCV (p. 175 below) are not satisfied, and in that case (by Prop. 3.74 below) RCV and CV are not satisfied, and hence (by Prop. 3.11 below) nor is CSO.

We shall exemplify this in terms of the example on p. 53 above. If A is "Albert goes to the party" and B is "Betty goes to the

[35] See Def 3.9 and p. 160 below.

[36] Pollock refers to this logic as C1, which was Lewis earlier name for it. See [Lewis, 1973b, footnote '*' on p. 130].

party" then it is clear that $\neg A$ is nomically implied by every nomic pyramid containing $\neg B$ and hence A undercuts $\neg B$. Now, this being so, by Pollock's criterion, similarity and dissimilarity with respect to $\neg B$ should be disregarded when worlds are compared with the purpose of evaluating conditionals with the antecedent A. To sum up: Albert's going undercuts Betty's not going and hence we should not try to be conservative with respect to her not going when finding closest A-worlds. Unfortunately, when put in this simple way, Pollock's criterion risks yielding the wrong truth values, and here's why: Although there will surely be worlds at which Betty goes because of Albert, among the closest A-worlds, there may also be such where she goes but not because of him, e.g., such where she goes before he passes by her window. These worlds will be dissimilar from the actual one in that Betty goes to the party, but this dissimilarity should be discounted since the fact that she doesn't go is undercut by the antecedent. But we don't want to allow that. It should not be true that had Albert gone then Betty might have gone before she knew he was going, and it should not be false that had he gone then she would have gone because of him.

To avoid such consequences, I think, one has to be more precise about exactly what is undercut in this example: It is not Betty's staying home *simpliciter* that is undercut by Albert's going, but rather her being home after the point in time at which he passes by her window. Her being home before that time should not be regarded as undercut by his going, and hence worlds where she parts before Albert shows up at her window should be regarded as unnecessarily unlike the actual world, and so should not be counted among the closest A-worlds. In this way we get rid of the worlds at which she goes independently of him. So it will not only be true that had Albert gone then Betty would also have, but also that had he gone then she would have gone with him.

On the other hand, it seems, the antecedent $A \wedge B$ undercuts $\neg B$ simpliciter, and there seems to be no reason to restrict our attention to worlds where Betty goes after the time at which Albert shows up at her window.

So it seems that Pollock's semantics doesn't support RCV (nor CSO) after all. On the other hand Pollock offers a kind of relative

closeness semantics that he thinks differs from that of Lewis (only) in two respects (see [Pollock, 1981, p. 251]):

Let

$w' <_w w''$ iff $w' \leq_w w''$ and $w'' \not\leq_w w'$.

1. His ordering relation is not (necessarily) connected, i.e., there may be worlds w, w', w'' such that $w' \not\leq_w w''$ and $w'' \not\leq_w w'$.

2. Instead of Lewis's definition

   $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} B$ iff $S_\leq(A) \cap R_w = \emptyset$ or there is $w' \in S_\leq(A)$ such that for all $w'' \leq_w w'$, $w'' \in S_\leq(A \rightarrow B)$,

   that he calls *weak*,[37] he offers something equivalent to the following:

   $w \models_{S_\leq} A \mathrel{\Box\!\!\rightarrow} B$ iff $w' \in S_\leq(B)$ whenever $w' \in S_\leq(A)$ and for all $w'' <_w w'$, $w'' \notin S_\leq(A)$,

   that he calls *strict*.[38]

If S is $S_\leq$ we define

$S[A]_w = \{w' : w' \in S(A)$ and for all $w'' <_w w'$, $w'' \notin S(A)\}$.

Then $w \models_S A \mathrel{\Box\!\!\rightarrow} B$ in the strict sense iff

$S[A]_w \subseteq S(B)$.

Now we can show

Proposition 2.39 *$w \models_S \Diamond A$ iff $S[A]_w \neq \emptyset$.*[39]

[37] Pollock's actual way of putting it is that, in this case '$\mathrel{\Box\!\!\rightarrow}$' is *weakly based on* the ordering relation $\leq$.

[38] In this case '$\mathrel{\Box\!\!\rightarrow}$' is *strictly based* on $\leq$, according to Pollock.

[39] For the definition of $\Diamond A$ see p. 158 below.

Proof. $w \models_S \Diamond A$ iff $w \models_S \neg(A \mathrel{\Box\!\!\rightarrow} \neg A)$ iff $w \not\models_S A \mathrel{\Box\!\!\rightarrow} \neg A$ iff $S[A]_w \not\subseteq S(\neg A)$ iff $S[A]_w \neq \emptyset$. ⊣

Even if $S(A) \neq \emptyset$ it might be that $S[A]_w = \emptyset$ since it might be the case that for every $w' \in S(A)$ there is some $w'' <_w w'$ such that $w'' \in S(A)$, i.e., the limit assumption might not be satisfied.

Now let us assume that our ordering satisfies the following strong version of the limit assumption:[40]

Condition 2.6 *For any $i \in I$ and $A \in L$ and $w'' \in S_{\leq}(A)$ there is one $w \in S_{\leq}(A)$ such that $w \leq_i w''$ and for every $w' \in I$ if $w' <_i w$ then $w' \notin S_{\leq}(A)$.*

We call this the non-connected limit assumption (NCL assumption for short).

We show that if $\leq$ satisfies the NCL assumption:

Proposition 2.40 *If $S[A]_w \subseteq S(B)$ and $S[B]_w \subseteq S(A)$ then $S[A]_w = S[B]_w$.*

Proof. Assume that $S[A]_w \subseteq S(B)$ and $S[B]_w \subseteq S(A)$ and for contradiction that there is $w' \in S[A]_w$ such that $w' \notin S[B]_w$. Now $w' \in S(B)$ and (since $w' \notin S[B]_w$) there is $w'' <_w w'$ such that $w'' \in S(B)$. By the NCL assumption there is $w''' \in S[B]_w$ such that $w''' \leq_w w''$, hence $w''' \in S(A)$, but $w''' <_w w'$ (by the transitivity of $\leq$), which is a contradiction. ⊣

Hence if $\leq$ is a non-total reflexive, transitive ordering of I that satisfies the NCL assumption and $S_{\leq}$ is obtained by the strict definition above then $S_{\leq}$ validates CSO. We have also seen that Pollock included RCV in his definition of SS.

On the other hand, as we have seen, there is reason to think that if we take undercutting seriously there are counterexamples to RCV and hence to CSO.

[40] It is true that Pollock explicitly rejects the limit assumption [1981, p. 251–2], but arguably the limit assumption is satisfied with respect to the sentences occurring in this example.

Pollock [1981, p. 250] offers an explicit definition of his ordering relation. Undercutting with respect to the antecedent plays a vital role in this definition, and thus the antecedent is part (as argument) of the expression defining this ordering relation. He notes [p. 251] that the ordering relation thus defined is not connected, but he fails to recognize, it seems to me, that it is also antecedent relative.

Chapter 3

Axiomatizations

3.1 Rules, Axioms and Logics

3.1.1 The Concept of Conditional Logic

A *conditional logic* Ψ is a subset of L containing all tautologies of propositional calculus and closed under modus ponens.

Definition 3.1 *A* rule of derivation *R is a subset of $P(L) \times L$ such that if $\langle \Gamma, \phi \rangle \in R$ then Γ is finite.*

If R is a rule of derivation we write

$$\Gamma \vdash_R \phi$$

for $\langle \Gamma, \phi \rangle \in R$.

If $\Theta \subseteq L$ and Ru is a set of rules of derivation for L then $\langle \Theta, Ru \rangle$ determines a conditional logic in the following way. For $\phi, \phi' \in L$, let

Definition 3.2 *$Us(\phi, \phi')$ iff there are $\psi, \psi' \in L$ such that ψ is atomic, ϕ contains ψ and ϕ' is the result of substituting ψ' for all occurrences of ψ in ϕ.*

Definition 3.3 $\Theta_{Ax} = \{\psi : Us(\phi, \psi) \;\&\; \phi \in \Theta\}$.

Let Tau be the set of all tautologies, and $MOPO$ the rule modus ponens. If Ru is a set of rules of derivation let

Definition 3.4 $\Gamma \vdash_{Ru} \phi$ *iff there is a sequence of sets* $\Gamma_0 \subseteq \Gamma_1 \subseteq \ldots \subseteq \Gamma_n$ *such that* $\Gamma = \Gamma_0$, $\phi \in \Gamma_n$ *and for each* $n \geq k > 0$ *there is* $\psi \in L$, $R \in Ru \cup \{MOPO\}$, *and* $\Delta \subseteq \Gamma_{k-1} \cup Tau$ *such that* $\Gamma_k = \Gamma_{k-1} \cup \{\psi\}$ *and* $\Delta \vdash_R \psi$.

Also let

Definition 3.5 $\vdash_{\langle\Theta,Ru\rangle} \phi$ *iff* $\Theta_{Ax} \vdash_{Ru} \phi$

Definition 3.6 $\Psi_{\langle\Theta,Ru\rangle} = \{\phi : \vdash_{\langle\Theta,Ru\rangle} \phi\}$

Proposition 3.1 $\Psi_{\langle\Theta,Ru\rangle}$ *is the smallest conditional logic containing* Θ_{Ax} *which is closed under the rules in* Ru.

If Ψ is a conditional logic, instead of $\phi \in \Psi$ we shall often write

$$\vdash_{\Psi} \phi.$$

hence

$$\vdash_{\Psi_{\langle\Theta,Ru\rangle}} \phi \text{ iff } \vdash_{\langle\Theta,Ru\rangle} \phi \text{ iff } \Theta_{Ax} \vdash_{Ru} \phi$$

Definition 3.7 *If* $Ru = \emptyset$ *then we write* $\Gamma \vdash \phi$ *for* $\Gamma \vdash_{Ru} \phi$. *And if* $\Theta = \{\alpha_1, \ldots, \alpha_n\}$ *and* $Ru = \{\rho_1, \ldots, \rho_m\}$, *instead of* $\Psi_{\langle\Theta,Ru\rangle}$ *we often write* $\langle\langle \rho_1, \ldots, \rho_m; \alpha_1, \ldots, \alpha_n \rangle\rangle$.

3.1.2 Rules

We shall consider the following rules of inference:[1]

MOPO From $A \to B$ and A infer B.

RCEA From $A \leftrightarrow B$ infer $(A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C)$.

[1] RCEA, RCEC, RCK and RCE are found under those names in the entry on Conditional Logic in *Handbook of Philosophical Logic* [Nute, 1984, p. 433]. MOPO, of course, is simply Modus Ponens, and RRCK is a restricted version of RCK.

RCEC From $A \leftrightarrow B$ infer $(C \mathrel{\Box\!\!\rightarrow} A) \leftrightarrow (C \mathrel{\Box\!\!\rightarrow} B)$.

RCK From $(A_1 \wedge A_2 \wedge \cdots \wedge A_n) \rightarrow B$ infer $(C \mathrel{\Box\!\!\rightarrow} A_1) \wedge C \mathrel{\Box\!\!\rightarrow} A_2 \wedge \cdots \wedge C \mathrel{\Box\!\!\rightarrow} A_n) \rightarrow (C \mathrel{\Box\!\!\rightarrow} B)$.

RCE From $A \rightarrow B$ infer $A \mathrel{\Box\!\!\rightarrow} B$.

RRCK From $A \rightarrow B$ infer $(C \mathrel{\Box\!\!\rightarrow} A) \rightarrow (C \mathrel{\Box\!\!\rightarrow} B)$.

I.e. MOPO=$\{\langle \Gamma, \phi \rangle : A, B \in L, \Gamma = \{A \rightarrow B, A\}$ and $\phi = B\}$, and RCEA=$\{\langle \Gamma, \phi \rangle : A, B, C \in L, \Gamma = \{A \leftrightarrow B\}$ and $\phi = (A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C)\}$, and so on for the others.

Soundness of Rules

For any semantics S, let

Definition 3.8 $Cn_S = \{\phi \in L : \models_S \phi\}$.

We say that a rule R of derivation is *sound with respect to* the semantics S iff

$$\psi \in Cn_S \text{ whenever } \Gamma \vdash_R \psi \text{ and } \Gamma \subseteq Cn_S.$$

Obviously

Proposition 3.2 *MOPO is sound with respect to every semantics.*

It is also straightforward that for all semantics S,

Proposition 3.3 *S satisfies CP (Condition 1.2 on page 50 above) iff RCK is sound with respect to S.*

Recall that all class selection semantics satisfy CP; indeed they even satisfy the stronger GCP (Prop. 1.6). Also all ARRC semantics (so also all relative closeness semantics) satisfy CP (Prop. 1.26). So RCK (and hence also RRCK) is sound with respect to these semantics. It is also obvious that

Proposition 3.4 *S is consequent intensional iff RCEC is sound with respect to S.*

Again class selection as well as ARRC semantics are consequent intensional so RCEC is sound with respect to them.

In the same way,

Proposition 3.5 *S is antecedent intensional (p. 115 above) iff RCEA is sound with respect to S*

Relative closeness semantics are intensional (Prop. 1.16) so they are antecedent intensional, but we remember that not all class selection semantics are. However, propositional class selection semantics are antecedent intensional, so RCEA is sound with respect to these.

Proposition 3.6 *S is intensional iff RCEC and RCEA are sound with respect to S.*

The hyperintensional semantics we shall consider will all be consequent intensional; indeed they will all satisfy CP, since they are class selection or ARRC semantics. This means that they will not be antecedent intensional and hence RCEA will not be sound with respect to them.

3.1.3 Axiom schemata

The following axiom schemata[2] will be considered (some but not all of them have been mentioned earlier).

SDA $((A \vee B) \mathrel{\Box\!\!\rightarrow} C) \rightarrow ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (B \mathrel{\Box\!\!\rightarrow} C))$

ID $A \mathrel{\Box\!\!\rightarrow} A$

MOD' $(A \mathrel{\Box\!\!\rightarrow} \neg A) \rightarrow (B \mathrel{\Box\!\!\rightarrow} \neg A)$

MOD $(\neg A \mathrel{\Box\!\!\rightarrow} A) \rightarrow (B \mathrel{\Box\!\!\rightarrow} A)$

CC $((A \mathrel{\Box\!\!\rightarrow} B) \wedge (A \mathrel{\Box\!\!\rightarrow} C)) \rightarrow (A \mathrel{\Box\!\!\rightarrow} (B \wedge C))$

[2]The axioms SDA, ID, MOD, MP, CC, RT, CSO, CV, CA, CS and CEM can be found under those names in [Nute, 1980, p. 128–9]. MOD' is similar to MOD. RCV (which is found in [Pollock, 1976, p. 43] under the name A4) can be seen as a weaker (or restricted) variant of CV.

RT $((A \mathrel{\Box\!\!\rightarrow} B) \wedge ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)) \rightarrow (A \mathrel{\Box\!\!\rightarrow} C)$

MP $(A \mathrel{\Box\!\!\rightarrow} B) \rightarrow (A \rightarrow B)$

CSO $((A \mathrel{\Box\!\!\rightarrow} B) \wedge (B \mathrel{\Box\!\!\rightarrow} A)) \rightarrow ((A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C))$

CV $((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Diamond\!\!\rightarrow} B)) \rightarrow ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$

RCV $((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Box\!\!\rightarrow} B)) \rightarrow ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$

CA $((A \mathrel{\Box\!\!\rightarrow} B) \wedge (C \mathrel{\Box\!\!\rightarrow} B)) \rightarrow ((A \vee C) \mathrel{\Box\!\!\rightarrow} B)$

CS $(A \wedge B) \rightarrow (A \mathrel{\Box\!\!\rightarrow} B)$

CEM $(A \mathrel{\Box\!\!\rightarrow} B) \vee (A \mathrel{\Box\!\!\rightarrow} \neg B)$

3.1.4 Some Conditional Logics

We shall consider the following logics:[3]

Ck $\langle\langle$RCK;$\rangle\rangle$

PR $\langle\langle$RCK; ID, MP$\rangle\rangle$

[3]Ck, VW, VC, SS, and C2 are found in [Nute, 1980, pp. 129–30]. Nute defines Ck, VW etc. as $\langle\langle$RCK, RCEC;$\rangle\rangle$, $\langle\langle$RCK, RCEC; ID, MOD, MP, CSO, CV$\rangle\rangle$ etc., but by Prop. 3.8 below RCEC is redundant in these definitions. VW and VC are defined by Lewis in somewhat different terms in [Lewis, 1973b, pp. 132–3]. Nute defines VW, VC and C2 as $\langle\langle$RCK, RCEC; ID, MOD, MP, CSO, CV$\rangle\rangle$, $\langle\langle$RCK, RCEC; ID, MOD, MP, CSO, CV, CS$\rangle\rangle$, and $\langle\langle$RCK, RCEC; ID, MOD, MP, CSO, CV, CEM$\rangle\rangle$ respectively, and says (by implication) [pp. 130–1] that the schema CA is contained in these logics. I have found no proof of this in [Nute, 1980]. There is indeed a proof of completeness [Theorem 7.3.22 on p. 144] for C2 with respect to a class of models that validate CA (by additivity [p. 139]), but in this proof it appears that CA is used in a way that indicates that it is already assumed to be a theorem of C2. Since I use CA in my proof of completeness for these logics and I have not managed to derive it myself, from the other axioms, I have chosen to add it (redundantly, I suppose) to the list of axioms, defining those logics. In [Pollock, 1976, pp. 42–3] the logic $\langle\langle$RRCK, RCEA, RCE; CC, RCV, MP, CA, CS$\rangle\rangle$ is given the name SS. Nute [1980, p. 130], on the other hand, defines SS as $\langle\langle$RCK; ID, MOD, MP, CSO, CA, CS$\rangle\rangle$. (Cf. Section 2.8.4 above for a short discussion of this). C2 is defined in different terms in [Stalnaker, 1968b, p. 48]. The appellations PR (Premise Revision), AR (Antecedent Relativity), AR', PAR (Propositional Antecedent Relativity) and CT (Causal Trail) are of my own invention.

AR $\langle\langle$RCK; ID, MOD, MP$\rangle\rangle$

AR' $\langle\langle$RCK; ID, MOD', MP$\rangle\rangle$

PAR $\langle\langle$RCK, RCEA; ID, MOD, MP$\rangle\rangle$

CT $\langle\langle$RCK, RCEA; ID, MOD, MP, RT, CA$\rangle\rangle$

VW $\langle\langle$RCK; ID, MOD, MP, CSO, CV, CA$\rangle\rangle$

VC $\langle\langle$RCK; ID, MOD, MP, CSO, CV, CS, CA$\rangle\rangle$

C2 $\langle\langle$RCK; ID, MOD, MP, CSO, CV, CEM, CA$\rangle\rangle$

SS $\langle\langle$RCK; ID, MOD, MP, CSO, CA, CS$\rangle\rangle$

Definition 3.9 *A conditional logic is* half-normal *iff it is closed under RCK,* half-classical *iff it is closed under RCEC,* normal *iff it is closed under RCEA and RCK,* classical *iff it is closed under RCEA and RCEC, and* entailment preserving *iff it is closed under RCE.*[4]

If Ψ is any logic, obviously

Proposition 3.7 *If Ψ is normal (classical) then it is half-normal (half-classical).*

Also

Proposition 3.8 *If Ψ is (half-)normal then it is (half-)classical.*

Proof. Assume that Ψ is closed under RCK and that $\vdash_\Psi A \leftrightarrow B$. Then $\vdash_\Psi A \rightarrow B$ and $\vdash_\Psi B \rightarrow A$, so for any $C \in L$, by RCK, $\vdash_\Psi (C \mathrel{\Box\!\!\rightarrow} A) \rightarrow (C \mathrel{\Box\!\!\rightarrow} B)$ and $\vdash_\Psi (C \mathrel{\Box\!\!\rightarrow} B) \rightarrow (C \mathrel{\Box\!\!\rightarrow} A)$. $\dashv$

[4]These terms which are found in [Nute, 1980] are taken from [Chellas, 1975]. According to Nute a logic is half-normal iff it is a half-classical extension of $\langle\langle$RCK,RCEC;$\rangle\rangle$, and entailment preserving if it is classical and closed under RCE. I have chosen the present definitions instead, because they serve my purposes better.

Proposition 3.9 *If Ψ is half-normal then Ψ is an extension of $\langle\langle;ID\ \rangle\rangle$ iff it is entailment preserving.*

Proof. Assume that Ψ is a half-normal extension of $\langle\langle;\mathrm{ID}\rangle\rangle$ and that $\vdash_\Psi A \to B$ By ID, $\vdash_\Psi A \mathrel{\Box\!\!\to} A$, so, by RCK, $\vdash_\Psi A \mathrel{\Box\!\!\to} B$.

Assume on the other hand that Ψ is entailment preserving. Since for every $A \in L$, $\vdash_\Psi A \to A$, by RCE, $\vdash_\Psi A \mathrel{\Box\!\!\to} A$. $\dashv$

Proposition 3.10 *Ψ is half-normal iff it is an extension of $\langle\langle;CC\rangle\rangle$ which is closed under RRCK.*

Proof. Assume that Ψ is half-normal. Since $\vdash_\Psi (B \wedge C) \to (B \wedge C)$ we get by RCK, that $\vdash_\Psi CC$.

Assume on the other hand that Ψ is an extension of $\langle\langle;\mathrm{CC}\rangle\rangle$ closed under RRCK, and that $\vdash_\Psi (A_1 \wedge A_2 \wedge \cdots \wedge A_n) \to B$. Then by CC, $\vdash_\Psi ((C \mathrel{\Box\!\!\to} A_1) \wedge (C \mathrel{\Box\!\!\to} A_2)) \to (C \mathrel{\Box\!\!\to} (A_1 \wedge A_2))$, $\vdash_\Psi ((C \mathrel{\Box\!\!\to} (A_1 \wedge A_2)) \wedge (C \mathrel{\Box\!\!\to} A_3)) \to (C \mathrel{\Box\!\!\to} (A_1 \wedge A_2 \wedge A_3)), \ldots,$ $\vdash_\Psi ((C \mathrel{\Box\!\!\to} (A_1 \wedge A_2 \wedge \cdots \wedge A_{n-1})) \wedge (C \mathrel{\Box\!\!\to} A_n)) \to (C \mathrel{\Box\!\!\to} (A_1 \wedge A_2 \wedge \cdots \wedge A_n))$. Hence $\vdash_\Psi ((C \mathrel{\Box\!\!\to} A_1) \wedge C \mathrel{\Box\!\!\to} A_2 \wedge \cdots \wedge (C \mathrel{\Box\!\!\to} A_n)) \to (C \mathrel{\Box\!\!\to} (A_1 \wedge A_2 \wedge \cdots \wedge A_n))$, so by RRCK, $\vdash_\Psi ((C \mathrel{\Box\!\!\to} A_1) \wedge C \mathrel{\Box\!\!\to} A_2 \wedge \cdots \wedge (C \mathrel{\Box\!\!\to} A_n)) \to (C \mathrel{\Box\!\!\to} B)$. $\dashv$

We also note that

Proposition 3.11 *If Ψ is a half-normal extension of $\langle\langle;CSO,\ ID\rangle\rangle$ then it is a normal extension of $\langle\langle;RT,\ RCV\rangle\rangle$.*

Proof. Assume that Ψ is a half-normal extension of $\langle\langle;\mathrm{CSO},\ \mathrm{ID}\rangle\rangle$. By RCK, $\vdash_\Psi (A \mathrel{\Box\!\!\to} B) \to (A \mathrel{\Box\!\!\to} (A \wedge B))$, and by ID and RCK $\vdash_\Psi (A \wedge B) \mathrel{\Box\!\!\to} A$. Hence for any $C \in L$, by CSO, $\vdash_\Psi (A \mathrel{\Box\!\!\to} B) \to ((A \mathrel{\Box\!\!\to} C) \leftrightarrow ((A \wedge B) \mathrel{\Box\!\!\to} C))$, and so $\vdash_\Psi RT$ and $\vdash_\Psi RCV$.

Assume that $\vdash_\Psi A \leftrightarrow B$. By ID and RCK, $\vdash_\Psi A \mathrel{\Box\!\!\to} B$ and $\vdash_\Psi B \mathrel{\Box\!\!\to} A$, hence by CSO, for any C, $\vdash_\Psi (A \mathrel{\Box\!\!\to} C) \leftrightarrow (B \mathrel{\Box\!\!\to} C)$, so Ψ is closed under RCEA, i.e., it is normal. $\dashv$

If Δ is a finite subset of L we let

$$\wedge\Delta$$

be the conjunction of all the elements of Δ

Definition 3.10 *If Ψ is a conditional logic, $\Gamma \subseteq L$ and $\psi \in L$ let $\Gamma \vdash_\Psi \phi$ iff there is finite $\Delta \subseteq \Gamma$ such that $\vdash_\Psi \wedge\Delta \rightarrow \phi$.*

(If $\psi \in L$ we write $\psi \vdash_\Psi \phi$ instead of $\{\psi\} \vdash_\Psi \phi$, for simplicity.)

In Section 1.5 above we defined semantics of different types: Class selection semantics, Facts revisions semantics, Relative closeness semantics and ARRC semantics. For each of these types of semantics we shall identify a conditional logic that corresponds to the semantics in question, in the sense of containing all and only those sentences that are valid in all the semantics of that kind. We start with some definitions:

If Σ is a set of semantics let

$$\models_\Sigma \phi \text{ iff for all } S \in \Sigma, \models_S \phi,$$

and

$$Cn_\Sigma = \{\phi \in L : \models_\Sigma \phi\}.$$

Definition 3.11 *A conditional logic Ψ is* sound *with respect to a set of semantics Σ iff $\Psi \subseteq Cn_\Sigma$.*

Definition 3.12 *Ψ is* complete *with respect to a Σ iff $Cn_\Sigma \subseteq \Psi$.*

Now we shall discuss the different kinds of semantics in turn. Completeness of the different logics with respect to their corresponding sets of semantics will be shown using standard methods.[5] Given any conditional logic, by Lindenbaum's lemma there is a maximally consistent superset of any consistent set of sentences. We construct a preferred semantics for the conditional logic in question: the *canonical semantics*. The domain of this semantics will consist of all maximally consistent sets of sentences, with respect

[5]See for instance [Lewis, 1973b, pp. 124–130].

to the logic in question. Truth at a world will be defined as membership of that world, hence every consistent set of sentences will be true at some world of the canonical semantics. Finally we will show that the canonical semantics for a certain conditional logic is indeed a semantics of the type corresponding to that logic. So if a sentence is not contained in a certain conditional logic there will be consistent sets containing its negation and hence it will not be valid with respect to the canonical semantics for this logic, which is a semantics of the type corresponding to the logic in question; hence this sentence is not valid in all semantics of this type. To put it the other way round: If a sentence is valid at all semantics of a certain type then it is also contained in the corresponding conditional logic, hence the logic in question is complete with respect to this set of semantics.

In this way we will show that the logics Ck, PR, AR' and VW are (sound and) complete with respect to the sets of class selection semantics, weak generalized facts revision semantics, ARRC-semantics and relative closeness semantics respectively.

The crucial step in these proofs are the proofs that the canonical semantics for each of these logics is of the corresponding type, i.e., that the canonical semantics for Ck is a class selection semantics, that the canonical semantics for AR' is an AARC-semantics, and so on. In Section 1.5 we discussed various extra conditions that may be put on class selection semantics, facts revisions semantics and ARRC-semantics, respectively, such as the condition of propositionality (among many other conditions), which can be put on each of them, hence defining the set of propositional (etc.) class selection semantics, propositional (etc.) facts revision semantics, and so forth. We shall see that there is a simple way of identifying corresponding conditional logics for those restricted sets of semantics.

We start by defining the canonical semantics for a given conditional logic Ψ.

3.2 The Canonical Semantics for Ψ

If Ψ is a conditional logic, we say that $\Gamma \subseteq L$ is Ψ-*consistent* if

there is no $\phi \in L$ such that $\Gamma \vdash_\Psi \phi \wedge \neg\phi$.

And Γ is *maximally Ψ-consistent* if

Γ is Ψ-consistent and for every $\phi \in L$, if $\phi \notin \Gamma$ then $\Gamma \vdash_\Psi \neg\phi$.

Proposition 3.12 *If $\Gamma \subseteq L$ is maximally Ψ-consistent then for every $\phi \in L$, $\Gamma \vdash_\Psi \phi$ iff $\phi \in \Gamma$ iff $\Gamma \nvdash_\Psi \neg\phi$.*

Proposition 3.13 *If Γ is maximally Ψ-consistent then for every $\phi, \psi \in L$, $\phi \wedge \psi \in \Gamma$ iff $\phi \in \Gamma$ and $\psi \in \Gamma$.*

Proposition 3.14 *If Γ is maximally Ψ-consistent then for every $\phi, \psi \in L$, $\phi \vee \psi \in \Gamma$ iff $\phi \in \Gamma$ or $\psi \in \Gamma$.*

Lemma 3.15 (Lindenbaum's lemma) *If $\Gamma \subseteq L$ is Ψ-consistent then there is $\Delta \subseteq L$ such that Δ is maximally Ψ-consistent and $\Gamma \subseteq \Delta$.*

For a proof of this lemma, see e.g. [Lewis, 1973b, p. 125].

Let Υ_Ψ be the set of maximally Ψ-consistent subsets of L. For each $\Gamma \in \Upsilon_\Psi$ if A is atomic we let

$$at_\Gamma(A) = \Gamma(A) = \begin{cases} 1 & \text{if } A \in \Gamma, \\ 0 & \text{otherwise.} \end{cases}$$

$\Upsilon_\Psi (= \langle \Upsilon_\Psi, at \rangle)$ is obviously a domain (p. 25). We define a function S_Ψ from L to $P(\Upsilon_\Psi)$ thus: For each $A \in L$ let

$$S_\Psi(A) = \{\Gamma \in \Upsilon_\Psi : A \in \Gamma\}.$$

Lemma 3.16 *S_Ψ is a semantics for Υ_Ψ.*

Proof.

1. If A is atomic then $S_\Psi(A) = \{\Gamma \in \Upsilon_\Psi : A \in \Gamma\} = \{\Gamma \in \Upsilon_\Psi : \Gamma(A) = 1\}$.

2. By Prop. 3.12, $S_\Psi(\neg A) = \{\Gamma \in \Upsilon_\Psi : \neg A \in \Gamma\} = \Upsilon_\Psi - \{\Gamma \in \Upsilon_\Psi : A \in \Gamma\} = \Upsilon_\Psi - S_\Psi(A)$.

3. By Prop. 3.13, $S_\Psi(A \wedge B) = \{\Gamma \in \Upsilon_\Psi : A \wedge B \in \Gamma\} = \{\Gamma \in \Upsilon_\Psi : A \in \Gamma \text{ and } B \in \Gamma\} = S_\Psi(A) \cap S_\Psi(B)$, and so on for the other connectives except " $\Box\!\!\rightarrow$ ".

$\dashv$

We call S_Ψ the *canonical semantics* for Ψ.[6] We note the following

Proposition 3.17 $S_\Psi(A) \subseteq S_\Psi(B)$ *iff* $\vdash_\Psi A \rightarrow B$.

Proof. From right to left is trivial.

Assume that $S_\Psi(A) \subseteq S_\Psi(B)$ and for contradiction that $\nvdash_\Psi A \rightarrow B$. Then $A \wedge \neg B$ is Ψ-consistent, so there is $\Delta \in \Upsilon_\Psi$ such that $A \wedge \neg B \in \Delta$, but then $\Delta \in S_\Psi(A) - S_\Psi(B)$, which is a contradiction. $\dashv$

Now we proceed to show that the very weak logic Ck is sound and complete with respect to the set of class selection semantics, and then discuss the result of putting the extra requirements FID, FMP, FMOD and so forth on the class selection function.

3.3 Class Selection Semantics

3.3.1 Soundness of Ck with respect to the Set of Class Selection Semantics

Since obviously for any semantics S (cf. Def. 1.1), $Tau \subseteq Cn_S$, by Prop. 3.2, 3.3 and 1.6 we obtain that

Proposition 3.18 *Ck is sound with respect to the set of Class Selection Semantics.*

3.3.2 Completeness of Ck with respect to the Set of Class Selection Semantics

We show that Ck is complete with respect to the set of class selection semantics Σ_{CS}. This is the case if for every ϕ such that $\models_{\Sigma_{CS}} \phi$,

[6] This corresponds to Lewis's *canonical interpretation* of [p. 125].

$\emptyset_{Ax} \vdash_{\{RCK\}} \phi$. We can prove this if we can find a class selection semantics $S \in \Sigma_{CS}$ such that if $\emptyset_{Ax} \nvdash_{\{RCK\}} \phi$ then there is some $w \in S(\neg\phi)$ and so $\not\models_{\Sigma_{CS}} \phi$.

Define a class selection function for Υ_Ψ as follows: For each $\Gamma \in \Upsilon_\Psi$ and $A \in L$, let

$$f_\Psi(A, \Gamma) = \{\Delta \in \Upsilon_\Psi : \text{ if } A \mathrel{\Box\!\!\rightarrow} C \in \Gamma \text{ then } C \in \Delta\}.$$

f_Ψ is obviously a class selection function for Υ_Ψ. We show that

Lemma 3.19 *If Ψ is a half-normal conditional logic, and $\Gamma \in \Upsilon_\Psi$ then $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$ iff $f_\Psi(A, \Gamma) \subseteq S_\Psi(C)$.*

Proof. From left to right is trivial. So assume that $A \mathrel{\Box\!\!\rightarrow} C \notin \Gamma$. Let $\Gamma_A = \{\phi \in L : A \mathrel{\Box\!\!\rightarrow} \phi \in \Gamma\}$. We show that $\Gamma_A \cup \{\neg C\}$ is Ψ-consistent. Assume otherwise. Then there is a finite $\Gamma'_A \subseteq \Gamma_A$ such that $\vdash_\Psi \wedge\Gamma'_A \rightarrow C$, but then (by RCK) $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$ which is a contradiction.

By Lemma 3.15 there is a maximally Ψ-consistent subset Δ of L, such that $\Gamma_A \cup \{\neg C\} \subseteq \Delta$. Since $\Gamma_A \subseteq \Delta$, $\Delta \in f_\Psi(A, \Gamma)$, and since also $\Delta \notin S_\Psi(C)$, $f_\Psi(A, \Gamma) \not\subseteq S_\Psi(C)$. $\dashv$

Hence (by induction on the complexity of formulas),

Theorem 3.20 *If Ψ is a half-normal conditional logic then S_Ψ is the class selection semantics S_{f_Ψ} (for Υ_Ψ) defined by f_Ψ.*

Since Ck is a half-normal conditional logic it follows that

Proposition 3.21 *S_{Ck} is a class selection semantics for Υ_{Ck}.*

It now follows that

Proposition 3.22 *Ck is complete with respect to the set of class selection semantics.*

Proof. We must show that $Cn_{\Sigma_{CS}} \subseteq Ck$, i.e. that

if $\models_{\Sigma_{CS}} \phi$ then $\vdash_{Ck} \phi$.

Assume that for some $\phi \in L$ $\nvdash_{Ck} \phi$. Then $\neg\phi$ is Ck-consistent. Let Λ be some maximally Ck-consistent set of sentences that contains $\neg\phi$. Then $\Lambda \in S_{Ck}(\neg\phi)$, so $\not\models_{S_{Ck}} \phi$, and hence $\not\models_{\Sigma_{CS}} \phi$. $\dashv$

3.3.3 Half-Normal Logics and various Class Selection Semantics

Let f be any class selection function for some domain I, and S_f a class selection semantics (for I) defined in terms of f. We shall relate different axioms to the requirements below. We say that S_f (or sometimes f) satisfies any of the following requirements respectively if for all $A, B, C \in L$ and every $w \in I$:

FID $f(A, w) \subseteq S_f(A)$.

FMOD If $f(\neg A, w) \subseteq S_f(A)$ then for all $B \in L$, $f(B, w) \subseteq S_f(A)$.

FRT' If $f(A, w) \subseteq S_f(B)$ then $f(A, w) \subseteq f(A \wedge B, w)$.

FRT If $f(A, w) \subseteq S_f(B)$ and $f(A \wedge B, w) \subseteq S_f(C)$ then $f(A, w) \subseteq S_f(C)$.

FMP' If $w \in S_f(A)$ then $w \in f(A, w)$.

FMP If $f(A, w) \subseteq S_f(B)$ then $w \in S_f(A \rightarrow B)$.

FCSO If $f(A, w) \subseteq S_f(B)$ and $f(B, w) \subseteq S_f(A)$ then $f(A, w) = f(B, w)$.

FCV' If $f(A, w) \not\subseteq S_f(\neg B)$ then $f(A \wedge B, w) \subseteq f(A, w)$.

FCV If $f(A, w) \subseteq S_f(C)$ and $f(A, w) \not\subseteq S_f(\neg B)$ then $f(A \wedge B, w) \subseteq S_f(C)$.

FRCV' If $f(A, w) \subseteq S_f(B)$ then $f(A \wedge B, w) \subseteq f(A, w)$.

FRCV If $f(A, w) \subseteq S_f(C)$ and $f(A, w) \subseteq S_f(B)$ then $f(A \wedge B, w) \subseteq S_f(C)$.

FCA' $f(A \vee B, w) \subseteq f(A, w) \cup f(B, w)$.

FCA If $f(A, w) \cup f(B, w) \subseteq S_f(C)$ then $f(A \vee B, w) \subseteq S_f(C)$.

FSDA' $f(A, w) \cup f(B, w) \subseteq f(A \vee B, w)$.

FSDA If $f(A \vee B, w) \subseteq S_f(C)$ then $f(A, w) \cup f(B, w) \subseteq S_f(C)$.

FCS' If $w \models_{S_f} A$ then $f(A, w) = \{w\}$.

FCS If $w \in S_f(A \wedge B)$ then $f(A, w) \subseteq S_f(B)$.

FCEM' For all $A \in L$ and $w \in I$ there is exactly one $w' \in I$ such that $f(A, w) = \{w'\}$.

FCEM For all $A, B \in L$ and $w \in I$ either $f(A, w) \subseteq S_f(B)$ or $f(A, w) \subseteq S_f(\neg B)$.

We say that a semantics S validates an axiom ϕ iff

$$\{\phi\}_{Ax} \subseteq Cn_S.$$

Clearly,

Proposition 3.23 *If ϕ is one of the axioms of the logic Ψ then S_Ψ validates ϕ.*

If S_f is the class selection semantics defined by f then by Def. 1.8 (p. 31), for $X \in \{$ID, MOD, RT, MP, CSO, CV, RCV, CA, SDA, CS, CEM$\}$:

Proposition 3.24 *S_f validates X iff f satisfies FX.*

It is also obvious that for $X \in \{$FRT, FMP, FCV, FRCV, FCA, FSDA, FCS, FCEM$\}$:

Proposition 3.25 *If f satisfies X', then it satisfies X.*

Proposition 3.26 *If f is either finite and selective (p. 34) or projective (p. 35) and for every $C \in L$, $f(A, w) \subseteq S_f(C)$ whenever $f(B, w) \subseteq S_f(C)$, then $f(A, w) \subseteq f(B, w)$.*

Proof. Assume that $C \in L$, $f(A, w) \subseteq S_f(C)$ whenever $f(B, w) \subseteq S_f(C)$, and for contradiction that $f(A, w) \not\subseteq f(B, w)$.

If f is projective we obtain a contradiction by the proof of Prop. 1.9, and hence $f(A, w) \subseteq f(B, w)$.

If on the other hand f is finite and selective instead we obtain a contradiction by the proof of Prop. 1.8, and hence also in this case

$f(A, w) \subseteq f(B, w)$. $\dashv$

By Prop. 3.26, for $X \in \{$FRT, FMP, FCV, FRCV, FCA, FSDA, FCS, FCEM$\}$:

Proposition 3.27 *If f is finite and selective or projective and f satisfies X, then it satisfies X'.*

Proposition 3.28 *If f is nonempty (p. 39) and*

1. *f satisfies FCS, then it satisfies FCS'.*
2. *f satisfies FCEM, then it satisfies FCEM'.*

Proof. Assume that f satisfies FCS and for contradiction that for some $A \in L$ and $w \in I$, $w \in S_f(A)$ and $f(A, w) \neq \{w\}$. Then if f is nonempty there is $w' \in I$ such that $w' \neq w$ and $w' \in f(A, w)$. Thus there is $B \in L$ such that $w \in S_f(B)$ and $w' \notin S_f(B)$, hence $w \in S_f(A \wedge B)$ and $f(A, w) \not\subseteq S_f(B)$, which is a contradiction. So FCS' holds.

Assume for contradiction that there are $w', w'' \in f(A, w)$ such that $w' \neq w''$. Then there is (atomic) $B \in L$ such that $w' \in S_f(B)$ and $w'' \notin S_f(B)$, so $f(A, w) \not\subseteq S_f(B)$ and $f(A, w) \not\subseteq S_f(\neg B)$, contradicting FCEM. Since f is nonempty it follows that FCEM' holds. $\dashv$

If S is a class selection semantics satisfying FID, FMP and FMOD we say that S is a *basic class selection semantics.*

We show that AR is sound and complete with respect to the set of basic class selection semantics Σ_{BCS}. By Propositions 3.18 and 3.24 AR is sound with respect to the set of basic class selection semantics. Since AR is a half-normal conditional logic, by Theorem 3.20

Proposition 3.29 *S_{AR} is a class selection semantics defined by f_{AR}*

Also,

Proposition 3.30 *S_{AR} satisfies FID, FMP and FMOD.*

Proof. By Prop. 3.23, S_{AR} validates ID, MP and MOD, and hence by Prop. 3.24, S_{AR} satisfies FID, FMP and FMOD. $\dashv$

Thus S_{AR} is a basic class selection semantics for Υ_{AR}, and hence it follows the usual way that

Proposition 3.31 *AR is complete with respect to the set of basic class selection semantics.*

In a similar way one can show that e.g. VW is sound and complete with respect to the set of class selection semantics satisfying FID, FMOD, FMP, FCSO, FCV and FCA, and similarly for other half-normal logics.

3.4 Facts Revision Semantics

We define , for each $A \in L$

$$\Box A = \neg A \mathrel{\Box\!\!\rightarrow} A$$

and

$$\Diamond A = \neg(A \mathrel{\Box\!\!\rightarrow} \neg A),^{7}$$

saying that A is *subjunctively necessary* and *possible* respectively.

Proposition 3.32 *If S is a weak or strong facts revision semantics then for every $\phi \in L$, $\models_S \Diamond\phi$ whenever $S(\phi) \neq \emptyset$.*

Proof. Assume that $S(\phi) \neq \emptyset$. If c is a premise contraction function for S then for every $w \in I_S$, $S(c(\phi, Th_S(w)) \cup \{\phi\}) \neq \emptyset$, and then of course $c(\phi, Th_S(w)) \cup \{\phi\} \not\models \neg\phi$. Hence, if S is a weak or strong facts revision semantics, $w \not\models_S \phi \mathrel{\Box\!\!\rightarrow} \neg\phi$, and so $w \models_S \neg(\phi \mathrel{\Box\!\!\rightarrow} \neg\phi)$, i.e., $w \models_S \Diamond\phi$. $\dashv$

It seems we cannot express this by means of axioms or rules of

[7] Cf. [Lewis, 1973b, p. 22].

inference, so it seems hopeless to find any conditional logic that is sound and complete with respect to the set of weak or strong facts revision semantics. However, we shall generalize the notion of strong facts revision semantics and then identify the logic that determines the larger class of semantics defined by this generalized notion. First we give the following definition:

Definition 3.13 *A* generalized premise contraction function *c is a total function from $L \times P(L)$ to $P(L)$ such that for any $A \in L$ and $\Gamma \in P(L)$, $c(A, \Gamma) \subseteq \Gamma$.*

It is easy to see that there are generalized premise contraction functions. Let

Definition 3.14 *If c is a generalized premise contraction function such that $S(A \mathrel{\Box\!\!\rightarrow} B) = \{w : c(A, Th_S(w)) \cup \{A\} \models_S B\}$, then S is a* weak generalized facts revision semantics *for I.*

There is at least the trivial weak generalized facts revision semantics defined by the generalized premise contraction function $c_{triv}(A, \Gamma) = \emptyset$ for every $A \in L$ and $\Gamma \subseteq L$. In this case $S(A \mathrel{\Box\!\!\rightarrow} B) = \{w : S(A) \subseteq S(B)\}$, so S exists and is unique.

By the proof of Theorem 1.12 we obtain that

Proposition 3.33 *If S is a weak generalized facts revision semantics for I then S is a class selection semantics (for I) satisfying FID and FMP'.*

3.4.1 Soundness of PR with respect to the Set of Weak Generalized Facts Revision Semantics

Let Σ_{WGFR} be the set of all weak generalized facts revision semantics.

Proposition 3.34 *PR is sound with respect to Σ_{WGFR}.*

Proof. Assume that $S \in \Sigma_{WGFR}$. By Prop. 3.33, 3.24, and 3.25 we conclude that S is a class selection semantics (so RCK is sound

with respect to S) validating ID and MP. $\dashv$

We say that the conditional logic Ψ' is an *extension* of the conditional logic Ψ iff

$$\Psi \subseteq \Psi'.$$

We note that

Proposition 3.35 *If Ψ is a half-normal extension of PR, then $\vdash_\Psi A \mathrel{\Box\!\!\rightarrow} C$ iff $\vdash_\Psi A \rightarrow C$.*

Proof. If $\vdash_\Psi A \mathrel{\Box\!\!\rightarrow} C$ then, by MP, $\vdash_\Psi A \rightarrow C$. If on the other hand $\vdash_\Psi A \rightarrow C$ then it follows by ID and RCK that $\vdash_\Psi A \mathrel{\Box\!\!\rightarrow} C$. $\dashv$

3.4.2 Completeness of PR with respect to the Set of Weak Generalized Facts Revision Semantics

For each $\Gamma \in \Upsilon_\Psi$ let

$$\Gamma^{\rightarrow}_{A} = \{A \rightarrow \phi : A \mathrel{\Box\!\!\rightarrow} \phi \in \Gamma\}.$$

Now define a generalized premise contraction function c_Ψ for S_Ψ as follows: For each $\Gamma \in L$ and $A \in L$, let

$$c_\Psi(A, \Gamma) = \begin{cases} \Gamma^{\rightarrow}_{A} & \text{if } \Gamma \in \Upsilon_\Psi, \\ \emptyset & \text{otherwise.} \end{cases}$$

Proposition 3.36 *If Ψ is a half-normal extension of PR then c_Ψ is a generalized premise contraction function for S_Ψ.*

Proof. We must show that

$$c_\Psi(A, \Gamma) \subseteq \Gamma.$$

Assume that $\psi \in c_\Psi(A, \Gamma)$. Then $\psi \in \Gamma^{\rightarrow}_{A}$ so there is some $\phi \in L$ such that $\psi = A \rightarrow \phi$ and $A \mathrel{\Box\!\!\rightarrow} \phi \in \Gamma$, but then by MP, $A \rightarrow \phi \in \Gamma$, so $\psi \in \Gamma$. $\dashv$

Definition 3.15 $r_\Psi(A,\Gamma) = Cn_{S_\Psi}(c_\Psi(A,\Gamma) \cup \{A\})$.

It is obvious that if for all $A, B \in L$,

$$S_\Psi(A \Box\!\!\rightarrow B) = \{\Gamma \in \Upsilon_\Psi : B \in r_\Psi(A,\Gamma)\},$$

then S_Ψ is a weak generalized facts revision semantics for Υ_Ψ.

Lemma 3.37 *If Ψ is a half-normal extension of PR and $\Gamma \in \Upsilon_\Psi$, then $A \Box\!\!\rightarrow C \in \Gamma$ iff $C \in r_\Psi(A,\Gamma)$.*

Proof. From left to right is trivial. Assume that $A \Box\!\!\rightarrow C \notin \Gamma$. We show that $\Gamma_A^{\rightarrow} \cup \{A\} \cup \{\neg C\}$ is Ψ-consistent. Otherwise there is a finite $\Gamma_A^{'\rightarrow} \subseteq \Gamma_A^{\rightarrow}$ such that $\vdash_\Psi \wedge\Gamma_A^{'\rightarrow} \rightarrow (A \rightarrow C)$. Let $\Gamma'_A = \{\phi : A \rightarrow \phi \in \Gamma_A^{'\rightarrow}\}$. Then $\vdash_\Psi \wedge\Gamma'_A \rightarrow (A \rightarrow C)$, but then (by RCK) $A \Box\!\!\rightarrow (A \rightarrow C) \in \Gamma$, and so by ID and RCK, $A \Box\!\!\rightarrow C \in \Gamma$, which is a contradiction. So $r_\Psi(A,\Gamma) \cup \{\neg C\}$ is Ψ-consistent, and hence $C \notin r_\Psi(A,\Gamma)$.

$\dashv$

It is now obvious that

Theorem 3.38 *If Ψ is a extension of PR then S_Ψ is a weak generalized facts revision semantics for Υ_Ψ.*

It now follows that S_{PR} is a weak generalized facts revision semantics for Υ_{PR}.

so

Proposition 3.39 *PR is complete with respect to the set of weak generalized facts revision semantics.*

3.4.3 Half Normal Extensions of PR and Weak Generalized Facts Revision Semantics

Let c be any generalized premise contraction function defining some weak generalized facts revision semantics S_c. We call the function $r_c(A,\Gamma) = Cn_{S_c}(c(A,\Gamma) \cup \{A\})$ the *generalized premise revision function* based on c. Also if I is the domain of S_c let $T_{S_c} = \{Th_{S_c}(w) : w \in I\}$. Obviously $T_{S_c} \subseteq C_{S_c}$ (p. 37), and if $\Gamma \in T_{S_c}$ then for every $\phi, \psi \in L$,

1. $\phi \in \Gamma$ iff $\neg\phi \notin \Gamma$, and
2. $\phi \wedge \psi \in \Gamma$ iff $\phi \in \Gamma$ and $\psi \in \Gamma$.

Consider the following conditions.

PID For all $A \in L$ and $\Gamma \in T_{S_c}$, $A \in r_c(A, \Gamma)$.

PMP' For all $A \in L$ and $\Gamma \in T_{S_c}$, if $A \in \Gamma$ then $r_c(A, \Gamma) \subseteq \Gamma$.

PMP For all $A \in L$ and $\Gamma \in T_{S_c}$, if $B \in r_c(A, \Gamma)$ then $A \to B \in \Gamma$.

PMOD For all $A, B \in L$ and $\Gamma \in T_{S_c}$, if $A \in r_c(\neg A, \Gamma)$ then $A \in r_c(B, \Gamma)$.

PRT' For all $A, B \in L$ and $\Gamma \in T_{S_c}$, if $B \in r_c(A, \Gamma)$ then $r_c(A \wedge B, \Gamma) \subseteq r_c(A, \Gamma)$.

PCSO For all $A, B \in L$ and $\Gamma \in T_{S_c}$, if $A \in r_c(B, \Gamma)$ and $B \in r_c(A, \Gamma)$ then $r_c(A, \Gamma) = r_c(B, \Gamma)$.

PCV' For all $A, B \in L$ and $\Gamma \in T_{S_c}$, if $\neg B \notin r_c(A, \Gamma)$ then $r_c(A, \Gamma) \subseteq r_c(A \wedge B, \Gamma)$.

PRCV' For all $A, B \in L$ and $\Gamma \in T_{S_c}$, if $B \in r_c(A, \Gamma)$ then $r_c(A, \Gamma) \subseteq r_c(A \wedge B, \Gamma)$.

PCA' For all $A, B \in L$ and $\Gamma \in T_{S_c}$, $r_c(A, \Gamma) \cap r_c(B, \Gamma) \subseteq r_c(A \vee B, \Gamma)$.

PSDA' For all $A, B \in L$ and $\Gamma \in T_{S_c}$, $r_c(A \vee B, \Gamma) \subseteq r_c(A, \Gamma) \cap r_c(B, \Gamma)$.

PCS' For all $A \in L$ and $\Gamma \in T_{S_c}$, if $A \in \Gamma$ then $r_c(A, \Gamma) = \Gamma$.

PCS For all $A \in L$ and $\Gamma \in T_{S_c}$, if $A \wedge B \in \Gamma$ then $B \in r_c(A, \Gamma)$.

PCEM For all $A, B \in L$ and $\Gamma \in T_{S_c}$ either $B \in r_c(A, \Gamma)$ or $\neg B \in r_c(A, \Gamma)$.

It is easy to see that

Proposition 3.40 *1. S_c validates ID iff r_c satisfies PID.*

2. *S_c validates MP iff r_c satisfies PMP.*

3. *S_c validates MOD iff r_c satisfies PMOD.*

4. *S_c validates RT iff r_c satisfies PRT'.*

5. *S_c validates CSO iff r_c satisfies PCSO.*

6. *S_c validates CV iff r_c satisfies PCV'.*

7. *S_c validates RCV iff r_c satisfies PRCV'.*

8. *S_c validates CA iff r_c satisfies PCA'.*

9. *S_c validates SDA iff r_c satisfies PSDA'.*

10. *S_c validates CS iff r_c satisfies PCS.*

11. *S_c validates CEM iff r_c satisfies PCEM.*

Proposition 3.41 *r_c satisfies PMP' iff it satisfies PMP.*

Proof. Assume PMP' and that $B \in r_c(A, \Gamma)$. If $A \notin \Gamma$ then $\neg A \in \Gamma$ so trivially $A \to B \in \Gamma$. If on the other hand $A \in \Gamma$ then $r_c(A, \Gamma) \subseteq \Gamma$ so $B \in \Gamma$, and hence $A \to B \in \Gamma$.

Assume on the other hand PMP and that $A \in \Gamma$ and $B \in r_c(A, \Gamma)$. Then $A \to B \in \Gamma$ so, since $\Gamma \in T_{S_c}$, $B \in \Gamma$. $\dashv$

Proposition 3.42 *r_c satisfies PCS' iff it satisfies PCS.*

Proof. Assume PCS' and that $A \wedge B \in \Gamma$. Then $A \in \Gamma$, so $r_c(A, \Gamma) = \Gamma$ and hence $B \in r_c(A, \Gamma)$.

Assume on the other hand PCS and that $A \in \Gamma$. By the definition of r_c, $r_c(A, \Gamma) \subseteq \Gamma$. If $B \in \Gamma$, then, since $\Gamma \in T_{S_c}$, $A \wedge B \in \Gamma$, and hence $B \in r_c(A, \Gamma)$. $\dashv$

3.5 Antecedent Relative Closeness Semantics

We shall show that the following logic is both sound and complete with respect to the set of ARRC semantics.

AR' ⟨⟨RCK; ID, MOD', MP⟩⟩

Let Σ_{AR} be the set of all ARRC semantics.

3.5.1 Soundness of AR' with respect to the Set of ARRC Semantics

Proposition 3.43 *AR' is sound with respect to the set of ARRC semantics.*

Proof. We must show that $AR' \subseteq Cn_{\Sigma_{AR}}$, i.e. that

$$\text{if } \{ID, MOD', MP\}_{Ax} \vdash_{\{RCK\}} \phi \text{ then } \models_{\Sigma_{AR}} \phi.$$

I.e., we must show that if there is a sequence of sets $\Gamma_0 \subseteq \Gamma_1 \subseteq \ldots \subseteq \Gamma_n$ such that $\{ID, MOD', MP\}_{Ax} = \Gamma_0$, $\phi \in \Gamma_n$ and for each $n \geq k > 0$ there is $\psi \in L$, $R \in \{RCK, MOPO\}$ and $\Delta \subseteq \Gamma_{k-1} \cup Tau$ such that $\Gamma_k = \Gamma_{k-1} \cup \{\psi\}$ and $\Delta \vdash_R \psi$, then $\models_{\Sigma_{AR}} \phi$. This is shown by induction over n, using Theorem 1.33 for the basic step, and Prop. 3.2, 1.26 and 3.3 for the induction step. ⊣

3.5.2 Completeness of AR' with respect to the Set of ARRC Semantics

The proofs of completeness for AR' with respect to the set of ARRC-semantics, and for VW with respect to the set of relative closeness semantics (Section 3.6.2 below), respectively, will utilize many of the same components: For each sentence A and each world $\Gamma \in \Upsilon_\Psi$ we shall divide $S_\Psi(A)$ into the set $(Near^A_\Gamma)$ of worlds that affirm all conditionals true at Γ with A as antecedent, and the set (Rem^A_Γ) of those that disaffirm some of them. Then we shall define two

different orderings $\leq_\Psi$ (which is antecedent-relative) and $\leq'_\Psi$ (which is not) of Υ_Ψ in terms of this division. We formulate a criterion of adequacy for arrc-assignments for (domains of) canonical semantics, and show that (whenever Ψ is a half normal extension of AR'), if an arrc assignment for Υ_Ψ is adequate in this sense then it defines an ARRC semantics which is identical to S_Ψ. We show that the two defined orderings $\leq_\Psi$ and $\leq'_\Psi$ are adequate arrc-assignments of $\Upsilon_{AR'}$ and of Υ_{VW}, respectively, from which it follows that $S_{AR'}$ is an ARRC-semantics and (since $\leq'_\Psi$ is not antecedent relative) S_{VW} is a relative closeness semantics.

We begin with AR' and show first that

Proposition 3.44 *If Ψ is a half-normal extension of AR' then Ψ is an extension of AR.*

Proof. Assume that Ψ is a half-normal extension of AR'. Since $\vdash A \rightarrow \neg\neg A$, we get $\vdash_\Psi (\neg A \mathrel{\Box\!\!\rightarrow} A) \rightarrow (\neg A \mathrel{\Box\!\!\rightarrow} \neg\neg A)$ (RCK), $\vdash_\Psi (\neg A \mathrel{\Box\!\!\rightarrow} \neg\neg A) \rightarrow (B \mathrel{\Box\!\!\rightarrow} \neg\neg A)$ (MOD'), $\vdash_\Psi (B \mathrel{\Box\!\!\rightarrow} \neg\neg A) \rightarrow (B \mathrel{\Box\!\!\rightarrow} A)$ (RCK), and hence $\vdash_\Psi (\neg A \mathrel{\Box\!\!\rightarrow} A) \rightarrow (B \mathrel{\Box\!\!\rightarrow} A)$. $\dashv$

Since, of course, AR' is a half normal extension of itself,

Proposition 3.45 *AR' is a half-normal extension of AR.*

I have not been able to determine whether the reverse is also true, but we shall see later (Prop. 3.68) that all normal extensions of AR are extensions of AR'.

Now we show that the canonical semantics for AR', $S_{AR'}$ is an ARRC semantics. Remember that

$$\Diamond A = \neg(A \mathrel{\Box\!\!\rightarrow} \neg A).$$

For each $\Gamma \in \Upsilon_\Psi$, let

$$R_\Gamma = \{\Delta \in \Upsilon_\Psi : \text{If } A \in \Delta \text{ then } \Diamond A \in \Gamma\}.$$

R_Γ is the set of those worlds that only contain sentences that are subjunctively possible at Γ. Later when we define an arrc-assignment for Υ_Ψ, we will use R_Γ as the set of $\Gamma' \in \Upsilon_\Psi$ accessible from Γ. Therefore, we must among other things prove the following.

Proposition 3.46 *If Ψ is an extension of AR' and $\Gamma \in \Upsilon_\Psi$ then $\Gamma \in R_\Gamma$.*

Proof. Assume that there is $\Gamma \in \Upsilon_\Psi$ such that $\Gamma \notin R_\Gamma$. Then there is $A \in \Gamma$ such that $\Diamond A \notin \Gamma$ and hence $A \mathrel{\Box\!\!\rightarrow} \neg A \in \Gamma$. By MP, $A \rightarrow \neg A \in \Gamma$, and hence $\neg A \in \Gamma$, which is a contradiction. $\dashv$

If we follow Lewis and say that that $A \in L$ is entertainable at a world w whenever there is a world w' accessible from w at which A is true, then it seems natural to require that ϕ be subjunctively impossible at w whenever ϕ is not entertainable at w. We show that this is the case with respect to the canonical interpretation, provided that Ψ is a half-normal extension of AR'. First we show

Proposition 3.47 *If Ψ is a half-normal extension of AR', $\Gamma \in \Upsilon_\Psi$ and $S_\Psi(A) = \emptyset$, then $A \mathrel{\Box\!\!\rightarrow} \neg A \in \Gamma$.*

Proof. Assume that Ψ is a half-normal extension of AR' and that $S_\Psi(A) = \emptyset$. Then (by Lemma 3.15) A is not Ψ-consistent, so $\vdash_\Psi A \rightarrow \neg A$. Thus (by Prop. 3.35) $\vdash_\Psi A \mathrel{\Box\!\!\rightarrow} \neg A$, and hence $A \mathrel{\Box\!\!\rightarrow} \neg A \in \Gamma$. $\dashv$

Proposition 3.48 *If Ψ is a half-normal extension of AR', $\Gamma \in \Upsilon_\Psi$ and $S_\Psi(A) \cap R_\Gamma = \emptyset$ then $A \mathrel{\Box\!\!\rightarrow} \neg A \in \Gamma$.*

Proof. Assume that Ψ is a half-normal extension of AR' and $S_\Psi(A) \cap R_\Gamma = \emptyset$. If $S_\Psi(A) = \emptyset$ the conclusion follows by Prop. 3.47.

If on the other hand $S_\Psi(A) \neq \emptyset$, assume for contradiction that $A \mathrel{\Box\!\!\rightarrow} \neg A \notin \Gamma$. Let $\Gamma_{nec} = \{\neg B : B \mathrel{\Box\!\!\rightarrow} \neg B \in \Gamma\}$, and $\Lambda' = \Gamma_{nec} \cup \{A\}$. Λ' is Ψ-consistent. For assume not, then there is a finite $\Gamma'_{nec} \subseteq \Gamma_{nec}$ such that $\vdash_\Psi \wedge\Gamma'_{nec} \rightarrow \neg A$. But for each $\neg B \in \Gamma'_{nec}$, $B \mathrel{\Box\!\!\rightarrow} \neg B \in \Gamma$, and therefore also $A \mathrel{\Box\!\!\rightarrow} \neg B \in \Gamma$ (by MOD'), so $A \mathrel{\Box\!\!\rightarrow} \neg A \in \Gamma$ (by RCK), which contradicts an assumption. So (under this assumption) Λ' is Ψ-consistent. Let Λ be any maximal Ψ-consistent set such that $\Lambda' \subseteq \Lambda$. $\Lambda \in S_\Psi(A)$. Assume that $\Lambda \notin R_\Gamma$. Then there is $B \in \Lambda$ such that $\Diamond B \notin \Gamma$, and hence

$B \mathrel{\Box\!\!\rightarrow} \neg B \in \Gamma$. But then $\neg B \in \Gamma_{nec}$ so $\neg B \in \Lambda$, and hence Λ is not Ψ-consistent, which is a contradiction. Hence $\Lambda \in R_\Gamma$, so $S_\Psi(A) \cap R_\Gamma \neq \emptyset$. But this contradicts the main assumption, so in either case $A \mathrel{\Box\!\!\rightarrow} \neg A \in \Gamma$. $\dashv$

We say that a world w *S-affirms* the conditional $A \mathrel{\Box\!\!\rightarrow} C$ if $w \models_S A \wedge C$, and that it *S-disaffirms* it if $w \models_S A \wedge \neg C$. For each $A \in L$ and $\Gamma \in \Upsilon_\Psi$, let

$$Near_\Gamma^A = \{\Delta \in \Upsilon_\Psi : \text{ if } A \mathrel{\Box\!\!\rightarrow} C \in \Gamma \text{ then } C \in \Delta\}.$$

Hence (if Ψ contains ID) $Near_\Gamma^A$ is the set of worlds that S_Ψ-affirm all conditionals having A as antecedent that are true at Γ. For $\Gamma \in \Upsilon_\Psi$ we let

$$\Gamma_A = \{\phi : A \mathrel{\Box\!\!\rightarrow} \phi \in \Gamma\}.$$

Thus

Proposition 3.49 $Near_\Gamma^A = \{\Delta \in \Upsilon_\Psi : \Gamma_A \subseteq \Delta\}$.

If Ψ contains ID then for each $A \in L$ and $\Gamma \in \Upsilon_\Psi$, $A \in \Gamma_A$. Hence,

Proposition 3.50 *If Ψ contains ID then $Near_\Gamma^A \subseteq S_\Psi(A)$*

Since $Near_\Gamma^A = f_\Psi(A, \Gamma)$, by Lemma 3.19:

Proposition 3.51 *If Ψ is a half-normal conditional logic and $\Gamma \in \Upsilon_\Psi$, then $Near_\Gamma^A \subseteq S_\Psi(B)$ iff $A \mathrel{\Box\!\!\rightarrow} B \in \Gamma$.*

Therefore,

Proposition 3.52 *If Ψ is a half-normal conditional logic and $\Gamma \in \Upsilon_\Psi$, then $S_\Psi(B) \cap Near_\Gamma^A \neq \emptyset$ iff $A \mathrel{\Diamond\!\!\rightarrow} B \in \Gamma$.*

Proof. $S_\Psi(B) \cap Near_\Gamma^A \neq \emptyset$ iff $Near_\Gamma^A \not\subseteq S_\Psi(\neg B)$ iff $A \mathrel{\Box\!\!\rightarrow} \neg B \notin \Gamma$ iff $\neg(A \mathrel{\Box\!\!\rightarrow} \neg B) \in \Gamma$ iff $A \mathrel{\Diamond\!\!\rightarrow} B \in \Gamma$. $\dashv$

Proposition 3.53 *If Ψ is a half-normal extension of AR', $\Gamma \in \Upsilon_\Psi$ and $S_\Psi(A) \cap R_\Gamma \neq \emptyset$, then $Near^A_\Gamma \neq \emptyset$.*

Proof. Assume that $S_\Psi(A) \cap R_\Gamma \neq \emptyset$. Then (since Γ is Ψ-consistent) $A \mathrel{\Box\!\!\rightarrow} \neg A \notin \Gamma$. If Γ_A is not Ψ-consistent, then there is a finite $\Gamma'_A \subseteq \Gamma_A$ such that $\vdash_\Psi \wedge\Gamma'_A \rightarrow \neg A$, and hence by RCK $A \mathrel{\Box\!\!\rightarrow} \neg A \in \Gamma$, which is a contradiction, so Γ_A is Ψ-consistent. Let Δ be any maximally consistent superset of Γ_A. By Prop. 3.49, $\Delta \in Near^A_\Gamma$, so $Near^A_\Gamma \neq \emptyset$. ⊣

Proposition 3.54 *If Ψ is an extension of AR' and $\Gamma \in \Upsilon_\Psi$ then $Near^A_\Gamma \subseteq R_\Gamma$.*

Proof. If $\Delta \in Near^A_\Gamma$ then Δ is a maximally Ψ-consistent superset of Γ_A. Assume for contradiction that there is $\psi \in \Delta$ such that $\Diamond\psi \notin \Gamma$, and hence $\psi \mathrel{\Box\!\!\rightarrow} \neg\psi \in \Gamma$. Then, by MOD', $A \mathrel{\Box\!\!\rightarrow} \neg\psi \in \Gamma$, but then $\neg\psi \in \Gamma_A$, which is a contradiction. It follows that $\Delta \in R_\Gamma$, so $Near^A_\Gamma \subseteq R_\Gamma$. ⊣

It follows from Prop. 3.53 and Prop. 3.54 that

Proposition 3.55 *If Ψ is a half-normal extension of AR' then $Near^A_\Gamma \neq \emptyset$ iff $S_\Psi(A) \cap R_\Gamma \neq \emptyset$.*

Let

$$Rem^A_\Gamma = S_\Psi(A) - Near^A_\Gamma.$$

We show that this is the set of elements of Υ_Ψ that S_Ψ-disaffirm at least one conditional with A as antecedent true at Γ.

Proposition 3.56 *$Rem^A_\Gamma = \{\Delta \in \Upsilon_\Psi :$ there is $C \in L$ such that $A \wedge \neg C \in \Delta$ and $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma\}$.*

Proof. Assume that $\Delta \in Rem^A_\Gamma$. Then $A \in \Delta$ and not for all $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$, $C \in \Delta$, i.e., there is $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$ such that $\neg C \in \Delta$, and since Δ is maximally Ψ-consistent, $A \wedge \neg C \in \Delta$. ⊣

Lemma 3.57 *If Ψ is a half-normal extension of AR' and $S_\Psi(A \wedge \neg C) \cap R_\Gamma \subseteq Rem^A_\Gamma$, then $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$.*

Proof. Assume that Ψ is a half-normal extension of AR' and $S_\Psi(A \wedge \neg C) \cap R_\Gamma \subseteq Rem^A_\Gamma$, and for contradiction that $A \mathrel{\Box\!\!\rightarrow} C \notin \Gamma$. Let $\Gamma_A = \{B : A \mathrel{\Box\!\!\rightarrow} B \in \Gamma\}$, and $\Gamma_{nec} = \{\neg B : B \mathrel{\Box\!\!\rightarrow} \neg B \in \Gamma\}$. By MOD', $\Gamma_{nec} \subseteq \Gamma_A$. $\Gamma_A \cup \{\neg C\}$ is Ψ-consistent, for otherwise there is a finite $\Gamma'_A \subseteq \Gamma_A$ such that $\vdash_\Psi \wedge\Gamma'_A \rightarrow C$, but then (by RCK) $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$, which is a contradiction. Let Λ be any maximally Ψ-consistent subset of L such that $\Gamma_A \cup \{\neg C\} \subseteq \Lambda$. $A \wedge \neg C \in \Lambda$. Also $\Lambda \in R_\Gamma$, for if not, there is $D \in \Lambda$ such that $\Diamond D \notin \Gamma$ and hence $D \mathrel{\Box\!\!\rightarrow} \neg D \in \Gamma$, whence $\neg D \in \Gamma_{nec} \subseteq \Lambda$, which is a contradiction. But since $A \wedge \neg C \in \Lambda$ it follows by our main assumption that $\Lambda \in Rem^A_\Gamma$, so there is $\phi \in L$ such that $A \wedge \neg\phi \in \Lambda$ and $A \mathrel{\Box\!\!\rightarrow} \phi \in \Gamma$. But then $\phi \in \Lambda$ (since $\Gamma_A \subseteq \Lambda$), which contradicts Λ being Ψ-consistent, so $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$. $\dashv$

A direct consequence of Lemma 3.57 is

Proposition 3.58 *If Ψ is a half-normal extension of AR', $\Gamma \in \Upsilon_\Psi$ and $S_\Psi(A \wedge \neg C) \cap R_\Gamma = \emptyset$, then $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$.*

And a direct consequence of Lemma 3.57 and Prop. 3.56 is:

Proposition 3.59 *If Ψ is a half-normal extension of AR', then $S_\Psi(A \wedge \neg C) \subseteq Rem^A_\Gamma$ iff $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$.*

We also show that

Lemma 3.60 *If Ψ is a half-normal extension of AR', $\Gamma \in S_\Psi(A \mathrel{\Box\!\!\rightarrow} C)$ and $S_\Psi(A) \cap R_\Gamma \neq \emptyset$, then there is $\Lambda \in \Upsilon_\Psi$ such that $\Lambda \in Near^A_\Gamma$ and $A \wedge C \in \Lambda$.*

Proof. Assume that Ψ is a half-normal extension of AR', $\Gamma \in S_\Psi(A \mathrel{\Box\!\!\rightarrow} C)$ and $S_\Psi(A) \cap R_\Gamma \neq \emptyset$. Then by Prop. 3.55, $Near^A_\Gamma \neq \emptyset$. Take any $\Lambda \in Near^A_\Gamma$. Since $\Gamma \in S_\Psi(A \mathrel{\Box\!\!\rightarrow} C)$, by Prop. 3.59, $A \wedge \neg C \notin \Lambda$, and hence $A \wedge C \in \Lambda$. $\dashv$

(In the following, if $\leq$ is an arrc-assignment, instead of $\leq (\Gamma, A) = \langle \Delta, \Delta' \rangle$ we write $\Delta \leq^A_\Gamma \Delta'$ as usual).

If Ψ is an extension of AR' and $\leq$ is an arrc-assignment for Υ_Ψ, we say that $\leq$ is *adequate* if

1. $\Delta \notin Rem^A_\Gamma$ whenever $\Delta' \in Near^A_\Gamma$ and $\Delta \leq^A_\Gamma \Delta'$,
2. $\Delta \in Rem^A_\Gamma$ whenever $\Delta, \Delta' \in S_\Psi(A)$ and $\Delta' <^A_\Gamma \Delta$.

We define a total function $\leq_\Psi$ from $\Upsilon_\Psi \times L$ to $P(\Upsilon_\Psi \times \Upsilon_\Psi)$ as follows

Definition 3.16 *For $\Gamma, \Delta, \Delta' \in \Upsilon_\Psi$ and $A \in L$, $\Delta \leq^A_\Gamma \Delta'$ iff [$\Delta \notin Rem^A_\Gamma$ or $\Delta' \in Rem^A_\Gamma$ or $\Delta' \notin R_\Gamma$] and [$\Delta \in R_\Gamma$ or $\Delta' \notin R_\Gamma$].*

i.e.

$\Delta' <^A_\Gamma \Delta$ iff [$\Delta \in Rem^A_\Gamma$ and $\Delta' \notin Rem^A_\Gamma$ and $\Delta' \in R_\Gamma$] or [$\Delta \notin R_\Gamma$ and $\Delta' \in R_\Gamma$].

Lemma 3.61 *If Ψ is a half-normal extension of AR' then $\leq_\Psi$ is an arrc-assignment for Υ_Ψ.*

Proof.

1. Assume that $\Delta \leq^A_\Gamma \Delta'$ and $\Delta' \leq^A_\Gamma \Delta''$. If $\Delta'' \notin R_\Gamma$ then trivially $\Delta \leq^A_\Gamma \Delta''$, so we assume that $\Delta'' \in R_\Gamma$. Then [$\Delta' \notin Rem^A_\Gamma$ or $\Delta'' \in Rem^A_\Gamma$] and [$\Delta' \in R_\Gamma$], and hence also [$\Delta \notin Rem^A_\Gamma$ or $\Delta' \in Rem^A_\Gamma$] and [$\Delta \in R_\Gamma$], and so $\Delta \leq^A_\Gamma \Delta''$. We have shown that $\leq^A_\Gamma$ is transitive.
2. Assume that for some $\Delta, \Delta' \in \Upsilon_\Psi$, $\Delta \not\leq^A_\Gamma \Delta'$. Then $\Delta' \in R_\Gamma$. If $\Delta \notin R_\Gamma$ then trivially $\Delta' \leq^A_\Gamma \Delta$, so we assume that $\Delta \in R_\Gamma$. Then not [$\Delta \notin Rem^A_\Gamma$ or $\Delta' \in Rem^A_\Gamma$] so [$\Delta \in Rem^A_\Gamma$ and $\Delta' \notin Rem^A_\Gamma$], and hence $\Delta' \leq^A_\Gamma \Delta$. We have shown that $\leq^A_\Gamma$ is strongly connected.
3. By Prop. 3.46, Γ is self accessible.
4. Assume that $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$. Then $A \rightarrow C \in \Gamma$ since $\{MP\}_{Ax} \subseteq \Gamma$, so $A \wedge \neg C \notin \Gamma$ and hence $\Gamma \notin Rem^A_\Gamma$. Since $\Gamma \in R_\Gamma$ it follows that for any $\Delta \in \Upsilon_\Psi$, $\Gamma \leq^A_\Gamma \Delta$, so Γ is $\leq^A_\Gamma$-minimal.

5. If $\Delta \notin R_\Gamma$ then it follows directly from the definition of $\leq$ that for any $\Delta' \in \Upsilon_\Psi$, $\Delta' \leq^A_\Gamma \Delta$, so inaccessible worlds are $\leq^A_\Gamma$-maximal.

6. Assume that $\Delta \in R_\Gamma$ and $\Delta' \notin R_\Gamma$. Since $[\Delta' \in R_\Gamma$ or $\Delta \notin R_\Gamma]$ is not satisfied, $\Delta' \not\leq^A_\Gamma \Delta$, i.e. $\Delta <^A_\Gamma \Delta'$, so accessible worlds are more similar to Γ than inaccessible worlds.

$\dashv$

Proposition 3.62 *If Ψ is a half-normal extension of AR' then $\leq_\Psi$ is an adequate arrc-assignment for Υ_Ψ.*

Proof. We must show that $\leq_\Psi$ is adequate. Assume that Ψ is a half-normal extension of AR' and that $\Delta \leq^A_\Gamma \Delta'$, $\Delta' \in Near^A_\Gamma$, and for contradiction that $\Delta \in Rem^A_\Gamma$. Then $\Delta' \in Rem^A_\Gamma$ or $\Delta' \notin R_\Gamma$, which by Prop. 3.54 is a contradiction. Hence $\Delta \notin Rem^A_\Gamma$.

Now instead assume that $\Delta, \Delta' \in S_\Psi(A)$ and $\Delta' <^A_\Gamma \Delta$. If $\Delta \notin R_\Gamma$ then by Prop. 3.54, $\Delta \notin Near^A_\Gamma$, hence $\Delta \in Rem^A_\Gamma$. If on the other hand $\Delta \in R_\Gamma$ then by the definition of $\leq_\Psi$, $\Delta \in Rem^A_\Gamma$. Thus, $\leq_\Psi$ is adequate. $\dashv$

Proposition 3.63 *If Ψ is a half-normal extension of AR' then $\leq_\Psi$ satisfies the ARL assumption (p. 58).*

Proof. Assume that Ψ is a half-normal extension of AR' and that $S_{\leq_\Psi}(A) \neq \emptyset$. Assume first that $S_{\leq_\Psi}(A) \cap R_\Gamma = \emptyset$. Take any $\Delta \in S_{\leq_\Psi}(A)$. Then if $\Delta' <^A_\Gamma \Delta$, $w' \in R_\Gamma$, and hence $\Delta' \notin S_{\leq_\Psi}(A)$.

Assume now instead that $S_{\leq_\Psi}(A) \cap R_\Gamma \neq \emptyset$. Then by Prop. 3.53, $Near^A_\Gamma \neq \emptyset$. Take any $\Delta \in Near^A_\Gamma$. By ID, $Near^A_\Gamma \subseteq S_{\leq_\Psi}(A)$, and hence $\Delta \in S_{\leq_\Psi}(A)$, but since $\Delta \in Near^A_\Gamma$ there is no Δ' such that $\Delta' <^A_\Gamma \Delta$. So in both cases, ARL holds. $\dashv$

Lemma 3.64 *If Ψ is a half-normal extension of AR', $\leq$ is an adequate arrc-assignment for Υ_Ψ, and $\Gamma \in \Upsilon_\Psi$, then $S_\Psi(A \mathrel{\Box\!\!\rightarrow} C) =$*

$\{\Gamma : S_\Psi(A) \cap R_\Gamma = \emptyset$ *or there is* $\Gamma' \in S_\Psi(A)$ *such that for all* $\Gamma'' \leq^A_\Gamma \Gamma'$, $\Gamma'' \in S_\Psi(A \to C)\}$.

Proof. First we assume that $\Gamma \in S_\Psi(A \Box\!\!\to C)$. $S_\Psi(A) \cap R_\Gamma = \emptyset$ or not. If the former the desired result is trivial, so we assume that $S_\Psi(A) \cap R_\Gamma \neq \emptyset$ By Lemma 3.60 there is $\Gamma' \in \Upsilon_\Psi$ such that $\Gamma' \in Near^A_\Gamma$ and $A \wedge C \in \Gamma'$. Now assume that $\Gamma'' \leq^A_\Gamma \Gamma'$, so $\Gamma'' \notin Rem^A_\Gamma$ (since $\leq$ is adequate), and hence by Prop. 3.56, $A \wedge \neg C \notin \Gamma''$, so $\Gamma'' \in S_\Psi(A \to C)$.

Assume on the other hand that $S_\Psi(A) \cap R_\Gamma = \emptyset$ or there is $\Gamma' \in S_\Psi(A)$ such that for all $\Gamma'' \leq^A_\Gamma \Gamma'$, $\Gamma'' \in S_\Psi(A \to C)$. If $S_\Psi(A) \cap R_\Gamma = \emptyset$ then $S_\Psi(A \wedge \neg C) \cap R_\Gamma = \emptyset$, so by Prop. 3.58, $A \Box\!\!\to C \in \Gamma$. If instead we assume that $S_\Psi(A) \cap R_\Gamma \neq \emptyset$, then there is $\Gamma' \in S_\Psi(A)$ such that for all $\Gamma'' \leq^A_\Gamma \Gamma'$, $\Gamma'' \in S_\Psi(A \to C)$. Again if $S_\Psi(A \wedge \neg C) \cap R_\Gamma = \emptyset$ then $A \Box\!\!\to C \in \Gamma$, so we assume that $S_\Psi(A \wedge \neg C) \cap R_\Gamma \neq \emptyset$. If $\Gamma'' \in S_\Psi(A \wedge \neg C) \cap R_\Gamma$, then $\Gamma' <^A_\Gamma \Gamma''$, hence (since $\leq$ is adequate) $\Gamma'' \in Rem^A_\Gamma$. Therefore, $S_\Psi(A \wedge \neg C) \cap R_\Gamma \subseteq Rem^A_\Gamma$, so (by Lemma 3.57) $A \Box\!\!\to C \in \Gamma$. $\dashv$

Now by Prop. 3.62 we get

Theorem 3.65 *If* Ψ *is a half-normal extension of AR' then* S_Ψ *is an ARRC semantics for* Υ_Ψ *defined by* $\leq_\Psi$.

Since AR' is a half-normal extension of itself it follows that $S_{AR'}$ is an ARRC-semantics for $\Upsilon_{AR'}$ defined by $\leq_{AR'}$, and hence as before that

Proposition 3.66 *AR' is complete with respect to the set of ARRC semantics.*

3.5.3 Half-Normal Extensions of AR' and various ARRC-Semantics

PAR We have seen Prop. 1.25 that all propositional ARRC semantics are intensional and hence the derivation rule RCEA is sound with respect to this set of semantics. We now show that PAR (AR'

+ RCEA) is complete with respect to the set Σ_{PAR} of propositional ARRC semantics.

First we note that

Proposition 3.67 *If Ψ is a normal extension of AR then Ψ is an extension of AR'.*

Proof. Assume that Ψ is a normal extension of AR. $\vdash A \leftrightarrow \neg\neg A$, so $\vdash_\Psi (A \mathrel{\Box\!\!\rightarrow} \neg A) \rightarrow (\neg\neg A \mathrel{\Box\!\!\rightarrow} \neg A)$ since Ψ is normal. By MOD, $\vdash_\Psi (\neg\neg A \mathrel{\Box\!\!\rightarrow} \neg A) \rightarrow (B \mathrel{\Box\!\!\rightarrow} \neg A)$, and hence $\vdash_\Psi (A \mathrel{\Box\!\!\rightarrow} \neg A) \rightarrow (B \mathrel{\Box\!\!\rightarrow} \neg A)$. ⊣

Hence, by Prop. 3.44,

Proposition 3.68 *Ψ is a normal extension of AR iff Ψ is a normal extension of AR'.*

Proposition 3.69 *If Ψ is a normal extension of AR then S_Ψ is a propositional ARRC semantics.*

Proof. Assume that Ψ is a normal extension of AR. By Lemma 3.64 (and Prop. 3.68), S_Ψ is an ARRC semantics. We assume that $S_{\leq_\Psi}(A) = S_{\leq_\Psi}(B)$. By Prop. 3.17, $\vdash_\Psi A \leftrightarrow B$, and hence by RCEA for any $C \in L$, and any $\Gamma \in \Upsilon_\Psi$, $\Gamma \models_{S_\Psi} A \mathrel{\Box\!\!\rightarrow} C$ iff $\Gamma \models_{S_\Psi} B \mathrel{\Box\!\!\rightarrow} C$. Thus $\Gamma_A = \Gamma_B$, and therefore $Near^A_\Gamma = Near^B_\Gamma$ and hence $Rem^A_\Gamma = Rem^B_\Gamma$, so for any $\Gamma', \Gamma'' \in \Upsilon_\Psi$, $\Gamma' \leq^A_\Gamma \Gamma''$ iff $\Gamma' \leq^B_\Gamma \Gamma''$. So $\leq_\Psi$ is propositional, and hence S_Ψ is a propositional ARRC semantics. ⊣

Since PAR is a normal extension of AR it follows by Prop. 3.69 that The canonical semantics for PAR, S_{PAR}, is a propositional ARRC semantics for Υ_{PAR}. and hence we have shown that

Proposition 3.70 *PAR is complete with respect to the set of propositional ARRC semantics.*

Other extension of AR' Let $S_{\leq}$ be an ARRC-Semantics for some I defined by the arrc-assignment $\leq$, and satisfying the ARL-assumption (p. 58).

Recall (p. 58 above) that for each $w \in I$ and $A \in L$,

$$S_{\leq}(A)_w = \{w' \in S_{\leq}(A) \cap R_w : \text{If } w'' <^A_w w' \text{ then } w'' \notin S_{\leq}(A)\}.$$

We note that

Proposition 3.71 *If Ψ is a half-normal extension of AR' then for every $\Gamma \in \Upsilon_\Psi$, $S_{\leq_\Psi}(A)_\Gamma = Near^A_\Gamma$.*

Proof. Assume that $\Delta \in S_{\leq_\Psi}(A)_\Gamma$. Then $\Delta \in S_{\leq_\Psi}(A) \cap R_\Gamma$, so by Prop. 3.53, $Near^A_\Gamma \neq \emptyset$. But if $\Delta \in Rem^A_\Gamma$, then there is $\Delta' <^A_\Gamma \Delta$, so $\Delta \in Near^A_\Gamma$.

Assume on the other hand that $\Delta \in Near^A_\Gamma$. Then by Prop. 3.54 and ID, $\Delta \in S_{\leq_\Psi}(A) \cap R_\Gamma$. There is no Δ' such that $\Delta' <^A_\Gamma \Delta$; hence $\Delta \in S_{\leq_\Psi}(A)_\Gamma$. ⊣

Proposition 3.72 *If $w' \in S_{\leq}(A)_w$, $w'' \leq^A_w w'$, and $w'' \in S_{\leq}(A)$, then $w'' \in S_{\leq}(A)_w$.*

Proof. Assume that $w' \in S_{\leq}(A)_w$, $w'' \leq^A_w w'$, $w'' \in S_{\leq}(A)$ and that $w''' <^A_w w''$. Then $w''' <^A_w w'$ (by the transitivity of $\leq$), and hence $w''' \notin S_{\leq}(A)$. Since $w' \in R_w$, also $w'' \in R_w$, and hence $w'' \in S_{\leq}(A) \cap R_w$. We have shown that $w'' \in S_{\leq}(A)_w$. ⊣

Now if $\leq$ satisfies the ARL assumption, we can show that

Proposition 3.73 $S_{\leq}(A \mathrel{\Box\!\!\rightarrow} C) = \{w \in I : S_{\leq}(A)_w \subseteq S_{\leq}(C)\}$.

Proof. Assume that $S_{\leq}(A)_w \subseteq S_{\leq}(C)$ and that $S_{\leq}(A) \cap R_w \neq \emptyset$. Then by the ARL assumption there is $w' \in S_{\leq}(A)$ such that if $w'' <^A_w w'$ then $w'' \notin S_{\leq}(A)$. $w' \in S_{\leq}(A) \cap R_w$, so $w' \in S_{\leq}(A)_w$.

Assume that $w'' \leq^A_w w'$ and that $w'' \in S_{\leq}(A)$ (since otherwise trivially $w'' \in S_{\leq}(A \to C)$). Then by Prop. 3.72, $w'' \in S_{\leq}(A)_w$, and hence $w'' \in S_{\leq}(C)$, so $w'' \in S_{\leq}(A \to C)$.

Assume on the other hand that $S_{\leq}(A) \cap R_w = \emptyset$ or there is $w' \in S_{\leq}(A)$ such that for all $w'' \leq_w^A w'$, $w'' \in S_{\leq}(A \to C)$. If $w''' \in S_{\leq}(A)_w$ then $S_{\leq}(A) \cap R_w \neq \emptyset$, so there is $w' \in S_{\leq}(A)$ such that for all $w'' \leq_w^A w'$, $w'' \in S_{\leq}(A \to C)$. If $w''' \notin S_{\leq}(C)$ then $w''' \notin S_{\leq}(A \to C)$, but then $w''' \not\leq_w^A w'$ (i.e., $w' <_w^A w'''$), which is a contradiction. Thus $S_{\leq}(A)_w \subseteq S_{\leq}(C)$. $\dashv$

If $S_{\leq}$ for I is an ARRC semantics we say that $S_{\leq}$ satisfies any of the following requirements respectively if for all $A, B, C \in L$ and every $w \in I$,

CID $S_{\leq}(A)_w \subseteq S_{\leq}(A)$.

CMOD' If $S_{\leq}(A)_w \subseteq S_{\leq}(\neg A)$ then $S_{\leq}(B)_w \subseteq S_{\leq}(\neg A)$.

CMOD If $S_{\leq}(\neg A)_w \subseteq S_{\leq}(A)$ then $S_{\leq}(B)_w \subseteq S_{\leq}(A)$.

CRT' If $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ then $S_{\leq}(A)_w \subseteq S_{\leq}(A \wedge B)_w$.

CRT If $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ and $S_{\leq}(A \wedge B)_w \subseteq S_{\leq}(C)$ then $S_{\leq}(A)_w \subseteq S_{\leq}(C)$.

CMP' If $w \in S_{\leq}(A)$ then $w \in S_{\leq}(A)_w$.

CMP If $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ then $w \in S_{\leq}(A \to B)$.

CCSO If $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ and $S_{\leq}(B)_w \subseteq S_{\leq}(A)$ then $S_{\leq}(A)_w = S_{\leq}(B)_w$.

CCV' If $S_{\leq}(A)_w \not\subseteq S_{\leq}(\neg B)$ then $S_{\leq}(A \wedge B)_w \subseteq S_{\leq}(A)_w$.

CCV If $S_{\leq}(A)_w \subseteq S_{\leq}(C)$ and $S_{\leq}(A)_w \not\subseteq S_{\leq}(\neg B)$ then $S_{\leq}(A \wedge B)_w \subseteq S_{\leq}(C)$.

CRCV' If $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ then $S_{\leq}(A \wedge B)_w \subseteq S_{\leq}(A)_w$.

CRCV If $S_{\leq}(A)_w \subseteq S_{\leq}(C)$ and $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ then $S_{\leq}(A \wedge B)_w \subseteq S_{\leq}(C)$.

CCA' $S_{\leq}(A \vee B)_w \subseteq S_{\leq}(A)_w \cup S_{\leq}(B)_w$.

CCA If $S_{\leq}(A)_w \cup S_{\leq}(B)_w \subseteq S_{\leq}(C)$ then $S_{\leq}(A \vee B)_w \subseteq S_{\leq}(C)$.

CSDA' $S_{\leq}(A)_w \cup S_{\leq}(B)_w \subseteq S_{\leq}(A \vee B)_w$.

CSDA If $S_{\leq}(A \vee B)_w \subseteq S_{\leq}(C)$ then $S_{\leq}(A)_w \cup S_{\leq}(B)_w \subseteq S_{\leq}(C)$.

CCS' If $w \models_{S_{\leq}} A$ then $S_{\leq}(A)_w = \{w\}$.

CCS If $w \in S_{\leq}(A \wedge B)$ then $S_{\leq}(A)_w \subseteq S_{\leq}(B)$.

CCEM' For all $A \in L$ and $w \in I$ there is exactly one $w' \in I$ such that $S_{\leq}(A)_w = \{w'\}$.

CCEM For all $A, B \in L$ and $w \in I$ either $S_{\leq}(A)_w \subseteq S_{\leq}(B)$ or $S_{\leq}(A)_w \subseteq S_{\leq}(\neg B)$.

By Prop. 3.73, if $\leq$ satisfies the ARL assumption we get for $X \in$ {ID, MOD, MOD', RT, MP, CSO, CV, RCV, CA, SDA, CS, CEM}:

Proposition 3.74 *$S_{\leq}$ validates X iff it satisfies CX.*

It is also obvious that for $X \in$ {CRT, CMP, CCV, CRCV, CCA, CSDA, CCS, CCEM}:

Proposition 3.75 *If $S_{\leq}$ satisfies X', then it satisfies X.*

If S is a propositional ARRC semantics satisfying the ARL-assumption, CRT and CCA we say that S is a *causal trail semantics*.

We show as a particular topic of interest that CT is sound and complete with respect to the set of causal trail semantics Σ_{CT}. By Theorem 1.33, Prop. 3.3, 1.25, 3.6 and Prop. 3.74, CT is sound with respect to the set of causal trail semantics.

Since CT is a half-normal extension of AR' (because it is a normal extension of AR, Prop. 3.68) it follows by Theorem 3.65 that S_{CT} is an ARRC-semantics for Υ_{CT} defined by $\leq_{CT}$ (p. 170).

Since CT is a normal extension of AR' we get by Prop. 3.69 that

Proposition 3.76 *S_{CT} is a propositional ARRC semantics.*

Proposition 3.77 *S_{CT} satisfies CRT and CCA.*

Proof. By Prop. 3.23, S_{CT} validates RT and CA, and by Prop. 3.63, $\leq_{CT}$ satisfies the ARL assumption. Hence by Prop. 3.74, $S_{\leq_{CT}} = S_{CT}$ satisfies CRT and CCA. ⊣

We have now shown that S_{CT} is a causal trail semantics for Υ_{CT}. and hence that

Proposition 3.78 *CT is complete with respect to the set of causal trail semantics.*

In a similar way one can show that e.g. VW is complete with respect to the set of ARRC semantics satisfying CID, CMOD, CMP, CCSO, CCV and CCA, and similarly for other half-normal extensions of AR'. However, it seems we cannot show this way that VW is also complete with respect to the set of relative closeness semantics. This will instead be achieved by a slightly different route in the next section.

3.6 Relative Closeness Semantics

3.6.1 Soundness of VW with respect to the Set of Relative Closeness Semantics

We have already shown (Theorem 1.33) that the logic VW, i.e., $\langle\langle$RCK; ID, MOD, CSO, CV, MP, C$\rangle\rangle$, is sound with respect to the set Σ_{RC} of relative closeness semantics. We now show that it is also complete with respect to the same set.

3.6.2 Completeness of VW with respect to the Set of Relative Closeness Semantics

First we observe that

Proposition 3.79 *Ψ is a half normal extension of VW iff Ψ is a normal extension of VW.*

Proof. From right to left is trivial. Assume that Ψ is a half normal extension of VW. Then it is a half normal extension of $\langle\langle$;ID, CSO

$\rangle\rangle$. Hence by Prop. 3.11 it is a normal extension of VW. $\dashv$

Hence by Prop. 3.68,

Proposition 3.80 *If Ψ is a half normal extension of VW then Ψ is a half normal extension of AR'.*

We shall need the following

Lemma 3.81 *If Ψ is a half-normal extension of VW, $\Gamma \in \Upsilon_\Psi$ and $(A \wedge B) \mathrel{\Box\!\!\rightarrow} D \in \Gamma$, then $A \mathrel{\Box\!\!\rightarrow} (B \rightarrow D) \in \Gamma$.*

Proof. Assume that Ψ is a half-normal extension of VW and $(A \wedge B) \mathrel{\Box\!\!\rightarrow} D \in \Gamma$. By ID and RCK, $(A \wedge \neg B) \mathrel{\Box\!\!\rightarrow} \neg B \in \Gamma$, so again by ID and RCK and Prop. 3.13, $((A \wedge B) \mathrel{\Box\!\!\rightarrow} (B \rightarrow D)) \wedge ((A \wedge \neg B) \mathrel{\Box\!\!\rightarrow} (B \rightarrow D)) \in \Gamma$. By CA, $((A \wedge \neg B) \vee (A \wedge B)) \mathrel{\Box\!\!\rightarrow} (B \rightarrow D) \in \Gamma$, so by ID, RCK and CSO, $A \mathrel{\Box\!\!\rightarrow} (B \rightarrow D) \in \Gamma$. $\dashv$

As we just noted it does not seem possible to obtain directly from the completeness proof in Section 3.5.2 that, in the case of the logic VW, $\leq^A = \leq^B$ for all A, B. However, we will be able to relate $Near^A_\Gamma$ and $Near^B_\Gamma$ (and thus Rem^A_Γ and Rem^B_Γ) for particular choices of A and B and this turns out to be enough.

Proposition 3.82 *If Ψ is a half-normal extension of VW and $S_\Psi(A) = S_\Psi(B)$ then $Near^A_\Gamma = Near^B_\Gamma$.*

Proof. Assume that $S_\Psi(A) = S_\Psi(B)$. Then $\vdash_\Psi A \leftrightarrow B$, and hence by Prop. 3.79 for any $C \in L$, $\vdash_\Psi (A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C)$, so for any $\Gamma \in \Upsilon_\Psi$, $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$ iff $B \mathrel{\Box\!\!\rightarrow} C \in \Gamma$. Hence $\Gamma_A = \Gamma_B$, and so $Near^A_\Gamma = Near^B_\Gamma$. $\dashv$

Proposition 3.83 *If Ψ is a half-normal extension of VW and $Near^A_\Gamma \cap S_\Psi(B) \neq \emptyset$ then $Near^{A \wedge B}_\Gamma = Near^A_\Gamma \cap S_\Psi(B)$.*

Proof. if $Near^A_\Gamma \cap S_\Psi(B) \neq \emptyset$ then by Prop. 3.52, $A \Diamond\!\!\rightarrow B \in \Gamma$. Assume that $\Delta \in Near^{A\wedge B}_\Gamma$. Then by Prop. 3.49, $\Gamma_{A\wedge B} \subseteq \Delta$. If $\phi \in \Gamma_A$ then $A \Box\!\!\rightarrow \phi \in \Gamma$, so by CV, $(A \wedge B) \Box\!\!\rightarrow \phi \in \Gamma$, and hence $\phi \in \Gamma_{A\wedge B}$. Thus $\Gamma_A \subseteq \Delta$, and hence $\Delta \in Near^A_\Gamma$, so $Near^{A\wedge B}_\Gamma \subseteq Near^A_\Gamma$ and by ID, $Near^{A\wedge B}_\Gamma \subseteq Near^A_\Gamma \cap S_\Psi(B)$.

In the other direction assume for contradiction that there is $\Delta \in Near^A_\Gamma \cap S_\Psi(B)$ such that $\Delta \notin Near^{A\wedge B}_\Gamma$. Then by ID, $\Delta \in Rem^{A\wedge B}_\Gamma$, and hence there is $\neg C \in \Delta$ such that $(A\wedge B) \Box\!\!\rightarrow C \in \Gamma$. But then by Lemma 3.81, $A \Box\!\!\rightarrow (B \rightarrow C) \in \Gamma$. Now, since $B \wedge \neg C \in \Delta$, by Prop. 3.52, $A \Diamond\!\!\rightarrow (B \wedge \neg C) \in \Gamma$, which is a contradiction. $\dashv$

Proposition 3.84 *If Ψ is a half-normal extension of VW and $Near^A_\Gamma \subseteq S_\Psi(B)$ then $Near^A_\Gamma = Near^{A\wedge B}_\Gamma = Near^{B\wedge A}_\Gamma$.*

Proof. If $Near^A_\Gamma \neq \emptyset$ this follows from Propositions 3.82 and 3.83. If on the other hand $Near^A_\Gamma = \emptyset$ then (by Lemma 3.15 and Prop. 3.49) Γ_A is not Ψ-consistent. Hence there is a finite $\Gamma'_A \subseteq \Gamma_A$ such that $\vdash_\Psi \wedge\Gamma'_A \rightarrow \neg A$. But then by RCK, $A \Box\!\!\rightarrow \neg A \in \Gamma$. It follows that $\Diamond A \notin \Gamma$, and hence that $S_\Psi(A) \cap R_\Gamma = \emptyset$, and so that $S_\Psi(A\wedge B) \cap R_\Gamma = \emptyset$. By Prop. 3.55, this means that $Near^{A\wedge B}_\Gamma = \emptyset$. $\dashv$

Proposition 3.85 *If Ψ is a half-normal extension of VW and $Near^A_\Gamma \cap Near^B_\Gamma \neq \emptyset$ then $Near^A_\Gamma \cap S_\Psi(B) = Near^B_\Gamma \cap S_\Psi(A)$.*

Proof. Assume that $Near^A_\Gamma \cap Near^B_\Gamma \neq \emptyset$. Then by Prop. 3.50, $Near^A_\Gamma \cap S_\Psi(B) \neq \emptyset$, so by Prop. 3.83, $Near^{A\wedge B}_\Gamma = Near^A_\Gamma \cap S_\Psi(B)$. But also $Near^{B\wedge A}_\Gamma = Near^B_\Gamma \cap S_\Psi(A)$ and hence by Prop. 3.82, $Near^A_\Gamma \cap S_\Psi(B) = Near^B_\Gamma \cap S_\Psi(A)$ $\dashv$

A direct consequence of this is the following

Proposition 3.86 *If Ψ is a half-normal extension of VW and $Near^A_\Gamma \subseteq S_\Psi(B)$ then either $Near^A_\Gamma \cap Near^B_\Gamma = \emptyset$ or $Near^A_\Gamma = Near^B_\Gamma \cap S_\Psi(A)$.*

Proposition 3.87 *If Ψ is a half-normal extension of VW then either $Near_\Gamma^{A\vee B} = Near_\Gamma^A$ or $Near_\Gamma^{A\vee B} = Near_\Gamma^B$ or $Near_\Gamma^{A\vee B} = Near_\Gamma^A \cup Near_\Gamma^B$.*

Proof.

1. Assume that $Near_\Gamma^{A\vee B} \subseteq S_\Psi(A)$. Then by Prop. 3.84 and Prop. 3.82, $Near_\Gamma^{A\vee B} = Near_\Gamma^{(A\vee B)\wedge A} = Near_\Gamma^A$.

2. If $Near_\Gamma^{A\vee B} \subseteq S_\Psi(B)$, then in the same way $Near_\Gamma^{A\vee B} = Near_\Gamma^B$.

3. Assume that $Near_\Gamma^{A\vee B} \not\subseteq S_\Psi(A)$ and $Near_\Gamma^{A\vee B} \not\subseteq S_\Psi(B)$. Then by Prop. 3.50, $Near_\Gamma^{A\vee B} \cap S_\Psi(B) \neq \emptyset$ and $Near_\Gamma^{A\vee B} \cap S_\Psi(A) \neq \emptyset$, and thus by Prop. 3.82 and 3.83, $Near_\Gamma^A = Near_\Gamma^{(A\vee B)\wedge A} = Near_\Gamma^{A\vee B} \cap S_\Psi(A)$. Hence $Near_\Gamma^A \subseteq Near_\Gamma^{A\vee B}$, and in the same way $Near_\Gamma^B \subseteq Near_\Gamma^{A\vee B}$, so $Near_\Gamma^A \cup Near_\Gamma^B \subseteq Near_\Gamma^{A\vee B}$.

   In the other direction, if $\Delta \in Near_\Gamma^{A\vee B}$, then by Prop. 3.49, $\Gamma_{A\vee B} \subseteq \Delta$. Assume for contradiction that $\Gamma_A \not\subseteq \Delta$ and $\Gamma_B \not\subseteq \Delta$. Then there are $\phi, \psi \in L$ such that $A \mathrel{\Box\!\!\rightarrow} \phi, B \mathrel{\Box\!\!\rightarrow} \psi \in \Gamma$ and $\phi, \psi \notin \Delta$. By RCK, $A \mathrel{\Box\!\!\rightarrow} (\phi \vee \psi) and B \mathrel{\Box\!\!\rightarrow} (\phi \vee \psi) \in \Gamma$, so by CA, $(A \vee B) \mathrel{\Box\!\!\rightarrow} (\phi \vee \psi) \in \Gamma$, and hence $\phi \vee \psi \in \Gamma_{A\vee B} \subseteq \Delta$, which is a contradiction (by Prop. 3.14). So $\Gamma_A \subseteq \Delta$ or $\Gamma_B \subseteq \Delta$ and hence $\Delta \in Near_\Gamma^A \cup Near_\Gamma^B$, and we have shown that $Near_\Gamma^{A\vee B} \subseteq Near_\Gamma^A \cup Near_\Gamma^B$.

$\dashv$

Let

$$Near_\Gamma = \bigcup_{A\in L} Near_\Gamma^A$$

By Prop. 3.54, $Near_\Gamma \subseteq R_\Gamma$. We define a total function $\leq'_\Psi$ from Υ_Ψ to $P(\Upsilon_\Psi \times \Upsilon_\Psi)$ as follows

$\Delta' <_\Gamma \Delta$ ($\Delta \not\leq_\Gamma \Delta'$) iff

$[\Delta \notin R_\Gamma$ and $\Delta' \in R_\Gamma]$ or $[\Delta \notin Near_\Gamma$ and $\Delta' \in Near_\Gamma]$ or there is $B \in L$ such that $[\Delta \in Rem_\Gamma^B$ and $\Delta' \in Near_\Gamma^B]$.

I.e.,

$\Delta \leq_\Gamma \Delta'$ iff

$[\Delta \in R_\Gamma$ or $\Delta' \notin R_\Gamma]$ and $[\Delta \in Near_\Gamma$ or $\Delta' \notin Near_\Gamma]$ and for all $B \in L$, $[\Delta \notin Rem_\Gamma^B$ or $\Delta' \notin Near_\Gamma^B]$.

The following lemmas collect some facts about this ordering.

Lemma 3.88 *If Ψ is a half-normal extension of VW and $\Delta, \Delta' \in Near_\Gamma^A$ then $\Delta \leq_\Gamma \Delta'$.*

Proof. Assume that $\Delta, \Delta' \in Near_\Gamma^A$. Then (by Prop. 3.54) $\Delta, \Delta' \in R_\Gamma$, and also obviously $\Delta, \Delta' \in Near_\Gamma$. We assume for contradiction that there is $B \in L$ such that $\Delta' \in Near_\Gamma^B$ and $\Delta \in Rem_\Gamma^B$. Then $Near_\Gamma^A \cap Near_\Gamma^B \neq \emptyset$, so by Prop. 3.85, $Near_\Gamma^A \cap S_\Psi(B) = Near_\Gamma^B \cap S_\Psi(A)$. $\Delta, \Delta' \in Near_\Gamma^A \cap S_\Psi(B)$ but $\Delta \notin Near_\Gamma^B \cap S_\Psi(A)$ which contradicts this, so $\Delta \leq_\Gamma \Delta'$. ⊣

Lemma 3.89 *If Ψ is a half-normal extension of VW then not [$\Delta <_\Gamma \Delta'$ and $\Delta' <_\Gamma \Delta$].*

Proof. Assume for contradiction that $\Delta <_\Gamma \Delta'$ and $\Delta' <_\Gamma \Delta$. If $\Delta' \notin R_\Gamma$ then by Prop. 3.54 for all $B \in L$, $\Delta' \notin Near_\Gamma^B$, hence $\Delta \leq_\Gamma \Delta'$, which contradicts our assumption. Thus $\Delta' \in R_\Gamma$, and in the same way $\Delta \in R_\Gamma$. Since $\Delta \in R_\Gamma$, if $\Delta' \notin Near_\Gamma$ then again $\Delta \leq_\Gamma \Delta'$, so $\Delta' \in Near_\Gamma$. In the same way $\Delta \in Near_\Gamma$, so there are $A, B \in L$ such that $\Delta \in Near_\Gamma^B$, $\Delta' \in Rem_\Gamma^B$, $\Delta' \in Near_\Gamma^A$, and $\Delta \in Rem_\Gamma^A$.

By Prop. 3.56 there are $C, D \in L$ such that $A \wedge \neg C \in \Delta$, $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$, $B \wedge \neg D \in \Delta'$, and $B \mathrel{\Box\!\!\rightarrow} D \in \Gamma$. Now by Lemma 3.81,

$$\text{if } A \mathrel{\Diamond\!\!\rightarrow} (B \wedge \neg D) \in \Gamma \text{ then } (A \wedge B) \mathrel{\Diamond\!\!\rightarrow} \neg D \in \Gamma.$$

Now $B \wedge \neg D \in \Delta'$, so $S_\Psi(B \wedge \neg D) \cap Near_\Gamma^A \neq \emptyset$, and hence by Prop. 3.52, $A \diamond\!\!\rightarrow (B \wedge \neg D) \in \Gamma$. Since $A \in \Delta \in Near_\Gamma^B$ it follows that $S_\Psi(A) \cap Near_\Gamma^B \neq \emptyset$, and hence $B \diamond\!\!\rightarrow A \in \Gamma$. By CV, $(B \wedge A) \Box\!\!\rightarrow D \in \Gamma$, and by ID, RCK and CSO, $(A \wedge B) \Box\!\!\rightarrow D \in \Gamma$, which contradicts Lemma 3.81. $\dashv$

Lemma 3.90 *If Ψ is a half-normal extension of VW, $\Delta \in S_\Psi(A)$, and $\Delta' \in Near_\Gamma^A$, then $\Delta' \leq_\Gamma \Delta$.*

Proof. Assume that $\Delta \in S_\Psi(A)$ and $\Delta' \in Near_\Gamma^A$. If $\Delta \in Rem_\Gamma^A$ then $\Delta' <_\Gamma \Delta$, and hence by Lemma 3.89, $\Delta' \leq_\Gamma \Delta$. If on the other hand $\Delta \in Near_\Gamma^A$, then by Lemma 3.88, $\Delta' \leq_\Gamma \Delta$. $\dashv$

Lemma 3.91 *If Ψ is a half-normal extension of VW and $\Delta <_\Gamma \Delta'$ then for every $\Lambda \in \Upsilon_\Psi$, $\Delta <_\Gamma \Lambda$ or $\Lambda <_\Gamma \Delta'$.*

Proof. The critical case is when $\Delta, \Delta' \in Near_\Gamma$, so we assume this and that $\Delta <_\Gamma \Delta'$. Then there is $A \in L$ such that $\Delta \in Near_\Gamma^A$ and $\Delta' \in Rem_\Gamma^A$. If $\Lambda \notin Near_\Gamma$, then $\Delta <_\Gamma \Lambda$ so we assume that also $\Lambda \in Near_\Gamma$. So there is $B \in L$ such that $\Lambda \in Near_\Gamma^B$. If $\Lambda \in Near_\Gamma^A$ then $\Lambda <_\Gamma \Delta'$ so we assume that $\Lambda \notin Near_\Gamma^A$, and hence that $Near_\Gamma^A \neq Near_\Gamma^B$. Now by Prop. 3.87, $Near_\Gamma^{A \vee B} = Near_\Gamma^A$ or $Near_\Gamma^{A \vee B} = Near_\Gamma^B$ or $Near_\Gamma^{A \vee B} = Near_\Gamma^A \cup Near_\Gamma^B$.

1. Assume that $Near_\Gamma^{A \vee B} = Near_\Gamma^A$. Then $\Delta \in Near_\Gamma^{A \vee B}$ and $\Lambda \notin Near_\Gamma^{A \vee B}$. Since $\Lambda \in Near_\Gamma^B$, by Prop. 3.50, $\Lambda \in S_\Psi(B) \subseteq S_\Psi(A \vee B)$. Hence $\Lambda \in Rem_\Gamma^{A \vee B}$, and so $\Delta <_\Gamma \Lambda$.

2. Assume next that $Near_\Gamma^{A \vee B} = Near_\Gamma^B$. Then $\Lambda \in Near_\Gamma^{A \vee B}$. If $\Delta \in Near_\Gamma^{A \vee B}$ then by Lemma 3.88, $\Delta' \notin Near_\Gamma^{A \vee B}$, and hence $\Lambda <_\Gamma \Delta'$. If on the other hand $\Delta \notin Near_\Gamma^{A \vee B}$, we assume for contradiction that $\Delta' \in Near_\Gamma^{A \vee B}$. Since $\Delta \in Near_\Gamma^A$, by Prop. 3.50, $\Delta \in S_\Psi(A \vee B)$, which means that $\Delta \in Rem_\Gamma^{A \vee B}$. But $\Delta <_\Gamma \Delta'$ and $\Delta' <_\Gamma \Delta$, which contradicts Lemma 3.89, so also in this case $\Lambda <_\Gamma \Delta'$.

3. Assume finally that $Near_\Gamma^{A\vee B} = Near_\Gamma^A \cup Near_\Gamma^B$. Since in that case $\Delta \in Near_\Gamma^{A\vee B}$, by Lemma 3.88 it follows that $\Delta' \notin Near_\Gamma^{A\vee B}$, and since $\Lambda \in Near_\Gamma^{A\vee B}$, that $\Lambda <_\Gamma \Delta'$.

$\dashv$

Lemma 3.92 *If Ψ is a half-normal extension of VW, $\leq'_\Psi$ is an rc-assignment for Υ_Ψ.*

Proof.

1. Assume that for some $\Delta, \Delta', \Lambda \in \Upsilon_\Psi$, $\Delta' \leq_\Gamma \Lambda$ and $\Lambda \leq_\Gamma \Delta$, and for contradiction that $\Delta' \not\leq_\Gamma \Delta$, i.e., that $\Delta <_\Gamma \Delta'$. But then by Lemma 3.91, $\Delta <_\Gamma \Lambda$ or $\Lambda <_\Gamma \Delta'$, which is a contradiction, so $\leq_\Gamma$ is transitive.

2. If $\Delta \not\leq_\Gamma \Delta'$ and $\Delta' \not\leq_\Gamma \Delta$, then $\Delta' <_\Gamma \Delta$ and $\Delta <_\Gamma \Delta'$, which contradicts Lemma 3.89. Thus $\leq_\Gamma$ is strongly connected.

3. By Prop. 3.46, Γ is self accessible.

4. Assume that $A \in \Gamma$. If $A \mathrel{\Box\!\!\rightarrow} C \in \Gamma$ then $C \in \Gamma$ since $\{MP\}_{Ax} \subseteq \Gamma$, so $\Gamma \in Near_\Gamma^A \subseteq Near_\Gamma$. If for some $\Delta \in \Upsilon_\Psi$, $\Delta <_\Gamma \Gamma$, then for some $A \in \Gamma$, $\Gamma \in Rem_\Gamma^A$, which is a contradiction. Hence for any $\Delta \in \Upsilon_\Psi$, $\Gamma \leq_\Gamma \Delta$, so Γ is $\leq_\Gamma$-minimal.

5. If $\Delta \notin R_\Gamma$ then it follows directly from the definition of $\leq$ that for any $\Delta' \in \Upsilon_\Psi$, $\Delta' \leq_\Gamma \Delta$, so inaccessible worlds are $\leq_\Gamma$-maximal.

6. If $\Delta \in R_\Gamma$ and $\Delta' \notin R_\Gamma$, then again directly by the definition of $\leq$, $\Delta <_\Gamma \Delta'$, so accessible worlds are more similar to Γ than inaccessible worlds.

$\dashv$

Lemma 3.93 *If Ψ is a half-normal extension of VW, $\leq'_\Psi$ is an adequate arrc-assignment for Υ_Ψ.*

Proof. Obviously by Lemma 3.92, $\leq'_\Psi$ is not only an rc-assignment but also an arrc-assignment for Υ_Ψ.

We must show that $\leq'_\Psi$ is adequate. Assume that Ψ is a half-normal extension of VW (and hence by Prop. 3.80 a half-normal extension of AR'), that $\Delta \leq^A_\Gamma \Delta'$ (i.e., $\Delta \leq_\Gamma \Delta'$) and $\Delta' \in Near^A_\Gamma$. If $\Delta \in Rem^A_\Gamma$, then not for all $A \in L$, $[\Delta \notin Rem^A_\Gamma$ or $\Delta' \notin Near^A_\Gamma]$, which is a contradiction. Hence $\Delta \notin Rem^A_\Gamma$.

Now instead assume that $\Delta, \Delta' \in S_\Psi(A)$ and $\Delta' <^A_\Gamma \Delta$ (i.e., $\Delta' <_\Gamma \Delta$). Thus $\Delta \not\leq_\Gamma \Delta'$, so by Lemma 3.90, $\Delta \notin Near^A_\Gamma$. Hence $\Delta \in Rem^A_\Gamma$. ⊣

Now using Lemma 3.64 we obtain

Theorem 3.94 *If Ψ is a half-normal extension of VW, then S_Ψ is a relative closeness semantics for Υ_Ψ (defined by $\leq'_\Psi$).*

Since VW is a half-normal extension of itself, S_{VW} is an relative closeness semantics for Υ_{VW}. It now follows in the usual way that

Proposition 3.95 *VW is complete with respect to the set of relative closeness semantics.*

3.6.3 Half-Normal Extensions of VW and various Relative Closeness Semantics

As was the case with respect to half-normal extensions of Ck, PR and AR' and their attendant sets of semantics, we can show that various half-normal extensions of VW are sound and complete with respect to different subsets of the set of relative closeness semantics. As an example of this we show that the logic VC is sound and complete with respect to the set of relative closeness semantics that have the following property:

> A relative closeness semantics S for the domain I defined by the rc-assignment $\leq$ is *centered*[8] if for every $w, w' \in I$ if $w \neq w'$ then $w <_w w'$.

[8] Cf. footnote 41 on p. 46 above.

By Theorem 1.33 and Prop. 2.13 (p. 82), the logic VC is sound with respect to the set of centered relative closeness semantics. It remains to show that.

Proposition 3.96 *S_{VC} is a centered relative closeness semantics.*

Proof. Since VC is a half-normal extension of VW, by Theorem 3.94, S_{VC} is a relative closeness semantics for Υ_{VC}. We must show that S_{VC} is centered.

Assume not, i.e., that there are $\Gamma, \Delta \in \Upsilon_{VC}$ such that $\Gamma \neq \Delta$ and $\Gamma \not<_{\Gamma} \Delta$. Since $\Gamma \neq \Delta$ there is $B \in \Gamma - \Delta$. Take any $A \in \Gamma \cap \Delta$. By MP, $\Gamma \in Near^A_\Gamma$, and since $\Gamma \not<_\Gamma \Delta$, $\Delta \notin Rem^A_\Gamma$. Hence $\Delta \in Near^A_\Gamma$. Therefore $Near^A_\Gamma \not\subseteq S_{VC}(B)$ and so (by Prop. 3.51), $A \mathrel{\Box\!\!\rightarrow} B \notin \Gamma$. Hence by CS, $A \wedge B \notin \Gamma$ which is a contradiction. $\dashv$

Hence

Proposition 3.97 *VC is complete with respect to the set of centered relative closeness semantics.*

In a similar way C2 and other half normal extensions of VW may be related to their specific sets of relative closeness semantics.

Summary

The subject matter of this thesis is the semantic and logical properties of subjunctive conditional sentences, or counterfactuals, as they are also called.

It is argued that current formal systems of conditional logic and semantics do not represent our intuitions about the logical properties of such sentences. In particular the principles

$$\text{(CSO)}\ ((A \mathrel{\Box\!\!\rightarrow} B) \wedge (B \mathrel{\Box\!\!\rightarrow} A)) \rightarrow ((A \mathrel{\Box\!\!\rightarrow} C) \leftrightarrow (B \mathrel{\Box\!\!\rightarrow} C))$$

$$\text{(CV)}\ ((A \mathrel{\Box\!\!\rightarrow} C) \wedge (A \mathrel{\Diamond\!\!\rightarrow} B)) \rightarrow ((A \wedge B) \mathrel{\Box\!\!\rightarrow} C)$$

are shown to be questionable from an intuitive point of view. The principle CV is contested, but CSO is a part of nearly every current axiomatization of conditional logic. It is argued that the inclusion of CSO stems from a certain conception of similarity or closeness between possible worlds inherent in those systems: Closeness between worlds is thought of as a relation fixed by the context of utterance. This assumption leads to the acceptance of the principle CSO. The main objective of this thesis is thus to show that this conception of closeness is unwarranted and that it leads to unintuitive results. Instead a certain *antecedent-relative* notion of closeness is proposed. This notion of closeness is defined in causal terms. The formal consequences of substituting this notion for the traditional non-relative notion of closeness are traced and the resulting formal semantics and its attendant axiomatic system are shown to possess desirable properties from an intuitive point of view.

Much of Chapter 1 is concerned with expressing different formal semantical frameworks in similar terminology with the purpose of

comparing them. This chapter starts with a short discussion on the methods used in this thesis, followed in Section 1.1, by a taxonomic discussion with the purpose of identifying the subject matter of the thesis. In Section 1.2 there is a short discussion of an alternative, epistemic approach to conditionals that is not pursued further. After those preliminary considerations, in Section 1.3 follows a discussion of the notion of *compositionality*. Here a framework for classifying sentences with respect to this notion is suggested. This framework receives a formal treatment in the first part of Section 1.5 (Subsections 1.5.1, 1.5.2 and 1.5.3). In Section 1.4 standard views on how conditionals should be classified with respect to this framework are presented, and discussed. This discussion is carried a bit further in Section 1.5.3.

Section 1.5 is entirely devoted to a presentation and comparison of different formal semantics for conditionals found in the literature. These are the *class* (or world set) *selection semantics* (Subsection 1.5.4), semantics based on an operation of *sentence set revision* typical of cotenability theory (Subsection 1.5.5), the well known *relative closeness semantics* (Subsection 1.5.6), and a generalization of those, the *antecedent relative relative closeness semantics*. Those different kinds of formal semantics are positioned with respect to the framework of compositionality defined in Subsections 1.5.1 to 1.5.3, and there is also a discussion of the conditions under which different formal semantics are equivalent.

The whole of Chapter 3 is concerned with relating the different types of semantics defined in Section 1.5 to different axiomatizations of conditional logic. Proofs of soundness and completeness for axiomatic systems with respect to those different semantic types are presented. Those proofs proceed by methods that are fairly standard within modal logic, hence no real novelty is claimed here. The purpose of this presentation is rather to provide a handy and unified framework in which the special semantics presented in Chapter 2 can be related to its proper axiomatization, and in which this axiomatization can be related to other kinds of semantics.

All this is meant to be preambulatory with respect to the discussion on time and conditional truth conducted in Chapter 2, which I regard as the main contribution of this thesis. Chapter 2 starts with

discussions of *historical conservativity*, and *historical possibility* in Sections 2.1 to 2.3 in which a formal framework for talking about possible worlds in their temporal dimension (temporal worlds) is presented. This framework is used in Section 2.4.1 to define a simple (preliminary) relation of closeness between such worlds (simple late departure), that is shown to define a relative closeness semantics. The rest of this section is devoted to a discussion of a theory that can be interpreted as a version of this kind of semantics, namely the semantics proposed in [Lewis, 1979].

In Section 2.5 it is argued that this relation of closeness fails on intuitive grounds and instead an alternative relation of closeness, defined in terms of causal notions, is proposed, and is shown to define an antecedent relative relative closeness semantics of a special kind (causal trail semantics). In Section 2.6 this semantics is shown to possess desirable properties with respect to what axioms and rules of inference that are supported, and in Sections 2.7 and 2.8 it is compared to certain semantics (proposed by other authors) with which it shares important traits.

Bibliography

Adams, E. (1970). Subjunctive and indicative conditionals. *Foundations of Language*, 6:39–94.

Alchourrón, C. E., Gärdenfors, P., and Makinson, D. (1985). On the logic of theory change: Partial meet contraction and revison functions. *Journal of Symbolic Logic*, 50:510–530.

Austin, J. L. (1956). Ifs and cans. In *Philosophical Papers*, pages 153–180. Oxford University Press. 1961.

Bennett, J. (1974). Counterfactuals and possible worlds. *Canadian Journal of Philosophy*, 4:381–402.

Bennett, J. (1982). Even if. *Linguistics and Philosophy*, 5:403–4018.

Bennett, J. (1984). Counterfactuals and temporal direction. *The Philosophical Review*, 93:57–91.

Bennett, J. (1988). Farewell to the phlogiston theory of conditionals. *Mind*, 97:509–527.

Bennett, J. (1995). Classifying conditionals: The traditional way is right. *Mind*, 104:331–354.

Bennett, J. (1998). On forward and backward counterfactual conditionals. Forthcoming article.

Bowie, G. L. (1979). The similarity approach to counterfactuals: Some problems. *Noûs*, 13:477–498.

Chellas, B. F. (1975). Basic conditional logic. *Journal of Philosophical Logic*, 4:133–153.

Chisholm, R. M. (1946). The contrary-to-fact conditional. *Mind*, 55(220):289–307.

Davis, W. A. (1979). Indicative and subjunctive conditionals. *The Philosophical Review*, 88:544–564.

Davis, W. A. (1980). Jackson on counterfactuals. *Australasian Journal of Philosophy*, 58:62–65.

Downing, P. B. (1959). Subjunctive conditionals, time order and causation. *Proceedings of the Aristotelian Society*, pages 125–140.

Dudman, V. H. (1981). *Tense and Time in English.* Macquarie University.

Dudman, V. H. (1983). Tense and time in English verb clusters of the primary pattern. *Australian Journal of Linguistics*, pages 25–44.

Dudman, V. H. (1984). Parsing 'if'-sentences. *Analysis*, 44.4:145–153.

Ellis, B., Jackson, F., and Pargetter, R. (1977). An objection to possible world semantics for counterfactuals. *Journal of Philosophical Logic*, 6:355–357.

Fine, K. (1975). Review of Lewis's *Counterfactuals. Mind*, 84:451–458.

Frege, G. (1891). Sense and reference. *The Philosophical Review*, 57:207–230. 1948.

Gärdenfors, P. (1978). Conditionals and changes of belief. In Niiniluoto, I. and Tuomela, R., editors, *The Logic and Epistemology of Scientific Change*, volume 30(2-4) of *Acta Philosophica Fennica*, pages 381–404. North-Holland.

Gärdenfors, P. (1988). *Knowledge in Flux.* The MIT Press.

Gibbard, A. (1978). Two recent theories of conditionals. In Harper et al. [1981], pages 211–247.

Goodman, N. (1946). The problem of counterfactual conditionals. In *Fact Fiction and Forecast*, chapter 1. Bobbs-Merrill Comp. 1965.

Grice, P. (1967a). Indicative conditionals. In *Studies in the Way of Words*, pages 58–85. Harvard University Press. 1989.

Grice, P. (1967b). Logic and conversation. In Cole, P. and Morgan, J. L., editors, *Syntax and Semantics*, volume 3. Academic Press. 1975.

Grove, A. (1986). *Two Modellings for Theory Change*, volume 13. Auckland Philosophy Papers.

Hansson, S. O. (1995). The emperor's new clothes: Some recurring problems in the formal analysis of counterfactuals. In Crocco, G., Fariñas del Cerro, L. F., and Herzig, A., editors, *Conditionals: from Philosophy to Computer Science*, pages 13–31. Clarendon Press.

Harper, W. L., Stalnaker, R. C., and Pearce, G., editors (1981). *Ifs*, volume 15 of *The University of Western Ontario Series in Philosophy of Science*. D. Reidel Publishing Company.

Jackson, F. (1977). A causal theory of counterfactuals. *Australasian Journal of Philosophy*, 55:3–21.

Jackson, F. (1987). *Conditionals*. Basil Blackwell.

Jackson, F. (1990). Classifying conditionals. *Analysis*, 50:134–147.

Lewis, C. I. (1912). Implication and the algebra of logic. *Mind*, 21:522–531.

Lewis, D. (1973a). Causation. In Lewis [1986a], pages 159–167.

Lewis, D. (1973b). *Counterfactuals*. Basil Blackwell.

Lewis, D. (1979). Counterfactual dependence ond time's arrow. In Lewis [1986a], pages 32–52.

Lewis, D. (1981). Ordering semantics and premise semantics for counterfactuals. *Journal of Philosophical Logic*, 10:217–234.

Lewis, D. (1986a). *Philosophical Papers*, volume 2. Oxford University Press.

Lewis, D. (1986b). Postscripts to "Counterfactual dependence ond time's arrow". In Lewis [1986a], pages 52–65.

Lindström, P. (1997). David Lewis' teori för kontrafaktiska satser. *Filosofisk Tidskrift*, nr. 4, 1997, pages 23–26.

Lindström, S. and Rabinowicz, W. (1997). The Ramsey test revisited. In *AGM in and out of focus*, chapter 6. Department of Philosophy, University of Uppsala. 9th European Summer School in Logic, Language and Information.

Loewer, B. (1976). Counterfactuals with disjunctive antecedents. *The Journal of Philosophy*, 73:531–536.

Loewer, B. (1979). Cotenability and counterfactual logics. *Journal of Philosophical Logic*, 8:99–115.

Mackie, J. L. (1973). *Truth, Probability and Paradox.* Clarendon Press.

McKay, T. and van Inwagen, P. (1977). Counterfactuals with disjunctive antecedents. *Philosophical Studies*, 31:353–356.

Mitchell, D. (1962). *An Introduction to Logic.* Hutchinson.

Moore, G. E. (1912). *Ethics.* Oxford University Press.

Nute, D. (1980). *Topics in Conditional Logic*, volume 20 of *Philosphical Studies Series in Philosophy.* D. Reidel Publishing Company.

Nute, D. (1981). Introduction. *Journal of Philosophical Logic*, 10:127–147.

Nute, D. (1984). Conditional logic. In Gabbay, D. and Guenthner, F., editors, *Handbook of Philosophical Logic*, volume 2, pages 387–439. D. Reidel Publishing Company.

Pollock, J. L. (1976). *Subjunctive Reasoning*, volume 8 of *Philosphical Studies Series in Philosophy*. D. Reidel Publishing Company.

Pollock, J. L. (1981). A refined theory of counterfactuals. *Journal of Philosophical Logic*, 10:239–266.

Ramsey, F. P. (1931). Law and causality. In Ramsey [1978], pages 128–151. Edited by D. H. Mellor.

Ramsey, F. P. (1978). *Foundations: Essays in Philosophy, Logic, Mathematics and Economics*. Routledge & Kegan Paul. Edited by D. H. Mellor.

Rescher, N. (1961). Belief-contravening suppositions. *The Philosophical Review*, 70:176–196.

Slote, M. A. (1978). Time in counterfactuals. *The Philosophical Review*, 87:3–27.

Stalnaker, R. C. (1968a). A theory of conditionals. In Resher, N., editor, *Studies in Logical Theory*, number 2 in American Philosophical Quarterly monograph series, pages 98–112. Basil Blackwell.

Stalnaker, R. C. (1968b). A theory of conditionals, 1968. In Harper et al. [1981], pages 41–55.

Thomason, R. H. and Gupta, A. (1978). A theory of conditionals in the context of branching time. In Harper et al. [1981], pages 299–322.

Tichý, P. (1976). A counterexample to the Stalnaker-Lewis analysis of counterfactuals. *Philosophical Studies*, 29:271–273.

van Fraassen, B. C. (1978). A temporal framework for conditionals and chance. In Harper et al. [1981], pages 323–340.

Weiner, J. (1979). Counterfactual conundrum. *Noûs*, 13:499–509.

Index

S-historical possibility, 78

Adams, Ernest, 10
adequacy of an arrc-assignment, 170
AGM theory, 14n
Alchourrón, C. E., 14n
antecedent intensional semantics, 115
antecedent relative relative closeness semantics, 57
ARL assumption, 58
ARRC semantics, 57
arrc-assignment, 56
Austin, J. L., 2n

base-time of a conditional, 128
basic class selection semantics, 157
Bennett, Jonathan, 11n, 47n, 63, 63n, 64, 69, 69n, 79n, 86, 87n, 89, 89n, 90, 91, 100, 101, 101, 102n, 106, 119–123
Bowie, G. Lee, 69n

c-function, 26
canonical semantics, 153
causal trail
  criterion of closeness, 103
  notion of, 102n
  semantics, 176
centered relative closeness semantics, 184
centering, 47n, 184
ceteris paribus presuppositions, 54n
Chellas, Brian F., 9, 24n, 148n
Chisholm, Roderick M., 13n, 35–38
class selection function, 31
  finite, 34
  projective, 35
  propositional, 33
  selective, 34
  uniform, 39
class selection semantics, 31
co-commencing, 73
compatibility, 37
completeness, 150
composition function, 26
compositionality, 26
conditional logic, 143
  classical, 148
  entailment preserving, 148
  half-classical, 148
  half-normal, 148
  normal, 148
consequent intensional seman-

tics, 32
consistency
with respect to a logic, 151
maximal, 152
with respect to a semantics, 37
contraposition, 24
convergence miracle, 87n
cotenability, 36n, 124
defined in terms of causal trails, 137
criterion of closeness
causal trail, 103
late departure
felonious, 96
simple, 80
MPF, 121
CT-criterion, 103

Davis, Wayne A., 22n, 64n, 69n
de re/de dicto, 76n
despiteness relation, 133
deterministic law, 74
distinctiveness of a precondition, 108
divergence miracle, 87n
divisibility of the antecedent, 105
domain, 25
Downing, P. B., 65–68, 70
Dudman, V. H., 11n

e-function, 26
Ellis, Brian, 30, 30n
exemption, 112
extension of a conditional logic, 160
extensional
complex, 16, 27
function, 26
semantics, 16, 27

felonious late departure, 96
Fine, Kit, 30n, 89, 89n, 91, 99
FLD-closeness, 96
Frege, Gottlob, 16–18

generalized premise contraction function, 159
generalized premise revision function, 161
Gibbard, Allan, 10n, 19, 64n
Goodman, Nelson, 4, 14n, 36–38, 102, 124, 125, 125n, 127, 131
Grice, Paul, 21
Grove, A., 45
Gupta, Anil, 63, 64, 86n
Gärdenfors, Peter, 14n, 44, 45n

Hansson, Sven Ove, 14n, 40, 41n
historic antecedent, 127
historical alternatives
felonious, 79
genuine, 78
historical conservativity, 68
modified, 71
refined, 84
hyperintensional
complex, 18
semantics, 29

intensional
complex, 18, 29
semantics, 18, 29

iterative facts revision semantics, 43

Jackson, Frank, 11n, 19, 20, 21n, 22, 22n, 30, 30n, 99, 102n, 105n

late departure
  felonious, 96
  simple, 80
lawful world, 82
Lewis, C. I., 22n
Lewis, David, 4n, 6, 8, 9, 10n, 19, 20, 22, 23, 29, 33, 45, 45n, 46, 46n, 47, 47n, 48, 49, 53n, 63, 64n, 72, 79n, 82, 86–99, 104, 109, 111, 119, 121, 138, 138n, 140, 147n, 150n, 152, 153n, 158n, 166, 189
limit assumption, 48, 51
Lindström, Per, 49
Lindström, Sten, 10n
Loewer, Barry, 30n, 53n

Mackie, J. L., 20–22
Makinson, D., 14n
maximality of inaccessible worlds, 47, 57, 80, 110
McKay, Thomas, 30n
minimality of the index-world, 46, 57, 80, 110
miracle, 72, 86
  convergence, 87n
  divergence, 87n
  quasi-miracle, 97
Mitchell, David, 22
modified historical conservativity, 71
Moore, George Edward, 2n
Morgenbesser, Sidney, 100n
MPF criterion of closeness, 121

nomic pyramid, 127
non distinguishedness, 112
nonempty class selection function, 39
Nute, Donald, 1n, 8, 9, 12, 18, 20, 22, 23, 24n, 27, 30, 31n, 85, 128, 137, 138, 144n, 146–148n

overlap of temporal worlds, 73

Pargetter, Robert, 30, 30n
Pollock, John L., 3, 19, 32, 53, 53n, 64n, 85, 99, 104, 111, 125–128, 133–141, 146, 147n
possible world, 25
precondition for the antecedent, 107
premise contraction function, 37
principle of conservativity, 68
principle of exportation, 65
propositional class selection semantics, 33
propositional semantics, 57

quasi-miracle, 97

Rabinowicz, Wlodek, 10n
ramp, 106
Ramsey, Frank P., 13–14

rc-assignment for a domain, 46
re-function, 28
reference-time of a conditional, 129
refined historical conservativity, 84
relative closeness semantics, 47
  centered, 184
Rescher, Nicholas, 11
rt-function, 28
rule of derivation, 143

self accessibility, 46, 56, 80, 110
semantics, 25
  antecedent intensional, 115
  antecedent relative relative closeness, 57
  canonical, 153
  causal trail, 176
  class selection, 31
    basic, 157
  consequent intensional, 32
  extensional, 27
  facts revision
    iterative, 43
    strong, 38
    weak, 38
    weak generalized, 159
  hyperintensional, 29
  intensional, 29
  propositional
    ARRC, 57
    class selection, 33
  relative closeness, 47
  subcase-adequate, 106
  temporally adequate, 77
  truth functional, 26
  world-relative truth functional, 28
similarity, 47
simple late departure, 80
SLD-closeness, 80
Slote, Michael A., 99, 100n, 128–132, 132n, 133–136
smooth departure, 104
soundness
  of a conditional logic, 150
  of rules, 145
Stalnaker, Robert C., 1, 8, 9, 13n, 14, 15, 18, 19, 22–24, 31n, 63n, 147n
strengthening of antecedent, 23
strict minimality of the index-world, 46n
strong connectedness
  of arrc-assignments, 56
  of LC-closeness, 110
  of rc-assignments, 46
  of SLD-closeness, 80
strong facts revision semantics, 38
sub-cases of the antecedent, 105
subcase-adequate semantics, 106
subhistoriy, 73
subjunctive necessity, 158
subjunctive possibility, 158

t-function, 26
temporal precisification, 105n
temporal world, 72
temporally adequate semantics, 77
theory
  of a world, 36

Thomason, Richmond H., 63, 64, 86n
Tichý, Pavel, 99
transitivity
  invalid rule, 22
  of arrc-assignments, 56
  of LC-closeness, 110
  of rc-assignments, 46
  of SLD-closeness, 80
truth functional semantics, 26

undercutting, 127
  defined in terms of causal trails, 136
unrealized facts, 72n

validity
  of sentences, 25
van Frassen, Bas C., 19
van Inwagen, Peter, 30n

weak S-compatibility, 37
weak centering, 47n
weak facts revision semantics, 38
weak generalized facts revision semantics, 159
Weiner, Joan, 69n
world-point, 72
world-relative truth functional semantics, 28